



FoLang Programming Language

FoLang is a general-purpose programming language designed to be **expressive, consistent, and extensible**, merging functional fluency with object-centric abstractions.

Normative Status

This document is the authoritative and normative definition of the FoLang programming language.

All FoLang syntax, semantics, type-system rules, name-resolution rules, execution behavior, and other language requirements are defined by this document.

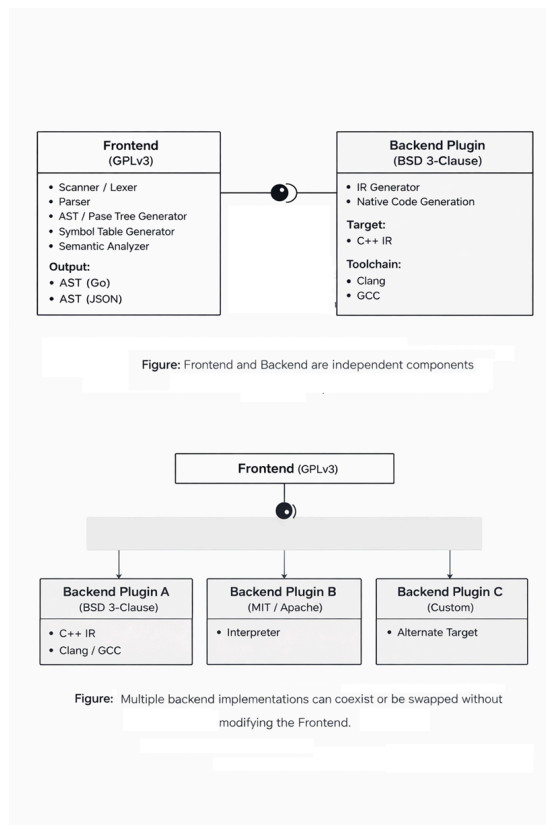
Every addition, correction, clarification, modification, or extension to the FoLang language must be recorded in this document before it becomes part of the language.

A compiler or other language implementation may use any internal architecture, parser, intermediate representation, runtime, or backend. However, to be identified as a conforming FoLang implementation, its externally observable behavior must adhere to the requirements defined in this document.

When an implementation conflicts with this specification, the specification governs unless the discrepancy is formally accepted and incorporated as a specification revision.

Implementation-specific extensions must be clearly identified as extensions and must not be represented as standard FoLang features unless they have been incorporated into this specification.

Design Overview



FoLang follows a deliberately different approach from conventional programming language designs. The system is structured to ensure **clear separation of concerns, license isolation, and extensibility through well-defined integration boundaries**.

Compiler and Backend

1. Frontend

The Frontend is responsible for source-level analysis and semantic processing of FoLang programs.

Components

- Scanner / Lexer
- Parser
- AST / Parse Tree Generator
- Symbol Table Generator

- Semantic Analyzer

Implementation

- Implemented in **Go**
- Uses Go structures internally for AST, symbol-table, and semantic processing.
- The externally consumable frontend output is serialized according to the selected backend's interchange contract.
- The frontend always writes that artifact under the reserved project-root `build/` directory.
- The backend-supplied contract identifies the supported FoLang/plugin protocol version, HIR schema version, and wire format.
- The backend installs/provides this interchange-contract file in the same installation directory as the FoLang compiler executable.
- The frontend reads that installed backend contract to determine the interchange representation required by the selected backend.
- The `build/` directory is generated output, never a source/package/library discovery root.
- If `build/` does not exist when compilation begins, the compiler creates it. If it already exists, it may be empty or contain previous generated output.

License

- **GNU General Public License v3 (GPLv3)**

2. Backend

The Backend is responsible for transforming validated frontend output into executable artifacts.

Default Backend should be downloaded/build separately. They are not bundled with Frontend Binary

Components

- Intermediate Representation (IR) Generator
- Native Binary Executable Generation

Implementation

A backend may be implemented in any language. It consumes the validated frontend artifact from the project-root `build/` directory. The artifact encoding is selected by the backend-supplied interchange contract.

During backend installation, that interchange-contract file is placed in the same installation directory as the FoLang compiler executable. The frontend reads the installed contract to determine the compatible FoLang/plugin protocol version, HIR schema version, and wire format required by the selected backend. The exact artifact basename and format-specific schema/version may therefore vary by backend contract, but the artifact location is always beneath

`<project-root>/build/`.

Default Backend

- Backend orchestration implemented in **Go**
- Code generation target is **C++**
- Uses **Clang** or **GCC** to generate native binaries from generated C++ IR

License

3rd Party Backends can have their own licensing terms and implementation choices. Default backend has the following license. Default backend is not part of the complete compiler binary and is separate; it must be downloaded or built separately.

- **BSD 3-Clause License** ***

Frontend Output Contract

Installed Backend Interchange Contract

The selected backend supplies this contract to tell the frontend which FoLang/plugin protocol, HIR schema, and wire format it accepts. During backend installation, the contract file is placed in the same installation directory as the FoLang compiler executable. The frontend reads it from that installation location. The default backend uses the same mechanism.

```
{
  "protocol": "foLang-plugin/1.0",
  "hir_schema": "foLang-hir/1",
  "wire": "protobuf",
}
```

The frontend has one fixed **location** contract and a backend-selected **encoding** contract:

```
<project-root>/build/
└─ <frontend-artifact>    encoding/version selected by backend contract
```

Rules:

- the output location is always the reserved root-level `build/` directory;
- the selected backend supplies the supported protocol version, HIR schema version, and wire format through its installed interchange-contract file;
- that contract file resides in the same installation directory as the FoLang compiler executable;

- `wire="protobuf"` in the example above means that particular backend contract requests Protocol Buffers; another supported backend contract may request a different compatible wire format/version;
- `build/` is compiler-generated and excluded from all source, package, source-library, operator-source, and packaged-library discovery;
- if `build/` is absent, the frontend creates it;
- if `build/` already exists, being empty or non-empty is not a project-layout error;
- `.fo1` source files placed under `build/` are invalid project layout and are never compiled as source;
- backends consume the validated frontend artifact from `build/`.

Licensing Summary

Layer	Responsibility	Implementation	License
Frontend	Parsing and semantic analysis	Go	GPLv3
Backend (default)	IR processing and native code generation	Go (orchestration) + C++ (target)	BSD 3-Clause

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3. Capability Security Model

FoLang’s compiler ships with all language features compiled in but **systems and FFI features are disabled by default**. The compiler has no hardcoded keys — capability configuration happens entirely at install time. This moves authorization from source code (developer-controlled) to the compiler installation (organization-controlled).

Feature Tiers

Tier	Features	Default State
<code>application</code>	All standard language features, <code>co.net</code> , <code>co.core</code> , <code>co.encoding</code> , <code>co.crypto</code> , etc.	✔ Always enabled
<code>system</code>	Raw pointers, pointer arithmetic, <code>co.sys.unsafe</code> , MMIO, heap allocators	🔒 Disabled — requires install-time configuration
<code>ffi</code>	<code>@co.dap.native</code> , <code>co.sys.ffi</code> , extern types, <code>co.lang.void</code> pointers, C ABI	🔒 Disabled — requires install-time configuration

Quick Start

Hello World

```
// src/appl.fo1 – fixed application entry file, no annotation needed
co.out.println("Hello FoLang!");
```

Or with an alias for shorter form:

```
@co.ddap.alias(co.out, as="out")

out.println("Hello FoLang!");
```

Variables

```
// typed
name co.lang.string = "Rao";
age co.lang.int = 30;

// inferred from value – := errors if already declared
name := "Rao";
age := 30;

// define infer and assign if not defined, otherwise assing new value
name ?= "Kumar";
```

`co.lang.string` and `co.lang.int` are Builtin Data types to know more about Builtin Data types, plese refer section [Builtin Data Types](#)

`=`, `:=` and `?=` are built in operators to know more about Built in operators please refer section [Builtin Operators](#)

Constants and Immutability

FoLang separates two properties that are often confused. One is about *when* a value is known. The other is about *whether* it can change.

```
// compile-time constant – value known while compiling, substitutable
@co.dap.const SIZE co.lang.int = 1024;

// immutable binding – cannot be reassigned, value need not be known early
@co.dap.final startedAt co.lang.int = co.sys.now();
```

Annotation	Guarantees	Value known at compile time	Usable as an index
<code>@co.dap.const</code>	value is fixed and known while compiling	✓	✓
<code>@co.dap.final</code>	binding cannot be reassigned	✗ not required	✗

Every `@co.dap.const` is also immutable. The reverse does not hold: a `@co.dap.final` binding may be initialised from a function call, a file, or user input, so the compiler cannot substitute a literal for it.

`@co.dap.final` is the declaration-site form of the immutable object kind described in the object mutation policy. `makeImmutable` applies the same property to a value at run time.

Only `@co.dap.const` may name an array size or a dependent type index, because those positions require a value the compiler can substitute. See section [Dependent Type Index Rules](#).

Single Source Application File

FoLang developers can create a complete executable program in one source file. A **single-source application** is an application whose entry file contains the complete program and does not depend on user package source files.

A single-source application file and an application entry file are the same fixed structural source, `src/app1.fo1`, and use the same entry-file grammar, context, and restrictions. A project is a single-source application when `src/app1.fo1` contains the complete program and there are no application package directories below `src/`. This section presents the allowed constructs, so a developer can start programming without first reading the complete specification.

For single source application layout please refer [Project Layout](#)

Allowed Constructs

The application file may contain:

- built-in directives that are valid for an entry file
- package and library imports
- import aliases declared with `as=`
- file-local aliases for `co.*` paths declared with `@co.ddap.alias`
- type aliases and ADTs declared with `co.lang.type`
- parameterized `co.lang.type` constructors such as `Option(T) co.lang.type = Some(T) | None()`
- new types declared with `co.lang.newtype`
- opaque types declared with `co.lang.opaquetype`
- dependent-type aliases and dependent-type usages that do not declare an ordinary or type-level function
- subtype declarations
- supertype declarations
- non-capturing entry-local function-pattern groups
- capturing entry-local `let` function-pattern groups
- variable declarations, initialization, assignment, and mutation
- calls to built-in methods and functions
- calls to imported package and library APIs
- ordinary expressions
- comprehensions
- ternary expressions
- conditions and loops
- containment operations such as `contains`
- iterators such as `each`
- pattern matching and destructuring

```
a co.lang.int = 10;
b co.lang.int = 20;

co.out.println( a + b );
```

`co` is a keyword/reservedword in `fo1ang` for more details please refer section [Reserved Words](#)

`co` is a built in package in `foLang` for more details please check [Built in Packages](#)

`a + b` is an expression in `foLang`

More about Expression rules please refer section [Expressions](#).

Built-in and Imported Names

All `co.*` paths are always available.

A developer may use the complete built-in path:

```
co.out.println("Hello");
co.core.list.of(1, 2, 3);
```

A developer may optionally create a file-local alias:

```
@co.ddap.alias(co.out, as="out")
@co.ddap.alias(co.core.list, as="list")

out.println("Hello");
values := list.of(1, 2, 3);
```

Creating an alias does not hide the complete `co.*` name; both forms remain valid in that file.

Third party packages User packages and libraries are not automatically available. They must be imported using `@co.ddap.import`. When `as=` is present, the imported API is accessed through that alias. When `as=` is omitted, the complete imported package or library path must be used.

```
@co.ddap.import(package="hr.employee", as="emp")
first := emp.EmployeeService.find(1001);

@co.ddap.import(package="finance.payroll")
second := finance.payroll.calculate(request);
```

A developer needs to import the package to use

`@co.ddap.import` and `@co.ddap.alias` are built in directives to know more about built in directives please refer section [Built-in Directives](#)

for more details on `foLang` import directive please check [Import details](#)

`println` is a built in method in `foLang` for more details please check [Built in Methods](#)

Variable Kinds

`foLang` supports different kind of variables for different purposes, even though language provides, developers cannot use at random places. For more information where they are supported refer section [Variable Kind support](#)

Simple Variable Declaration

```
someVar co.lang.int;
someString co.lang.string;
```

With Initialization

```
someBool co.lang.bool = co.const.true;
someInt co.lang.int = 42;
```

With Type Inference

```
someVal := "Hello, World!";
someNum := 3.14; // if not defined, define and initialize; else throws error
someR ?= "Kamesh"; // if not defined, define and initialize; else reassign
```

Alpha Character and String Literals

The alpha release accepts only the basic literal subset:

```
message := "hello";
letter := 'A';
```

A string is unprefixed, remains on one source line, and cannot contain a backslash or an embedded double quote. A character literal contains exactly one non-backslash character.

Encoding prefixes, escaped characters, raw strings, and universal character names are reserved for a later release. The scanner recognizes the complete spelling and reports an unsupported-feature error instead of tokenizing it as several unrelated expressions. For example, all of these are rejected:

```
x := "quoted: \"text\"";
x := '\n';
x := u8"hello";
x := R"(raw text)";
x := '\N{LATIN CAPITAL LETTER A}';
```

Pointer Declaration

```
somePtr    co.lang.int->>(*);
someDb1Ptr co.lang.int->(**);
someDeepPtr co.lang.int->(****);
```

The number of consecutive `*` characters is the pointer degree. Any positive degree is permitted; `*` and `**` above are common examples, not a maximum.

Array Declaration

```
someArray      co.lang.int->([5]); // single dimension
someDb1Array    co.lang.int->([2,3]); // multi dimension
someJaggedArray co.lang.int->([2][3]); // jagged
someVLArray     co.lang.int->([...]); //varialbe length
someZeroLA      co.lang.int->([0]); //zero length array
someZeroDimA    co.lang.int->([.]); // zerodimension array
```

Array Declaration with Initialization

```
someInitializedArray  co.lang.int->([3]) = [1, 2, 3];
someInitializedArray1 co.lang.int->([])   = [1, 2, 3];
someInitializedDb1Array co.lang.int->([,]) = [[1, 2], [3, 4]];
```

Reference Declaration

```
someRef      co.lang.int->(&); // reference
someLValueRef co.lang.int->(&&); // LValue reference
someHpRef     co.lang.int->(~); // heap allocated reference
someAddr      co.lang.int->(@); // address
someThunk     co.lang.int->(^); // thunk
someSlice     co.lang.int->[:]); // slice
```

Range Declaration

```
// Typed range variable declaration
someRange co.lang.int->(..);

// Inferred range declarations
rangeI := 1 .. 10; // [1, 10] ExcludeStart=false, ExcludeEnd=false
rangeS := 0 <.. 5; // (0, 5] ExcludeStart=true, ExcludeEnd=false
rangeL := 0 ..< 100; // [0, 100) ExcludeStart=false, ExcludeEnd=true
rangeB := 0 <..< 100; // (0, 100) ExcludeStart=true, ExcludeEnd=true
rangeE := .. 100; // open lower bound (_, 100]
rangeF := 1 ..; // open upper bound [1, _)
```

Auto and Dynamic Variable Declaration

```
someAutoVar    co.lang.auto    = "Hello"; // type inferred from value; initialization required
someDynamicVar co.lang.dynamic; // dynamic typing
```

Lazy

```
@co.dap.lazy
x = add(1, 2); //on doing some thing on x calls add(1,2) till that time add function on right hand side is not invoked
```

Bind Variables

`$_[0-9]*`

Discard / Wildcard Variable

`_`

Comma and Grouping

```
// Comma
x co.lang.int = 10, y co.lang.string = "Hello", z co.lang.bool = co.const.true;

// Grouping
(x co.lang.int = 10, y co.lang.string = "Hello", z co.lang.bool = co.const.true);

// Tuple/grouping expression
pair := (x, y);
```

A comma inside a parenthesized expression or grouped declaration must introduce another expression or declarator. Therefore `(x,)`, `(x, y,)`, and `(x co.lang.int = 10,)` are invalid. Collection literals have their own rules and may permit a trailing comma where their production says so. Record patterns also require another field after every comma; `Employee{id: value,}` is invalid.

Fat Pointers

```
x co.lang.int->(*, kind="", meta={});

co.lang.int->(*, meta={});

y co.lang.int->(*, meta={len:co.lang.usize, vtab:somepkg.VTable->(*)});

z co.lang.int->(*,kind=region, meta={});
```

```
Pointer
├─ base_type: T
├─ kind: <FatKind>
│   ├── thin
│   ├── slice
│   ├── relative
│   ├── trait
│   ├── buffer
│   ├── view
│   ├── opaque
│   ├── custom
│   ├── mem
│   ├── nullptr
│   ├── sptr
│   ├── uptr
│   ├── ptrdiff
│   ├── usize
│   ├── ssize
│   └── (region) ← optional syntactic sugar
└─ meta:
    ├── region: heap | stack | global | numa(N) | mmio | constant | ...
    └── len, cap, vtab, bits, endian, ...
```

Pointers for address manipulation

```
y co.lang.word->(repr=intptr);
z co.lang.word->(sign=unsigned, repr=uintptr);
p co.lang.word->(repr=ptrdiff);
n co.lang.word->(sign=unsigned, repr=usize);
m co.lang.word->(repr=isize);
o co.lang.void->(repr=nullptr);
```

Relative Pointers

```
z co.lang.int->(*,kind=relative, meta={});
```

Note Other than normal variable declaration rest have restrictions

Conditions, Loops and Iterators

Conditions

```
(boolean truth).do({
}).otherwise(boolean truth).do({
}).otherwise.do({
});
```

Loops

```
(boolean truth).loop({
}).otherwise(boolean truth).loop({
}).otherwise.loop({
});
```

Condition and Loop Mix

```
(boolean truth).do({
}).otherwise(boolean truth).loop({
}).otherwise(boolean truth).do({
}).otherwise.loop({
});
```

Ternary Operator

```
s = (boolean truth).then(some var/value).default(some val/var);
s = (boolean truth).then(some var/val).otherwise(boolean truth).then(some var/val).default(some var/val);
```

Looping Arrays / Lists / Maps / Ranges

```
arr co.lang.int->([5]) = [6,7,8,9,10];
arr.each(idx, val).do({
  co.out.print(idx);
  co.out.print(" :: ");
  co.out.println(val);
});

arr.each(_, val).do({
  co.out.println(val);
});
```

Array / List / Map / Range — Contains Element

```
arr co.lang.int->([5]) = [35,57,96,81,31];
k co.lang.int = 31;
arr.contains(k).do({
  co.out.println(k);
}).otherwise.do({
  co.out.println("Not Found");
});
```

Comprehensions (*planned*)

```
k := (1 .. 10).filter(|x| => x % 2 == 0).map(|x| => x * x);

result := for (x <- List(1,2,3)).yield(x * 2);           // List(2, 4, 6)
result := for (x <- Set(1,2,3)).yield(x * 2);           // Set(2, 4, 6)
result := for (x <- Some(5)).yield(x * 2);              // Some(10)
result := for (x <- fetchData()).yield(x.process());   // Future

ages := {"A":30,"B":40,"C":66,"e":88};
upper := for ((name, age) <- ages).yield(name.toUpperCase, age);
```

Pattern Matching

```
x co.lang.int = 10;

x.match.case(n: n > 10 => { n = n+100; "GT" }).case( n: n < 10 => "LT").default("EQ");
x.match.case(n: n > 10 => { n = n+100; "GT" }).case( n: n < 10 => "LT").case(_=>"EQ");
x.match(co.pattern.Type).case(co.lang.int => ...).case(co.lang.float => ...);
x.match(co.pattern.Value).case(0 => ...).case(1 => ...);
x.match(co.pattern.Instance).case(xx.CAT => ...).case(xx.DOG => ...).default("Animal");
x.match(co.pattern.Object).case(xx.Ball => "Ball").case(xx.CAT => "CAT").default("Unknown");
x.match(co.pattern.Shape).case(Point{x, y} => ...).default(...);

x.match(co.pattern.Any).case(co.lang.int => ...).case(co.lang.float => ...).case(0 => ...).default( ...);

x.match(PositiveEvenMatcher).case(0 => "Neither even nor odd").case(2 => "First Even Prime").default(...);
```

A match chain contains one or more `.case(...)` arms followed by at most one terminal `.default(...)` arm. `.otherwise` is not a match arm; it belongs to conditional and ternary then chains. A case or default result may itself be a ternary then expression, and that nested expression may consequently contain `.default(...)`.

`_` is a special discard/wildcard variable. In a call it is permitted only as the first, index/key argument of a receiver-qualified `.each`, as in `items.each(_, value)`. Transparent grouping around the member callee, for example `(items.each)(_, value)`, does not change this rule. The value argument of `each` cannot be discarded, and `contains(_)` and `containsVal(_)` are invalid because containment must compare a real value. Patterns and the filename-derived declaration-name form give `_` their own explicitly described meanings; it is not a general expression identifier.

`PositiveEvenMatcher` is custom matcher for more details about creating custom matcher please refer to section [Custom Matcher](#)

Type Declarations

```
// Alias
x co.lang.type = co.lang.int;

// New
x co.lang.newtype = co.lang.int;

// Opaque
x co.lang.opaquetype = co.lang.int; //opaque types act like alias with base type from which they declared and act like distinct type
(new types) for others even when two opaque types derived from same base types

// ADT (tagged union)
y co.lang.type = co.lang.int | co.lang.char;

// Subtype / covariant
test co.lang.subtype = co.lang.int;

// Supertype / contravariant
test co.lang.supertype = co.lang.int;
```

Let Bindings

```
y co.lang.int = let({x = 10}).in({x + 1});
y co.lang.int = let({$ = 10}).in({$ + 1}); // $ refers to the value being defined

x co.lang.int = (x + 1).where(x = 10);
x co.lang.int = ($ + 1).where($ = 10);

offset := 100;

let adjust(0) = offset;
let adjust(n) = n + offset;
```

`$` is a special identifier usable in ordinary `let` binding expressions for recursive or self-referential expressions.

Ordinary `let` value-binding expressions remain available in language contexts that permit them, but they are forbidden directly in the application entry file. In the entry file, `let` is reserved exclusively for a named function-pattern group that captures at least one surrounding runtime binding. It cannot introduce an anonymous function, a general closure value, or a curried function.

Function Pattern

```
Option(T) co.lang.type= Some(T) | None();

f(Some(x)) => { this.return x + 1; }
f(None()) => { this.return 0; }

// desugars to:
f(v Option(co.lang.int))->(co.lang.int) = {
  this.return v.match()
    .case(x: Some(x) => x + 1)
    .case(_: None() => 0);
}
```

`=>` introduces a bare function-pattern clause. `=>>` is the distinct function-delegation operator and `==>>` is the closure/curry expression and does not introduce a function pattern.

Function-pattern groups are permitted in the application entry file as restricted entry-local dispatch helpers. A bare group cannot capture surrounding runtime variables. A `let` function-pattern group must capture at least one already initialized entry-file runtime binding and is the only entry-file construct that permits such capture. Neither form permits ordinary function declarations, anonymous functions, general closure values, currying, partial application, or escape as a function value.

More about Let and function patterns please refer section [Let and Function Patterns](#)

Single source application file is for testing and getting feel of `fo1ang` Real world applications contain more than single source application file they use concepts of abstraction, encapsulation, inheritance and polymorphism at the core with many other features. Also these applications depend on external libraries, packages to achieve clear boundaries with above features. To develop these kind of applications we need following features at minimum.

1. [package source files](#) under [packages](#)
2. [Entry File](#)
3. [Libraries](#) and [Library surface files](#)
4. [imports](#)

FoLang Supports many features to develop enterprise application with [intent](#) and [FoLang Philosophy — Uniform Object Model](#) are listed below

Complete Feature list:

1. [Packages](#)
2. [UDT](#)
3. [Functions](#)
4. [Units](#)
5. [Imports](#)
6. [Macros](#)
7. [Templates](#)
8. [Annotations and Decorators](#)
9. [Type Classes](#)
10. [Types](#)
11. [Generics](#)
12. [Matchers](#)
13. [Lambdas](#)
14. [Execution Models and Control Abstractions](#)
15. [Extensions](#)
16. [Native code](#)
17. [Indexers](#)
18. [Dependent Types and Type-Level Functions](#)
19. [Dynamic Runtime](#)
20. [Local/Nested Types and Functions](#)
21. [Libraries](#)
22. [Operators](#)
23. [Forward / Extern Declarations](#)
24. [Labels and Named Blocks](#)
25. [Reflections](#)
26. [Comprehensions](#)

In FoLang, file-backed primary declarations use their own `<Name>.fo1` files. Package functions and non-UDT type declarations are grouped in any number of `*.unit.fo1` files, while struct-associated behavior is placed in `<StructName>.comp.unit.fo1`. These are all [package source files](#). Lets discuss packages before going to UDTs and Functions

Packages

Package Identity

FoLang projects use four standardized project-root domains: `src/`, `src1ib/`, `lib/`, and `build/`. These directory names are compiler-defined filesystem domains, not packages, and never contribute namespace components.

Only `src/` is mandatory before compilation. `src1ib/` and `lib/` are optional and must be omitted when unused; if either is physically present, it must contain valid content for that domain and cannot be empty. `build/` is compiler-managed generated output: it may be absent before compilation, in which case the frontend creates it, and if already present it may be empty or non-empty without causing a layout error.

Ordinary project package discovery occurs only below `src/`:

- `src/` itself is not a package.
- `src/` must exist before compilation.
- an application project has the fixed direct surface file `src/appl.fo1` ;
- a standalone packaged-library project has the fixed direct surface file `src/library.fo1` ;
- exactly one of `src/appl.fo1` or `src/library.fo1` must exist;
- if both exist, project classification is ambiguous and compilation fails;
- if neither exists, no valid project surface exists and compilation fails;
- no other file may occur directly in `src/` ;
- every other entry directly under `src/` must be a non-empty package directory containing valid FoLang source;
- package dot paths are relative to `src/` ;

- nested package directories may form an arbitrarily deep package hierarchy;
- `src/lib/`, `lib/`, and `build/` never participate in the project's ordinary package index.

Examples:

```
/appl/src/hr/          -> package "hr"
/appl/src/hr/employee/ -> package "hr.employee"
/appl/src/auth/        -> package "auth"
```

Neither the application root nor `src/` is a package.

Multi-File Packages

Multiple `.fol` files in the same package directory automatically belong to that package:

```
src/hr/employee/
├─ Employee.fol      -> hr.employee
├─ EmpService.fol    -> hr.employee
└─ EmpValidator.fol  -> hr.employee
```

Project Layout

FoLang uses one project-root architecture for applications and standalone packaged-library projects. The compiler is invoked with the **project root**, not with an individual source surface file. From that root it deterministically classifies the project and discovers every relevant domain.

```
folang compiler <project-root>
|
+-- src/appl.fol      -> application
|
+-- src/library.fol   -> standalone packaged library
|
+-- src/lib/          -> optional project-local source libraries/operators
|
+-- lib/              -> optional packaged .folenc dependencies
|
+-- build/            -> generated output; create when absent
```

The project kind changes the meaning/grammar of the fixed direct surface file under `src/`; it does not create a different filesystem architecture.

The build command therefore needs only the project root. The developer does not identify an entry/surface file separately, and the compiler does not search for an arbitrary candidate filename: it checks the reserved structural paths `src/appl.fol` and `src/library.fol`, then discovers `src/`, optional `src/lib/`, optional `lib/`, and compiler-managed `build/` relative to that same root.

Project Identity and Output Basename

The logical project name is derived from the basename of the canonical project root directory. The fixed structural source filenames `appl.fol` and `library.fol` do **not** contribute the application or library name.

```
/projects/payroll/
├─ src/
│  └─ appl.fol

project kind = application
project name = payroll
output basename = payroll
```

```
/libraries/hrlib/
├─ src/
│  └─ library.fol

project kind = standalone packaged library
library name = hrlib
compiled library artifact = hrlib.folenc
```

Therefore:

```
project-root basename
  -> project/library identity and default output basename

src/appl.fol
  -> application project kind only

src/library.fol
  -> standalone packaged-library project kind only
```

`appl.fol` never implies an application named `appl`, and `library.fol` never implies a library named `library`.

For an application, the executable uses the project name as its default basename; the platform/backend determines the final executable file convention or suffix. For example, a project rooted at `payroll/` has executable basename `payroll`, not `appl`.

For a standalone packaged library, the standard compiled FoLang library artifact uses the derived library name:

```
<hrlib project root>/      -> hrlib.folenc
```

This identity rule avoids redundant source-level naming and therefore avoids directory-name versus declared-name mismatch states.

The following application tree illustrates all available domains. `srclib/` and `lib/` are shown to document where those domains live; in a real project they must be omitted entirely when unused and cannot exist as empty directories. `build/` is compiler-managed and may be absent or empty before compilation.

```
/appl/
├── src/                                <- REQUIRED project source domain; NOT a package
│   ├── appl.fol                      <- fixed application entry / single-source application surface
│   ├── hr/                           <- package hr
│   │   ├── employee/                 <- package hr.employee
│   │   │   ├── Employee.fol
│   │   │   └── EmpService.fol
│   │   └── payroll/                  <- package hr.payroll
│   │       └── Payroll.fol
│   └── auth/                          <- package auth
│       └── Auth.fol
├── srclib/                            <- OPTIONAL; omit when unused; if present, non-empty
│   ├── ffi/
│   │   ├── library.fol
│   │   └── <internal packages>/
│   ├── system/
│   │   ├── library.fol
│   │   └── <internal packages>/
│   ├── advanced/
│   │   ├── library.fol
│   │   └── <internal packages>/
│   ├── dynamicvmrt/
│   │   ├── library.fol
│   │   └── <internal packages>/
│   └── operators/
│       └── library.fol                <- exactly one file; no subdirectories
├── lib/                               <- OPTIONAL; omit when unused; if present, one or more *.folenc
│   ├── postgres.folenc
│   └── crypto.folenc
└── build/                             <- OPTIONAL before compile; compiler creates/manages it
    └── <frontend-artifact>           <- encoding/version selected by backend contract
```

A standalone packaged-library project uses the same root domains:

```
/hrlib/
├── src/                                <- REQUIRED
│   ├── library.fol                  <- fixed standalone-library surface; @co.dap.library(type=...)
│   ├── emp/                         <- internal package emp
│   │   ├── Employee.fol
│   │   └── EmpService.fol
│   └── auth/                         <- internal package auth
│       └── AuthService.fol
├── srclib/                            <- OPTIONAL; omit when unused; if present, non-empty
├── lib/                               <- OPTIONAL; omit when unused; if present, non-empty
└── build/                             <- compiler-managed; may be absent/empty before compilation
```

The empty `srclib/` and `lib/` entries in the second diagram are structural illustration only. An actual project must not contain either directory empty.

Reserved-Domain Rules

`src/`, `srclib/`, `lib/`, and `build/` are standardized project-root directory names. They are filesystem/compiler domains only; none is a package. Only `src/` must exist before compilation.

`src/` rules:

- `src/` is the only project directory that must exist before compilation and it cannot be empty.
- `src/appl.fol` is the fixed application project surface.
- `src/library.fol` is the fixed standalone packaged-library project surface.
- exactly one of those two structural files must exist.
- `src/appl.fol` classifies the project as an application; `src/library.fol` classifies it as a standalone library.
- both files present is a compile-time project-layout error.
- neither file present is a compile-time project-layout error.
- no other file may occur directly in `src/`.

- all reusable project source must reside in package subdirectories below `src/`.
- a package directory physically present below `src/` must contain valid FoLang source; empty package directories are invalid.
- package names begin below `src/`; therefore `src/hr/employee/` is `hr.employee`, never `src.hr.employee`.

`src/lib/` rules:

- `src/lib/` is optional and must be omitted when the project uses no project-local special source libraries or operator bootstrap.
- if `src/lib/` is physically present, it must contain at least one valid standardized child domain and cannot be empty.
- `src/lib/` itself is not a package or a library.
- only the standardized immediate child directories `ffi/`, `system/`, `advanced/`, `dynamicvmrt/`, and `operators/` are permitted.
- because every special kind has one fixed directory name, a project can contain at most one project-local source library of each kind.
- `ffi/`, `system/`, `advanced/`, `dynamicvmrt/`, and `operators/` are source-domain/library roots, not packages, and their names do not become package namespace components.
- `src/lib/ffi/`, `src/lib/system/`, `src/lib/advanced/`, and `src/lib/dynamicvmrt/`, when present, must each contain exactly one direct file named `library.fol`; every other entry must be an internal package directory.
- `library.fol` is a reserved structural filename. It does not create a library named `library`; the fixed enclosing directory determines both the project-local source-library identity and its kind.
- no nested `library.fol` is permitted. Nested source-library boundaries are forbidden.
- implementation package paths begin below the source-library root and may be arbitrarily deep. For example, `src/lib/ffi/native/marshal/` is internal package `native.marshal` of the FFI source library.
- source-library implementation packages never enter the owning project's ordinary `src/` package index and cannot be bypass-imported as ordinary project packages.
- `src/lib/operators/`, when present, must contain exactly one file named `library.fol`, with no additional files or subdirectories.
- `src/lib/operators/library.fol` defines the project operator table used only while parsing and resolving that project's `src/` domain. That operator table is not inherited by `src/lib/ffi/`, `src/lib/system/`, `src/lib/advanced/`, or `src/lib/dynamicvmrt/`.
- project-local `ffi`, `system`, and `advanced` source libraries cannot declare custom operators or operator overload implementations. Their source is parsed with the language-owned operator set only.
- for every standardized child of `src/lib/`, `library.fol` is the fixed structural surface filename. In the `operators/` slot it is parsed as the operator bootstrap surface rather than as an importable API surface.
- `src/lib/`, `operators/`, and `library.fol` create no package namespace.
- a FoLang project is closed: every source package, non-operator source library, and `.folenc` artifact physically present in the project must be reachable through the applicable import/dependency graph. Orphan project content is a compile-time error.

`lib/` rules:

- `lib/` is optional and must be omitted when the project has no packaged `.folenc` dependencies.
- if `lib/` is physically present, it must contain one or more valid compiled FoLang library artifacts named `*.folenc`; an empty `lib/` directory is a project-layout error.
- `.folenc` is the standard Protocol Buffers binary library artifact format.
- no alternate non-binary compiled-library artifact format is defined.
- `.fol` source files are invalid in `lib/`, and `lib/` never participates in source discovery.
- multiple separately built FFI, system, advanced, dynamic-runtime, or application libraries may coexist in `lib/`; the one-per-kind rule applies only to project-local source libraries under `src/lib/`.
- `lib/` is not a repository for unused artifacts. Every `.folenc` physically present in a project's `lib/` directory must be reachable through an effective transitive library dependency created by actual exported-symbol use; an unreachable artifact is a compile-time error.

`build/` rules:

- `build/` is compiler-managed generated output and never participates in source discovery.
- `build/` need not exist before compilation; the frontend creates it when absent.
- an existing `build/` directory may be empty or non-empty without causing a project-layout error.
- the frontend emits its validated external representation under project-root `build/` using the wire format/version selected by the backend interchange contract.
- `.fol` source files placed under `build/` are invalid project layout.

Project-Domain Presence Invariant

FoLang distinguishes **absence** from an **empty declared domain**:

```

src/ absent          -> error
src/appl.fol only    -> application project
src/library.fol only  -> standalone library project
both structural surfaces present -> error
neither structural surface present -> error
src/ otherwise empty/invalid -> error

src/lib/ absent       -> valid
src/lib/ present but empty -> error

lib/ absent           -> valid
lib/ present but empty -> error

build/ absent         -> frontend creates it
build/ present and empty -> valid
build/ present and non-empty -> valid generated-output state

```

For developer-owned input/dependency domains, physical presence expresses intent to participate in the project. Therefore `src/lib/` and `lib/` must contain valid domain content when present. `build/` is exempt because it is compiler-owned generated state.

Structural Surface and Metadata Filenames

FoLang standardizes structural source positions so the project context is determined from the source domain rather than from arbitrary package placement:

Structural source	Valid location	Meaning	Does the containing directory define a package?
<code>appl.fol</code>	application project's <code>src/</code>	fixed application entry / single-source application surface	no; <code>src/</code> is not a package
<code>library.fol</code>	standalone packaged-library project's <code>src/</code>	fixed packaged-library API surface; kind declared explicitly by <code>@co.dap.library(type=...)</code>	no; <code>src/</code> is not a package
<code>library.fol</code>	<code>src/lib/ffi/</code> , <code>src/lib/system/</code> , <code>src/lib/advanced/</code> , <code>src/lib/dynamicvmrt/</code>	project-local source-library API surface	no; each fixed <code><kind>/</code> directory is a library root, not a package
<code>library.fol</code>	<code>src/lib/operators/</code>	project-local operator bootstrap surface	no; <code>operators/</code> is not a package and permits no subdirectories
<code>package.fol</code>	inside an ordinary package directory	package metadata/aliasing	yes; the directory is already the package

The direct source surface under `src/` is both position- and filename-structural. `src/appl.fol` classifies the root as an application project, while `src/library.fol` classifies it as a standalone packaged-library project. Neither filename creates a package or an `App1 / Library` declaration.

`library.fol` is therefore consistently used as the library-surface filename: at `src/library.fol` for a standalone library project and at `src/lib/<kind>/library.fol` for a project-local source-library or operator surface. `package.fol` remains the fixed package metadata filename and does not create its enclosing package.

Package Aliasing

If there is a folder `/appl/src/hr/empl` and under that there is a fol file called `Employee.fol` then the import statement as we know will be

```
@co.ddap.import(package="hr.empl", as="emp")
```

where `as` is not a mandatory attribute

An import names a package, never a declaration inside it. The package is the folder, so `Employee.fol` under `/appl/src/hr/empl` belongs to package `hr.empl`; the file name is not part of the path. Once the package is imported, the declaration is reached as `emp.Employee`.

Now we want to change `empl` to `emp`, simple way is `change the folder name`, but we want to keep the `physical folder name` as is.

For example, `/appl/src/hr/empl` may be given the logical package name `hr.emp` while the physical folder remains `/appl/src/hr/empl`.

```
// /appl/src/hr/empl/package.fol
_ co.lang.package = {
  name: "emp"
};
```

`package.fol` is a reserved source filename. At most one may exist in a package folder, and its only top-level declaration must be `_ co.lang.package`. In this special source form, `_` occupies the declaration-name position but does not derive a package named `package` from the filename. The required `name` member supplies the logical name of the current package segment. Parent package segments are unchanged.

The import will be as below:

```
@co.ddap.import(package="hr.emp", as="emp")
```

Note: This is a **Planned** Feature not finalized to be part of initial release.

UDT (User defined Data types)

FoLang provides the following user-defined data declaration kinds:

1. cstructs
2. structs
3. unions
4. enums
5. classes
6. modules
7. interfaces
8. signatures

Each ordinary file-backed primary declaration is placed in its own `<Name>.fol` file. The declaration name is never repeated in source; `_` occupies the declaration-name position and the compiler derives the name from the filename.

For more information about UDTs, see [Built In Kinds](#).

Struct Declaration

```
// Employee.fol
_ co.lang.struct = {
  id co.lang.int;
  name co.lang.string;
}
```

More about structs: [Structs in detail](#).

C-Struct Declaration

`co.lang.cstruct` is a C-like value type: it is passed by value, has a simple memory layout, and is safe to cross supported ABI boundaries.

```
// Point.fol
_ co.lang.cstruct = {
  x co.lang.int;
  y co.lang.int;
}
```

```
// Rect.fol
_ co.lang.cstruct = {
  origin Point;
  width co.lang.int;
  height co.lang.int;
}
```

Enum Declaration

```
// Status.fol
_ co.lang.enum = {
  Active,
  Inactive
}
```

Union Declaration

```
// NumberOrText.fol
_ co.lang.union = {
  intValue co.lang.int;
  strValue co.lang.string;
}
```

Class Declaration

```
// Employee.fol
_ co.lang.class = {
  getEmployeeDetails()->(Employee) = empmodule.getEmployeeDetails;

  getEmployeeInfo()->(Employee) =>> empmodule.getEmployeeDetails();
  // delegating – internally redirecting the call to module function
}

// $1, $2, $3 ... are previous results captured as bind variables
//Emp.fol
_ co.lang.class={
  dosomething(a co.lang.int, b co.lang.int)->(co.lang.int)=>>somePack.someMethod(a)=>>someOthPack.someOtherMeth($1, b);
}
```

More about classes: [Classes in detail](#).

For unused-symbol validation, use of the class declaration does not implicitly use its methods. The class declaration itself must participate in the live semantic graph, and every user-declared class method must independently be used. Class fields are excluded from unused-symbol validation.

Employee class	USED
└─ name field	not checked
└─ age field	not checked
└─ getName()	USED
└─ debugDump()	UNUSED -> compile-time error

Interface vs Signature

```
// EmployeeContract.fol
_ co.lang.signature = {
  ...
}
```

More about signatures: [signatures in detail](#).

```
// EmployeeApi.fol
_ co.lang.interface = {
  ...
}
```

More about interfaces: [interfaces in detail](#).

Module Declaration

```
// Employee.fol
_ co.lang.struct = {
  ...
}
```

```
// EmployeeModule.fol
_ co.lang.signature = {
  ...
}
```

```
// EmployeeModImpl.fol
@co.dap.module(signature=EmployeeModule)
_ co.lang.module->{
  signature=EmployeeModule,
  matches=EmployeeModule
} = {
  ...
}
```

More about modules: [Modules in detail](#).

Units

A unit is a stateless source container. A package may contain any number of ordinary unit files, and all their members are consolidated directly into the package namespace.


```
arithmetic.unit.fol
conversion.unit.fol
optional.unit.fol
```

Each ordinary unit file contains:

```
_ co.lang.unit = {
  ...
}
```

The filename is organizational only. It does not create a unit symbol or another qualification level. For package `math`, a function `abs` declared in any ordinary unit is accessed as `math.abs(...)`, not `math.Arithmetic.abs(...)`.

A companion unit uses the reserved filename form `<StructName>.comp.unit.fol`. Its members attach to the matching same-package struct:

```
Employee.fol
Employee.comp.unit.fol
```

Only `co.lang.struct` supports a companion unit. See [Units in detail](#) and [Struct Companion Units](#).

Matchers

Custom Matcher

```
// PositiveEvenMatcher.fol
@co.dap.matcher
_ co.lang.matcher->(type=co.lang.int) = {
  matchCase(
    value co.lang.int,
    pattern co.lang.untyped
  )->(co.lang.int, co.lang.MatchBindings) = {
    // user logic
    // 0 = no match, >0 = match
  }
}
```

A matcher declaration supports exactly one matched-subject type, specified by `type=`. It must declare exactly one protocol function named `matchCase`; overloading `matchCase` inside a matcher is not permitted. A matcher for another subject type must be declared as a separate `<Name>.fol` primary declaration.

At compile time, the compiler resolves the type named by `type=` and the type of the first `matchCase` parameter. The two resolved types must be equivalent. Type aliases are compared after alias resolution. The second parameter must have type `co.lang.untyped`, and the result must be exactly `(co.lang.int, co.lang.MatchBindings)`.

```
// InvalidMatcher.fol
_ co.lang.matcher->(type=co.lang.int) = {
  matchCase(
    value co.lang.string,
    pattern co.lang.untyped
  )->(co.lang.int, co.lang.MatchBindings) = {
    ...
  }
  // compiler error: declared matcher type and first parameter type differ
}
```

The parameter names themselves are not significant; their order and resolved types define the protocol shape.

Comprehensions (*planned*)

```
k := (1 .. 10).filter(|x| => x % 2 == 0).map(|x| => x * x);

result := for (x <- List(1,2,3)).yield(x * 2);           // List(2, 4, 6)
result := for (x <- Set(1,2,3)).yield(x * 2);           // Set(2, 4, 6)
result := for (x <- Some(5)).yield(x * 2);              // Some(10)
result := for (x <- fetchData()).yield(x.process());   // Future

ages := {"A":30,"B":40,"C":66,"E":88};
upper := for ((name, age) <- ages).yield(name.toUpperCase, age);
```

Extensions

Extension functions may be declared in an ordinary package unit:

```
// string-extension.unit.fo1
_ co.lang.unit = {

  @co.dap.extension(fortype=co.lang.string, what=extends)
  upperCase()->(string) = {
    this.return this.upper();
  }

  @co.dap.extension(fortype=[co.lang.string], what=overrides)
  equals(str co.lang.string)->(co.lang.bool) = {
    this.return this == str;
  }
}
```

Because ordinary unit filenames create no symbol, extension activation identifies the contributing package, not the unit file.

For the current package, omit `from`:

```
@co.ddap.use(methods=[equals, upperCase])
k.upperCase();
```

For another package, use its alias or complete package path:

```
@co.ddap.import(package="text.util", as="tu")

@co.ddap.use(from="tu", methods=[upperCase])
@co.ddap.use(from="text.util", methods=[upperCase])
```

See [Activating Instance Methods](#) for activation of typeclass instances.

Reflections

```
@co.dap.reflection(enable=True, package="co.meta")

x co.lang.int = 10;
x.reflect().getType(); //co.lang.int
x.reflect().getValue(); //10;
x.reflect().getKind(); // value
```

Type Classes

Monads, Applicatives, Functors, Monoids and Transformers

`@co.dap.typeclass(kind=...)` is the single annotation for all typeclass definitions. `kind` specifies the algebraic structure — `Functor`, `Applicative`, `Monad`, `Monoid`, `Transformer`, or any user-defined kind. Instances of any typeclass always use `co.lang.instance`.

In a file-backed typeclass declaration, `_` is the filename-derived declaration-name placeholder and the following parenthesized clause declares the typeclass parameters. They are separate grammar components, so the canonical spelling includes a space: `_ (F(_))`, not `_(F(_))`. A parameter such as `T` denotes an ordinary type, while `F(_)` denotes a unary type constructor and `G(_, _)` denotes a binary type constructor. Otherwise-unbound type variables introduced in an operation signature, such as `A` and `B`, are implicitly universally quantified within that operation.

Functor

```
//Functor.fo1
@co.dap.typeclass(kind=Functor)
_ (F(_)) co.lang.typeclass = {
  map(value F(A), f (A)->B) -> (F(B));
}

// ListFunctor.fo1
_ co.lang.instance->(for=Functor, type=List) = {
  map(value List(A), f (A)->B)->(List(B)) = {
    result = List(B){};
    value.each(_, item).do({ result.append(f(item)) });
    this.return result;
  }
}
```

Applicative

```
//Applicative.fol
@co.dap.typeclass(kind=Applicative)
- (F(_)) co.lang.typeclass = {
  pure(x A) -> (F(A));
  apply(fab F(A->B), fa F(A)) -> (F(B));
}

// OptionApplicative.fol
- co.lang.instance->(for=Applicative, type=Option) = {
  pure(x A)->(Option(A)) = { this.return Some(x); }
  apply(fab Option(A->B), fa Option(A))->(Option(B)) = {
    this.return (fab, fa)
      .match
      .case((Some(f), Some(x)) => Some(f(x)))
      .default(None());
  }
}
```

Monad

```
//Monad.fol
@co.dap.typeclass(kind=Monad)
- (F(_)) co.lang.typeclass = {
  pure(x A) -> (F(A));
  flatMap(fa F(A), f (A)->F(B)) -> (F(B));
}

// OptionMonad.fol
- co.lang.instance->(for=Monad, type=Option) = {
  pure(x A)->(Option(A)) = { this.return Some(x); }
  flatMap(fa Option(A), f (A)->Option(B))->(Option(B)) = {
    this.return fa.match().case(Some(x) => f(x)).default(None);
  }
}
```

Monoid

```
//Monoid.fol
@co.dap.typeclass(kind=Monoid)
- (T) co.lang.typeclass = {
  empty() -> (T);
  combine(a T, b T) -> (T);
}

// IntMonoid.fol
- co.lang.instance->(for=Monoid, type=co.lang.int) = {
  empty()->(co.lang.int) = { this.return 0; }
  combine(a co.lang.int, b co.lang.int)->(co.lang.int) = { this.return a + b; }
}
```

Transformer

```
//Transformer.fol
@co.dap.typeclass(kind=Transformer)
- (F(_), G(_)) co.lang.typeclass = {
  map(value F(A), f (A)->B) -> (G(B));
}

// ListToSetTransformer.fol
- co.lang.instance->(for=Transformer, types=[List, Set]) = {
  map(value List(A), f (A)->B)->(Set(B)) = {
    result = Set(B){};
    value.each(_, item).do({ result.insert(f(item)) });
    this.return result;
  }
}
```

Using an Instance

An instance is selected **by name**. There is no implicit search.

```
@co.ddap.import(package="abc.tc", as="tc")

xs List(co.lang.int) = [1, 2, 3];
double(x co.lang.int)->(co.lang.int) = { this.return x * 2; }

ys := tc.ListFunctor.map(xs, double);
```

`map` takes the container as its first argument, exactly as the typeclass declares it. The call names `ListFunction`, so the compiler resolves it the same way it resolves any other imported declaration.

FoLang does **not** search visible packages for an instance that happens to match a type. A typeclass is a contract, an instance is a named implementation of that contract, and the caller names the one it means. Nothing is inferred, so an unresolved call reports a name the developer actually wrote rather than a failed search they never saw.

Method syntax such as `xs.map(f)` is unrelated to typeclasses. It calls an associated function declared in a companion unit for the type. Companion units live in the type's own package, so method calls are unambiguous by construction. A companion function and a typeclass instance may both be named `map`; they are different declarations and are reached differently.

Activating Instance Methods

An instance may also be **activated**, which makes its functions callable as methods on the receiver. Activation uses the same directive that activates extensions.

```
@co.ddap.import(package="abc.tc", as="tc")
@co.ddap.use(from="tc.ListFunction", methods=[map, reduce])

ys := xs.map(double);           // resolves to tc.ListFunction.map(xs, double)
```

Activation is explicit and block-scoped. Importing a package does **not** activate anything in it, so adding an import can never change how an existing call resolves.

methods and from

`methods` selects the functions to activate.

For extension functions contributed by ordinary package units:

- omit `from` to select the current package
- use an imported package alias to select another package
- use the complete package path when no alias exists

```
@co.ddap.use(methods=[upperCase]);           // current package
@co.ddap.use(from="tu", methods=[upperCase]); // imported package alias
@co.ddap.use(from="text.util", methods=[upperCase]); // complete package path
```

Ordinary unit filenames are not accepted by `from`, because they do not create symbols.

For a typeclass instance, `from` continues to name the instance declaration:

```
@co.ddap.use(from="tc.ListFunction", methods=[map, reduce]);
```

Listing names is optional. Omit `methods` to activate every eligible method from the selected package or instance; provide it to activate a subset. Conflict detection remains receiver-aware and block-scoped.

How a method call resolves

For `xs.map(f)`, where `xs` has type `List(A)`:

1. a class method or companion-unit function on `List`
2. an activated extension for `List`
3. an activated instance function whose typeclass declares `map` with the receiver as its first parameter
4. otherwise, an error

The first match wins. A type's own declarations therefore always take precedence over anything activated into scope, and no activation can silently replace behaviour the type already defines.

Within one scope a given method name may be activated at most once for a given receiver type. Activating `map` for `List` from two sources is an error at the second `@co.ddap.use`, which names both. The conflict is reported where the activation is written, never at a distant call site.

The parser's built-in-method classification is only a lookup candidate. If the frontend recognizes a control-chain shape before receiver-aware lookup, it must retain the complete original member/call chain. A receiver-owned, companion, extension, or activated-instance declaration that wins the order above restores/remains the ordinary method call represented by that chain. An early dedicated control-flow node is therefore only a lowering candidate and becomes final only when the built-in meaning wins.

Activation never affects generic code. A function polymorphic over a typeclass takes the instance as an ordinary parameter and calls it by name.

A Typeclass Is a Type

A typeclass may be used wherever a type is expected. Its values are its instances. This is the same relationship a signature has to a module:

```
mm EmployeeModule = EmployeeModImpl;    // signature as type, module as value
f  Functor(List)  = tc.ListFunction;    // typeclass as type, instance as value
```

An instance is therefore an ordinary first-class value. It can be held in a variable, stored in a collection, returned from a function, and chosen at run time.

```
inst := (useCache).return(cachedFunc).otherwise.return(plainFunc);
result := inst.map(xs, transform);
```

Nothing about an instance is special to the compiler, which is why FoLang needs no instance resolution algorithm and no coherence rules.

Writing Code Generic Over a Typeclass

Activation makes `xs.map(f)` work for a **known** container. Code that must work for **any** container cannot activate anything, because the container type is not known until the caller supplies it. Such code takes the instance as an ordinary parameter.

```
@co.dap.generic(type={F:{typename}, A:{typename}, B:{typename}})
mapAll(inst Functor(F), value F(A), fn (A)->B)->(F(B)) = {
  this.return inst.map(value, fn);
}
```

Parameter	What it is
<code>F</code>	the container kind — <code>List</code> , <code>Option</code> , <code>Tree</code>
<code>inst</code>	the instance, an implementation of <code>Functor</code> for <code>F</code>
<code>value</code>	the container itself, an ordinary value

The caller supplies the instance:

```
ys := mapAll(tc.ListFunc, xs, double);
opt2 := mapAll(tc.OptionFunc, opt, double);
```

The function never learns what `F` is. It knows only that `inst` provides `map`, which is the whole contract. One definition therefore serves every container that has a `Functor` instance.

A type is never “a Functor” in FoLang. `List` does not become a Functor; an instance implements Functor operations for `List`, and the list stays a plain list. The Functor-ness lives entirely in `inst`.

When a wrapper is worth writing

`mapAll` above forwards directly to `inst.map`, so it adds nothing a caller could not write themselves. A wrapper earns its place when it does more than forward — when it combines several typeclass operations, adds logic, or fixes some parameters and leaves others open.

```
@co.dap.generic(type={F:{typename}})
doubleAll(inst Functor(F), value F(co.lang.int))->(F(co.lang.int)) = {
  this.return inst.map(value, (x co.lang.int)->(co.lang.int) = {
    this.return x * 2;
  });
}
```

This one fixes the element type and the operation, so it says something a bare `inst.map` call does not.

Where an Instance Is Declared

An instance is declared in the package that defines the typeclass, or in the package that defines the type. That exact package, not a sub-package.

```
abc.tc.ListFunc      for=Func, type=List      // OK  typeclass's package
myapp.ab.TreeFunc    for=Func, type=myapp.ab.Tree // OK  type's package
other.util.ListFunc  for=Func, type=List      // ERR neither is theirs
```

A typeclass is an ordinary declaration and may live in any package; `abc.tc` above is a user package, not a built-in one. Sub-packages are distinct packages, so an instance for `myapp.ab.Tree` belongs in `myapp.ab`, not in `myapp` or `myapp.ab.instances`. This matches the rule for companion units, which also sit in their type’s own package. Because a package spans every `.fo1` file in its folder, each instance still gets its own `<InstanceName>.fo1` file and uses `co.lang.instance` in source.

The rule is permissive on both sides on purpose. Requiring the typeclass’s package alone would make a typeclass usable only by its own author, since nobody could implement it for their own types. Requiring the type’s package alone would stop a library from shipping instances for common built-in types.

For library authors. If you define a type and want it usable with someone else’s typeclass, declare the instance in your package. If you define a typeclass, you may ship instances for types you do not own, including built-in ones — `IntMonoid` above sits beside `Monoid`, which is exactly this case. If you need an instance for a typeclass and a type you both do not own, wrap the type in a `co.lang.newtype` you do own and declare the instance for the wrapper.

This placement rule is semantic, not syntactic. A misplaced instance parses correctly and is reported during name resolution, so the diagnostic can name the typeclass, the type, and the two packages in which the instance would have been legal.

Labels and Named Blocks

```
// Label
outer:{
    // statements
}

//Blocks Anonymous block
{
    // statements
}

// Named Block
labelBlock co.lang.block={

}

labelBlock.expand();
```

Blocks have their own scope and context for variables, a variable pre declared outside the block will be accessible inside the block, a block can have its own variable with same name and different type or same type which overrides/shadows parent or outer blocks variables, and the scope of such variables are limited to that block it is very similar to C/C++

Blocks cannot live outside functions they must be inside functions or methods only

Inner blocks for class, struct, typeclass, module or any other construct other than functions/methods are prohibited. // throws compiler error

```
somefun (a co.lang.int, b co.lang.int)->(co.lang.int)={

    some_other co.lang.float = 20.1f;
    {
        some_other co.lang.char = 'c';
        co.out.println( some_other);    //prints c
    }

    co.out.println(some_other); //prints 20.1;

    {
        some_other co.lang.float = 11.1f;
        co.out.println(some_other); // prints 11.1f
    }

    co.out.println(some_other); // still prints 20.1f;

    {
        some_other = some_other + 1.1f;
        co.out.println(some_other); // prints 21.2
    }

    co.out.println(some_other); // prints 21.2; as this was changed in third block
}
```

imports

FoLang supports three import forms. User packages and libraries must be imported before use. When an import declares `as=`, symbols are accessed through that alias. When `as=` is omitted, the complete imported package or library path is used.

An import declaration makes the target context available for name resolution but does **not** by itself make that dependency used. The compiler records the imported context by reference in the importing context and treats the import as provisional until at least one symbol is successfully resolved through it. A zero-use import is removed from the effective imported-context/dependency graph and is reported as a compile-time error during final unused-content validation.

1. Normal Package Import

Use this for ordinary project packages.

```
@co.ddap.import(package="hr.employee", as="emp")

e := emp.getEmployee(1);
co.out.println(e.name);
```

Resolution:

```
package="hr.employee" -> /appl/src/hr/employee/
```

2. Source Library Import

Use this for same-owner project-local libraries whose source is available under the reserved `src/lib/` root.

```
@co.ddap.import(library="ffi", src-library=true, as="ffilib")
```

Resolution:

```
library="ffi", src-library=true  
-> <project-root>/src/lib/ffi/library.fol
```

For `src-library=true`, the `library` value selects one standardized project-local source-library slot under `src/lib/`. The permitted values are `ffi`, `system`, `advanced`, and `dynamicvmrt`. These names are fixed by FoLang and identify both the source library and its library kind.

The resolved file must be the fixed source-library surface file:

```
// <project-root>/src/lib/ffi/library.fol  
_ co.lang.library={  
  
}
```

Meaning:

- `library` selects one of the standardized project-local source-library slots under `src/lib/`
- `src-library=true` switches lookup from the ordinary project `src/` package index to the source-library index
- `library.fol` is the mandatory and only direct source file at that source-library root
- the enclosing fixed directory determines the library kind; a `@co.dap.library(type=...)` annotation is neither required nor permitted on a project-local source-library surface
- only the projected surface API is visible to the owning project's `src/` domain
- source in the owning project's `src/` domain can import the source library only through this `library.fol` surface; it cannot directly import any implementation package beneath the source-library root
- implementation packages beneath the source-library root are private to that source-library compilation domain and cannot be imported through ordinary project-package imports
- inside the source-library compilation domain, `library.fol` and other internal packages may import internal packages according to ordinary private package rules, but those imports are still provisional and become live only when at least one symbol from the imported package is actually used

3. Packaged Library Import

Use this for third-party or prebuilt libraries.

```
@co.ddap.import(library="hrlib", as="hr")  
@co.ddap.import(library="paylib", as="pay")
```

Resolution:

```
library="hrlib" -> <project-root>/lib/hrlib.folenc
```

Only the packaged library's projected surface API is visible to the consumer.

Import Directive Fields

Field	Required	Default	Meaning
<code>package</code> or <code>library</code>	one required	—	logical package path
<code>src-library</code>	✗	<code>false</code>	when <code>true</code> , <code>library=</code> selects <code>ffi</code> , <code>system</code> , <code>advanced</code> , or <code>dynamicvmrt</code> under reserved <code>src/lib/</code> and resolves to its <code>library.fol</code> surface or packaged <code>.folenc</code> library name
<code>as</code>	✗	none — full dot path required when omitted	local alias; valid FoLang identifier

Notes:

- `as` is optional — when omitted, no short alias is created and the full imported package path must be used to access symbols
- dots are not allowed in `as`
- for an project-local source library, the fixed `src/lib/<kind>/` path is the source of truth for library kind
- `@co.dap.library(type=...)` is not used on project-local `library.fol` surfaces

- packaged libraries may retain kind metadata in their compiled `.fo1enc` projection

Examples:

```
@co.ddap.import(package="hr.employee", as="emp")
em emp.Employee;

@co.ddap.import(package="hr.employee")
em hr.employee.Employee;
```

Valid `as` Values

<code>as="hr"</code>	✓
<code>as="v1_hr"</code>	✓
<code>as="v1.hr"</code>	✗
<code>as="123hr"</code>	✗

Cycles

Compiler error if any cycle exists through:

- package imports

Examples:

- `packageA` imports `packageB`, and `packageB` imports `packageA`

Symbol Resolution

Resolution Order

For symbols without a prefix:

1. current package symbol table
2. parent package chain upward
3. error if not found

For symbols with an alias prefix:

1. current package imported-context map
2. parent package chain imported-context maps
3. error if not found

For imports declared **without** `as`:

1. use the full imported package path directly
2. resolve it through imported-context lookup using that full path
3. error if not found

Example Context Graph

```
pp1 (grandparent package)
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs

p1 (parent package)
├─ Parent -> &pp1
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs

c1 (current package)
├─ Parent -> &p1
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs
```


Resolution Examples

```
lookup "OwnType"
  -> current symbol table

lookup "ParentType"
  -> parent chain

lookup "hr.Employee"
  -> imported-context lookup for alias "hr"

lookup "hr.employee.Employee"
  -> imported-context lookup for full imported package path when no alias exists

lookup "unknown.Type"
  -> imported-context lookup fails -> compiler error
```

Short Summary

- FoLang projects standardize the root domains `src/`, `src/lib/`, `lib/`, and `build/`; none is a package; only `src/` is mandatory before compilation
- project kind is determined structurally from `src/`: `src/appl.fo1` means application and `src/library.fo1` means standalone packaged library; exactly one must exist and no other direct file is permitted
- project/application/library identity is derived from the canonical project-root directory basename; `appl.fo1` and `library.fo1` are structural filenames only and never determine the output name
- package dot paths begin below `src/`
- `src/lib/` permits only `ffi/`, `system/`, `advanced/`, `dynamicvmrt/`, and `operators/` as immediate children
- the `ffi`, `system`, `advanced`, and `dynamicvmrt` source-library roots are not packages and each contains exactly one direct file named `library.fo1`; their remaining source lives in arbitrarily deep internal package subdirectories
- the fixed `src/lib/<kind>` directory determines the project-local source-library identity and kind; no `@co.dap.library(type=...)` annotation is used there
- `src/lib/operators/` contains exactly one `library.fo1`, has no subdirectories, and creates no package namespace
- project-local source-library internals never enter the owning project's ordinary `src/` package index
- at most one project-local source library exists for each standardized kind; additional independently built libraries of any kind are consumed as `.fo1enc` artifacts through `lib/`
- `lib/` contains zero or more Protocol Buffers binary `.fo1enc` libraries; no alternate non-binary compiled-library format is defined
- `src/lib/` and `lib/` are optional by absence but invalid when present and empty; `build/` is compiler-managed, may be absent or empty before compilation, and carries the backend-contract-selected frontend artifact
- each ordinary package source file has exactly one primary top-level declaration
- package functions, templates, macros, and non-UDT type declarations must be enclosed in ordinary `*.unit.fo1` files
- all ordinary unit members are consolidated directly into the package namespace
- struct companion behavior must be declared in `<StructName>.comp.unit.fo1`
- `src/appl.fo1` is the fixed non-package application surface and `src/library.fo1` is the fixed non-package standalone-library surface
- `co.*` is always available and never imported
- `@co.ddap.import(package="...")` imports normal packages
- `@co.ddap.import(library="ffi|system|advanced|dynamicvmrt", src-library=true, ...)` imports the corresponding project-local source library through its fixed `library.fo1` surface
- `@co.ddap.import(library="...")` imports a packaged `.fo1enc` library from `lib/`
- imports reference canonical prepared contexts/symbol tables through `ImportedContexts`; imported symbol tables are never copied
- an import is provisional until at least one symbol is actually resolved through it; zero-use imports are pruned from the effective dependency graph and reported as errors
- source in the owning project's `src/` domain can access a project-local source library only through its `library.fo1` projected surface; source-library internal packages cannot be directly imported from the owning project's ordinary `src/` package domain
- unused-symbol validation is common to applications, project-local source libraries, and standalone library builds: every class method, module function/method, companion-unit function, and usage-checkable ordinary-unit symbol must independently be used; fields/state values are excluded
- application projects are closed: every package below `src/`, every existing non-operator `src/lib` domain, every internal package of an project-owned source library, and every `.fo1enc` in `lib/` must be reachable through effective dependencies created by actual symbol use; orphan content is a compile-time error
- every ordinary library surface exports only permitted boundary contracts and public function signatures; implementation packages remain hidden

Let and Function Patterns

Bare Function-Pattern Group

A bare function-pattern group does not capture surrounding runtime bindings:

```
classify(0) => { this.return "zero"; }
classify(n).where(n > 0) => { this.return "positive"; }
classify(_) => { this.return "negative"; }
```

The formal clause shape is:

```
name(patterns) [.where(guard)] => result
```

`=>` is mandatory. `=>>` belongs to ordinary function delegation, `==>>` belongs to closure/curry expression and are not accepted here.

A bare group may use:

- its parameters and names introduced by its patterns
- its own name for recursion
- entry-local type names
- `co.*` APIs
- imported package and library APIs
- compile-time symbols that do not require runtime capture

It may not reference an entry-file runtime variable from the surrounding context:

```
offset := 100;

adjust(n) => { this.return n + offset; }
// compiler error: bare function-pattern groups cannot capture `offset`
```

Capturing `let` Function-Pattern Group

The `let` form exists only to declare an entry-local function-pattern group that captures one or more surrounding entry-file runtime bindings:

```
offset := 100;

let adjust(0) = offset;
let adjust(n) = n + offset;

result := adjust(10);
```

The captured names must resolve to surrounding runtime bindings that are already declared and definitely initialized before the first clause of the group. Built-in names, imported declarations, type names, parameters, and the function's own recursive name are not captures.

A `let` function-pattern group must capture at least one surrounding runtime binding. When no capture is required, the bare form must be used:

```
let fib(0) = 1;
let fib(1) = 1;
let fib(n) = fib(n - 1) + fib(n - 2);
// compiler error: this group captures nothing; remove `let`

fib(0) => { this.return 1; }
fib(1) => { this.return 1; }
fib(n) => { this.return fib(n - 1) + fib(n - 2); }
```

The `let` marker is therefore not an optional spelling of a bare function-pattern group. It explicitly requests restricted lexical capture.

Similarities

Bare and capturing `let` function-pattern groups both:

- group compatible clauses by function name and arity
- dispatch by parameter patterns and optional `.where(...)` guards
- evaluate clauses in normal pattern-selection order
- may call themselves recursively
- must maintain compatible parameter and result types across all clauses
- follow the normal pattern ordering, reachability, overlap, and exhaustiveness rules
- are private to the entry file
- cannot be imported, exported, or annotated with package visibility
- cannot be converted to, assigned as, passed as, or returned as function values
- cannot be partially applied or curried

Clause Syntax and Termination

Both forms accept the same pattern list and optional guard:

```
classify(0) => "zero";
classify(n).where(n > 0) => "positive";
classify(_) => { this.return "negative"; }

offset := 10;
let adjust(0) = offset;
let adjust(n).where(n > 0) = n + offset;
let adjust(_) = { this.return offset; }
```

Annotations may precede either clause form and are retained on the clause for later checking. Because every function-pattern group is private to the entry file, package visibility/export annotations are invalid even though other entry-local metadata annotations may be used.

An expression-bodied clause is a simple statement and must end with `;`. A block-bodied clause ends at its closing `}` and must not be followed by `;`. Newlines never terminate clauses.

Supported parameter patterns are:

- `_` wildcard patterns;
- literals, including signed numeric literals;
- binding names;
- qualified identity names;
- constructor patterns such as `Some(value)`;
- record patterns such as `Employee{id: value, name: _}`;
- tuple patterns with at least two elements, such as `(left, right)`.

`.where(expression)` is evaluated only after the clause's parameter patterns match. Names bound by those patterns are available to the guard and result. The guard expression must have type `co.lang.bool`.

Clauses are considered in source order. A clause is eligible when its arity matches, all parameter patterns match, and its optional guard evaluates to `co.const.true`. The compiler rejects incompatible arities or result types, unreachable clauses, invalid overlaps, and non-exhaustive groups where exhaustiveness is required by the declared result contract.

Differences

Form	Surrounding runtime capture	Intended use
<code>name(pattern) => result</code>	No	Entry-local pattern dispatch that depends only on its arguments, recursion, built-ins, imports, and compile-time names
<code>let name(pattern) = result</code>	Yes, at least one capture required	Entry-local pattern dispatch that also depends on existing entry-file runtime bindings

Clauses of the same name and arity must all use the same form. Mixing bare and `let` clauses in one function-pattern group is a compiler error.

A capturing `let` function-pattern group is still not a general closure facility. It is named, entry-local, non-first-class, and non-escaping. The compiler may internally retain a capture environment, but FoLang does not expose that environment as a closure object.

`let` cannot be used directly in the entry file for value bindings, anonymous functions, general closure expressions, or curried functions:

```
let base = 10;           // compiler error: entry-file let value binding
let result = base + 1;   // compiler error: entry-file let value binding
let add = (a, b) => a + b; // compiler error: anonymous function
let counter = closure { ... }; // compiler error: general closure expression
let add = a => b => a + b;  // compiler error: curried function
```

The compiler may lower either function-pattern form to a private entry helper, but neither source construct is a general-purpose function declaration.

Package in detail

Package Identity

Ordinary project packages exist only in subdirectories below the reserved `src/` domain. `src/` itself is not a package and, for an application project, contains exactly one direct file: the fixed `app1.fo1` surface. No other direct file is permitted there.

- package dot paths start below `src/`
- the application root and `src/` are not packages
- `src1ib/`, `lib/`, and `build/` are separate reserved domains and are excluded from ordinary project package discovery
- source-library roots under `src1ib/` are not packages; only their descendant implementation directories define packages within that source-library compilation domain

Examples:

```
/apl/src/hr/           -> package "hr"
/apl/src/hr/employee/  -> package "hr.employee"
/apl/src/auth/         -> package "auth"
```

Multi-File Packages

Multiple `.fo1` files in the same package folder below `src/` automatically belong to the same package:

```
src/hr/employee/
├─ Employee.fol      -> hr.employee
├─ EmpService.fol   -> hr.employee
└─ EmpValidator.fol -> hr.employee
```

Application Project Layout

See the canonical [Application Project Layout](#). The key ownership boundaries are:

```
/appl/
├─ src/                -> appl.fol + application package directories
├─ src/lib/            -> standardized project-local special source libraries
├─   └─ ffi/           -> library.fol + internal package directories
├─   └─ system/        -> library.fol + internal package directories
├─   └─ advanced/      -> library.fol + internal package directories
├─   └─ dynamicvmrt/   -> library.fol + internal package directories
├─   └─ operators/     -> library.fol only
├─ lib/                -> zero or more *.folenc Protocol Buffers binary libraries
└─ build/              -> generated compiler output
```

Only paths below `src/` enter the ordinary project `src/` package index. Package discovery inside a source library begins only below its fixed `src/lib/<kind>/` root and remains private to that library.

Package Access Rules

Four access levels control visibility across package boundaries:

Annotation	Same Package	Parent	Sub-Package	Unrelated	Entry / Surface
<code>@co.dap.public</code>	✓	✓	✓	✓	✓
<code>@co.dap.package</code>	✓	✓	✓	✗	✗
<code>@co.dap.protected</code>	✓	✗	✓	✗	✗
<code>@co.dap.private</code>	✓	✗	✗	✗	✗

Meaning:

- `@co.dap.public` -> visible everywhere
- `@co.dap.package` -> visible across the package family
- `@co.dap.protected` -> visible downward into subpackages only
- `@co.dap.private` -> visible only inside the declaring package

Example:

```
// hr/employee-access.unit.fol - package "hr"

@co.dap.public
_ co.lang.unit = {

  @co.dap.public
  getEmployee(id co.lang.int)->(Employee) = { ... }    // anyone can call

  @co.dap.package
  validateId(id co.lang.int)->(co.lang.bool) = { ... } // hr package family

  @co.dap.protected
  baseQuery()->(co.lang.string) = { ... }              // visible to subpackages

  @co.dap.private
  normalizeId(id co.lang.int)->(co.lang.int) = { ... } // private declaration
}
```

`co.*` Paths and Aliases

`co.*` Is Always Available

All `co.*` paths are part of the language and are always in scope. They are never imported through `@co.ddap.import`.

```
co.out.println("hello");
co.in.readLine();
x co.lang.int = 42;
```

Being **in scope** does not automatically mean **permitted in every context**.

A package rule, library kind, or other compiler restriction may still forbid use of a particular `co.*` facility. In such cases the name is still known to the compiler, but its use is rejected at compile time.

@co.ddap.alias

Use aliases only to shorten `co.*` paths.

```
@co.ddap.alias(co.out, as="out")
@co.ddap.alias(co.core.list, as="list")
@co.ddap.alias(co.encoding, as="enc")

out.println("hello");
list.of(1, 2, 3);
enc.json.serialize(data);

// full form still works alongside
co.out.println("world");
```

Rules:

- target must be a `co.*` path
- alias must be a valid identifier
- alias scope is file-local
- aliasing user packages is not allowed; use `as=` in `@co.ddap.import`
- duplicate aliases in the same file are compiler errors
- when no alias is declared, the complete `co.*` path is used
- declaring an alias does not disable or hide the complete `co.*` path ***

Package Source Files

A package folder may contain three ordinary source-file categories plus the reserved `package.fo1` metadata form. The fixed project surfaces `src/appl.fo1` and `src/library.fo1` are not ordinary package-source files and are invalid inside a package. Likewise, `src/lib/<kind>/library.fo1` is a structural source-library surface and is invalid inside an ordinary package. The compiler classifies ordinary package files from filenames before parsing, using the longest recognized suffix first:

```
<Name>.comp.unit.fo1  -> companion unit
<Fragment>.unit.fo1   -> ordinary package unit
package.fo1           -> reserved package metadata/alias form
<Name>.fo1            -> file-backed primary declaration
```

Reserved structural filenames are recognized before the generic `<Name>.fo1` rule, and a filename ending in `.comp.unit.fo1` is never classified as an ordinary `.unit.fo1` file.

File-Backed Primary Declarations

A primary declaration file has the form `<Name>.fo1` and contains exactly one primary top-level declaration. The source must use `_` in the declaration-name position:

```
// Employee.fo1
_ co.lang.struct = {
  id co.lang.int;
  name co.lang.string;
}
```

The compiler derives the declaration name `Employee` from the filename. An explicit primary name is invalid:

```
// Employee.fo1
Employee co.lang.struct = { ... }
// compiler error: file-backed primary declarations must use `_`
```

This rule applies to ordinary file-backed declarations that use a `<name> co.lang.<kind>` primary form, including classes, structs, cstructs, enums, unions, interfaces, signatures, modules, instances, matchers, objects, and other ordinary `co.lang.*` primary kinds. A `co.lang.library` declaration is not an ordinary `<Name>.fo1` primary: it is valid only in the reserved `library.fo1` surface form described in [Libraries](#). Declaration families with a different surface grammar define their filename binding in their own sections.

The following declaration forms are stated exceptions and keep an explicit name in the head, because filename derivation cannot express what they need:

Form	Why
struct and cstruct declared inside <code>library.fo1</code>	one library surface may carry several boundary declarations
<code>co.lang.data</code> algebraic data type	the head names the variants
parameterized <code>co.lang.type</code>	a filename cannot carry <code>(T)</code>
type declarations in <code>src/appl.fo1</code>	the entry file is not filename-backed as an ordinary package declaration

File-level directives, imports, aliases, annotations, and decorators may appear before the primary declaration. They do not count as additional primary declarations.

FoLang permits the following top-level declaration kinds across ordinary package source files and their reserved special source forms:

1. struct
2. cstruct
3. class
4. module
5. signature
6. interface
7. enum
8. union
9. typeclass
10. instance
11. matcher
12. annotation or object declaration
13. package declaration
14. library declaration
15. unit declaration

Their source-file placement is part of the grammar:

- ordinary filename-backed primary declarations use `<Name>.fol`
- a package declaration uses the reserved `package.fol` source form
- an ordinary unit uses `<Fragment>.unit.fol`
- a companion unit uses `<StructName>.comp.unit.fol`
- a project-local special source-library declaration uses the reserved `src/lib/<kind>/library.fol` surface form
- a standalone packaged-library declaration uses the fixed direct `src/library.fol` project-surface form and must declare its kind explicitly with `@co.dap.library(type=...)`

A source file is invalid when it contains multiple unrelated primary declarations, places a package declaration outside `package.fol`, places a unit declaration outside a recognized unit filename, places a library declaration outside one of its context-valid surface positions, or places project/library metadata outside its dedicated source form.

Package-level functions and non-UDT type declarations belong inside ordinary unit files.

Filename Canonicalization

Filename-derived declaration identity is independent of filesystem case sensitivity. The compiler normalizes and case-folds the filename stem to construct the duplicate-detection and lookup key:

```
canonical file key = caseFold(normalize(filename stem))
```

The language-level declaration is then represented using FoLang's canonical declaration spelling. Therefore:

```
employee.fol
Employee.fol
EMPLOYEE.fol
```

all denote the canonical declaration name `Employee`. They conflict rather than declaring separate types, even on a case-sensitive filesystem.

The same rule applies to companion owners:

```
employee.comp.unit.fol -> companion of Employee
```

This rule is specific to filename-derived primary declarations and companion-owner identity; it does not make every FoLang identifier case-insensitive. The compiler must not rely on operating-system filename comparison.

The filename stem must form a valid FoLang declaration identifier after normalization. Case variants and canonically equivalent spellings produce the same package-index key.

Ordinary Package Units

An ordinary package-unit file has the form `<Fragment>.unit.fol`:

```
// arithmetic.unit.fol
_ co.lang.unit = {
  abs(value co.lang.int)->(co.lang.int) = {
    ...
  }
}
```

The fragment name is organizational only. It does not create a public `Arithmetic` symbol. All ordinary unit members in the package are merged into one package namespace:

```
math/  
├─ arithmetic.unit.fol  
├─ comparison.unit.fol  
└─ optional.unit.fol
```

```
math.abs(-10);  
math.max(10, 20);  
value math.Option(co.lang.int);
```

The following qualification is invalid:

```
math.Arithmetic.abs(-10); // compiler error
```

Any number of ordinary unit files may exist in one package. During package indexing, the compiler shallow-parses their declarations, merges their members with filename-derived primary declarations, and reports duplicate names or duplicate function signatures according to FoLang overload rules.

Companion Unit Files

A companion unit has the form `<StructName>.comp.unit.fol` and must contain `_ co.lang.unit`. The canonical owner name comes from the filename, not from the source body.

```
Employee.fol  
Employee.comp.unit.fol
```

The compiler can detect the following before parsing either body:

- duplicate primary declarations after filename canonicalization
- duplicate companion files for the same canonical owner
- an orphan companion file for which no same-package primary declaration exists

After the owner header is known, the compiler also verifies that the owner is a `co.lang.struct`. Other primary declaration kinds cannot own companion units.

Package Indexing Levels

```
Level 1: filename and package indexing  
  classify primary, ordinary-unit, and companion-unit files  
  canonicalize primary names and companion owners  
  detect duplicate primary files  
  detect duplicate or orphan companion files  
  shallow-parse ordinary units  
  merge ordinary-unit members into the package namespace  
  report package-namespace conflicts  
  
Level 2: companion validation  
  parse companion declarations  
  validate explicit receivers against the filename-derived owner  
  merge companion members into the owner's companion namespace  
  report duplicate companion members and receiver mismatches  
  
Level 3: full semantic resolution  
  resolve declaration bodies, expressions, calls, patterns, and types
```

This organization lets an imported source package build most of its symbol index cheaply before full parsing. Compiled packages may store the same index as metadata so imports need no source parsing.

Application Entry File

The application entry file is the fixed **special executable source form** `src/appl.fol`. It is not a package, does not create an importable namespace, and is not subject to the ordinary package-file rule requiring exactly one primary declaration.

The compiler creates a dedicated **entry-file context** for it:

```
ApplicationEntryContext  
├─ file directives, imports, and aliases  
├─ entry-local non-structural type declarations  
├─ non-capturing function-pattern groups  
├─ capturing `let` function-pattern groups  
└─ executable statements and expressions
```

Everything declared directly in this context is private to the entry file. Entry-local symbols cannot be imported, exported, or referenced by package or library source files.

Allowed Entry-File Constructs

The application entry file uses exactly the same grammar and allowed-construct rules described in [Single Source Application File](#). The earlier section provides the developer-facing overview; this section defines the entry file's formal context, privacy, and dependency direction.

Entry-Local Function Patterns

Function-pattern groups are allowed as a special entry-file construct even though ordinary function declarations are forbidden. FoLang provides two entry-file forms with the same pattern-dispatch model but different capture semantics.

Entry-File Dependency Direction

The entry file may depend on packages and libraries, but packages and libraries may never depend on the entry context:

```
application entry file
  ↓ uses
packages and libraries
```

This allows the entry file to coordinate application startup while preserving package and library independence.

Libraries

Library Project Layout

A standalone packaged-library project uses the same project-root architecture as an application. Only `src/` must exist before compilation. `src/lib/` and `lib/` are optional and must be omitted when unused; when present they cannot be empty. `build/` is compiler-managed and may be absent or empty before compilation.

```
/hrlib/
├── src/
│   ├── library.fol          <- fixed standalone-library surface; @co.dap.library(type=...)
│   ├── emp/                <- internal package emp
│   │   ├── Employee.fol
│   │   └── EmpService.fol
│   └── auth/               <- internal package auth
│       ├── Auth.fol
│       └── AuthService.fol
├── src/lib/                <- optional; omit when unused
├── lib/                    <- optional; omit when unused
└── build/                  <- compiler-generated output; created when absent
    └── <frontend-artifact>
```

The empty `src/lib/` and `lib/` entries above are diagrammatic placeholders only; an actual project cannot contain those directories empty.

Consumers see only what the direct library surface under `src/` projects. All package subfolders below the library project's `src/` are internal. The root-level `build/` directory is generated output and is not an internal package.

Library Surface file

FoLang uses surface files in two situations:

1. Packaged library project surfaces
2. Application-workspace source library surfaces

A library surface contains `co.lang.library` and defines the public boundary data contracts and boundary-adaptor functions through which consumers call the library. Its valid structural location depends on the compilation domain. For an project-local source library, the fixed `src/lib/<kind>/library.fol` path supplies identity and kind. For a standalone packaged-library project, the fixed `src/library.fol` file is the library surface and must declare its kind explicitly with `@co.dap.library(type=...)`.

```
src/appl.fol          -> application entry
src/library.fol       -> standalone packaged-library surface
src/lib/ffi/library.fol -> application/project-local source-library surface
```

A library surface is not an ordinary package file. It may contain multiple boundary data declarations and public functions inside one `co.lang.library` declaration.

Packaged Library Project Surface

A packaged-library project has no application entry file. Its mandatory `src/` domain contains exactly one direct FoLang source file named `library.fol`. That fixed file is the packaged-library surface. All other source below `src/` resides in internal package directories.


```

/hrlib/
├── src/
│   ├── library.fol          <- fixed direct library surface; explicit project-level kind declaration
│   └── emp/
│       ├── Employee.fol
│       └── EmployeeService.fol
│   └── auth/
│       └── AuthService.fol
├── srclib/                  <- optional; omit when unused
├── lib/                     <- optional; omit when unused
└── build/                   <- compiler-generated output
    └── <frontend-artifact>

```

Rules:

- exactly one direct FoLang source file named `library.fol` exists under the packaged-library project's `src/` directory
- that direct surface file must contain the project-level `@co.dap.library(type=...)` declaration followed by `_ co.lang.library = { ... }`
- the explicit library kind is required because a standalone packaged-library project has no enclosing application `srclib/<kind>/` path from which the compiler can infer the kind
- the compiler records the declared kind on the library surface context/symbol table before validating the surface, then applies the declaration, boundary-type, public-signature, adapter-body, capability, and other restricted-construct rules for that library kind
- a missing, unsupported, or conflicting standalone library-kind declaration is a compile-time error
- the supported packaged-library kinds are `application`, `dynamicvmrt`, `advanced`, `system`, and `ffi`; `operator` is not a packaged-library kind and operator metadata is never exported through `.folenc`
- the project is compiled into the standard packaged library artifact `.folenc`, which is Protocol Buffers binary
- consumers import it with `@co.ddap.import(library="...")`
- internal package folders are compiled into the library but are not directly visible to consumers
- the frontend output is emitted under the library project's root-level `build/` directory using the wire format/version selected by the backend interchange contract
- `build/` is generated output and never an internal package or source-discovery root

Project-Local Source Library Surface

Project-local special source libraries live only under the fixed `srclib/` domain. `srclib/` itself is neither a package nor a library. FoLang standardizes the only permitted immediate children:

```

<project-root>/srclib/
├── ffi/
│   ├── library.fol
│   └── <internal packages>/
├── system/
│   ├── library.fol
│   └── <internal packages>/
├── advanced/
│   ├── library.fol
│   └── <internal packages>/
├── dynamicvmrt/
│   ├── library.fol
│   └── <internal packages>/
└── operators/
    └── library.fol

```

For `ffi`, `system`, `advanced`, and `dynamicvmrt`:

- the immediate directory name is fixed by FoLang and determines both library identity and library kind;
- the directory itself is not a package;
- exactly one direct file is permitted: `library.fol`;
- `library.fol` contains `_ co.lang.library = { ... }` and does not use `@co.dap.library(type=...)`;
- all implementation source must reside in descendant package directories;
- descendant package names are relative to the source-library root and may be arbitrarily deep;
- no nested `library.fol` or nested source-library boundary is permitted;
- internal packages remain private to that source-library compilation domain and never enter the ordinary project `src/` package index;
- source in the owning project's `src/` domain accesses the library only through the projected `library.fol` API surface.
- These libraries don't generate `.folenc` they are embedded into the application and built as single application protobuf file for backend.

Example:

```

<project-root>/srclib/ffi/
├── library.fol          <- FFI public surface; ffi/ is not a package
├── native/              <- internal package native
│   └── marshal/         <- internal package native.marshal
│       └── Marshal.fol
└── database/            <- internal package database
    └── Postgres.fol

```

```
// <project-root>/src/lib/ffi/library.fo1
_ co.lang.library = {
  ...
}
```

Import:

```
@co.ddap.import(library="ffi", src-library=true, as="ffilib")
```

Resolution:

```
library="ffi", src-library=true
-> <project-root>/src/lib/ffi/library.fo1
```

The same structural rule applies to `system`, `advanced`, and `dynamicvmrt`.

The one-per-kind restriction applies only to **project-local source libraries within one project**. Projects may consume any number of separately built libraries of any library kind as `.fo1enc` artifacts from `lib/`.

`src/lib/operators/` is the special bootstrap slot described in [Operators](#). It uses the same fixed structural surface filename as every other `src/lib/` slot, so it contains exactly `library.fo1` and no package subdirectories. Its filesystem position selects the dedicated operator-source grammar; it is not an ordinary importable source-library API surface.

Unified Surface Model

Every ordinary library kind uses the same conceptual surface shape. The `src/lib/operators/` bootstrap slot is deliberately excluded because it is parser metadata rather than an importable library API:

```
library surface
├─ boundary data contracts
└─ public boundary-adapter functions
```

Library kind changes the permitted boundary representation and transfer semantics, not the conceptual form of the API.

Library kind	Boundary data	Transfer semantics
<code>application</code>	<code>co.lang.struct</code>	automatic deep snapshot
<code>dynamicvmrt</code>	<code>co.lang.struct</code>	automatic deep snapshot
<code>advanced</code>	<code>co.lang.struct</code>	automatic deep snapshot
<code>system</code>	<code>co.lang.cstruct</code>	system ABI value
<code>ffi</code>	<code>co.lang.cstruct</code>	C ABI value

`co.lang.struct` and `co.lang.cstruct` solve different problems:

```
struct -> FoLang semantic data contract
cstruct -> physical ABI-compatible value contract
```

Value transfer must not be confused with C-compatible layout. Application-family libraries therefore use expressive `struct` contracts transferred as deep snapshots, while system and FFI libraries use restricted `cstruct` contracts.

Allowed Surface Declarations

A library surface may contain only:

- file- or library-level imports needed by its adapter implementations
- `co.lang.struct` boundary declarations for `application`, `dynamicvmrt`, and `advanced`
- `co.lang.cstruct` boundary declarations for `system` and `ffi`
- public free-function API declarations with boundary-adapter definitions

The following declaration kinds are forbidden directly in every library surface:

- classes
- modules
- interfaces
- signatures
- units and companion units
- associated functions
- operator functions
- enums, unions, and other ADTs
- type aliases, newtypes, and opaque types
- objects, instances, type classes, and dependent types
- macros, templates, annotations, and decorators

- global variables, pointers, references, addresses, or mutable surface state

Surface `struct` and `cstruct` declarations are data contracts only. They cannot have companion units, associated functions, operators, methods, or other behavior on the surface.

Declarations directly inside the `co.lang.library` body are exported by default. Imports and implementation details are never exported.

Public Signature Type Closure

Every public surface function signature must be closed over the library's exported boundary type set.

For `application`, `dynamicvmrt`, and `advanced` surfaces, parameters and results may use only:

- approved built-in types
- `co.lang.struct` types declared in the same library surface

For `system` and `ffi` surfaces, parameters and results may use only:

- ABI-safe built-in types
- `co.lang.cstruct` types declared in the same library surface

The same closure rule applies recursively to fields of surface boundary types:

- a surface `struct` field may use an approved built-in type or another surface-declared `struct`
- a surface `cstruct` field may use an ABI-safe built-in type or another surface-declared `cstruct`
- an internal package type may never appear in a public function signature or surface boundary-type field
- pointers, references, and addresses may never cross any public library surface

A built-in type is not automatically surface-safe merely because it belongs to `co.lang`. An approved surface built-in must be concrete, fully resolved, and transferable under the library kind's boundary semantics.

The following categories are forbidden in public surface fields and signatures:

- inference-only types such as `co.lang.auto` and `co.lang.infer`
- dynamically typed or unconstrained carriers such as `co.lang.dynamic`, `co.lang.any`, `co.lang.typed`, and `co.lang.untyped`
- function, closure, delegate, loader, AST, reflection, or runtime implementation values
- pointer, reference, address, thunk, and implementation-handle types
- any type whose reachable representation contains a forbidden type

For application-family surfaces, managed built-ins such as `co.lang.string` are permitted when the compiler defines deep-snapshot reconstruction for them. For system and FFI surfaces, only built-ins with a defined ABI representation are permitted; for example, `co.lang.string` is not directly `cstruct`-compatible.

Valid:

```
// Employee.fol
_ co.lang.struct = {
  id      co.lang.int;
  name    co.lang.string;
  address Address;
}

// Address.fol
_ co.lang.struct = {
  city co.lang.string;
}
```

Invalid:

```
getEmployee(id co.lang.int)->(emp.internal.Employee);
// compiler error: an internal type escapes through the public signature
```

Boundary-Adapter Functions

A public surface function may contain a definition, but its definition is restricted to boundary adaptation.

A boundary-adapter function may:

- read its input parameters and their fields
- declare temporary local values used only for conversion
- construct internal input values
- perform deterministic field-to-field or representation conversion
- invoke exactly one internal implementation operation
- receive that operation's result
- construct and return an allowed surface `struct`, surface `cstruct`, or built-in value
- use a direct delegation expression when no conversion is required

A boundary-adapter function may not:

- implement business rules or domain calculations
- perform business validation or authorization decisions
- orchestrate multiple implementation operations
- call another surface function
- perform persistence, networking, file I/O, retries, caching, or transactions
- start concurrency, scheduling, asynchronous work, processes, or continuations
- contain loops, recursion, workflow branching, or externally observable state mutation
- expose an internal value directly without converting it to an allowed boundary type

The compiler enforces the restricted adapter statement set. The restriction is structural: arbitrary executable logic is not accepted merely because it is placed in a surface file.

A direct delegate is the simplest adapter form:

```
health()->(co.lang.bool)
=>> health.internal.Service.health();
```

A converting adapter may map between the public contract and an internal model.

Standalone Packaged Application-Library Surface Example

```
// library.fol - standalone packaged application library
@co.dap.library(type="application")
_ co.lang.library = {

  @co.ddap.import(package="emp", as="emp")

  Employee co.lang.struct = {
    name co.lang.string;
    id co.lang.int;
  }

  getEmployee(empId co.lang.int)->(Employee) = {
    internalEmployee := emp.EmployeeService.getEmployee(empId);

    this.return Employee{
      name: internalEmployee.name,
      id: internalEmployee.id
    };
  }
}
```

The surface owns conversion from the internal `emp.Employee` representation to the public `hrLib.Employee` contract. The `emp` package owns validation, business rules, persistence, and workflow.

The consumer sees the equivalent API contract:

```
// Employee.fol
_ co.lang.struct = {
  name co.lang.string;
  id co.lang.int;
}

getEmployee(empId co.lang.int)->(Employee);
```

The consumer does not see the body of `getEmployee` or the `emp` package.

Standalone Packaged System-Library Surface Example

The following example is a separately authored packaged-library project. Project-local `srcLib/system/library.fol` does not repeat the kind annotation because its fixed path already establishes `system`.

```
// library.fol - standalone packaged system library
@co.dap.library(type="system")
_ co.lang.library = {

    @co.ddap.import(package="driver.internal", as="impl")

    Point co.lang.cstruct = {
        x co.lang.int32;
        y co.lang.int32;
    }

    getOrigin()->(Point) = {
        internalPoint := impl.DriverService.getOrigin();

        this.return Point{
            x: internalPoint.x,
            y: internalPoint.y
        };
    }
}
```

Pointers, addresses, native bindings, and hardware state belong in `driver.internal`. They are forbidden in the public surface and cannot appear in the exported signature.

Boundary Transfer Semantics

For `application`, `dynamicvmrt`, and `advanced` libraries:

```
consumer boundary value
  ↓ automatic deep snapshot of the complete reachable value graph
library-local built-in or surface struct
  ↓ surface conversion
internal implementation value
  ↓ surface conversion
library-local built-in or surface struct
  ↓ automatic deep snapshot
consumer-local boundary value
```

The snapshot rule applies to both surface structs and approved built-in arguments/results. The library never receives the consumer's live object identity, and the consumer never receives a live internal-library object.

For `system` libraries, `cstruct` values cross according to the selected backend/platform system ABI.

For `ffi` libraries, `cstruct` values cross according to the declared C ABI, including its layout, alignment, and calling-convention requirements.

Surface-to-Internal Dependency Direction

The source-level dependency is one-way:

```
library surface
  ↓ imports and invokes
internal packages
```

Internal packages:

- do not import the library surface
- do not use surface-declared `struct` or `cstruct` types
- define their own implementation and domain types
- return internal values to the surface adapter
- contain all business logic, validation, workflow, I/O, and state management

This prevents a surface/internal compilation cycle and keeps the public contract independent from the internal domain model.

An internal implementation symbol may be visible to its own library surface without becoming part of the consumer API. Library encapsulation takes precedence over ordinary package visibility: consumers cannot resolve internal package symbols even when those symbols are callable from the surface during library compilation.

Consumer API Projection

The compiler does not expose the complete surface compilation context to a consumer. It creates a projected imported symbol table.

The projected API contains:

- the library identity and kind
- complete public surface `struct` or `cstruct` definitions, including field names and permitted field types
- public function names
- public function parameter names and types
- public function result types
- calling-convention, effect, error, and linkage metadata where applicable

The projected API does not contain:

- function bodies or implementation AST/HIR
- imports used by the surface
- local conversion variables
- delegate or implementation targets
- internal package paths or symbols
- private compiler-generated helpers
- business classes, modules, units, or internal data types

Therefore, a function definition may exist in the surface source and compiled artifact while the consumer's symbol table contains only its signature.

The backend may retain implementation bodies for code generation, linking, or whole-program optimization. Backend possession of implementation code does not make that code semantically visible to the consumer.

Source libraries follow the same rule. Availability of source code does not weaken the projected API boundary.

Library Compilation Order

A library uses the same import-usage and symbol-usage validator as an application. It is processed in these logical stages:

1. parse the surface header, boundary data declarations, and public function signatures;
2. parse/prepare internal packages without depending on surface types;
3. register internal imports as provisional context references rather than copying symbol tables;
4. type-check and link surface boundary-adapter bodies against internal package symbols, activating an import only when a symbol from it is actually used;
5. complete semantic resolution and mark exact usage-checkable symbols live;
6. prune and report zero-use imports;
7. reject unused usage-checkable symbols and unreachable internal source/packages;
8. emit the compiled implementation and projected consumer API.

The declarations intentionally exported by `library.fol` are semantic roots of the library build. This order permits the surface to call internal packages while preventing internal packages from depending on the surface, and it applies to both project-local source-library compilation and standalone packaged-library compilation.

Library Kinds

For project-local source libraries, the library kind is structural and comes from the fixed path:

```
src/lib/ffi/      -> ffi
src/lib/system/   -> system
src/lib/advanced/ -> advanced
src/lib/dynamicvmrt/ -> dynamicvmrt
src/lib/operators/ -> operator bootstrap context (not importable; not packaged)
```

All project-local `src/lib/<kind>/library.fol` surfaces therefore do **not** repeat the kind with `@co.dap.library(type=...)`. This includes `src/lib/operators/library.fol`; the enclosing `operators/` slot establishes the dedicated operator bootstrap context.

A separately authored packaged-library project is not classified by an enclosing `src/lib/<kind>/` slot. Its fixed `src/library.fol` surface must therefore contain the project-level `@co.dap.library(type=...)` declaration. The compiler uses that explicit kind to create/tag the library surface context and symbol table, then verifies that the surface contains only constructs allowed for that kind before producing the `.folenc` projection. The `application` kind is used for an ordinary safe packaged library.

Shared Meaning

- `application` -> ordinary safe library implementation
- `dynamicvmrt` -> application features plus full `co.meta` dynamic-runtime support
- `advanced` -> macro, template, concurrency, transformation, and runtime-machinery implementation
- `system` -> unsafe low-level runtime, pointer, process, native, and platform-control implementation
- `ffi` -> foreign-interface implementation

Library kind controls internal capabilities. All kinds still expose only boundary data contracts and public boundary-adapter functions.

`application` Libraries

Internally allowed:

- ordinary safe language features
- classes, modules, interfaces, signatures, units, and ordinary structs
- normal application-level packages and libraries

Internally forbidden:

- pointers, references, and addresses
- native functions annotated with `@co.dap.native`

- macros and templates
- low-level concurrency machinery
- advanced transformation or dynamic-runtime machinery

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

`dynamicvmrt` Libraries

Internally allowed:

- all `application` capabilities
- full `co.meta` support
- runtime reflection, instrumentation, dynamic loading, patching, and eval-based facilities

Internally forbidden:

- system and FFI capabilities unless reached through an imported public library surface
- raw pointers, addresses, and native functions in its own implementation

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

`advanced` Libraries

Internally allowed:

- all ordinary safe language features
- macros and templates
- `async`, `parallel`, `continuation`, `scheduling`, and `transformation` machinery
- compiler/runtime behavior definitions permitted to advanced libraries

Internally forbidden:

- raw pointers
- references and addresses
- native functions

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

`system` Libraries

Internally allowed:

- pointers, references, addresses, and word-level types
- native functions
- structs and cstructs
- units containing free functions
- basic value-typed functions
- type aliases
- basic threading and process primitives

Internally forbidden:

- classes
- modules
- interfaces and signatures
- overloading and overriding
- new operator definitions
- generics and templates
- macros
- reflection and dynamic runtime
- dependent types and type classes
- function patterns
- closures and currying

- advanced concurrency machinery such as continuations, defer, lazy evaluation, and thunks
- variadic, named, optional, and default parameters

Public boundary:

- surface `cstruct` contracts
- ABI-safe built-in types
- system ABI value transfer

`ffi` Libraries

Internally allowed:

- native and extern declarations
- pointers, references, addresses, and C ABI types
- cstructs and restricted implementation structs
- units containing free functions
- foreign calling-convention and symbol-linkage metadata

Internally forbidden:

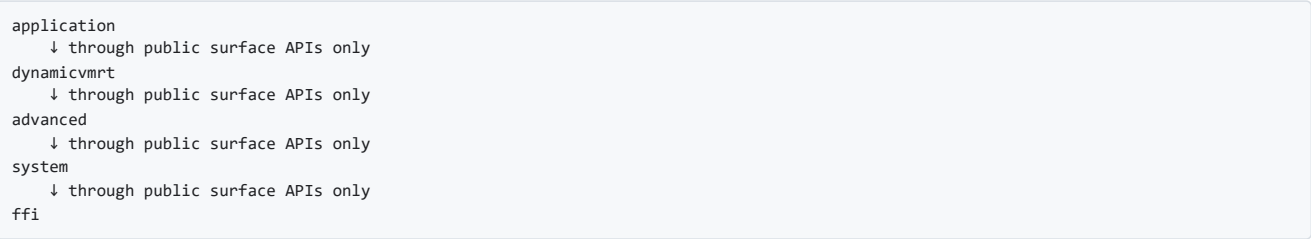
- classes
- modules
- interfaces and signatures
- overloading and overriding
- new operator definitions
- generics and templates
- macros
- reflection and dynamic runtime
- dependent types and type classes
- closures, currying, and advanced concurrency machinery

Public boundary:

- surface `cstruct` contracts
- C-ABI-safe built-in types
- C ABI value transfer

Dependency Direction

Allowed dependency flow is one-way:



A library may depend on a library at the same or a lower level only when the dependency does not create a cycle. Reverse dependencies are compiler errors.

Cross-Library Communication

From	To	Allowed?	Notes
<code>application</code>	<code>dynamicvmrt</code>	✓	projected public surface only
<code>application</code>	<code>advanced</code>	✓	projected public surface only
<code>application</code>	<code>system</code>	✓	projected public surface only
<code>application</code>	<code>ffi</code>	✓	projected public surface only
<code>dynamicvmrt</code>	<code>advanced</code>	✓	projected public surface only
<code>dynamicvmrt</code>	<code>system</code>	✓	projected public surface only
<code>dynamicvmrt</code>	<code>ffi</code>	✓	projected public surface only
<code>dynamicvmrt</code>	<code>application</code>	✗	reverse dependency
<code>advanced</code>	<code>system</code>	✓	projected public surface only
<code>advanced</code>	<code>ffi</code>	✓	projected public surface only

advanced	application	✗	reverse dependency
advanced	dynamicvmrt	✗	reverse dependency
system	ffi	✓	projected public surface only
system	application	✗	reverse dependency
system	dynamicvmrt	✗	reverse dependency
system	advanced	✗	reverse dependency
ffi	application	✗	reverse dependency
ffi	dynamicvmrt	✗	reverse dependency
ffi	advanced	✗	reverse dependency
ffi	system	✗	reverse dependency

Units in detail

A `co.lang.unit` is a stateless file-level declaration container. It is not instantiable and does not create an object, runtime scope, or public unit namespace.

FoLang has two unit-file forms:

1. **ordinary package unit** — `<Fragment>.unit.fol`
2. **struct companion unit** — `<StructName>.comp.unit.fol`

Both forms use `_ co.lang.unit`; explicit unit names are invalid.

Ordinary Package Units

Example package layout:

```
math/
├─ arithmetic.unit.fol
├─ comparison.unit.fol
└─ optional.unit.fol
```

```
// arithmetic.unit.fol
_ co.lang.unit = {
  abs(value co.lang.int)->(co.lang.int) = { ... }

  max(
    a co.lang.int,
    b co.lang.int
  )->(co.lang.int) = { ... }
}
```

```
// optional.unit.fol
_ co.lang.unit = {
  Option(T) co.lang.type =
    Some(T) | None();

  isSome(value Option(co.lang.int))->(co.lang.bool) = {
    ...
  }
}
```

All declarations are contributed directly to the `math` package namespace:

```
math.abs(-10);
math.max(10, 20);
value math.Option(co.lang.int);
```

The unit filenames never appear in qualified names:

```
math.Arithmetic.abs(-10); // compiler error
math.Optional.Option(co.lang.int); // compiler error
```

Within the same package, the functions and types may be referenced without the package prefix according to ordinary package name-resolution rules.

An ordinary unit may contain:

- receiverless functions
- `co.lang.type` aliases and ADT/type-constructor declarations
- newtype and opaque-type declarations
- subtype and supertype declarations

- macros and template declarations
- other non-instantiable type declarations explicitly permitted by their sections
- public, package, protected, or private members

An ordinary unit may not contain:

- fields or mutable unit state
- executable loose statements
- classes, structs, cstructs, enums, unions, modules, interfaces, or signatures
- lifecycle declarations for the unit
- a declaration that creates an independent nested primary namespace

A unit file may group closely related declarations or merely organize a convenient set of package-level functions and types. Semantic cohesion is recommended but not enforced.

Unused-Symbol Validation for Units

FoLang validates ordinary units at symbol granularity. Importing the package or using one declaration from a unit does not make the other declarations in that unit used.

For every ordinary `<Fragment>.unit.fol`, every usage-checkable declaration originating from that unit must independently participate in the live semantic graph. This includes:

- receiverless functions
- aliases
- `co.lang.type` declarations
- ADT/type-constructor declarations
- newtypes
- opaque types
- subtypes and supertypes
- macros and templates
- every other declaration kind that the unit grammar exposes as a package symbol

For example:

```
utility.unit.fol
├─ Id alias           USED
├─ Result type        USED
├─ parse function     USED
└─ format function    UNUSED -> compile-time error
```

The unit file is not considered wholly used merely because `Id`, `Result`, or `parse` is used. Every usage-checkable symbol is validated independently.

Fields do not exist in ordinary units and therefore do not participate in this rule.

Package-Namespace Consolidation

All ordinary unit files in one package are shallow-parsed and consolidated into one package symbol table. Their order and filenames have no semantic effect.

```
package members =
  filename-derived primary declarations
+ declarations from every ordinary unit file
```

A duplicate is reported as soon as the package index is built. Conflicts include:

- the same type or value name contributed by two ordinary units
- an ordinary-unit member colliding with a filename-derived primary declaration
- duplicate function signatures not permitted by overload rules
- declarations that normalize to the same canonical package symbol

Diagnostics should identify both source files.

Companion Units

A companion unit is governed by the filename and owner-validation rules in [Struct Companion Units](#). Its members do not merge directly into the package namespace; they attach to the owner struct's companion namespace.

The owner struct is validated for use as an ordinary filename-backed primary declaration. If the struct has a companion unit, every user-declared function in that companion unit must additionally be used independently. Use of the owner struct does not implicitly use any companion function, and use of one companion function does not use its siblings.

Employee struct	USED
Employee.comp.unit.fol	
├ calculateTax()	USED
├ serialize()	USED
└ debugDump()	UNUSED -> compile-time error

Fields of the owner struct are excluded from unused-symbol validation.

Units have no fields, identity, instances, inheritance, polymorphic dispatch, or independent construction.

CStructs

`co.lang.cstruct` is a C-like value type — passed by value, simple memory layout, safe to cross zone boundaries. Unlike `co.lang.struct` which is passed by reference, `co.lang.cstruct` is always copied on pass.

```
// Point.fol
_ co.lang.cstruct = {
  x co.lang.int;
  y co.lang.int;
}

// Rect.fol
_ co.lang.cstruct = {
  origin Point;
  width  co.lang.int;
  height co.lang.int;
}
```

cstruct Rules

always passed by value — never by reference
 simple memory layout — no metadata
 can contain only simple types and other cstructs

cannot contain <code>co.lang.struct</code>	✗	has metadata
cannot contain <code>co.lang.string</code>	✗	heap allocated
cannot contain <code>co.lang.dynamic</code>	✗	runtime type info
cannot contain classes	✗	vtable, metadata
cannot contain modules	✗	
cannot contain any heap allocated type	✗	
cannot have methods		
cannot have associated functions		
cannot embed <code>co.lang.struct</code>		
safe to cross direct ABI and zone boundaries		

allowed field types:

<code>co.lang.int</code> , <code>co.lang.uint</code> , <code>co.lang.float</code>	✓	primitives
<code>co.lang.bool</code> , <code>co.lang.char</code> , <code>co.lang.byte</code>	✓	primitives
<code>co.lang.int->[N]</code>	✓	fixed size arrays
<code>co.lang.cstruct</code>	✓	other cstructs

Packed cstruct — no padding, exact memory layout

Used for hardware registers, binary protocols, exact memory mapped formats:

```
// Register.fol
@co.dap.packed
_ co.lang.cstruct = {
  flags  co.lang.uint8;
  status co.lang.uint8;
  data   co.lang.uint16;
}
```

SIMD cstruct — aligned for vector operations

Used for math, graphics, signal processing:

```
// Vec4.fol
@co.dap.simd(aligned=16)
_ co.lang.cstruct = {
  x co.lang.float;
  y co.lang.float;
  z co.lang.float;
  w co.lang.float;
}
```

Both together

```
// AVXVec.fol
@co.dap.packed
@co.dap.simd(align=32)
_ co.lang.cstruct = {
    data co.lang.float;
}
```

`@co.dap.packed` and `@co.dap.simd` are specialisations of `co.lang.cstruct` — same rules, same zone boundary safety. They are not separate types.

Structs

```
// myStruct.fol
_ co.lang.struct={
    field1 co.lang.int;
    field2 co.lang.string;
    field3 co.lang.bool;
}
```

Struct Rules

```
structs cannot extend other structs
structs cannot contain methods directly
structs can compose other structs
structs cannot have default values to fields/members
structs can embed other structs (Go lang like)
structs can have a same-package companion file named `<StructName>.comp.unit.fol`
```

The struct declaration remains pure data. Associated behaviour is declared separately in `<StructName>.comp.unit.fol`.

Struct Embedding

Embedding promotes fields of an embedded struct directly into the outer struct — they act as the outer struct's own fields at construction and access sites. This is distinct from composition where the embedded struct is a named field.

```
// E.fol
_ co.lang.struct = {
    id co.lang.int;
    name co.lang.string;
}

// ✅ No conflict – id and name promoted as B's own fields
// B.fol
_ co.lang.struct = {
    age co.lang.float;
    E; // embedded – id and name promoted
}

b := B{ age: 25.0, id: 1, name: "Rao" }; // all fields at same level
b.id // direct – no b.E.id needed
b.name // direct – no b.E.name needed
b.age // direct
```

```
// ❌ Compiler error – name conflict between B.name and E.name
// B.fol
_ co.lang.struct = {
    name co.lang.string; // conflicts with E.name
    E;
    age co.lang.float;
}
// Fix 1 – rename B's conflicting field
// Fix 2 – use explicit composition instead: e E;
```

```
// Explicit composition – no promotion, always qualified access
// B.fol
_ co.lang.struct = {
    name co.lang.string;
    e E; // named field – no conflict, no promotion
    age co.lang.float;
}

b.name ; // B's own name
b.e.id ; // E's id – always explicit
b.e.name; // E's name – always explicit
```

Embedding Rules

Situation	Behavior
Embedded field, no conflict	Promoted — acts as the outer struct’s own field
Embedded field, name conflict with outer	✗ Compiler error — rename or use composition
Multiple embeds, no conflict between them	All fields promoted
Multiple embeds, conflict between embedded structs	✗ Compiler error
Explicit composition (<code>e E</code>)	No promotion — always accessed via <code>b.e.field</code>

FoLang does **not** silently shadow conflicting fields. Any name conflict is a compiler error — the programmer must make a conscious decision to rename or switch to explicit composition.

Struct Declaration Relationships

A struct cannot physically declare another struct, enum, class, module, function, signature, interface, or other named declaration inside its body. A struct body remains a pure data declaration containing fields and permitted embeddings only.

Use one of the following instead:

- **composition** through a named field
- **embedding** where the declaration kind’s embedding rules permit it
- a separately declared target-local declaration restricted with `@co.dap.local`

```
// EmployeeAddress.fol
@co.dap.local(for=hr.employee.Employee)
_ co.lang.struct = {
  _ street co.lang.string;
  _ city  co.lang.string;
}
```

```
// Employee.fol
_ co.lang.struct = {
  _ id      co.lang.int;
  _ name    co.lang.string;
  _ address EmployeeAddress; // composition
}
```

The following physical nesting is invalid:

```
// Employee.fol
_ co.lang.struct = {
  _ Address co.lang.struct = { // ✗ nested declaration
    _ city co.lang.string;
  }

  _ address Address;
}
```

structs cannot declare inner structs	✗ compiler error – only through <code>@co.dap.local</code>
structs can declare inner enums/ADTs	✗ compiler error – only through <code>@co.dap.local</code> or <code>@co.dap.nested</code>
structs cannot declare inner classes	✗ compiler error – struct is pure data
structs cannot declare inner modules	✗ compiler error – struct is pure data

`@co.dap.local` controls declaration visibility only. It does not automatically compose or embed the annotated declaration into the target struct.

Struct Companion Units

A struct companion unit is declared in a file named `<StructName>.comp.unit.fol`. The matching struct remains in `<StructName>.fol`, and both files must belong to the same package.

```
// Vector.fol
_ co.lang.struct = {
  _ x co.lang.float;
  _ y co.lang.float;
}
```

```
// Vector.comp.unit.fol
_ co.lang.unit = {
  distance(
    left Vector,
    right Vector
  )->(co.lang.float) = {
    ...
  }

  zero()->(Vector) = {
    this.return Vector{x: 0.0, y: 0.0};
  }

  (value Vector) magnitude()->(co.lang.float) = {
    this.return co.math.sqrt(
      value.x * value.x + value.y * value.y
    );
  }

  (Vector) create(
    x co.lang.float,
    y co.lang.float
  )->(Vector) = {
    this.return Vector{x: x, y: y};
  }
}
```

The filename establishes ownership. No companion member needs an ordinary parameter merely to prove association.

Call forms:

```
d      := Vector.distance(first, second);
origin := Vector.zero();
length := value.magnitude();
point  := Vector.create(10.0, 20.0);
```

A companion unit may declare:

- receiverless companion functions, including zero-parameter functions and factories
- instance-associated functions with an explicit value receiver
- type-associated functions with an explicit type receiver
- operator functions permitted for the owner struct
- non-UDT type declarations associated with the owner when their own sections permit them

```
receiverless non-operator companion function
-> has no explicit receiver
-> ordinary parameters may have any legal types
-> no owner-typed parameter is required merely for association

instance-associated function
-> explicit receiver is a value of the matching struct

type-associated function
-> explicit receiver is the matching struct type itself
```

These are the general companion rules. Operator functions have the additional operand requirement defined in the Operator Functions subsection.

Companion declarations do not make the struct a class and do not add object identity, inheritance, virtual dispatch, lifecycle methods, or unit-level state.

Filename and Owner Validation

The compiler classifies companion files before parsing:

```
Vector.comp.unit.fol -> canonical owner Vector
```

At package-indexing level, the compiler verifies:

- a canonical primary declaration named `Vector` exists in the same package
- no second companion file resolves to the same canonical owner
- the companion is not orphaned

After the owner declaration header is available, the compiler verifies that the owner is a `co.lang.struct`. A class, cstruct, enum, union, module, interface, signature, or imported declaration with the same short name cannot become the owner.

Receiver Validation

The companion filename determines the required receiver root. Any explicit receiver must match that owner.

```
// Employee.comp.unit.fol
- co.lang.unit = {
  create(id co.lang.int)->(Employee) = { ... } // valid: receiverless

  (emp Employee) isValid()->(co.lang.bool) = { ... } // valid

  (Employee) empty()->(Employee) = { ... } // valid

  (dept Department) isValid()->(co.lang.bool) = { ... }
  // compiler error: receiver Department does not match companion owner Employee

  (Department) create()->(Department) = { ... }
  // compiler error: type receiver Department does not match companion owner Employee
}
```

Ordinary parameters may have any legal types. They are not used to establish companion ownership.

For a generic struct owner, receiver validation compares the canonical root declaration and generic arity. For example, a receiver based on `Box(T)` may belong to `Box.comp.unit.fol`; `Box(T, E)` or an unrelated root does not.

Companion Namespace and Conflicts

Companion members are resolved through the owner:

```
employee := Employee.create(1);
valid    := employee.isValid();
```

The complete companion namespace is formed from all legal owner-associated declarations and the single companion file. Duplicate names and normalized duplicate signatures are compile-time errors.

The following are invalid:

```
Employee.comp.unit.fol without Employee.fol
Employee.comp.unit.fol and employee.comp.unit.fol in the same package
Employee.comp.unit.fol when Employee.fol declares a class or cstruct
an explicit receiver whose canonical root is not Employee
```

Operator Functions

Operator functions associated with a struct belong in its companion unit. Ownership comes from the companion filename; receiver validation follows the same rule as other companion functions.

```
// Employee.comp.unit.fol
- co.lang.unit = {
  @co.dap.operator(symbol="+")
  (emp Employee) add(other Employee)->(Employee) = {
    ...
  }

  @co.dap.operator(symbol=="")
  equals(
    left Employee,
    right Employee
  )->(co.lang.bool) = {
    ...
  }

  @co.dap.operator(symbol">")
  (Employee) greater(
    left Employee,
    right Employee
  )->(co.lang.bool) = {
    ...
  }
}
```

Companion ownership always comes from `<StructName>.comp.unit.fol`. An operator function has an additional operand rule: a value receiver already supplies the owner instance, but a receiverless operator function or a type-receiver operator function must declare the companion owner type as its first ordinary parameter. This allows operator lowering to pass the actual owner instance to the function. The compiler compares the resolved first-parameter type with the companion owner type at compile time. This requirement establishes the operator operand, not companion ownership. An operator declaration that does not operate on the owner struct is invalid.

Unions

Unions are untagged ADTs

```
// myUnion.fol
- co.lang.union={
  intValue co.lang.int;
  strValue co.lang.string;
}
```

Enums

```
// myEnum.fol
- co.lang.enum={
  Variant1,
  Variant2,
  Variant3
}
```

An enum value's constant expression may use a registered custom operator at any declared precedence. Runtime assignment is forbidden everywhere in that constant-expression subtree, including inside grouping, arguments, and collection or object elements. Whether the resulting expression can actually be evaluated at compile time—including whether a custom operator is foldable—is checked after parsing.

classes

```
// Employee.fol
- co.lang.class = {
  getEmployeeDetails()->(Employee) = empmodule.getEmployeeDetails;
  // assigning module function to class's method

  getEmployeeInfo()->(Employee) =>> empmodule.getEmployeeDetails();
  // delegating – internally redirecting the call to module function
}

// $1, $2, $3 ... are previous results captured as bind variables
// Emp.fol
- co.lang.class={
  dosomething(a co.lang.int, b co.lang.int)->(co.lang.int)=>>>somePack.someMethod(a)=>>>someOthPack.someOtherMeth($1, b);
}
```

Classes with Operator methods

```
// Employee.fol
- co.lang.class = {
  @co.dap.operator(symbol="+")
  add(other Employee)->(Employee) = {
    // implicit instance method: operands are this and other
  }

  @co.dap.operator(symbol=="")
  @co.dap.static
  equals(
    left Employee,
    right Employee
  )->(co.lang.bool) = {
    // static operator implementation
  }

  @co.dap.operator(symbol=">")
  @co.dap.class
  greater(
    left Employee,
    right Employee
  )->(co.lang.bool) = {
    // class-associated operator implementation
  }
}
```

An unannotated operator function in a class is an implicit instance method; the hidden `this` value contributes the first operand. `@co.dap.instance` is the explicit spelling of the same category. `@co.dap.static` and `@co.dap.class` do not contribute an implicit operand, so their declared parameters are the complete operator operand list. Operator signature normalization includes these method categories so equivalent declarations are diagnosed as duplicates.

Class Declaration Relationships

A class cannot physically contain named class, struct, enum, module, function, signature, interface, or other declaration definitions as nested declarations. Class bodies contain fields, lifecycle declarations, and methods permitted by the class model.

Types and helper declarations used only by one class are declared in their ordinary legal source locations and restricted to that class with `@co.dap.local`:


```
// EmployeeAddress.fol
@co.dap.local(for=hr.employee.Employee)
- co.lang.struct = {
    street co.lang.string;
    city   co.lang.string;
}
```

```
// EmployeeStatus.fol
@co.dap.local(for=hr.employee.Employee)
- co.lang.enum = {
    Active,
    Inactive,
    Pending
}
```

```
// Employee.fol
- co.lang.class = {
    address EmployeeAddress;
    status  EmployeeStatus;

    getAddress()->(EmployeeAddress) = {
        this.return this.address;
    }
}
```

The local declarations are visible while compiling `hr.employee.Employee`, but they do not become nested names such as `Employee.EmployeeAddress` or `Employee.EmployeeStatus`.

The following is invalid:

```
// Employee.fol
- co.lang.class = {
    Address co.lang.struct = { // ❌ physical nested declaration
        city co.lang.string;
    }
}
```

Ordinary visibility annotations do not widen a target-local declaration beyond the declaration or closed target set named by `@co.dap.local` or `@co.dap.nested`.

Method Types

```
// Employee.fol
- co.lang.class = {

    @co.dap.static
    getEmployee()->(Employee) = {}

    @co.dap.instance
    getEmployee()->(Employee) = {}

    @co.dap.class
    getEmployee()->(Employee) = {}

    @co.dap.object
    getEmployee()->(Employee) = {}
}

@co.dap.oops(
    A: { inherit:true, virtual:true },
    B: { implements:true },
    C: { inherits:true, abstract=true },
    D: { inherits:true },
    E: { uses:true },
    F: { composes:true },
    G: { extends:true },
    H: { with:true },
    I: { associate:true },
)

// test.fol
- co.lang.class = {
    getTest(id int)->(test) = {}
}
```

The @@new and @@init Methods

`@@new` and `@@init` are lifecycle methods — compiler-owned, not user-definable outside the class. `@@` signals they are restricted lifecycle symbols, not regular methods.

Lifecycle names are valid only as class members. A unit (including a struct companion unit), module, interface, signature, local block, or package declaration cannot declare a lifecycle-named function.

```
// Employee.fol
@co.dap.generic(type={T:{typename}, R:{typename}})
_ co.lang.class = {

    id T;
    name R;

    // @@new is provided by default even if not overridden.
    // Override only when you genuinely need to change allocation behavior.

    @co.dap.class
    @co.dap.private
    @@new()->(co.lang.uninit) = { self.return co.const.none }

    @co.dap.class
    @co.dap.public
    @@new(a co.lang.typevalue, b co.lang.typevalue)->(co.lang.uninit) = {
        // Manual type aliasing - @co.dap.generic handles this automatically
        // Override @@new only when you need custom allocation logic
        T co.lang.type = a;
        R co.lang.type = b;

        // self keyword is allowed only in class methods
        self.parent.new();

        // uninit instance method internally calls new and init
        self.return co.lang.uninit.instance(Employee, self);
    }

    @co.dap.override
    @co.dap.constructor(access=private)
    @@init() = {}

    @co.dap.override
    @co.dap.constructor(access=public)
    @@init(id T, name R) = {
        this.parent.init();
        this.id = id;
        this.name = name;
    }

    getEmployee(id T)->(Employee) = {}
}
```

Anonymous Classes/Types

```
emp := co.lang.class{};

empObj := emp.init();

empobj1 := co.lang.class{
    name string;
}.init();
```

Interfaces

```
// IEmployee.fol
_ co.lang.interface = {
    storeEmployee(emp Employee)->(Employee);
}
```

Signatures

```
// Employee is an ordinary package-level declaration.
// MEmployee.fol
_ co.lang.signature = {
    storeEmployee(emp Employee)->(Employee);
}
```

Structurally they look similar — both are lists of contracts. The difference is **who implements them and how**.

	<code>co.lang.signature</code>	<code>co.lang.interface</code>
Implemented by	module via <code>matches=</code>	class via <code>implements=[]</code>
Number of implementations	Any number of distinct modules may match one signature	Any number of classes may implement one interface
Runtime cardinality of one implementation declaration	One module object per loaded module identity	Any number of independently constructed class objects
State model	One shared state for all references to the same module	Separate per-instance state for each class object
May specify required values	✓	✗ — methods only
May reference package-level types	✓	✓
May declare abstract/fixed type components	✓	✗
May require generic type constructors	✓	✗
May declare physical nested/local types	✗	✗
May use <code>@co.dap.local</code>	✗	✗
Instantiation involved	Module is declared once, not constructed	Class objects are constructed
Reference use	Compatible module references may be used through the signature type without creating another module	Interface references may refer to any implementing object instance
OOP dispatch	✗	✓ virtual/dynamic
Contract style	module values, functions, and type components	behavioral methods on object instances
Practical analogy	singleton component contract	object-instance behavioral contract
Origin	ML/OCaml-inspired modules	Java/C#/Go interfaces

- A `signature` is a **module contract** over values, functions, existing package-level types, and abstract or fixed type components. A type-component specification is a contract slot, not a physical nested type definition. Multiple modules may match one signature, but each module declaration denotes one module object with shared module state.
- An `interface` is a **behavioral contract** tied to class dispatch and polymorphism. It cannot declare module type components or own nested type definitions. A class implementing an interface may create any number of independent runtime objects.
- The approximation `module + signature ≈ singleton object + interface` is useful for understanding cardinality and shared state, but a module is a language-level component rather than a class-based singleton pattern.

Modules

A module is an ML/OCaml-style abstraction governed by an optional signature. A module may use package-level types and may satisfy type components declared by its signature, but it does not physically own or nest arbitrary type declarations. A module should not be introduced merely to prevent functions from appearing loose in a file; use `co.lang.unit` for that simpler structural purpose.

```
// Employee.fol – ordinary package-level type
_ co.lang.struct = {
  Id co.lang.int;
  Name co.lang.string;
}

// EmployeeModule.fol
_ co.lang.signature = {
  getEmployee(id co.lang.int)->(Employee);
}

// EmployeeModImpl.fol
@co.dap.module(signature=EmployeeModule)
_ co.lang.module->(signature=EmployeeModule, matches=EmployeeModule) = {

  getEmployee(id co.lang.int)->(Employee) = {
    this.return Employee{
      Id: 10,
      Name: "Rao"
    };
  }
}

mm EmployeeModule = EmployeeModImpl;
v Employee = mm.getEmployee(10);
```

Module Unused-Symbol Rule

A module declaration becoming live does not implicitly make all of its callable members live. The module declaration itself must participate in the semantic graph, and every user-declared module function/method must independently be used. Module fields/state values are excluded from unused-symbol validation.

```

EmployeeModImpl module      USED
├─ shared state             not checked
├─ getEmployee()            USED
└─ debugDump()              UNUSED -> compile-time error

```

This rule applies whether the module is compiled as application source, inside a project-local source library, or inside a standalone library project.

Module Cardinality and Singleton Analogy

A module declaration defines exactly one named module object for that loaded module identity. It is not an instantiable blueprint and does not create a new module object each time its name is referenced.

```

first EmployeeModule = EmployeeModImpl;
second EmployeeModule = EmployeeModImpl;

```

`first`, `second`, and `EmployeeModImpl` refer to the same module object. These bindings copy or retain the module reference; they do not clone or instantiate the module. Consequently, values declared directly by the module represent one shared module state for all references to that module.

A signature does not itself create a module object. It is a contract, and any number of separately declared modules may conform to the same signature:

```

DatabaseBackend signature
├─ PostgreSQLBackend module -> one named module object
├─ MySQLBackend module     -> one named module object
└─ TestDatabaseBackend module -> one named module object

```

Each conforming module declaration contributes its own single module object and its own module state. The signature does not restrict the number of distinct conforming module declarations.

A useful analogy is:

```

signature      ≈ interface contract
conforming module ≈ singleton object implementing that contract
class          ≈ instantiable object type

```

The analogy is intentionally limited. A FoLang module is a compiler-recognized named component, not a class made singleton through a private constructor, static field, or runtime pattern. It cannot be repeatedly constructed. Because module references are first-class in FoLang, they may be bound and used through a compatible signature type, but every reference to the same module declaration still denotes the same module object.

Modules are also broader than ordinary interface implementations. A matching module may provide module values, functions, and abstract, fixed, or generic type-component bindings required by its signature. An interface constrains object behaviour; it does not provide the same module type-component abstraction.

By contrast, a class declaration defines an instantiable type. Every class construction creates a distinct object with independent identity, state, and lifetime:

```

PostgreSQLBackend module
├─ one shared module object and module state

PostgreSQLConnection class
├─ connection1 -> independent object and state
├─ connection2 -> independent object and state
└─ connection3 -> independent object and state

```

Formal mental model: A FoLang module is a single named implementation component that may conform to a signature. It is comparable to a singleton object implementing an interface, but it is not instantiated from a class. Multiple distinct modules may conform to the same signature, while each module declaration denotes one module object. Unlike an ordinary singleton-interface implementation, a module may also satisfy abstract, fixed, and generic type components required by its signature.

Module instantiation A FoLang class or struct declaration introduces an instantiable type but does not create an instance. A FoLang module declaration introduces one named module component directly into its package. The module name acts as the binding for that component, so no separate construction expression is required. The module's runtime state is initialized once according to the language's module-initialization rules.

A module declaration is a singleton component declaration and binding, rather than merely an object definition.

Module Signature Contents

A `co.lang.signature` is a declarative contract for a module. It may specify required module values, functions, and type components. A signature does not allocate storage, initialize variables, execute statements, or provide function bodies.

A signature may contain:

- value specifications
- function signatures
- references to already existing accessible types
- abstract type-component specifications

- fixed or manifest type-component specifications
- abstract generic type-constructor specifications

A signature may not contain:

- value initializers
- executable statements
- function bodies
- concrete class, struct, enum, module, interface, or signature definitions
- arbitrary nested or target-local declarations

Type-component specifications are part of module conformance. They are not Java-, C++-, or C#-style inner types and do not participate in

`@co.dap.local`.

Value Specifications

A declaration such as:

```
// Counter.fol
_ co.lang.signature = {
  count co.lang.int;
  increment(amount co.lang.int)->();
}
```

requires a matching module to provide a value named `count` of type `co.lang.int` and a compatible `increment` function. The signature does not initialize `count` and does not define `co.lang.int`; the built-in type already exists.

```
// CounterImpl.fol
_ co.lang.module->(
  signature=Counter,
  matches=Counter
) = {
  count co.lang.int = 0;

  increment(amount co.lang.int)->() = {
    count.value = count + amount;
  }
}
```

The same rule applies when a value or function specification uses an existing accessible package type:

```
// EmployeeRepository.fol
_ co.lang.signature = {
  current hr.employee.Employee;
  find(id co.lang.int)->(hr.employee.Employee);
}
```

The matching module must provide `current` and `find`. It does not redefine `hr.employee.Employee`.

Abstract Type Components

An abstract type component declares that every matching module must supply a type binding for that component:

```
// Repository.fol
_ co.lang.signature = {
  Entity co.lang.type;

  current Entity;
  find(id co.lang.int)->(Entity);
}
```

`Entity co.lang.type;` does not define the representation of `Entity`. It defines a required module type component. A matching module binds it to a compatible existing type:

```
// EmployeeRepositoryImpl.fol
_ co.lang.module->(
  signature=Repository,
  matches=Repository
) = {
  Entity co.lang.type = hr.employee.Employee;

  current Entity = ...;
  find(id co.lang.int)->(Entity) = { ... }
}
```

Within a matching module, `Entity co.lang.type = ...` is a **signature-component binding**, not an arbitrary nested type declaration. Its name must correspond to an abstract type component declared by the matched signature. A module cannot use this form to introduce unrelated module-local types.

An abstract type component differs from `forward` and `extern` declarations:

```

abstract signature type component
  -> each matching module supplies its own compatible type binding

forward declaration
  -> one specific declaration is completed later

extern declaration
  -> one specific declaration is defined in another linkage or compilation unit

```

Fixed or Manifest Type Components

A signature may fix a type component to an already known type:

```

// IntegerRepository.fol
_ co.lang.signature = {
  Id co.lang.type = co.lang.int;

  find(id Id)->(co.lang.bool);
}

```

Here `Id` is predetermined as `co.lang.int`. A matching module uses that type component but does not choose or redefine it.

```

Entity co.lang.type;          -> abstract; implementor supplies the binding
Id co.lang.type = co.lang.int; -> fixed; signature supplies the type equality

```

Abstract Generic Type Constructors

A signature may require a generic type constructor without defining its representation:

```

// StackSignature.fol
_ co.lang.signature = {
  Stack(T) co.lang.type;

  empty(T)->(Stack(T));
  push(value T, stack Stack(T))->(Stack(T));
  pop(stack Stack(T))->(T, Stack(T));
}

```

`Stack(T) co.lang.type;` declares an **abstract generic type component**, also described as an abstract type constructor of arity one. The signature specifies that `Stack` accepts one type argument, but it does not define what `Stack(T)` is.

A matching module must provide a compatible type-constructor binding with the same name, arity, and declared constraints:

```

// ListStackModule.fol
_ co.lang.module->(
  signature=StackSignature,
  matches=StackSignature
) = {
  Stack(T) co.lang.type = co.core.list(T);

  empty(T)->(Stack(T)) = { ... }
  push(value T, stack Stack(T))->(Stack(T)) = { ... }
  pop(stack Stack(T))->(T, Stack(T)) = { ... }
}

```

Another matching module may choose another representation:

```

// ArrayStackModule.fol
_ co.lang.module->(
  signature=StackSignature,
  matches=StackSignature
) = {
  Stack(T) co.lang.type = collections.ArrayStack(T);
  ...
}

```

Therefore:

```

StackSignature
  -> requires a generic type constructor Stack(T)

ListStackModule
  -> binds Stack(T) to co.core.list(T)

ArrayStackModule
  -> binds Stack(T) to collections.ArrayStack(T)

```

A signature-component binding does not permit physical type nesting. When an implementation needs a new concrete struct, class, or enum representation, that declaration remains an ordinary package declaration and may be restricted to the implementing module with `@co.dap.local`; the module then binds the signature component to that declaration.

Signature Conformance Rules for Type Components

For every type component in a matched signature:

- an abstract component must receive exactly one compatible module binding
- a fixed component must retain the type equality declared by the signature
- a generic component binding must preserve generic arity, parameter kinds, bounds, variance, and other declared constraints
- component names must be unique within the signature
- component bindings cannot contain executable code
- extra module-local type bindings that do not correspond to signature components are compiler errors
- types referenced by value and function specifications must resolve after applying the module's type-component bindings

Module Declaration Relationships

A module cannot physically declare nested structs, enums, classes, modules, signatures, interfaces, or other arbitrary named declarations. It references ordinary package-level declarations through its functions and signature. The only type-like declarations permitted directly in a matching module are signature-component bindings that satisfy abstract type components declared by its matched signature; such bindings do not create independent nested declarations.

A declaration intended only for one module may be restricted with `@co.dap.local`:

```
// EmployeeModuleConfig.fol
@co.dap.local(for=hr.employee.EmployeeModImpl)
_ co.lang.struct = {
  timeout co.lang.int;
  retries co.lang.int;
}
```

```
// EmployeeModImpl.fol
_ co.lang.module = {
  connect(cfg EmployeeModuleConfig)->(co.lang.bool) = {
    ...
  }
}
```

The following remains invalid:

```
// EmployeeModImpl.fol
_ co.lang.module = {
  Config co.lang.struct = { // ❌ physical nested declaration
    timeout co.lang.int;
  }
}
```

A target-local declaration does not automatically become a module member name and is not projected through the module's signature. It becomes part of the signature view only when an explicit signature type component is bound to it or a signature value/function specification references it through an allowed type component.

Structs vs Classes vs Modules vs Units vs Packages

	Struct	CStruct	Class	Module	Unit	Package
Purpose	Pure data shape	C-like value type	Behaviour + data	Signature-backed ML-style abstraction	Stateless package-fragment or struct-companion container	Folder-based grouping
Fields	✅	✅ simple only	✅ per instance	❌	❌	❌
Module-level values	❌	❌	❌	✅ when declared directly or required by a signature	❌	❌
Functions / methods	Companion functions through <code><StructName>.comp.unit.fol</code> ; explicit receivers must match the struct	❌	✅ methods	✅ module functions	✅ package functions in ordinary units; companion functions in companion units	❌
Lifecycle (<code>@@new</code> / <code>@@init</code>)	❌	❌	✅	❌	❌	❌

<code>this</code> / <code>self</code>	✗	✗	✓	✗	✗	✗
Value/literal construction	✓	✓	✗	✗	✗	✗
Lifecycle instantiation (<code>new</code> / <code>init</code>)	✗	✗	✓ multiple objects	✗ — one module object per declaration	✗	✗
Runtime state cardinality	Per bound struct object	Per value	Per class object	One shared state for the module declaration	—	—
First class	✓	✓	✓	✓ module reference; referencing does not instantiate	✗	✗
Pass by	Reference	Value	Reference	Reference to the same module object	—	—
Contract	—	—	<code>interface</code> via <code>implements=[]</code>	<code>signature</code> via <code>matches=</code>	none	—
OOP / inheritance	✗	✗	✓	✗	✗	✗
Physically nested independent named type/container declarations	✗	✗	✗	✗	✗	N/A — packages contain separate source declarations
May be an <code>@co.dap.local</code> target	✓	✗	✓	✓	✗	✗
Pattern matching	✓	✓	✓	✗	✗	✗
Direct ABI / zone boundary safe	✗ — library boundaries require snapshots	✓	✗	✗	✗	✗
Associated functions	✓ through <code><StructName>.comp.unit.fol</code>	✗	—	—	✓ only in a struct companion unit	✗
Embedding	✓	✗	—	—	✗	✗
Declared with	<code>co.lang.struct</code>	<code>co.lang.cstruct</code>	<code>co.lang.class</code>	<code>co.lang.module</code>	<code>co.lang.unit</code>	folder path
C++ backend analogy	struct without methods	plain C struct	class	struct/class abstraction	package namespace fragment or static companion scope	namespace
Closest mental model	Rust struct	C struct	Java/C# class	singleton implementation component with ML-style type members	source fragment merged into a package, or a filename-bound struct companion	filesystem namespace

Mental model:

```

reach for struct → pure data; use `<StructName>.comp.unit.fol` for associated behaviour
reach for cstruct → physical ABI-compatible value data crossing direct zone or native boundaries
reach for class → behaviour, lifecycle, multiple instances
reach for module type components → one named implementation component with shared state, governed by an optional signature and capable of satisfying
reach for unit → package fragment (`*.unit.fol`) or struct companion (`*.comp.unit.fol`)
reach for package → folder-based grouping only, not a value

```

Declaration scoping rule: FoLang does not permit physical nesting of independent file-backed primary declarations. Classes, structs, cstructs, enums, unions, modules, interfaces, signatures, instances, matchers, and other package-owned primary declarations remain in their own `<Name>.fol` files. Ordinary and companion unit files are explicit package containers: they may contain functions and the non-UDT type declarations permitted by the unit rules, but they may not contain independent primary declarations such as classes, structs, enums, modules, interfaces, or signatures. Ordinary local functions and anonymous expressions remain the other explicit nesting exceptions. Supported package declarations may restrict visibility to exact same-package targets with `@co.dap.local`; the annotation changes visibility, not physical ownership. Signature type components and matching module bindings are contract slots rather than arbitrary nested package declarations. ***

Local and/or Nested types and functions

FoLang does not provide Java-, C++-, or C#-style physical nesting of independent named type and container declarations. Such declarations remain in their ordinary legal source locations. Ordinary local functions and anonymous expressions are explicit exceptions governed by the rules below.

Physical Nesting Rules

Prohibited Independent Named Declarations

Independent file-backed primary declarations cannot be physically declared inside another class, struct, cstruct, enum, union, module, unit, interface, signature, function, or executable block. This includes:

- classes, structs, cstructs, enums, and unions;
- modules, interfaces, signatures, and additional units;
- instances, matchers and other file-backed primary declarations.

Non-UDT type declarations are the deliberate unit exception. macros, templates, decorators , Type aliases, `co.lang.type` ADTs and type constructors, newtypes, opaque types, subtypes, and supertypes may be declared directly inside an ordinary unit, and inside a companion unit where their own rules permit association with the owner. They are not permitted loose at package-file scope or physically inside classes, structs, modules, functions, or executable blocks unless another section explicitly grants that context.

File-backed primary declarations retain package-owned identity and follow their normal `<Name>.fo1` placement rules. An association or visibility annotation such as `@co.dap.local`, `@co.dap.nested`, or `@co.dap.inner` does not physically move a separately declared declaration inside its target.

Local-Function Exception

A function body may contain an ordinary named local or inner function where the grammar permits a local-function declaration:

```
outer()->() = {
  value co.lang.int = 10;

  inner()->() = {
    co.out.println(value);
  }

  inner();
}
```

The local function has block-local declaration identity. Its free runtime names are resolved from its lexical declaration context, and its lifetime and escape behavior follow the ordinary inner-function rules. It is not a package member and cannot be independently imported or exported.

This exception permits local functions only. It does not permit a named class, struct, enum, module, unit, interface, signature, or another named type/container declaration inside a function body.

Anonymous-Expression Exception

The physical-nesting restriction does not apply to anonymous constructs that are expressions or type expressions rather than independent named declarations. They may be nested wherever their specific grammar category is permitted.

These include:

- anonymous function expressions;
- lambdas and callback blocks;
- anonymous class or anonymous type expressions;
- permitted anonymous polymorphic type expressions introduced by `forall`;
- ordinary nested block, object-construction, map, collection, and other value-producing expressions.

```
process()->() = {
  operation := (value co.lang.int)->(co.lang.int) = {
    this.return value * 2;
  };

  worker := co.lang.class {
    run(value co.lang.int)->(co.lang.int) = {
      this.return operation(value);
    }
  }.init();
}

transformer co.lang.type = forall(T).(T)->(T);
```

An anonymous construct has no independently addressable package declaration identity. Its scope, capture, lifetime, type, and escape behavior are determined by the rules for that specific construct. Syntactic containment of an anonymous expression does not create a Java-, C++-, or C#-style named nested declaration and does not violate the one-primary-declaration-per-package-file rule.

Relationship to Association Annotations

The exceptions above are distinct from FoLang's association annotations:

- an ordinary local function is physically declared inside an enclosing function and is lexically scoped;

- an anonymous construct is an expression without independent declaration identity;
- `@co.dap.local`, `@co.dap.nested`, and `@co.dap.inner` apply to separately declared declarations that retain their normal source identity.

The remainder of this section defines those separately declared association forms.

```
@co.dap.local(for=<declaration-reference>)
```

or:

```
@co.dap.local(
  for=[
    <declaration-reference>,
    <declaration-reference>
  ]
)
```

The annotated declaration is called a **target-local declaration**. The declarations named by `for` form its **local target set**.

Supported Declaration and Target Kinds

`@co.dap.local` may be applied only to these declaration kinds:

- `co.lang.class`
- `co.lang.struct`
- `co.lang.enum`
- `co.lang.module`
- a named function declared in a context that normally permits functions

Every entry in the local target set must resolve to exactly one:

- class
- struct
- enum
- module
- function overload

A target list may contain any combination of supported target kinds.

Signatures and interfaces do not support target-local or physically nested declarations. A `co.lang.signature` or `co.lang.interface` cannot be annotated with `@co.dap.local` and cannot appear in a local target set. A signature may declare abstract, fixed, or generic **type-component specifications** as part of its module contract; these are contract requirements rather than local or nested type definitions. Interfaces do not declare signature type components.

Other declaration kinds are not target-local unless a later section explicitly permits them.

Target-Set Syntax

The `for` argument accepts either:

1. one declaration reference; or
2. a non-empty list of declaration references.

The scalar form is equivalent to a singleton target list:

```
@co.dap.local(for=hr.employee.Employee)
```

```
@co.dap.local(for=[hr.employee.Employee])
```

The scalar form is canonical when only one target is required. Use the list form when two or more declarations require access:

```
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService,
    hr.employee.EmployeeValidator
  ]
)
// EmployeeState.fol
_ co.lang.enum = {
  Active,
  Inactive
}
```

Rules:

- the target list must contain at least one declaration reference
- each reference must resolve independently and unambiguously
- duplicate references to the same resolved declaration are a compiler error

- list order does not affect visibility or declaration identity
- the target list is a closed set; access is not granted to declarations omitted from it
- repeated `@co.dap.local` annotations are not used to accumulate targets; use one annotation with one complete target list

Invalid:

```
// State.fol
@co.dap.local(for=[])
_ co.lang.struct = { ... }
// ❌ empty target list
```

```
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.Employee
  ]
)
// State.fol
_ co.lang.struct = { ... }
// ❌ duplicate resolved target
```

Declaration-Reference Syntax

Each `for` entry is a compiler-resolved declaration reference, not a string, runtime expression, function call, or `co.meta.symbol` value.

For a non-function target, use only its complete qualified declaration name:

```
@co.dap.local(for=hr.employee.Employee)
@co.dap.local(for=hr.employee.EmployeeStatus)
@co.dap.local(for=hr.employee.EmployeeRules)
```

For a function target, use its complete qualified function signature because FoLang permits overloads:

```
@co.dap.local(
  for=hr.employee.Employee.calculate(co.lang.decimal)->()
)
```

Function references in a target list follow the same rule:

```
@co.dap.local(
  for=[
    hr.employee.Employee.calculate(co.lang.decimal)->(),
    hr.employee.Employee.calculate(co.lang.int)->(),
    hr.employee.Employee.validate()->(co.lang.bool)
  ]
)
// CalculationState.fol
_ co.lang.struct = { ... }
```

Parameter names are not part of the reference:

```
// ✅ canonical
hr.employee.Employee.calculate(co.lang.decimal)->()

// ❌ parameter names are not declaration identity
hr.employee.Employee.calculate(amount co.lang.decimal)->()
```

The parameter and return types must match one exact overload. An abbreviated function name, unresolved target, or ambiguous overload is a compiler error.

```
@co.dap.local(for=hr.employee.Employee.calculate)
// ❌ ambiguous when calculate is overloaded
```

Each listed overload is an independent target. Listing one overload does not grant access to sibling overloads with the same function name.

Same-Package Requirement

The target-local declaration and **every declaration in its local target set** must belong to the same exact package. The compiler compares their complete folder-derived package identities; matching only a parent package, package family, import alias, library, or source root is not sufficient.

For a function target, the target package is the package that owns the function's enclosing class, struct companion unit, module, unit, or other legal function container. A target-local function is checked in the same way: the package containing its legal function-owning declaration must match the package of every listed target.

The invariant is:

```
package(target-local declaration)
== package(target 1)
== package(target 2)
== ...
```

```
// hr/employee/Employee.fol
_ co.lang.class = { ... }

// hr/employee/EmployeeService.fol
_ co.lang.class = { ... }

// hr/employee/EmployeeState.fol
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
// EmployeeState.fol
_ co.lang.enum = { Active, Inactive }
// ✓ all declarations belong to package hr.employee
```

A declaration in another package cannot participate in the local target set, even when ordinary package visibility, imports, aliases, parent/subpackage relationships, or library membership would otherwise make the declaration resolvable.

```
// EmployeeState is declared in package hr.internal
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.internal.EmployeeService
  ]
)
// EmployeeState.fol
_ co.lang.enum = { Active, Inactive }
// ✗ local declaration and every target must have the same package
```

```
// CalculationState is declared in package hr.employee.internal
@co.dap.local(
  for=hr.employee.Employee.calculate(co.lang.decimal)->()
)
// CalculationState.fol
_ co.lang.struct = { ... }
// ✗ subpackages are distinct packages; both sides must be exactly hr.employee
```

The same-package rule prevents `@co.dap.local` from becoming a cross-package friendship or visibility-bypass mechanism. A local declaration is a selectively shared implementation detail inside one package, not an imported helper attached from elsewhere.

Visibility Domain

A target-local declaration may be resolved only from:

1. each exact declaration in its local target set; and
2. lexical scopes contained within each listed target's implementation.

For targets `A`, `B`, and `C`, the visibility domain is the union of their implementation scopes:

```
visibility(local declaration)
= scope(A) ∪ scope(B) ∪ scope(C)
```

It is not an intersection, and code does not need to be simultaneously inside every target.

Consequences:

- a declaration local to a class is available to that class body and its methods
- a declaration local to a module is available to that module body and its functions
- a declaration local to a struct is available while resolving that struct declaration
- a declaration local to an enum is available while resolving that enum declaration
- a declaration local to a function is available only in the body of the exact targeted overload and its nested statement scopes
- when several targets are listed, each listed target receives access independently
- sibling declarations and sibling overloads not listed cannot resolve it
- subclasses, companion units, extensions, callees, callers, and related declarations do not gain access automatically
- visibility is not inherited, transitive, or propagated through calls, composition, embedding, imports, or other local declarations
- it cannot be imported, exported, or projected through a library surface

```

@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
// EmployeeState.fol
_ co.lang.enum = {
  Active,
  Inactive
}

// Employee.fol
_ co.lang.class = {
  state EmployeeState; // ✅ listed target
}

// EmployeeService.fol
_ co.lang.class = {
  state EmployeeState; // ✅ listed target
}

// Payroll.fol
_ co.lang.class = {
  state EmployeeState; // ❌ not listed
}

```

Interaction with Ordinary Visibility

`@co.dap.local` establishes the final visibility boundary. Ordinary visibility annotations such as `@co.dap.public`, `@co.dap.package`, `@co.dap.protected`, or `@co.dap.private` cannot widen the declaration beyond its closed local target set.

```

@co.dap.public
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
// EmployeeState.fol
_ co.lang.enum = { Active, Inactive }

```

`EmployeeState` is still visible only to the listed targets. The ordinary visibility annotation is redundant for external resolution and may produce a compiler warning, but it never overrides `@co.dap.local`.

When most or all declarations in a package require access, ordinary package visibility should be used instead of maintaining a large target list.

Source Placement and Identity

A target-local declaration remains in the source position normally required for its declaration kind:

- a class, struct, enum, or module remains a normal primary declaration and follows the one-primary-declaration-per-package-file rule
- a function remains inside a unit, class, module, or another declaration context that normally permits functions
- `@co.dap.local` does not permit a free function to appear loose at package-file scope

The declaration retains a stable package-owned compiler identity. Its package identity must be exactly the same as the package identity of every target. It does not become physically nested and is not addressed as `Target.LocalName`.

```

hr.employee.EmployeeState
  local-for [
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]

```

The normalized local target set is declaration metadata. Target-list order does not create a different declaration identity.

No Implicit Relationship or Privilege

`@co.dap.local` changes visibility only. It does not imply:

- composition
- embedding
- inheritance
- lifecycle ownership
- memory ownership
- friendship or privileged private-member access
- automatic membership in any listed target
- shared runtime state among the targets

Composition must still be written explicitly through a field or another supported relation. Embedding remains a separate facility: only struct and enum declarations are eligible for embedding, and each use must follow the embedding rules defined for that declaration kind.

Escape Restrictions

A target-local type must not leak outside its complete visibility domain.

It cannot appear in:

- a public, package, or protected API visible outside the local target set
- a library surface
- the parameter or return types of a function to which it is local
- a field or method signature that exposes it beyond the local target set
- an exported generic specialization or type alias

A value whose static type is target-local must be converted to an externally visible type before leaving the union of the listed targets' visibility domains.

Usage

```
@co.dap.public
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
// EmployeeState.fol
_ co.lang.enum = { Active, Inactive }

// Employee.fol
_ co.lang.struct={

  state EmployeeState;
}
```

Invalid Physical Nesting

```
// Employee.fol
_ co.lang.class = {
  Address co.lang.struct = { ... } // ❌
}

// EmployeeModule.fol
_ co.lang.module = {
  Config co.lang.struct = { ... } // ❌
}

process()->() = {
  State co.lang.enum = { Ready, Done } // ❌ named type declaration
}

// EmployeeContract.fol
_ co.lang.signature = {
  Employee co.lang.struct; // ❌
}

// EmployeeApi.fol
_ co.lang.interface = {
  Result co.lang.struct; // ❌
}
```

The `process` example rejects the named enum declaration; it does not reject ordinary local functions or anonymous expressions. Those are permitted only under the explicit exceptions in [Physical Nesting Rules](#).

Use separately declared package declarations, composition, embedding where allowed, and `@co.dap.local` when a closed set of declarations requires selective access.

There is another annotation `@co.dap.nested` which is similar to local but captures target state.

`@co.dap.nested(target=hr.emp.Employee)`

Instead of `for` we use `target` and `target` is always a single fully qualified type/function.

`@co.dap.nested` behaves exactly like nested or inner classes or functions

Comparison Table

Annotation	attribute	multiple targets	capture the target state
@co.dap.local	for	✅ as list for single target can mention without list syntax	❌

@co.dap.nested	target	✗	✓
----------------	--------	---	---

@co.dap.inner Declarations

`@co.dap.inner` is an association annotation. It does **not** create a physically nested type, method, or function. The annotated declaration remains in its ordinary legal source location, retains its package-owned identity, and must satisfy the same exact-package rule that applies to `@co.dap.local` and `@co.dap.nested`.

Unlike `@co.dap.local` and `@co.dap.nested`, `@co.dap.inner` does not contain a `for` or `target` attribute. Its target relationship is established at the use site by explicitly embedding or attaching the declaration to a legal target. An `@co.dap.inner` declaration cannot be used as a standalone declaration.

Usage

```
@co.dap.public
// EmployeeState.fol
@co.dap.inner
_ co.lang.enum = {
  Active,
  Inactive
}

// Employee.fol
_ co.lang.struct = {
  EmployeeState;
  state EmployeeState;
}
```

The use of `EmployeeState;` establishes the inner association for this target. The annotation itself does not name the target.

Scope of @co.dap.inner Executable Declarations

For an `@co.dap.inner` function or method, parameters and local declarations resolve normally. A free runtime name is resolved from the **lexical context of the active call or attachment site**, not from the declaration site of the separately declared `@co.dap.inner` function.

The lookup order is:

1. parameters and local declarations of the `@co.dap.inner` function or method
2. the innermost lexical scope at the active call or attachment site
3. enclosing lexical scopes of that call or attachment site
4. statically resolved types, annotations, imports, and built-in `co.*` names
5. compiler error

This model is called **call-site lexical-context resolution**.

It is distinct from an ordinary inner function. An ordinary inner function is physically declared inside another function and resolves free runtime names from its enclosing lexical **declaration** context. An `@co.dap.inner` function is declared separately and therefore obtains its runtime context from the lexical scope in which it is actively attached or called.

It is also distinct from `@co.dap.dynamicscope`. Call-site lexical-context resolution does not search arbitrary runtime caller frames beyond the lexical scope chain of the active call site, and it does not use mixed-scope fallback. At every statically known use or call site, the compiler validates that each required runtime binding exists, is definitely initialized, has a compatible type, and permits the requested access or mutation.

For `@co.dap.inner` types and other non-executable declarations, ordinary static name and type resolution continues to apply. Call-site lexical-context resolution concerns only free runtime names used by executable function or method bodies.

An implementation may lower an `@co.dap.inner` declaration by copying, inlining, specialization, or another equivalent mechanism. Such lowering is not observable language semantics; the association and scope rules above are normative.

Exception

`@co.dap.inner`, `@co.dap.local`, and `@co.dap.nested` cannot be used with `co.lang.cstruct`.

Statements

A statement is a complete executable or declarative instruction. It may contain one or more expressions and may change program state, control execution, introduce declarations, or produce observable effects.

Common statement categories in `FoLang`

1. Declaration Statement
2. Initialization Statement
3. Expression Statement
4. Conditional Statement
5. Loop Statement etc.,

Expressions

An expression is a construct that evaluates to a value, produces an effect, or both, and may be contained within a larger expression or statement.

Expression Evaluation Order

ForLang defines a deterministic evaluation order for expressions.

Expression structure is determined first by:

1. explicit grouping with parentheses;
2. operator precedence;
3. operator associativity.

After the expression structure has been determined, operands and subexpressions are evaluated from left to right in their source-code order, except where a construct explicitly defines conditional, lazy, asynchronous, concurrent, or otherwise specialized evaluation behavior.

Ordinary Expressions

For an ordinary expression:

```
result = first() + second();
```

the evaluation order is:

1. evaluate `first()`;
2. evaluate `second()`;
3. apply the `+` operator;
4. assign the resulting value to `result`.

Precedence and Evaluation Order

Left-to-right evaluation does not override operator precedence.

For example:

```
result = first() + second() * third();
```

operator precedence determines the expression structure as:

```
result = first() + (second() * third());
```

The function calls are evaluated in source order:

1. evaluate `first()`;
2. evaluate `second()`;
3. evaluate `third()`;
4. multiply the results of `second()` and `third()`;
5. add the multiplication result to the result of `first()`;
6. assign the final result to `result`.

Therefore, precedence determines how values are combined, while evaluation order determines when each operand or subexpression is evaluated.

Function Calls

For a function call, the callable expression is evaluated first, followed by the arguments from left to right.

```
result = calculate(first(), second(), third());
```

The evaluation order is:

1. evaluate the callable expression `calculate`;
2. evaluate `first()`;
3. evaluate `second()`;
4. evaluate `third()`;
5. invoke `calculate` with the resulting argument values;
6. assign the returned value to `result`.

Name resolution performed at compile time does not constitute runtime evaluation.

Method Calls

For a method call, the receiver expression is evaluated first, followed by the arguments from left to right.

```
result = getEmployee().calculate(first(), second());
```


The evaluation order is:

1. evaluate `getEmployee()`;
2. resolve the resulting object as the method receiver;
3. evaluate `first()`;
4. evaluate `second()`;
5. invoke `calculate`;
6. assign the returned value to `result`.

The receiver expression must be evaluated exactly once.

Member Access

For member access, the expression producing the containing object is evaluated before the member is accessed.

```
value = getEmployee().address.city;
```

The evaluation order is:

1. evaluate `getEmployee()`;
2. access `address` on the resulting object;
3. access `city` on the resulting address;
4. assign the resulting value to `value`.

Each intermediate receiver is evaluated only once.

Indexing

For an indexing expression, the indexed object is evaluated first, followed by index expressions from left to right.

```
value = getMatrix()[row()][column()];
```

The evaluation order is:

1. evaluate `getMatrix()`;
2. evaluate `row()`;
3. perform the first indexing operation;
4. evaluate `column()`;
5. perform the second indexing operation;
6. assign the resulting value to `value`.

Assignment

For assignment, FoLang evaluates the assignment target first, then evaluates the right-hand-side expression, and finally performs the assignment.

```
array[index()] = calculate();
```

The evaluation order is:

1. evaluate `array`;
2. evaluate `index()`;
3. determine the destination location;
4. evaluate `calculate()`;
5. assign the resulting value to the destination.

Determining the assignment target does not read the previous value stored at that location unless the assignment operation explicitly requires it.

Simple Variable Assignment

For assignment to a simple variable:

```
result = first() + second();
```

the evaluation order is:

1. determine the binding represented by `result`;
2. evaluate `first()`;
3. evaluate `second()`;
4. apply the `+` operator;
5. assign the resulting value to `result`.

Compound Assignment

A compound assignment evaluates its target exactly once.

```
array[index()] += calculate();
```

The evaluation order is:

1. evaluate `array` ;
2. evaluate `index()` ;
3. determine the destination location;
4. read the current value from that location;
5. evaluate `calculate()` ;
6. apply the `+` operator;
7. assign the resulting value to the same destination.

The expression above must not behave as though it were expanded into a form that evaluates `array` or `index()` more than once.

Multiple Assignment

When an assignment contains multiple right-hand-side expressions, those expressions are evaluated completely from left to right before values are assigned to their targets.

```
a, b = first(), second();
```

The evaluation order is:

1. determine the target for `a` ;
2. determine the target for `b` ;
3. evaluate `first()` ;
4. evaluate `second()` ;
5. assign the first resulting value to `a` ;
6. assign the second resulting value to `b` .

Right-hand-side evaluation must complete before any target receives its new value.

This permits value exchange without an explicit temporary variable:

```
a, b = b, a;
```

Operator Expressions

Built-in and user-defined operators follow the same operand-evaluation rules.

For a binary operator:

```
left() + right()
```

FoLang evaluates `left()` before `right()` .

For a prefix unary operator:

```
-operand()
```

FoLang evaluates `operand()` before applying the operator.

For a postfix unary operator:

```
operand() !
```

FoLang evaluates `operand()` before applying the operator.

Operator overloading must not change the specified evaluation order of operands.

Short-Circuit Boolean Operations

Short-circuit Boolean operators evaluate the left operand first.

The right operand is evaluated only when it is required to determine the result.

For logical AND:

```
left() && right()
```

the evaluation order is:

1. evaluate `left()` ;
2. when the result is false, return false without evaluating `right()` ;
3. otherwise, evaluate `right()` and use its Boolean result.

For logical OR:

```
left() || right()
```

the evaluation order is:

1. evaluate `left()`;
2. when the result is true, return true without evaluating `right()`;
3. otherwise, evaluate `right()` and use its Boolean result.

A skipped operand produces no value, mutation, side effect, or error.

Conditional Expressions

A conditional expression evaluates its condition first and then evaluates exactly one selected result expression.

```
result = condition()
    .return(whenTrue())
    .otherwise.return(whenFalse());
```

The evaluation order is:

1. evaluate `condition()`;
2. when the condition is true, evaluate `whenTrue()` only;
3. otherwise, evaluate `whenFalse()` only;
4. assign the selected result to `result`.

The unselected expression must not be evaluated.

Conditions and Branches

For a condition-and-branch construct, conditions are evaluated sequentially from left to right.

```
firstCondition().do({
    firstBranch();
}).otherwise(secondCondition()).do({
    secondBranch();
}).otherwise.do({
    finalBranch();
});
```

The evaluation order is:

1. evaluate `firstCondition()`;
2. when true, execute `firstBranch()` and skip the remaining conditions and branches;
3. otherwise, evaluate `secondCondition()`;
4. when true, execute `secondBranch()` and skip the final branch;
5. otherwise, execute `finalBranch()`.

Only the selected branch is evaluated.

Pattern Matching

A pattern-matching subject is evaluated exactly once.

Cases are examined in source order unless a particular matcher explicitly defines another ordering rule.

```
getValue().match
    .case(firstPattern => firstResult())
    .case(secondPattern => secondResult())
    .default(defaultResult());
```

The evaluation order is:

1. evaluate `getValue()` exactly once;
2. test `firstPattern`;
3. when it matches, evaluate `firstResult()` and stop;
4. otherwise, test `secondPattern`;
5. when it matches, evaluate `secondResult()` and stop;
6. otherwise, evaluate `defaultResult()`.

Results belonging to unmatched cases are not evaluated.

Pattern guards are evaluated only after their corresponding structural pattern has matched.

Collection Literals

Elements of an array, list, tuple, set, map, or other collection literal are evaluated from left to right as they appear in the source.

```
values = [first(), second(), third()];
```

The evaluation order is:

1. evaluate `first()`;
2. evaluate `second()`;
3. evaluate `third()`;
4. construct the collection from the resulting values.

For a map entry, the key expression is evaluated before its corresponding value expression.

```
values = {  
  firstKey(): firstValue(),  
  secondKey(): secondValue()  
};
```

The evaluation order is:

1. evaluate `firstKey()`;
2. evaluate `firstValue()`;
3. evaluate `secondKey()`;
4. evaluate `secondValue()`;
5. construct the map.

Range Expressions

For a bounded range, the lower-bound expression is evaluated before the upper-bound expression.

```
range = lower() .. upper();
```

The evaluation order is:

1. evaluate `lower()`;
2. evaluate `upper()`;
3. construct the range.

An omitted bound is not evaluated and does not produce an implicit function call or side effect.

Comprehensions and Iteration

A comprehension or iteration construct follows its own defined iteration semantics.

Within each iteration:

- source expressions are evaluated in the order declared;
- filter conditions are evaluated before result expressions;
- a result expression is evaluated only when its filters succeed;
- individual operand evaluation remains left to right.

The language does not implicitly evaluate comprehension iterations concurrently unless concurrency is explicitly requested.

Lazy Expressions

An expression declared lazy is not evaluated when the lazy binding is created.

Its evaluation occurs only when demanded according to the rules of the corresponding lazy construct.

Once evaluation begins, the expression itself follows the normal FoLang evaluation-order rules unless the lazy construct specifies otherwise.

The specification for each lazy construct must also state whether its result is:

- evaluated once and cached;
- evaluated again for every demand;
- safe for concurrent demand;
- permitted to propagate errors more than once.

Asynchronous and Concurrent Expressions

Left-to-right evaluation determines the order in which asynchronous or concurrent operations are created, invoked, or submitted.

It does not necessarily determine the order in which independently executing operations complete.

```
submit(first());  
submit(second());
```

FoLang evaluates and submits `first()` before `second()`. However, completion order depends on the execution-model rules unless explicit ordering is requested.

Concurrent execution never arises implicitly from an ordinary expression. It must be introduced by an explicit FoLang concurrency, parallelism, asynchronous-execution, scheduling, or execution-model construct.

Errors During Evaluation

When evaluation of a subexpression produces an error that prevents further evaluation, later subexpressions are not evaluated.

```
result = first() + failing() + third();
```

When `failing()` terminates evaluation with an error:

1. `first()` has already been evaluated;
2. `failing()` produces the error;
3. `third()` is not evaluated;
4. no final value is assigned to `result`.

The applicable error-handling rules determine whether the error is propagated, matched, converted, recovered from, or terminates execution.

Side Effects

All observable side effects occur according to the defined evaluation order.

Observable effects include:

- object mutation;
- variable rebinding;
- input and output;
- file-system operations;
- network operations;
- synchronization;
- exception or error generation;
- interaction with foreign code;
- interaction with the runtime environment.

An implementation must not reorder operations when doing so could change any observable effect.

Compiler Optimizations

A compiler may optimize, combine, eliminate, or internally reorder operations only when the transformation preserves all behavior required by this specification.

In particular, an optimization must not change:

- the resulting value;
- the order or presence of observable side effects;
- the identity or aliasing behavior of objects;
- the timing or presence of specified errors;
- the selected conditional or pattern-matching branch;
- the number of times an effectful expression is evaluated;
- synchronization or concurrency guarantees.

Pure internal operations may be reordered when no conforming FoLang program can observe the difference.

Implementation Requirement

Every conforming FoLang implementation must preserve the evaluation order defined in this section.

An implementation may use any parser, intermediate representation, optimizer, runtime, or backend, but those implementation choices must not alter the externally observable behavior required by these rules.

Operators

FoLang distinguishes a **symbolic spelling** from an **operator**. A sequence of symbol characters is not automatically an operator merely because it resembles one. Its syntactic role is determined by the grammar context in which the whole spelling occurs.

For example:

```
a co.lang.int->(**);
```

In this declaration, `->` is the structural type-derivation marker and `**` is pointer-degree metadata inside the derivation. Neither occurrence is parsed as an expression operator. The same `**` spelling in `left ** right` is parsed as the registered power operator because it occurs in an operator-expression position.

FoLang uses one uniform implementation model for every expression operator. The difference between built-in, pre-declared, and project-local custom operators is only who registers the symbol and its parse properties.

built-in operator
symbol and parse properties registered by the language
one or more implementations may already exist
pre-declared operator glyph
symbol and parse properties registered by the language
no implementation is required to exist
project-local custom operator
symbol and parse properties registered once in src/lib/operators/library.fo1
no implementation is contained in the operator declaration
all three categories
implementations are ordinary mode=override operator functions
duplicate normalized operand signatures are errors

Symbolic Runs, Classification, and Boundaries

After comments, literals, and closed scanner-known composite spellings such as `@@new` and `@@init` are recognized, the lexer consumes each remaining complete contiguous run of symbol characters as one symbolic token candidate. It does not backtrack or split an unrecognized run into shorter valid tokens. Whitespace, comments, and delimiters such as `(`, `)`, `[`, `]`, `{`, and `}` end a symbolic run. Comment openers are recognized before ordinary symbolic-run scanning.

The complete run is then classified by its grammar context as one of the following:

- 1. a fixed structural or declaration spelling such as `->`, `=>`, `:=`, or `?=`;
- 2. a contextual metadata spelling, such as a contiguous run of one or more `*` characters inside `->(…)` to express pointer degree;
- 3. a registered expression operator; or
- 4. an unrecognized symbolic token, which is a compilation error.

There is no fallback splitting. Therefore, when `++` is not registered:

```
++ a;      // error: unrecognized symbolic token `++`
a ++ b;    // error: unrecognized symbolic token `++`
a++b;      // error: the contiguous run is still the single candidate `++`

+ +a;      // valid: two `+` operators separated by whitespace
+(+a);     // valid: `(` creates the token and operand boundary
a + +b;    // valid: binary `+` followed by unary `+`
```

A registered expression operator whose spelling contains more than one symbol character must have an explicit boundary on every operand-facing side. A boundary may be whitespace, a comment, or an applicable delimiter. Boundary presence is checked from the original source before whitespace and comments are discarded. The rule is based on the operator’s fixity:

- an infix multi-symbol operator requires a boundary before and after it;
- a prefix multi-symbol operator requires a boundary after it;
- a postfix multi-symbol operator requires a boundary before it.

Suppose `+-` is registered as an infix operator:

```
a +- b;      // valid
(a)+-(b);    // valid: the parentheses provide both operand boundaries
a+-b;        // error: missing boundaries
a +-b;       // error: missing the right boundary
a+- b;       // error: missing the left boundary
a + -b;      // valid: separate `+` and unary `-` operators
a + (-b);    // valid: the parenthesis separates the operators
```

This boundary requirement applies uniformly to built-in, pre-declared, and custom multi-symbol expression operators. It does not apply merely because a structural token or metadata spelling contains multiple symbols. Thus `co.lang.int->(**)` remains valid without spaces around `->` or inside the pointer metadata.

The statement-level definition spellings `:=` and `?=` are not expression operators, but they deliberately use the analogous two-sided boundary rule. Write `name := value` and `name ?= value`; compact forms such as `name:=value` and `name?=value` are invalid.

Operator `mode=override` and `mode=extends` are unsupported. Ordinary class method overriding through `@co.dap.override` remains a separate class feature.

Operator Implementations

An operator implementation is a normal function with operator metadata. It cannot be declared loose at package scope. Every implementation must be in one of these legal function-owning locations:

Operand owner	Required implementation location
---------------	----------------------------------

built-in type	an <code>@co.dap.extension</code> function inside a unit
<code>co.lang.struct</code>	the struct's same-package companion unit
<code>co.lang.class</code>	an operator method declared by the class
module, enum, union, interface, signature, <code>co.lang.cstruct</code>	unsupported

An implementation uses `mode=overload`, or omits `mode` because omission is shorthand for `mode=overload`. A one-character symbol may use a character literal; a multi-character symbol must use a string literal:

```
@co.dap.operator(symbol='+', mode=overload)
add(left Employee, right Employee)->(Employee) = {
    ...
}

@co.dap.operator(symbol="+-", mode=overload)
combine(left Employee, right Employee)->(Employee) = {
    ...
}
```

FoLang does not attempt to prove that an overload follows the conventional meaning of its symbol. A developer may overload `+` with any implementation. Using one symbol consistently for one recognizable concept is recommended for readability, but it is not a compiler-enforced semantic restriction.

For a receiverless struct-companion operator function, the first declared operand must have the matching struct type. A matching struct type in a later operand is insufficient. For a unary operator, the sole operand must have the matching struct type.

A matching instance receiver contributes the first operand. A matching type receiver establishes ownership but contributes no operand; its ordinary parameter list is the complete operator signature. In a class, an ordinary or `@co.dap.instance` operator method has an implicit `this` first operand, while `@co.dap.static` and `@co.dap.class` operator methods use only their declared parameters and require the first declared operand to have the enclosing class type.

After receiver normalization, the operand count must match the registered arity of the operator. Each distinct normalized operand signature is an overload. A second implementation with the same normalized signature is a duplicate and is rejected. This rule applies equally to built-in operators, pre-declared glyphs, and project-local custom operators.

Language-Owned Operators

Language-owned operators already have a registered symbol, fixity, precedence, associativity, and arity. They must not be redeclared in the project operator source. They receive additional implementations through `mode=overload`.

This category includes:

- 1. ordinary built-in operators such as `+`, `-`, `*`, and `==`;
- 2. pre-declared operator glyphs whose parse properties are fixed by the language but which may initially have no implementation.

Pre-Declared Operator Glyphs

The language pre-declares the documented mathematical and modifier glyph set, including `λ`, `∀`, `∃`, `∪`, `⇒`, `∂`, `⊥`, `↓`, `⋈`, `∘`, `∏`, and `ℱ`, with fixed parse properties. These symbols are reserved against project-local re-declaration, but they are available for implementation through ordinary operator overloads:

```
// ∪ is already registered by the language; only its implementation is supplied.
@co.dap.operator(symbol='∪', mode=overload)
union(left Set, right Set)->(Set) = {
    ...
}
```

Until a visible implementation matches the operand types, an expression using a pre-declared glyph parses successfully and then fails during operator resolution.

Hard-reserved spellings such as `::=`, `->>`, `<->`, backtick, backslash, `#`, and comment openers are different: they are not overloadable or declarable unless a later language revision explicitly assigns them operator semantics.

Project-Local Custom Operator Source

A custom operator is a symbol that is neither language-owned nor hard-reserved. Its symbol and parse properties are registered only in the fixed project-local operator bootstrap surface:

```
<application-root>/src/lib/operators/library.fo1
```

`src/lib/` and `src/lib/operators/` are not packages. `operators/` is the one standardized operator slot under `src/lib/`. If it is absent, the application introduces no project-local custom operator symbols. If it is present, it must contain exactly one file named `library.fo1` and no additional files or subdirectories. The fixed `library.fo1` name is shared by all `src/lib/` library slots; the enclosing `operators/` directory selects the dedicated operator bootstrap semantics.

The fixed file is parsed by the dedicated operator-source lexer and parser before the ordinary FoLang lexer and parser run. Its filesystem position already establishes the operator bootstrap context, so no library-kind annotation is used:

```
// src/lib/operators/library.fol
_ co.lang.library = {

  ⊗ co.lang.operator = {
    fixity: co.operator.fixity.infix,
    precedence: 60,
    associativity: co.operator.associativity.left,
    arity: co.operator.arity.binary,
    commutative: co.const.false,
    idempotent: co.const.false,
    identity: co.const.none,
    foldable: co.const.false,
    vectorizable: co.const.false,
    distributes_over: [],
    desugar: "intrinsic:tensor_product"
  };

  +- co.lang.operator = {
    fixity: co.operator.fixity.infix,
    precedence: 60,
    associativity: co.operator.associativity.left,
    arity: co.operator.arity.binary
  };
}
```

`_ co.lang.library` identifies the structural operator bootstrap surface. It is not an ordinary importable library surface and does not produce a `.folenc` artifact. Its body may contain only `co.lang.operator` declarations. Imports, functions, types, variables, expressions, implementation packages, and nested libraries are forbidden.

`co.lang.operator` is valid only in this dedicated source grammar. It is not an ordinary FoLang declaration kind and cannot appear in package source, `src/appl.fol`, or an ordinary library surface.

Operator declaration attributes

Every custom declaration must contain each required parse attribute exactly once. Optional metadata may appear at most once. Duplicate and unknown attributes are errors.

Attribute	Required	Accepted value	Meaning
<code>fixity</code>	yes	<code>infix</code> , <code>prefix</code> , <code>postfix</code> , or a reserved future fixity name (<code>circumfix</code> , <code>postcircumfix</code> , <code>precircumfix</code> , etc)	parser placement
<code>precedence</code>	yes	decimal integer from <code>0</code> through <code>100</code>	binding strength
<code>associativity</code>	yes	<code>left</code> , <code>right</code> , or <code>none</code>	grouping for equal precedence
<code>arity</code>	yes	<code>unary</code> , <code>binary</code> , <code>ternary</code> , or a positive decimal integer	normalized operand count
<code>commutative</code>	no	<code>co.const.true</code> or <code>co.const.false</code>	semantic/optimization metadata
<code>idempotent</code>	no	<code>co.const.true</code> or <code>co.const.false</code>	semantic/optimization metadata
<code>identity</code>	no	a FoLang literal, including <code>co.const.none</code>	identity-value metadata
<code>foldable</code>	no	<code>co.const.true</code> or <code>co.const.false</code>	compile-time folding metadata
<code>vectorizable</code>	no	<code>co.const.true</code> or <code>co.const.false</code>	vectorization metadata
<code>distributes_over</code>	no	a list of quoted operator symbols	distributivity metadata
<code>desugar</code>	no	a string literal	intrinsic or lowering hook metadata

In the alpha profile, only `infix`, `prefix`, and `postfix` are implemented. Consequently, an alpha `prefix` or `postfix` declaration must be unary and an alpha `infix` declaration must be binary. Other fixity names remain reserved for future delimiter and slot grammars and are rejected by the alpha conformance profile.

The four required parse attributes construct the tokenizer and precedence table. Optional semantic/optimization attributes do not change tokenization or parsing.

Declaration and implementation are separate

The operator library registers only the symbol and metadata. It contains no implementation. Implementations use the same `mode=overload` form as built-in and pre-declared operators:


```
// vector/Vector.comp.unit.fol
_ co.lang.unit = {

  @co.dap.operator(symbol='⊗', mode=overload)
  tensorProduct(left Vector, right Vector)->(Vector) = {
    ...
  }
}
```

A custom symbol has exactly one declaration in the operator library, but it may have any number of distinct implementation overloads in legal owners:

```
one ⊗ declaration
  Vector ⊗ Vector -> Vector
  Matrix ⊗ Matrix -> Matrix
  Tensor ⊗ Tensor -> Tensor
```

The declaration cannot be duplicated, aliased, merged, selected, or remapped. The implementations participate in ordinary operator overload resolution.

Symbol registration and exact recognition

The dedicated operator-source lexer reads the declaration name as one maximal contiguous symbol run. After the operator table is built, the ordinary lexer uses the same whole-run rule. A registered custom symbol is recognized only when the complete run matches that symbol; an unknown run is rejected without being split into shorter operators.

Multi-symbol operator uses must also satisfy the operand-boundary rule defined in [Symbolic Runs, Classification, and Boundaries](#). Consequently, registering `+-` permits `a +- b`, not `a+-b`. Unicode is not required for custom operators.

A custom symbol:

- must not be language-owned;
- must not be a hard-reserved spelling;
- must not contain `//` or `/*`, because a comment opener terminates the preceding symbolic run and is recognized before further operator matching;
- must have exactly one declaration in the operator library.

Symbols are global; implementations are resolved by scope and type

Once a custom symbol is registered—or a language-owned symbol is loaded—the ordinary tokenizer recognizes it throughout that compilation. The parser can therefore build the same operator expression in every package.

Whether an implementation is available is determined later through ordinary operator resolution. Receiver-owned class and companion implementations follow their normal lookup rules. Extension implementations require normal `@co.ddap.use` activation. A symbol with no matching visible implementation produces a resolution error, not a syntax error.

Using one symbol for one recognizable concept across its overloads is strongly recommended for readability, but the compiler does not and cannot enforce that recommendation. The type/signature rules, not the conventional meaning of the symbol, determine whether an overload is legal.

Operators Do Not Cross a Library Boundary

Operator notation is compilation-domain-local syntax, not a library API element. An ordinary library surface exports named function signatures and boundary data contracts. It exports no operator table.

```
geometry.folenc
├─ projected symbol table
└─ compiled implementation/linkage      (no operator table)
```

The project-local table produced from `src/lib/operators/library.fol` belongs only to the owning project's `src/` compilation domain. This is true whether the project is an application or a standalone packaged library. The table is never reused while parsing or resolving `src/lib/ffi/`, `src/lib/system/`, `src/lib/advanced/`, or `src/lib/dynamicvmrt/`.

Project-local `ffi`, `system`, and `advanced` source libraries cannot declare custom operators or operator overload implementations. They are compiled against the language-owned operator set.

A standalone packaged library of kind `application` may have its own optional project-local `src/lib/operators/library.fol`, exactly as an application project does. For a standalone library project, the frontend shallow-reads the fixed `src/library.fol` surface header early enough to obtain `@co.dap.library(type=...)`; `src/lib/operators/` is accepted only when that declared library kind permits project-local custom operators. For the current rules, the standalone `application` library kind may use this operator bootstrap.

That operator table belongs only to that standalone library project's `src/` compilation domain. Its operator expressions are fully resolved and lowered before the library is emitted. The resulting `.folenc` does not contain an importable operator table, and importing it cannot add symbols or parse properties to the consumer.

A project whose `src/` code wants operator notation for an imported boundary type must register the symbol in its own `src/lib/operators/library.fol` when necessary and provide a legal local implementation that delegates to the library's named public function.

Parallel Frontend Preparation, Imports, and Reachability

FoLang projects use eager preparation for performance and explicit import edges for semantics. Eagerly parsing or deserializing a project component does **not** make its symbols visible in another context.

Eager parallel preparation

After fixed-layout validation, the frontend may prepare independent domains in parallel:

```
project discovery
|
+-- srclib/operators/library.fol
|   -> dedicated operator parser
|   -> ProjectOperatorTable
|
+-- srclib/ffi/**
|   -> ASTs + isolated contexts/symbol tables
|
+-- srclib/system/**
|   -> ASTs + isolated contexts/symbol tables
|
+-- srclib/advanced/**
|   -> ASTs + isolated contexts/symbol tables
|
+-- srclib/dynamicvmrt/**
|   -> ASTs + isolated contexts/symbol tables
|
+-- lib/*.folenc
|   -> parallel protobuf deserialization
|   -> projected/exported context and symbol metadata
```

The non-operator `srclib` domains do not depend on the owning project's custom operator table, so they may be parsed concurrently with the operator bootstrap. Once the project-local operator table is complete when applicable, the fixed project surface under `src/` and package source below `src/` may be parsed, including in parallel where implementation scheduling permits.

Source parsing produces ASTs and canonical context/symbol-table entries. Deserializing a `.folenc` does not reparse source; it reconstructs the compiled library's projected/exported context and symbol metadata required by the frontend.

One canonical symbol table; imports store references

The frontend keeps one canonical context/symbol-table representation for each prepared package or library. An import does not copy or merge the target symbol table into the importer.

Conceptually:

```
ContextMap
├─ application-entry
├─ hr.employee
├─ ffi-surface
├─ crypto-folenc-surface
└─ ...

application-entry.ImportedContexts
├─ "ffi"    -> ffi-surface ContextId
└─ "crypto" -> crypto-folenc-surface ContextId
```

When the compiler resolves an import declaration it:

```
resolve import identity
  -> locate the already-prepared ContextId / SymbolTableId
  -> add a provisional reference to the importing context's ImportedContexts
```

The target symbol table remains canonical in the global maps. Import visibility is therefore represented by context references, not by symbol copying.

The import becomes semantically live only when name/type resolution actually consumes at least one symbol through that imported context:

```
resolve symbol through imported context
  -> mark exact SymbolId used
  -> mark the import used
  -> activate the dependency edge
  -> make the target context reachable
```

Using one imported symbol does not mark sibling symbols in that target context used. At finalization, every provisional import with zero successfully used symbols is removed from the effective `ImportedContexts`/dependency graph and reported as an unused-import compile-time error.

Common Symbol-Usage Validation

The same unused-symbol validator applies to all FoLang source compilation domains:

```

application
  root/surface = src/appl.fol

project-local source library
  root/surface = src/lib/<kind>/library.fol

standalone library project
  root/surface = src/library.fol

```

The roots and permitted constructs differ, but the usage algorithm is common.

For ordinary filename-backed primary declarations, the primary declaration must be live according to its normal semantic use. In addition:

Construct	Independently usage-checked symbols
class	every user-declared method; fields are ignored
struct	the struct declaration itself; fields are ignored
struct companion unit	every user-declared companion function
module	every user-declared function/method; fields/state values are ignored
ordinary unit	every exposed unit declaration: types, aliases, functions, macros/templates, and every other usage-checkable package symbol

Container liveness never propagates sideways or downward merely because the container was used:

```

class used
  != all class methods used

struct used
  != all companion functions used

module used
  != all module functions used

one unit symbol used
  != all symbols in that unit used

```

A symbol is considered used through successful semantic dependency, not merely because its declaration was parsed or because its containing package was imported. Compiler-required roots, surface exports, protocol/conformance requirements, operator implementations, and other language-defined semantic requirements may establish legitimate liveness without a direct source-level call/reference.

For a library build, declarations intentionally exported by the library surface are surface roots of that library compilation; they do not need an internal consumer to remain valid. Their implementation dependencies, and all other source declarations behind the surface, are still subject to the same symbol-level usage rules. For an project-local source library, the application must separately consume at least one projected surface symbol for the source-library import itself to be live.

A consuming compilation does not re-run private symbol-usage analysis inside a prebuilt third-party `.folenc`. That validation belongs to the producer's library build. The consumer validates only that its `.folenc` import actually contributes at least one used exported symbol.

Closed-project reachability

Physical presence in a FoLang project means participation in that project. FoLang does not permit orphan project source, orphan project-local source libraries, or unused `.folenc` artifacts. This closed-project rule applies to both application projects and standalone packaged-library projects.

An application dependency graph is rooted at:

```
src/appl.fol
```

Every package physically present below `src/` must be transitively reachable from that root through **effective** dependency edges created by actual symbol use. A written import with zero used symbols does not establish reachability.

Every existing non-operator project-local source library must also be reachable from the owning project's `src/` dependency graph through at least one actually consumed projected surface symbol. `src/lib/operators/` is the exception: its existence activates it automatically because it establishes the operator table for `src/`.

Ordinary source in the owning project's `src/` domain can never establish reachability by directly importing an internal package of `src/lib/<kind>/`; the only edge from the ordinary project source domain into a project-local source library terminates at its projected `library.fol` surface.

Each project-owned source library has its own private effective dependency graph rooted at its surface:

```

src/lib/ffi/library.fol
src/lib/system/library.fol
src/lib/advanced/library.fol
src/lib/dynamicvmrt/library.fol

```

Within that private compilation domain, `library.fo1` and internal packages may import internal packages. Those imports are still provisional: an internal package becomes reachable only when one or more of its symbols are actually consumed through the import. Because these sources belong to the same project owner, unreachable internal packages and unused usage-checkable symbols are compile-time errors.

Standalone library projects use the same rule, rooted at their fixed `src/library.fo1` project surface.

Third-party/prebuilt `.fo1enc` internals are different. Their producer already validated their private source graph and usage-checkable source symbols when the artifact was built. The consuming application validates only that each `.fo1enc` physically present in its `lib/` directory is itself reachable through an import that contributes at least one actually used exported symbol.

Reachability state is separate from symbol resolution state. A context may be fully parsed/deserialized and have a complete symbol table while still being unreachable. After semantic resolution reaches a fixed point, the compiler:

1. prunes provisional imports with zero used symbols
2. reports those imports as unused
3. constructs the effective dependency graph from surviving imports
4. validates every usage-checkable source symbol
5. compares discovered project entities with reachable entities
6. reports every orphan/unreachable package, `src/lib`, or `.fo1enc`

```
CandidateImport -- no used SymbolId --> prune + unused-import error  
CandidateImport -- used SymbolId ----> effective dependency edge
```

```
DiscoveredProjectEntities - ReachableProjectEntities = Orphans
```

This permits eager parallel preparation without weakening import isolation, without allowing unused project content, and without allowing an otherwise unused import to manufacture reachability.

Bootstrap Order

1. The compiler receives the project root as the build input.
2. Validate the project-root domains:
 - src/ is mandatory;
 - src/lib/ and lib/ are optional but cannot be empty when present;
 - build/ is compiler-managed and is created when absent.
3. Derive the project identity from the canonical project-root directory basename, then classify the project from the fixed src surface:
 - project-root basename -> project/library name and default output basename;
 - src/appl.fol -> application project;
 - src/library.fol -> standalone packaged-library project;
 - both present -> error;
 - neither present -> error.
4. For a standalone packaged-library project, shallow-read src/library.fol sufficiently to obtain and validate the project-level @co.dap.library(type=...) kind before kind-dependent bootstrap decisions.
5. From the same project root:
 - discover all package source below src/;
 - discover standardized srclib slots when srclib/ is present;
 - discover every lib/*.folenc artifact when lib/ is present;
 - prepare build/.
6. Start independent preparation work permitted by the classified project/kind:
 - a. parse srclib/operators/library.fol when present and permitted;
 - b. parse each existing non-operator srclib domain in its own isolated source-library context;
 - c. deserialize all lib/*.folenc artifacts into isolated projected library contexts/symbol tables.
7. For srclib/operators/library.fol:
 - validate the operator-only surface and build ProjectOperatorTable.
 - Its table belongs only to the owning project's src/ domain.
8. After ProjectOperatorTable is ready when applicable, parse the fixed project surface (src/appl.fol or src/library.fol) and all package source below src/, producing ASTs and canonical context/symbol-table entries.
9. Resolve import declarations by reference:
 - locate the prepared ContextId / SymbolTableId and add a provisional reference to the importing context's ImportedContexts.
 - Never copy imported symbol tables and do not mark the dependency live merely because the import was written.
10. During semantic/name/type resolution, when a symbol is successfully resolved through a provisional imported context:
 - mark that exact SymbolId used;
 - mark the import used;
 - activate the dependency edge;
 - make the target context reachable.Using one symbol never marks sibling symbols used.
11. Resolve each project-local source library's private internal dependencies independently from its srclib/<kind>/library.fol surface. Ordinary source under src/ can enter that domain only through the projected surface, never through a direct import of an internal package.
12. Continue semantic/name/type/operator resolution until dependency, usage, and resolution state reaches a fixed point.
13. Run common unused-content validation:
 - prune every import that has zero used symbols and report it as unused;
 - for an application, require every src package to be reachable from src/appl.fol through the surviving effective graph;
 - for a standalone library, require every internal src package to be reachable from src/library.fol through the surviving effective graph;
 - require every existing non-operator srclib domain to have at least one projected surface symbol consumed by the owning project's src graph;
 - within each project-owned srclib, require every internal package to be reachable from library.fol through actual symbol use;
 - require every lib/*.folenc artifact to contribute at least one used exported symbol;
 - srclib/operators is structurally active when present and permitted;
 - validate usage-checkable source symbols:
 - class: every declared method;
 - struct: primary declaration, plus every companion-unit function;
 - module: every declared function/method;
 - ordinary unit: every declared usage-checkable symbol;
 - fields/state values are excluded from unused-symbol checks.

Report all unused imports, unused symbols, and orphan/unreachable entities together as compile-time errors.

14. Complete final semantic validation.

15. Serialize the validated frontend result under <project-root>/build/ using the wire format/version selected by the backend interchange contract.

Imports contribute no operator metadata. Project-local operator bootstrap, source-library preparation, and packaged-library metadata loading therefore do not depend on import order. Visibility is established later by explicit `ImportedContexts` references. Those references are provisional until semantic resolution consumes at least one symbol through them; zero-use imports are pruned and rejected during final validation.

Forward / Extern Declarations

Variables extern declaration

// someUnit.unit.fol

```
_ co.lang.unit={
  @co.dap.declare(extern)
  someBool co.lang.bool;
}
```

Functions forward declaration

// someOtherUnit.unit.fol

```
_ co.lang.unit={

  @co.dap.declare(forward)
  getEmployee(id co.lang.int)->(somepack.Employee);

  // or - @co.dap.declare is optional for functions
  getEmployee(id co.lang.int)->(somepack.Employee);

}
```

Types external declaration

```
// Employee.fol
_ co.lang.class{

  @co.dap.declare(extern)
  Dept co.lang.struct;

}
```

For functions and types `@co.dap.declare` is optional. For variables it is required.

Functions

FoLang does not allow free-flowing package functions. Package functions must be declared inside an ordinary `<Fragment>.unit.fol` file. Their public identity is the package member name, not the unit filename.

Normal

```
// general.unit.fol
_ co.lang.unit = {

  fun1(k co.lang.int, b co.lang.char)->(co.lang.int, co.lang.char) = {
    // function body
  }

}
```

`foLang` function can return multiple values

Default Parameters

// somefununit.unit.fol

```

- co.lang.unit {
  fun1 (k co.lang.int, b co.lang.char = 10)->(co.lang.int, co.lang.char)={
    }
}

```

Variadic Functions

Curried functions are not allowed to be variadic, and vice versa.

//someCurried.unit.fol

```

- co.lang.unit{
  fun1 (k co.lang.int, ...b co.lang.char)->(co.lang.int, co.lang.char)={
    }
}

```

Optional Parameters

//someOptional.unit.fol

```

- co.lang.unit={
  fun1(k? co.lang.int)->()={
    k.omitted.do{

    }.otherwise{

    }
  };
}

```

Named Parameters

// someNamedParam.unit.fol

```

- co.lang.unit={
  fun1(~k co.lang.int)->()={

  }
}

```

Named Returns

//someNamedResults.unit.fol

```

- co.lang.unit={

  doManythings(a co.lang.int, b co.lang.int->(&, meta={type=out}))->(r co.lang.int, e co.lang.exception)={}

}

```

Function Delegates

// someFunDeletage.unit.fol

```

- co.lang.unit={
  @co.dap.delegate someDelegate co.lang.delegate = (a co.lang.int, b co.lang.int) -> (co.lang.int, co.lang.int);
}

```

Function Chaining

// somefunctionChaining.unit.fol

```

- co.lang.unit={
  fetchEmployee(empId co.lang.string)->(Employee)=>>empMod.getEmployee(this, empId);
}

```

Anonymous Functions

// someAnonymousFun.unit.fol

```

_ co.lang.unit={
  add := (a int, b int) -> (int) {
    this.return a + b;
  };

  res := (a int, b int) -> (int) {
    this.return a * b;
  }(10, 20);
}

```

for more details on functions please refer section [Functions in Details](#)

Functions in detail

Inline

```

// math-functions.unit.fol
_ co.lang.unit = {
  @co.dap.inline
  add(a co.lang.int, b co.lang.int)->(co.lang.int)={
    this.return a + b;
  }
}

```

Anonymous Functions

// someOtherAnonymousfun.unit.fol

```

_ co.lang.unit={
  add := (a int, b int) -> (int) {
    this.return a + b;
  };

  res := (a int, b int) -> (int) {
    this.return a * b;
  }(10, 20);
}

```

Lambda

Only allowed as an inline callback argument to receiver-qualified collection operations (e.g. `map`, `filter`, `reduce`, `forEach`, `sortBy`, `groupBy`).

Transparent grouping around the member callee is permitted, so `(nums.map)(|x| => x*x)` is equivalent to the ordinary call spelling. A bare function call such as `map(|x| => x)` is not a collection-method context. Using `|...|` anywhere else is a syntax/lint error.

// somelanbda.unit.fol

```

// Callback literal syntax, shown only in its valid collection-call context
_ co.lang.unit={

  nums.map(|x| => x*x);
  words.filter(|s| => s.len() > 3);
  pairs.reduce(|acc, e| => acc + e, 0);
  dict.map(|k, v| => v * 10);
  list.sortBy(|a, b| => a.score - b.score);

}

```

The lambda must be a direct argument of the allowed collection call. That call may itself be nested, for example `consume(nums.map(|x| => x*x))`; the enclosing call does not make the lambda an argument of `consume`.

Inner Function

//someInnerFun.unit.fol

```

_ co.lang.unit={
  myfun(a co.lang.int, b co.lang.int)->(co.lang.int)={
    p co.lang.int = 10;
    someother()->()={
      co.out.println(p);
    }
    someother();
    p = 20;
    someother();
  }
}

```


The function `someother` is an ordinary inner function because it is physically declared inside `myfun`. Its free runtime names are resolved from the lexical declaration context of `myfun`; therefore `p` denotes the binding declared in `myfun` regardless of where `someother` is called within its permitted lifetime.

`@co.dap.local`, `@co.dap.nested`, and `@co.dap.inner` apply to separately declared functions and do not change the lexical-scope rule for an ordinary inner function. They provide visibility, target-state capture, or call-site association without requiring the function body to be physically placed inside the target declaration.

The permitted local-function form above is useful when the inner function is small and belongs naturally to one enclosing function. This is the named local-function exception described in [Physical Nesting Rules](#).

Curried

// someOtherCurried.unit.fol

```
_ co.lang.unit={
  add(first co.lang.int)(second co.lang.int)->(co.lang.int)={
    this.return first + second;
  }
}
```

Closure

//someClosure.unit.fol

```
_ co.lang.unit={
  adder() -> ((co.lang.int) -> co.lang.int) = {
    sum co.lang.int = 0;
    this.return (x co.lang.int) -> (co.lang.int) = {
      sum += x;
      this.return sum;
    }
  }
}
```

Functions Taking and Returning Functions

Syntax 1 — Inline signature

//someInlineSignature.unit.fol

```
_ co.lang.unit={
  someFunction (r (co.lang.int, co.lang.int)->(co.lang.int))->((co.lang.int)->(co.lang.int))={
  }
}
```

Syntax 2 — Named type alias

//sommeNamedTypeAliases.unit.fol

```
_ co.lang.unit={
  someFArg co.lang.type = (co.lang.int, co.lang.int)->(co.lang.int);
  someFRet co.lang.type = (co.lang.int)->(co.lang.int);

  someFunction (r someFArg)->(someFRet)={
  }
}
```

Syntax 3 — Function objects

//someFunObject.unit.fol

```
_ co.lang.unit={
  someFArg co.lang.function = (a co.lang.int, b co.lang.int) -> (co.lang.int)={
    this.return a + b;
  }

  someFRet co.lang.function = (a co.lang.int) -> (co.lang.int)={
    this.return a * 2;
  }
}
```

Other ways to declare closures/function objects and types/ curried functions

//someAdditionalleg.unit.fol

```

_ co.lang.unit={
  myobj co.lang.function = (a co.lang.int, b co.lang.int)->(co.lang.int)={
    this.return a + b;
  }

  add (a co.lang.int, b co.lang.int)->(co.lang.int){ this.return a + b; }
  oObj co.lang.function = add;

  funtype co.lang.type = (a co.lang.int, b co.lang.int)->(co.lang.int);

  closure=(factor co.lang.int, val co.lang.int) ==>> factory * val;

  curry = (factor co.lang.int) (x co.lang.int) ==>> x * factor;
}

```

Associated Functions

For a user-defined struct, associated functions must be declared inside the same-package companion unit whose name matches the struct. For more details on associated function please refer section [Associated Functions in a Companion Unit](#)

Some Restrictions on Special Functions

1. Special functions

- Curried functions
- Functions with named arguments
- Functions with optional arguments
- Functions with default arguments
- Variadic functions
- Functions that take or return functions or function types
- Dynamically scoped and mixed-scoped functions

2. Restrictions

- They cannot be overloaded.
- They cannot be used as callbacks.
- They cannot participate in [Execution Models and Control Abstractions](#).

Scoping Rules for Functions

All functions in FoLang have a defined scope — the set of variables a function can access.

Default — Lexical / Static Scope

All of the following are **always** lexically scoped — this cannot be changed:

```

methods
inner methods
free functions
inner functions
target-local named functions
closures
lambdas
Generic Functions
Anonymous functions
Curried functions

```

Lexical scope means a function resolves names from its **declaration site**, not from the scope of its caller. Unit-level variables are forbidden, so a unit-scoped free function receives runtime values through parameters or introduces them locally. An ordinary inner function captures from the enclosing lexical declaration context. Calling that inner function does not replace its captured context with the caller's runtime scope.

```

// scope-example.unit.fo1
_ co.lang.unit = {

  foo()->() = {
    x co.lang.int = 10;

    printX()->() = {
      co.out.println(x); // ✅ x from the enclosing lexical scope
    }

    printX();
  }
}

```

@co.dap.inner — Call-Site Lexical Context

@co.dap.inner is not the syntax for an ordinary inner function. It annotates a separately declared function or method that becomes associated with a target at an explicit use or attachment site.

Its free runtime names use **call-site lexical-context resolution**:

1. @co.dap.inner parameters and local declarations
2. the innermost lexical scope at the active call or attachment site
3. enclosing lexical scopes of that call or attachment site
4. statically resolved types, annotations, imports, and co.* names
5. compiler error

The compiler validates these requirements at every statically known use or call site. This is a fixed property of @co.dap.inner; it is not selected through @co.dap.lexicalscope, @co.dap.dynamicscope, or @co.dap.mixedscope.

This differs from ordinary lexical inner functions, whose free runtime names are fixed by their declaration site, and from dynamically scoped associated functions, which may search outward through the runtime caller activation chain.

Associated Functions — Additional Scope Options

Only associated functions support non-lexical scoping via annotations:

The examples below are members of the same-package Employee companion unit.

@co.dap.lexicalscope — default, explicit declaration

```
// Employee.comp.unit.fol
_ co.lang.unit = {
  @co.dap.lexicalscope
  (emp Employee) process()->() = {
    co.out.println(emp.name); // ✅ declaration scope
  }
}
```

@co.dap.dynamicscope — accesses caller's scope

```
// Employee.comp.unit.fol
_ co.lang.unit = {
  @co.dap.dynamicscope
  (emp Employee) process()->() = {
    co.out.println(name); // name comes from caller's scope
  }
}
```

@co.dap.mixedscope — accesses both scopes

```
// Employee.comp.unit.fol
_ co.lang.unit = {
  // caller scope takes priority – shadows declaration scope on conflict
  @co.dap.mixedscope
  (emp Employee) process()->() = {
    co.out.println(name); // caller scope takes priority
    co.out.println(emp.id); // falls back to declaration scope
  }
}
```

Callback Scope Inside Dynamically Scoped Associated Functions

A callback block or lambda does not independently select lexical, dynamic, or mixed scope. The associated function that executes the callback determines how the callback's free runtime names are resolved.

Callback parameters and declarations made inside the callback always belong to the callback's local scope. Names that are not callback parameters or callback-local declarations follow the executing associated function's scope policy. //someOperation.unit.fol

```
_ co.lang.unit={
  someFun ()-> ()={
    nums.reduce(|acc, e| => {
      total.value += e;
      acc + e;
    }, 0);
  }
}
```

For this example:

acc, e	-> callback parameters
temporary locals	-> callback-local context
total	-> reduce's dynamic caller context
co.* and imports	-> normal built-in/import resolution

The callback syntax remains uniform. `reduce` is dynamically scoped, so unresolved runtime names in the callback are resolved through the active caller-context chain. A lexically scoped associated function would instead resolve free runtime names through its declaration context.

Why Dynamic Scope Exists — `.do`, `.loop`, `.each`, and Collections

FoLang's control-flow model is built on dynamically scoped associated functions. The executing function supplies the scope policy, so blocks do not require separate capturing and non-capturing forms.

//someScopeEg1.unit.fo

```
_ co.lang.unit={
  someFun()->()={
    x    co.lang.int = 10;
    total co.lang.int = 0;
    arr  co.lang.int->([5]) = [1, 2, 3, 4, 5];

    // .do reads and modifies the caller's x
    (x > 5).do({
      x.value = 20;
      co.out.println(x);
    });

    // .loop modifies the caller's x
    (x > 0).loop({
      x.value -= 1;
    });

    // .each modifies the caller's total
    arr.each(_, val).do({
      total.value += val;
    });

    // .filter, .map, and .reduce use the same dynamic caller context
    nums.filter(|x| => x > 10);
    nums.reduce(|acc, e| => {
      total.value += e;
      acc + e
    }, 0);
  }
}
```

The compiler does not create a separate capture description for each control block. It resolves names according to the scope mode of the associated function executing that block.

FoLang Control Flow Uses Dynamic Scope

no if/else keywords	-> <code>.do</code> / <code>.otherwise</code>	- dynamic scope
no for/while keywords	-> <code>.loop</code>	- dynamic scope
no foreach keywords	-> <code>.each</code>	- dynamic scope
no in keyword	-> <code>.contains</code>	- dynamic scope
no map/filter keywords	-> <code>.map</code> / <code>.filter</code>	- dynamic scope

all control flow is implemented as associated functions
 control associated functions are dynamically scoped
 caller variables are accessible and mutable under the dynamic lookup rules
 block syntax requires no separate capture mode

Dynamic and Mixed Lookup Order

For `@co.dap.dynamicscope`, runtime variable lookup follows this order:

1. function/callback parameters and local declarations
2. immediate caller activation context
3. caller activation contexts outward through the dynamic call chain
4. built-in `co.*` names and imported declarations visible to the source file
5. compiler error if no compatible declaration is available

For `@co.dap.mixedscope`, lookup follows this order:

1. function/callback parameters and local declarations
2. immediate caller activation context
3. caller activation contexts outward through the dynamic call chain
4. lexical declaration context
5. built-in `co.*` names and imported declarations visible to the source file
6. compiler error if no compatible declaration is available

Local declarations shadow caller declarations. Caller declarations shadow declarations from the lexical declaration context in mixed scope.

Types, annotations, imports, and other compile-time declarations continue to use ordinary static resolution. Dynamic lookup applies to runtime bindings, not to the identity of types or imported APIs.

Call-Site Validation of Dynamic Requirements

A dynamically or mixed-scoped associated function may reference a caller-provided runtime name that is not declared in its lexical context:

//someDynScope1.unit.fo1

```
_ co.lang.unit={
  @co.dap.dynamicscope
  (Employee) addToTotal(value co.lang.int)->() = {
    total.value += value;
  }
}
```

At every statically known call site, the compiler validates that the active caller-context chain provides a definitely initialized binding named `total` with a compatible type and permitted mutability. A call is rejected when the required binding cannot be proven to exist.

Because non-lexically scoped functions are non-first-class and non-escaping, their call sites remain statically identifiable. This avoids runtime string-based name lookup and allows the compiler to represent dynamic requirements using context and symbol-table references.

Scope Rules Summary

Function Type	Lexical	Dynamic	Mixed
methods	✓ declaration-site only	✗	✗
ordinary inner methods	✓ declaration-site only	✗	✗
free functions	✓ declaration-site only	✗	✗
ordinary inner functions	✓ enclosing declaration context only	✗	✗
target-local named functions	✓ declaration-site only	✗	✗
<code>@co.dap.inner</code> functions/methods	call-site lexical context	✗ no unrestricted caller-chain lookup	✗
anonymous functions	✓ declaration-site only	✗	✗
closures	✓ declaration-site capture	✗	✗
lambdas/callback blocks	determined by executing associated function	determined by executing associated function	determined by executing associated function
associated functions	✓ default	✓ opt-in	✓ opt-in
<code>.do</code> / <code>.loop</code> / <code>.each</code>	✗	✓ built-in	✗
<code>.map</code> / <code>.filter</code> / <code>.reduce</code>	✗	✓ built-in	✗

Additional Restrictions

- dynamically or mixed-scoped functions cannot be returned
- dynamically or mixed-scoped functions cannot be stored as values
- dynamically or mixed-scoped functions cannot be passed as ordinary callbacks
- dynamic or mixed caller contexts cannot cross thread or process boundaries
- dynamic or mixed execution cannot continue after the caller frame ends
- dynamic or mixed-scoped functions cannot participate in [Execution Models and Control Abstractions \(library type=advanced\)](#)
- dynamically or mixed-scoped functions cannot be curried
- named, optional, variadic, and default-parameter forms follow the same non-escaping and call-site-validation rules

Types

In ordinary package source, `co.lang.type`, type aliases, newtypes, opaque types, subtypes, supertypes, and parameterized `co.lang.type` constructors must be declared inside an ordinary `*.unit.fo1` file. They are contributed directly to the package namespace. Entry files, signatures, modules satisfying signature type components, and dedicated library surfaces follow their own explicitly stated rules.

Examples in this section that show only a type declaration are fragments from inside a legal unit or other legal enclosing declaration.

The three axes — each adds one new power:

Axis 1: Polymorphism (terms depend on types)

```
// "Give me a type, I'll give you a value"

// Without: write separate functions
//sometypes1.unit.fol
_ co.lang.unit={

    identityInt(x int) → int={}
    identityStr(x string) → string={}

    // With: one function works for all types
    identity(T)(x T) → T={}

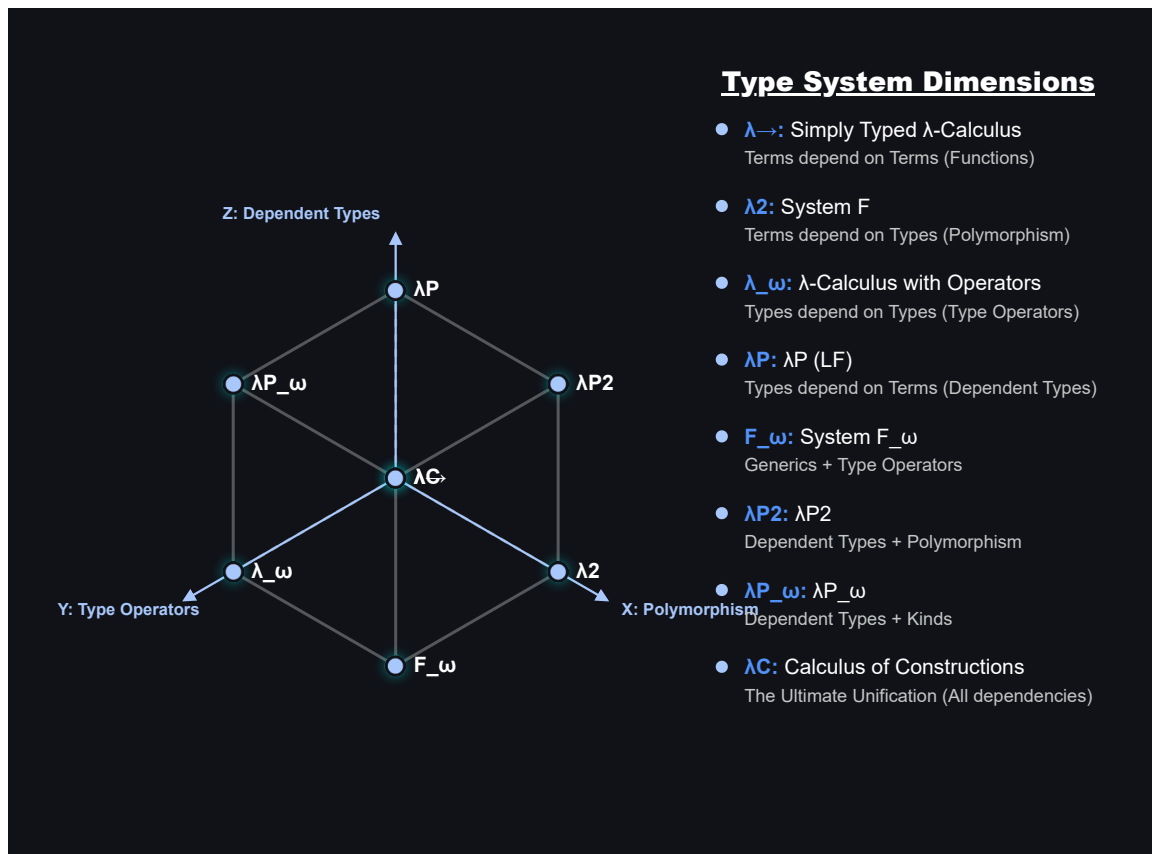
}
// This is generics / parametric polymorphism
// System F, Java generics, your @co.dap.generic
```

Axis 2: Type operators (types depend on types)

```
// "Give me a type, I'll give you a type"
//sometypes2.unit.fol
_ co.lang.unit={
    List(Int) → List ={}          // List of ints type → type
    Map(String, Int) → Map ={}    // type → type → type
    Option(T) → Some(T) | None;   // type constructor
}
// This is kinds / higher-kinded types
// Your folang: Option(T) co.lang.type = Some(T) | None()
```

Axis 3: Dependent types (types depend on values)

```
// "Give me a value, I'll give you a type"
//sometypes3.unit.fol
_ co.lang.unit={
    Vector(3) → array of exactly 3 elements
    Matrix(2, 3) → 2x3 matrix
    NonZero(n) → proof that n ≠ 0
}
// The type changes based on a runtime value
divide(a int, b NonZero(int)) → int={}
// compiler PROVES b can't be zero
```



Dependent Types

Type-Level Functions — Functions That Return Types

A function that accepts a type or value and returns a type is a **type-level function**. When its result depends on a value argument, it defines or selects a dependent type. This is distinct from a parameterized `co.lang.type` constructor such as `Option(T)`, whose declaration directly defines a family of types.

// sometypes4.unit.fol

```
_ co.lang.unit={
  // Vector – value-indexed type-level function
  // takes   → co.lang.int (size)
  // returns → co.lang.dependentType (a type)
  Vector(n co.lang.int)→(co.lang.dependentType) =
    co.lang.int→([n]);

  // calling Vector(3) returns a TYPE at compile time
  // that type is co.lang.int→([3])
  v3 Vector(3) = [1, 2, 3]; // type is Vector(3)
  v4 Vector(4) = [1, 2, 3, 4]; // type is Vector(4) – different type!

  // Vector(3) ≠ Vector(4) – completely different types
  // size is part of the type – compiler knows at compile time
}
```

More About Type

```
Name(T) co.lang.data = variants;
  → concrete ADT type-constructor definition
  → right-hand-side definition is mandatory

Name(T) co.lang.type;
  → abstract type-constructor requirement
  → permitted only inside a signature

Name(T) co.lang.type = ExistingType(T);
  → concrete type alias or signature-component binding
  → permitted in modules and ordinary type declarations
```

A Type-Level Function Returns a Type

```
Vector      → type-level function
Vector(3)   → function call → returns type co.lang.int→([3])
Vector(4)   → function call → returns type co.lang.int→([4])

just like:
  add(1, 2) → returns a value (3)
  Vector(3) → returns a type (int[3])
```

Compiler Enforced Size Safety

//sometypes5.unit.fol

```
_ co.lang.unit={
  // dot product – only valid for same size vectors
  // compiler enforces this via dependent types
  dotProduct(a Vector(n), b Vector(n))→(co.lang.int) = {
    // n is same for both – compiler verified
  }

  v3 Vector(3) = [1, 2, 3];
  v4 Vector(4) = [1, 2, 3, 4];

  dotProduct(v3, v3); // ✅ same type Vector(3)
  dotProduct(v3, v4); // ❌ compiler error – Vector(3) ≠ Vector(4)
}
```

Matrix — Two-Parameter Type-Level Function

//sOMEMATRIX.unit.fol

```

_ co.lang.unit={
  // Matrix – takes rows and cols, returns dependent type
  Matrix(r co.lang.int, c co.lang.int)->(co.lang.dependentType) =
    co.lang.int->([r, c]);

  m34 Matrix(3, 4) = [[1,2,3,4],[5,6,7,8],[9,10,11,12]];
  m45 Matrix(4, 5) = ...;

  // matrix multiply – cols of A must equal rows of B
  // n must match – compiler verified
  multiply(a Matrix(r, n), b Matrix(n, c))->(Matrix(r, c)) = {
    // compiler ensures dimensions are compatible
  }

  multiply(m34, m45); //  Matrix(3,4) × Matrix(4,5) = Matrix(3,5)
  multiply(m34, m34); //  compiler error – 4 ≠ 3
}

```

Stack — Value and Type Parameter

//somestack.unit.fo1

```

_ co.lang.unit={
  // Stack – takes size and element type
  Stack(n co.lang.int, T co.lang.type)->(co.lang.dependentType) =
    T->([n]);

  s Stack(10, co.lang.int) = ...; // stack of max 10 ints
  t Stack(5, co.lang.string) = ...; // stack of max 5 strings
}

```

Type Is Value + Kind Combined

```

Vector(3):
  kind = Vector    (what it is)
  value = 3        (how many)
  type = Vector(3) (both together – the dependent type)

Vector(3) ≠ Vector(4) ← different types entirely
Vector(3) = Vector(3) ← same type

```

Type Constructors and Type-Level Functions

```

// option.unit.fo1
_ co.lang.unit = {
  // Parameterized type declaration: Option is a type constructor.
  Option(T) co.lang.type =
    Some(T) | None();

  // Value-indexed type-level function: Vector computes a dependent type.
  Vector(n co.lang.int)->(co.lang.dependentType) =
    co.lang.int->([n]);
}

```

`Option` and `Vector` both operate at the type level, but they are different declaration categories:

```

Option(T) co.lang.type
  -> type-constructor declaration
  -> substitution produces Option(T)

Vector(n)->(co.lang.dependentType)
  -> type-level function
  -> computation produces a type

```

Simple Dependent Type

//someiden.unit.fo1

```

_ co.lang.unit={
  identity(x co.lang.int)->(x.type) ={ this.return x; }
}

```


Compile-Time and Runtime Values in Type-Related Computation

FoLang distinguishes value-indexed dependent types, compile-time type computation, runtime type descriptors, and runtime values whose concrete types may differ. These mechanisms are related, but they are not interchangeable.

1. Value-Indexed Dependent Types

A dependent type may contain a value as part of its type identity. The value index may be a compile-time constant, a symbolic type-level value, or a runtime function parameter whose relationship to the result is tracked by the compiler. `//someIndex.unit.fol`

```
_ co.lang.unit = {  
  readVector(n co.lang.int) -> (Vector(n)) = {  
    ...  
  }  
}
```

The compiler does not need to know the concrete value of `n` while compiling `readVector`. It records the relationship:

```
input index = n  
result type = Vector(n)
```

Likewise: `// sometypes6.unit.fol`

```
_ co.lang.unit = {  
  dotProduct(  
    left Vector(n),  
    right Vector(n)  
  ) -> (co.lang.int) = {  
    ...  
  }  
}
```

requires both vectors to have the same value index. Dependent typing means that a type contains or is constrained by a value; it does not mean that an arbitrary runtime branch can silently change the static type of an already compiled variable.

2. Compile-Time Type Computation

A function may compute and return a type when it is guaranteed to execute during compilation. `// someFuncomp.unit.fol`

```
_ co.lang.unit = {  
  @co.dap.compiletime  
  @co.dap.eager  
  chooseType(value co.lang.int) -> (co.lang.type) = {  
    (value < 100)  
    .return(co.lang.string)  
    .otherwise.return(co.lang.bool);  
  }  
}
```

The arguments must be compile-time evaluable when the result is used in a static type position: `//someTest1.unit.fol`

```
_ co.lang.unit = {  
  someFun() -> () = {  
    Selected co.lang.type = chooseType(10);  
    value Selected = "Hello";  
  }  
}
```

Invalid: `//someTest2.unit.fol`

```
_ co.lang.unit = {  
  someFun() -> () = {  
    input := co.in.readInt();  
    Selected co.lang.type = chooseType(input);  
    // compiler error: `input` is not compile-time evaluable  
  }  
}
```

Conceptually:

```
compile-time value  
-> compiler executes the type function  
-> one concrete static type is available before ordinary type checking completes
```

3. Built in compile type computation

A function may compute and return a type when it is guaranteed to execute during compilation.

`decltype` built in method

The arguments must be compile-time evaluable when the result is used in a static type position: `// someFun5.unit.fol`

```
_ co.lang.unit={
  someFun()->()={
    someIntVar co.lang.int ;
    someVar co.hokrlt.type.decltype(someIntVar) = 200;
  }
}
```

4. Runtime Type Descriptors

An ordinary function returning `co.lang.type` produces a runtime type descriptor when it is not executed at compile time. `//someruntime2.unit.fol`

```
_ co.lang.unit={
  selectType(value co.lang.int)->(co.lang.type) = {
    (value < 100)
      .return(co.lang.string)
      .otherwise.return(co.lang.bool);
  }

  selectedType co.lang.type = selectType(input);
}
```

Here, `selectedType` is a runtime object that describes a type. It may be used for reflection, runtime validation, dynamic loading, metadata inspection, or dynamic object creation where those capabilities are permitted.

A runtime type descriptor cannot ordinarily be used as the static type of a declaration: `//someruntime3.unit.fol`

```
_ co.lang.unit={
  someFun()->()={
    value selectType(input);

    // compiler error: runtime type descriptor used in a static type position
  }
}
```

Conceptually:

```
runtime value
  -> runtime type-selection function
  -> co.lang.type descriptor object
```

A value that represents a type is not automatically a compile-time-resolved static type. ***

5. `@co.dap.typefromvalue`

`@co.dap.typefromvalue` derives a type from the type of a returned compile-time value. In the initial FoLang specification, it is permitted only for compile-time evaluation. `//sometypeval.unit.fol`

```
_ co.lang.unit={
  @co.dap.compiletime
  @co.dap.typefromvalue
  inferType(value co.lang.int)->(co.lang.type) = {
    (value < 100)
      .return("Hello")
      .otherwise.return(co.const.true);
  }
}
```

The compiler derives:

```
"Hello"      -> co.lang.string
co.const.true -> co.lang.bool
```

Every argument must be compile-time evaluable when the result is used in a type position. Runtime value-based type selection must instead use an explicit runtime representation.

6. Runtime Values with Different Concrete Types

When a runtime branch may produce unrelated concrete value types, the function must return one stable outer type. FoLang may use a tagged value, an ADT, `co.lang.dynamic` where permitted, or another explicitly packaged representation.

Using `co.lang.tag`: `//someruntime1.unit.fol`

```

_ co.lang.unit={
  selectValue(value co.lang.int)->(co.lang.tag) = {
    (value < 100)
      .return(co.lang.tag(co.lang.string, "Hello"))
      .otherwise.return(
        co.lang.tag(co.lang.bool, co.const.true)
      );
  }
}

```

The outer static type is always `co.lang.tag`:

```

co.lang.tag
├ runtime type descriptor
└ value compatible with that descriptor

```

An ADT provides a more strongly typed alternative:

```

SelectedValue co.lang.data =
  StringValue(co.lang.string)
| BoolValue(co.lang.bool);

selectValue(value co.lang.int)->(SelectedValue) = {
  (value < 100)
    .return(StringValue("Hello"))
    .otherwise.return(BoolValue(co.const.true));
}

```

Summary

```

Vector(n)
  -> value-indexed dependent type

@co.dap.comptime function returning co.lang.type
  -> compile-time type computation

ordinary runtime function returning co.lang.type
  -> runtime type descriptor

runtime branch returning unrelated value types
  -> tag, ADT, dynamic value, or another explicit package

```

A runtime type descriptor is a value that represents a type. It must not be confused with a statically resolved dependent type.

Parameterized Type Declarations and Type-Level Functions

Two declaration families produce types from parameters. The spelling depends on whether the declaration directly defines a type family or computes a type through a function body. `//someEg7.unit.fol`

```

_ co.lang.unit={
  // all parameters are types -> parameterized co.lang.type declaration
  Option(T) co.lang.type = Some(T) | None();
  someAlias(F) co.lang.type = Functor(F);

  // a value parameter is present -> type-level function syntax
  Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);
  Stack(n co.lang.int, T co.lang.type)->(co.lang.dependentType) = T->([n]);
}

```

A parameterized `co.lang.type` declaration is a type constructor. Its type parameters appear directly in the declaration head and it does not use `@co.dap.generic`.

A function that accepts values or type values and returns `co.lang.dependentType` is a type-level function. `Stack` demonstrates why the function form exists: it can mix value parameters and type-valued parameters and compute the resulting type.

`co.lang.dependentType` is both a type-producing return kind and a direct type-declaration kind. A type-level function uses it when a value parameter determines the produced type. A direct declaration may use it when no parameter list is required:

```

LengthBound co.lang.dependentType = co.lang.int;

```

The kind is also usable in a declarator. If a function returns `co.lang.dependentType`, a binding receiving that result may therefore be declared `co.lang.dependentType`.

A type-level function has exactly one unnamed type-producing result. That result may be a union using `|`, but comma-separated multiple results are invalid: `//somebadeg1.unit.fol`

```

_ co.lang.unit={
  Choice(n co.lang.int)->(co.lang.dependentType | co.lang.type) = co.lang.int;
  Bad(n co.lang.int)->(co.lang.dependentType, co.lang.type) = co.lang.int; // invalid
}

```

Parameterized aliases are transparent

An alias declared with `co.lang.type` names the same type, not a new one. `//someParameg1.unit.fo1`

```

_ co.lang.unit={
  someFun()->()={
    someAlias(F) co.lang.type = Functor(F);

    someAlias(List); // the same type as Functor(List)
    someAlias(Option); // the same type as Functor(Option)
  }
}

```

Because the alias adds no identity, it creates no separate instance slot: an instance cannot be declared for `someAlias` as distinct from one for `Functor`. Declaring one would be a duplicate.

Named parameters also allow reordering and partial application, which a positional placeholder could not express. `//someTypeClz1.unit.fo1`

```

_ co.lang.unit={
  Pair(F, G) co.lang.type = Transformer(F, G);
  Flip(F, G) co.lang.type = Transformer(G, F);
  Fixed(F) co.lang.type = Transformer(F, Set);
}

```

Use `co.lang.newtype` instead when a **distinct** identity is wanted, such as the wrapper described in [Where an Instance Is Declared](#).

Dependent Type Index Rules

An **index** is an argument to a dependent type, such as the `n` in `Vector(n)`, or a dimension in an array derivation, such as the `n` in `co.lang.int->([n])`. Both positions obey the same rules.

An index is a literal or a name

An index is an integer literal or a name. Arithmetic, function calls, indexing and every other operator are rejected. `// someIdxEG1.unit.fo1`

```

_ co.lang.unit={
  someFun()->()={
    @co.dap.const SIZE co.lang.int = 1024;

    v Vector(3); // ✓ literal
    v Vector(SIZE); // ✓ @co.dap.const name
    buf co.lang.int->([SIZE]); // ✓ same rule for array sizes

    v Vector(n + 1); // ✗ arithmetic is not permitted in an index
    v Vector(computeSize()); // ✗ a call is not permitted in an index
    buf co.lang.int->([n * 2]); // ✗ same rule for array sizes
  }
}

```

This restriction applies only to the **size** of an array, never to element access. Indexing an array is an ordinary expression and arithmetic is fine.

```

buf co.lang.int->([SIZE]); // size – restricted
buf[i + 1] = 42; // access – unrestricted
buf[compute(x)] = 7; // access – unrestricted

```

What a named index may resolve to

A name used as an index resolves in exactly one of two ways.

A parameter bound by the enclosing signature. A type constructor or a function signature introduces the name, and every use of it inside that signature and its body refers to the bound parameter. The name is not a constant; it stands for whatever value the caller supplies. `//someEG2.unit.fo1`

```

_ co.lang.unit={
  // n is introduced here, and bound for the whole declaration
  Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);

  // n is introduced by this signature, and both parameters must share it
  dotProduct(a Vector(n), b Vector(n))->(co.lang.int) = {
    // ...
  }

  // n is introduced as a value parameter and reused in the return type
  readVector(n co.lang.int)->(Vector(n)) = {
    // ...
  }
}

```

A `@co.dap.const` **compile-time constant**. Outside a signature that binds it, a name has nothing to bind to, so it must be a constant the compiler can substitute.

```

@co.dap.const SIZE co.lang.int = 1024;
buf co.lang.int->([SIZE]);      // ✓ SIZE substitutes to 1024
v Vector(SIZE);                // ✓ same rule for dependent types

```

Nothing else qualifies. `@co.dap.final` marks an immutable binding, and an immutable value need not be known while compiling, so it cannot be substituted.

```

@co.dap.final n co.lang.int = readInput();
bad Vector(n);                // ✗ immutable, but not known at compile time

m co.lang.int = 10;
alsoBad Vector(m);            // ✗ an ordinary variable is not an index

```

So in a plain variable declaration, where no signature is binding anything, the only legal names are `@co.dap.const` constants.

An index is non-negative

Zero is permitted; a negative index is not.

```

empty co.lang.int->([0]);      // ✓ zero-length array

buf co.lang.int->([-1]);        // ✗ rejected while parsing
v Vector(-1);                 // ✗ rejected while parsing

@co.dap.const OFFSET co.lang.int = -1;
buf co.lang.int->([OFFSET]);    // ✗ rejected after substitution

```

A negative literal cannot be written at all, because no prefix operator is reachable in an index position. A negative constant is rejected when the compiler substitutes it. Both are compile-time errors.

When two dependent types are equal

Two dependent types are equal when their constructors are the same and their indices are pairwise equal. An index comparison has exactly three cases.

Index form	Compared by
integer literal	value
<code>@co.dap.const</code> name	substituted literal value
parameter	name identity

```

Vector(3)   vs Vector(3)      // equal
Vector(n)   vs Vector(n)      // equal
Vector(n)   vs Vector(m)      // NOT equal – rejected
Vector(SIZE) vs Vector(1024)  // equal when @co.dap.const SIZE = 1024

```

Rejecting `Vector(n)` against `Vector(m)` is the point of the feature. It is what lets the compiler catch a size mismatch without the developer writing a single check.

What FoLang deliberately does not do

FoLang does not decide index equality up to arithmetic. `Vector(n+1)` and `Vector(1+n)` are not merely unequal — they cannot be written.

Accepting them would require symbolic reasoning, and there is no partial version of it. Once `n+1 == 1+n` is accepted, the next reasonable request is `2*n == n+n`, and the type checker becomes a theorem prover by accretion. That is the complexity FoLang is built to avoid.

The cost is narrow. Length-arithmetic signatures such as `concat(Vector(n), Vector(m)) -> Vector(n+m)` are out of scope; return a dynamically sized type and check at run time instead. Everything that needs only same-parameter identity still works, and that covers the common cases.

```
multiply(a Matrix(r, n), b Matrix(n, c)) -> (Matrix(r, c))
dotProduct(a Vector(n), b Vector(n))    -> (co.lang.int)
zip(a Vector(n), b Vector(n))           -> (Vector(n))
```

Matrix multiplication, the usual demonstration of dependent types, needs only that the shared `n` matches.

Dependent types are checked, never inferred

Every dependent type appears in a written signature. FoLang never infers one.

This is what keeps checking decidable without a constraint solver, and it is why FoLang does not adopt Hindley-Milner style whole-program inference. Inferring a dependent type would mean inferring the index **value**, not merely the type, which is the step that makes checking undecidable in general.

Indexer

Indexer functions for a struct are associated functions and must be declared inside `<StructName>.comp.unit.fol`.

```
// MyList.fol
_ co.lang.struct = {
  eles co.lang.int->([...]);
}

// MyList.comp.unit.fol
_ co.lang.unit = {

  @co.dap.indexer(symbol="[]")
  (g MyList) get(index co.lang.int)->(co.lang.int) = {
    this.return g.eles[index];
  }

  @co.dap.indexer(symbol="[]=")
  (g MyList) set(index co.lang.int, value co.lang.int)->() = {
    g.eles[index] = value;
  }
}

1st MyList;
co.out.println(1st[0]);
1st[1] = 22;
```

Generics

```
@co.dap.generic(
  at=runtime,
  types=[
    {name= T, variance=invariant, bound=Number, kind=param},
    {name= R ,variance=invariant, bound=Number, kind=result}
  ],
  impredicative=false,
  resolution=compiletime
)
add(a T, b T)->(R) = { this.return a + b; }
```

Generic annotation fields:

Attribute	Values
types	map with type and other details
requires	
resolution	<code>runtime</code> , <code>compiletime</code>
refied	<code>true</code> or <code>false</code>
where	<code>usesite</code> or <code>callsite</code>
specializable	<code>true</code> or <code>false</code>
impredicative	<code>true</code> or <code>false</code>

types attributes |Attribute| Values| |—| |name|| |constraints|| |upper-bound|| |lower-bound|| |default|| |variance| `covariant`, `invariant`, `contravariant` | |nullable|| |inference| `param`, `result`, `arg`, `var` | |capabilities|| |where-rules|| |typekind| `type`, `class`, `function`, `struct`, `typeconstructor` | |inclusive||

The above ones will be `types` attribute's sub attributes in a map format

E.g., `@co.dap.generic(types={ T: {variance:..., bound:..., ....} })`

These are not independent attributes these are depend on type

Generic Functions — Parameters and Return Values

Rank-1: Outer function is generic; parameter uses the same type variable

`T` is fixed at the call site before the function parameter is used. The passed function is already monomorphic inside the body.

Syntax 1 — Inline signature `//somGen1.unit.fol`

```
_ co.lang.unit={
  @co.dap.generic(types=[{name=T}])
  someFunction(f (T, T)->(T), a T)->(T) = {}
}
```

Syntax 2 — Named type alias `//somGen2.unit.fol`

```
_ co.lang.unit={
  @co.dap.generic(types=[{ name=T}])
  someFArg co.lang.type = (T, T)->(T);

  @co.dap.generic(types=[{ name=T}])
  someFunction(f someFArg, a T, b T)->(T) = {}
}
```

Rank-2: The function parameter is itself polymorphic (higher-rank)

The passed function stays generic **inside the callee**. The callee decides what `T` is. Uses existing `forall`.

Syntax 1 — Inline signature `//someGen3.unit.fol`

```
_ co.lang.unit={
  someFunction(f forall(T).(T, T)->(T))->(co.lang.int) = {
    this.return f(1, 2);
  }
}
```

Syntax 2 — Named type alias `//someGen4.unit.fol`

```
_ co.lang.unit={
  someFArg co.lang.type = forall(T).(T, T)->(T);

  someFunction(f someFArg)->(co.lang.int) = {}
}
```

`//someGen5.unit.fol`

```
_ co.lang.unit={
  // Correct — Syntax 2 with co.lang.type
  someFArg co.lang.type = forall(T).(T, T)->(T);

  someFunction(f someFArg)->(co.lang.int) = {}
}
```

Returning Generic Functions

Rank-1 return `//someGen6.unit.fol`

```
_ co.lang.unit={
  @co.dap.generic(types=[{name=T}])
  makeAdder(a T)->((T)->(T)) = {
    this.return (b T)->(T) = { this.return a + b; };
  }
}
```

Rank-2 return — returning a polymorphic function `//somGen7.unit.fol`

```
_ co.lang.unit={
  makeIdentity()->( forall(T).(T)->(T) ) = {
    this.return forall(T).(x T)->(T) = { this.return x; };
  }
}
```

Rank-3: A Parameter is Itself a Rank-2 Function

Rank-3 works naturally in FoLang via `forall` nesting. No new constructs needed.

Syntax 1 — Inline `//someGen8.unit.fol`

```
_ co.lang.unit={
  // f takes a Rank-2 function as its argument – that is Rank-3
  applyRank2(
    f (forall(T).(T, T)->(T)) -> (co.lang.int)
  ) -> (co.lang.int) = {
    this.return f(1, 1);
  }
}
```

Syntax 2 — Named type aliases (cleaner) `//someGen9.unit.fol`

```
_ co.lang.unit={
  rank2FnType co.lang.type = forall(T).(T, T)->(T);
  rank3ArgType co.lang.type = (rank2FnType) -> (co.lang.int);

  applyRank2(f rank3ArgType) -> (co.lang.int) = {
    this.return f(1, 1);
  }
}
```

Rank-3 return `//somGen10.unit.fol`

```
_ co.lang.unit={
  makeRank2Consumer() -> ((forall(T).(T)->(T)) -> (co.lang.int)) = {
    this.return (f forall(T).(T)->(T)) -> (co.lang.int) = {
      this.return f(42);
    };
  }
}
```

Impredicativity — Instantiating `T` with a `forall` Type

Impredicativity is when a type variable `T` in a generic is itself instantiated with a `forall` type. Example of what this means: `//somGen11.unit.fol`

```
_ co.lang.unit={

  @co.dap.generic(types=[{name=T}])
  box(x T) -> (Box(T)) = {}

  someFun()->()={
    // Impredicative call – T being set to forall(U).(U)->(U)
    result := box(forall(U).(U)->(U)); // ❌ not legal without explicit opt-in
  }
}
```

Most type systems reject this by default. FoLang takes an opt-in approach.

v1 Workaround — Option C: Wrapping with `co.lang.type`

Not true impredicativity but solves 90% of practical cases: `//somGen12.unit.fol`

```
_ co.lang.unit={
  polyId co.lang.type = forall(U).(U)->(U);

  // box takes co.lang.type – no impredicative unification needed
  box(x co.lang.type) -> (Box(co.lang.type)) = {}

  someFun()->()={
    result := box(polyId); // ✅ works – x is co.lang.type, not a forall type
  }
}
```

v2 — Option A: `impredicative:true` in `@co.dap.generic`

Explicit opt-in via the existing annotation. The compiler only permits `forall` instantiation where declared: `//somGen13.unit.fol`


```

_ co.lang.unit={
  @co.dap.generic(
    types=[{name=T,variance=invariant}],
    impredicative=true
  )
  box(x T) -> (Box(T)) = {}

  polyId co.lang.type = forall(U).(U)->(U);

  someFun()->()={
    result := box(polyId); // ✅ legal – impredicative:true explicitly opts in
  }
}

```

Generic Function Rank Support Matrix

Scenario	Allow?	Notes
Rank-1 generic param (Syntax 1, 2, 3)	✅ Yes	Natural extension, no new concepts
Rank-1 generic return (Syntax 1, 2, 3)	✅ Yes	Same as above
Rank-2 param via <code>forall</code> (Syntax 1, 2)	✅ Yes	<code>co.lang.type</code> naturally holds polymorphic types; no new kind needed
Rank-2 param via Syntax 3 <code>co.lang.function</code>	❌ Compiler error	Function objects are concrete values; use <code>co.lang.type = forall(T).(T)->(T)</code> instead
Rank-2 return via <code>forall</code> (Syntax 1, 2)	✅ Yes	Same reasoning as param
Rank-3 via <code>forall</code> nesting (Syntax 1, 2)	✅ Yes	No new constructs — <code>forall</code> nesting works naturally
Rank-3 return	✅ Yes	Same reasoning as Rank-3 param
Rank-3 via Syntax 3 <code>co.lang.function</code>	❌ Compiler error	Same rule as Rank-2; function objects are concrete
Impredicative — v1 workaround (Option C)	✅ Yes	Wrap <code>forall</code> type in <code>co.lang.type</code> ; solves 90% of real cases
Impredicative — true opt-in (Option A)	➡️ soon v2	<code>impredicative:true</code> in <code>@co.dap.generic</code> ; explicit opt-in

`@co.dap.generic` remains the declaration annotation for constraints, variance, reification, and the other generic metadata described below. A direct generic clause supplies the named parameters and their arity; the two forms may be used together when that metadata is required.

```
// LinkedList.fol
@co.dap.generic(types=[{name=T}])
_ co.lang.struct={
    value T;
    next LinkedList;
    prev LinkedList;
}

k := LinkedList.new(co.lang.int); // when we call new it returns an object of type co.lang.uninit
actualList := k.init(); // this is what create a fully formed object of type class
```

```
// Employee.fol
@co.dap.generic(types=[{name=T},{name= R}])
_ co.lang.class = {
    id T;
    name R;

    @co.dap.override
    @co.dap.constructor(access=private)
    @@init() = {}

    @co.dap.override
    @co.dap.constructor(access=public)
    @@init(id T, name R) = {
        this.parent.init();
        this.id = id;
        this.name = name;
    }

    getEmployee(id T)->(Employee)={}
}

a := Employee.new(co.lang.int, co.lang.string);
b := a.init(1, "Rao");
```

Normally we need not use @@new and @@init it is special case only applicable for Generics when doing something really different,

Normal conditions to create/instantiate object of class we just call init which internally call new
In specific cases as above we need to do two calls or use call chain like below

```
c := Employee.new(co.lang.int,co.lang.string).init(1,"Rao");
```

This new and init methods will be available without you overriding/overloading new and init

```
// Employee.fol
@co.dap.generic(types=[{name=T},{name= R}])
_ co.lang.class = {
    id T;
    name R;
}

c := Employee.new(co.lang.int,co.lang.string).init(1,"Rao");
```

This works folang automaticall provides all type parameters new and all field init implementations.

Generics Inheritances and Types

This is in conceptual stage not supported.

- A) Abstract vs concrete type members
- B) Path-dependent types
 1. Type-level projection
 2. Path-dependent In folang how it would be

forall

What `forall` Is — and Is Not

`forall` is **not** a general-purpose generic declaration keyword. It is a **type-level expression only**, restricted to contexts where a polymorphic type must appear as an anonymous value inline — specifically Rank-2 and Rank-3 parameter and return positions, and `co.lang.type` aliases.

Named generic structs, classes, functions, and methods use `@co.dap.generic` as their sole generic-parameter declaration mechanism. `forall` is not a declaration mechanism; `forall` at declaration level is a **compiler error**.

Where forall Is Allowed — Type Expression Form Only

forall(T). followed by an anonymous type body. The . is the syntactic signal that what follows is a type body, not a declaration name.

Pattern:

```
forall(T). <anonymous type body>

// co.lang.type alias — naming a polymorphic type for reuse
someFArg co.lang.type = forall(T).(T, T)->(T);

// Rank-2 inline parameter — callee decides what T is
someFunction(f forall(T).(T)->(T)) -> (co.lang.int) = {}

// Rank-2 return type — returning a polymorphic function
makeIdentity() -> (forall(T).(T)->(T)) = {}

// Rank-3 inline parameter — f takes a Rank-2 function
applyRank2(f (forall(T).(T, T)->(T)) -> (co.lang.int)) -> (co.lang.int) = {}
```

Where forall Is Banned — Use @co.dap.generic Instead

```
// ❌ compiler error — forall at declaration level
forall(T) identity(x T)->(T) = {}

// ✅ correct
@co.dap.generic(types=[{name=T,variance=invariant}])
identity(x T)->(T) = {}

// ❌ compiler error
// LinkedList.fol
forall(T)
_(T) co.lang.struct = { value T; next LinkedList; }

// ✅ correct
// LinkedList.fol
@co.dap.generic(types=[{name=T}])
_ co.lang.struct = { value T; next LinkedList; }

// ❌ compiler error — Rank-1 generics belong to @co.dap.generic
forall(T) someFunction(f (T,T)->(T), a T)->(T) = {}

// ✅ correct
@co.dap.generic(types=[{name=T,variance=invariant}])
someFunction(f (T,T)->(T), a T)->(T) = {}
```

Quick Reference

Form	Status	Context
forall(T) name ...	❌ Compiler error	Declaration level — use @co.dap.generic instead
forall(T).(T)->(T)	✅ Allowed	Type level only — Rank-2/3 param, return, co.lang.type alias

The rule in one sentence: forall(T). forms an anonymous polymorphic type expression; it is never a declaration keyword or a file-backed declaration-name mechanism.

Generic declarations are supported only for structs, classes, functions, and methods. Their type parameters are introduced exclusively by @co.dap.generic.

The following declaration-head generic forms are invalid:

```
// Cache.fol
_(T) co.lang.module = {}           // compiler error
// operations.unit.fol
_(F(_)) co.lang.unit = {}         // compiler error
Callback(T) co.lang.delegate = (T)->(T); // compiler error
```

A parameterized co.lang.type declaration is the separate type-constructor form and does not use @co.dap.generic:

```
// option.unit.fol
_ co.lang.unit = {
  Option(T) co.lang.type =
    Some(T) | None();
}
```

Generic structs and classes remain file-backed primary declarations. Their names come from filenames:

```
// LinkedList.fol
@co.dap.generic(types=[{name=T}])
_ co.lang.struct = {
  value T;
  next LinkedList;
  prev LinkedList;
}

myIntList LinkedList = LinkedList.withTypes(co.lang.int);
```

```
// Employee.fol
@co.dap.generic(types=[{name=T}, {name=R}])
_ co.lang.class = {
  id T;
  name R;
}

emp Employee = Employee.new(co.lang.int, co.lang.string).init;
```

Generic functions use the same annotation but are declared inside a legal function-owning context such as an ordinary unit, class, or companion unit:
//sommGen1.unit.fol

```
_ co.lang.unit={

  @co.dap.generic(types=[{name=T}, {name=R}])
  add(a T, b T)->(R) = {
    ...
  }
  someFun()->()={
    add_int_int := add.withTypes(co.lang.int,co.lang.int);

    // or

    add_int_int co.lang.function = add.withTypes(co.lang.int,co.lang.int);

    k := add_int_int(12,10);
  }
}
```

Specialization

`@co.dap.specialize` to specialize generics for specific types upfront //sommGen2.unit.fol

```
_ co.lang.unit={

  @co.dap.generic(
    types=[
      {name=T}
    ],
    requires=[
      co.lang.Add(left=T, right=T, result=T)
    ]
  )
  add(a T, b T)->(T) = {
    this.return a + b;
  }
}
```

for the above generic want to specialize for `co.lang.int` //sommGen5.unit.fol

```

_ co.lang.unit={
  @co.dap.specialize(
    target=add,
    types=[
      {name=T, type=co.lang.int}
    ]
  )
  addInt(a co.lang.int, b co.lang.int)->(co.lang.int) = {
    this.return co.intrinsic.intAdd(a, b);
  }
}

```

`foLang` provides partial specialization below is the example for partial specializationn `//sommGen7.unit.foI`

```

_ co.lang.unit={
  @co.dap.generic(
    types=[
      {name=T},
      {name=R}
    ]
  )
  transform(value T)->(R) = {
    ...
  }
}

```

`//sommGen8.unit.foI`

```

_ co.lang.unit={
  @co.dap.specialize(
    target=transform,
    types=[
      {name=T, type=co.lang.string},
      {name=R}
    ]
  )
  transformString(value co.lang.string)->(R) = {
    ...
  }
}

```

*fields of specialize

Attribute	Values
target	the generic fully qualified name includes package name if omitted it is current package
types	resoulution types
priority	
strategy	intrinsic

Generic Declarations and Type Constructors

FoLang distinguishes annotation-based generic declarations from parameterized `co.lang.type` constructors.

Generic Structs, Classes, Functions, and Methods

`@co.dap.generic` is the sole mechanism for declaring generic structs, classes, functions, and methods. Generic parameters for these declaration kinds must not appear in the declaration head.

```

// Box.foI
@co.dap.generic(types=[{name=T}])
_ co.lang.struct = {
  value T;
};

```

```

// conversion.unit.foI
_ co.lang.unit = {
  @co.dap.generic(types=[{name=T}, {name=R}])
  convert(value T)->(R) = {
    ...
  }
}

```

These forms are invalid:

```
// Box.fol
_(T) co.lang.struct = { ... }      // compiler error
// Container.fol
_(T) co.lang.class = { ... }      // compiler error
```

co.lang.type Constructors

A `co.lang.type` constructor does not use `@co.dap.generic`. Its type parameters appear directly in the type declaration head, and the declaration must be inside an ordinary unit or another explicitly legal type-declaration context.

```
// option.unit.fol
_ co.lang.unit = {
  Option(T) co.lang.type =
    Some(T) | None();
}
```

`Option` denotes a unary type constructor:

```
Option : Type -> Type
```

Applying it produces a type: `//applyingEg1.unit.fol`

```
_ co.lang.unit={
  someFun()->()={
    value Option(co.lang.int);
    employeeOption Option(Employee);
  }
}
```

`Some(T)` and `None()` are value constructors declared by the `Option` definition itself. They do not require separate function implementations.

`@co.dap.generic` is invalid on `co.lang.type`, and declaration-head type parameters are invalid on structs, classes, functions, methods, signatures, interfaces, modules, enums, unions, cstructs, units, and other declaration kinds unless a later specification version explicitly adds support.

No Type-Constructor or Type-Function Annotation

FoLang requires neither `@co.dap.typeconstructor` nor `@co.dap.typefunction`.

```
Option(T) co.lang.type = ...
  -> recognized syntactically as a type-constructor declaration

ElementType(container co.lang.type)->(co.lang.type) = ...
  -> recognized syntactically as a type-level function
```

The declaration form already determines the category unambiguously.

Templates

Typed

```
// mytypedtemplate.unit.fol

_ co.lang.unit={
  @co.dap.template
  add(a co.lang.int, b co.lang.int)->(co.lang.int) ={
    this.return a + b;
  }
}
```

Untyped

```
// MyTemplate.unit.fol

_ co.lang.unit={

  @co.dap.template
  add(a, b)->(co.lang.untyped) ={
    this.return a + b;
  }
}
```

Annotations and Decorators

```
// Annotation – static object, can carry data
```

```
// myAnnotation.fol
```

```
_ co.lang.object->(for=annotation) = {  
    value  co.lang.string;  
    enabled co.lang.bool;  
}
```

```
// Decorator – function, transforms target, returns
```

```
// Decorator.unit.fol
```

```
_ co.lang.unit={  
  
    @co.dap.decorator  
    myDecorator(target co.lang.function)->(co.lang.function) = { }  
}
```

Note Directives and Pragmas are not allowed to create as they are language internals

Macros

```
// a. Basic macro
//macro1.unit.fol

_ co.lang.unit={

  @co.dap.macro
  say()->()={ this.return co.macro.quote({ println("Line 1") println("Line 2") }); }

  // b. Escape assign
  @co.dap.macro
  yes_esc_assign()->(co.lang.untyped)={
    this.return co.macro.quote({
      co.macro.esc(y) = 42;
      co.out.println("Inside macro: y = ", y);
    });
  }

}

// c. Debug macro with gensym
// macro2.unit.fol

_ co.lang.unit={
  @co.dap.macro
  debug(expr)->(co.lang.untyped)={
    tmp := co.macro.gensym(co.lang.var, "tmp");
    this.return co.macro.quote({
      tmp = co.macro.esc(expr);
      co.out.println("Result: ", tmp);
      tmp;
    });
  }

  // d. if/else condition macro
  @co.dap.macro(
    group = {items:["if","else"], chain:true},
    sugarform={forms:["if expr block"]},
    bind={vars:["x"]},
    isolate={vars:["temp", "index"]},
    gensym={prefix:"tmp_"},
    hygienic=true,
    argtransform={param:"body", wrap:"lambda", whentype:"block"},
    desugar={exprs:["if($cond) { $block }" => "if($cond,$block)"]},
    mode="inject"
  )
  if(condition expr, body block)->()={

    blockormacro co.lang.kind = block | macro

    @co.dap.macro(
      group={items:["if","else"], chain:true},
      sugarform={forms:["else block","else if"]},
      chainswith={macro:"if", position:"immediate", required:true},
      argtransform={param:"body", wrap:"lambda", whentype:"block"},
      standalone=false,
      desugar={exprs:[
        "else if($cond) { $block }" => "else(if($cond, $block))",
        "else { $elseblock }" => "else($elseblock)"
      ]},
    )
    else(body blockormacro)->()={
  }
}
```

Other macro utilities: 1. `@co.dap.compose(using=["base_if", "blockify"])` 2. `@co.dap.guard(expr="is_bool_expr(expr)")` 3. Quasiquote macros use `co.macro.quote` and `co.macro.unquote`

Execution Models and Control Abstractions (library type=advanced)

FoLang executes code sequentially by default. It also provides a uniform execution model for concurrency, parallelism, asynchronous execution, coroutines, continuations, scheduling, and structured task execution.

Developers express the intended execution semantics by applying annotations such as `@co.dap.thread`, `@co.dap.task`, or `@co.dap.process` to a method. When the method is submitted through facilities such as `co.cpcsa.submitToPool`, `co.cpcsa.submitThread`, or `co.cpcsa.submitToEventLoop`, the FoLang runtime selects and manages the appropriate execution mechanism.

Depending on the annotation, submission operation, runtime environment, and execution policy, FoLang may use a thread pool, virtual or green threads, an event loop, a dedicated operating-system thread, or a separate process. Communication operations such as sending and receiving values are also handled through the `co.cpa` package. Developers therefore describe the required execution behavior without directly managing the underlying threads, processes, pools, or event loops.

The `@co.dap.continuation` annotation enables continuation support for a function. An annotated function can use constructs provided by the `co.cpa` package to suspend execution, yield control or a value, preserve its execution state, and later resume from the suspension point.

Native Code (Library type system/ffi)

The `@co.dap.native` annotation enables access to the `co.native` package. Through this package, developers can write assembly and machine-level code using facilities such as `co.native.asm` and `co.native.inline`, providing low-level capabilities similar to those available in C++.

Native Functions

// native.unit.fo

```
_ co.lang.unit={
  @co.dap.native
  nativeMethod(a co.lang.int, b co.lang.int)->(co.lang.int) ={
    // native implementation
  }
}
```

Dynamic Runtime (library type=dynamicvmrt)

The `@co.ddap.dynamicruntime` annotation enables full access to the `co.meta` package. Through this package, developers can use dynamic class and type loading, monkey patching, runtime reflection, instrumentation, eval-based code execution, and other advanced metaprogramming capabilities.

Variable Kinds Support

Kind	Where
Normal	All
Pointers	Library of type <code>system</code> and <code>ffi</code>
Arrays	All
References Heap, Lvalue, Rvalue	Library of type <code>system</code> and <code>ffi</code>
Addresses	Library of type <code>system</code> and <code>ffi</code>
Thunks	Library <code>application</code> or <code>system</code> or <code>ffi</code> or <code>advanced</code> or <code>dynamicvmrt</code>
Ranges	All
Slices	All

Builtin Data Types

Type	Kind
<code>co.lang.string</code>	
<code>co.lang.int</code>	
<code>co.lang.bit</code>	
<code>co.lang.double</code>	
<code>co.lang.float</code>	
<code>co.lang.long</code>	
<code>co.lang.byte</code>	
<code>co.lang.char</code>	
<code>co.lang.any</code>	
<code>co.lang.dynamic</code>	
<code>co.lang.auto</code>	
<code>co.lang.infer</code>	
<code>co.lang.bool</code>	
<code>co.lang.void</code>	

co.lang.data	
co.lang.value	
co.lang.typed	
co.lang.untyped	
co.mem.region	
co.lang.nothing	
co.lang.word	
co.lang.MatchBindings	
co.lang.tag	
co.lang.typevalue	
co.lang.uninit	
co.lang.literal	
co.lang.pointer	
co.lang.address	
co.lang.reference	
co.lang.thunk	
co.lang.array	
co.lang.slice	
co.lang.range	
co.lang.just	
co.lang.operator	operator-source-only declaration kind; invalid in ordinary FoLang source
co.lang.typeclass	
co.lang.typeconstructor	reserved for a future language version; currently unsupported in source declarations
co.lang.typefunction	reserved for a future language version; currently unsupported in source declarations

A name appearing in this registry is not necessarily an enabled source-language feature. A built-in kind is usable only when this specification defines its declaration syntax and semantics. An undocumented or explicitly reserved kind remains unavailable and must produce an unsupported-feature diagnostic when used.

Built-in Directives

Kind		
PRAGMA	"@co.pdap.compiler", "@co.pdap.scale"	
DIRECTIVE	"@co.ddap.movetotop", "@co.ddap.import", "@co.ddap.dynamicruntime", "@co.ddap.use", "@co.ddap.alias"	
ANNOTATION	"@co.dap.template", "@co.dap.macro", "@co.dap.operator", "@co.dap.annotation", "@co.dap.library", "@co.dap.module", "@co.dap.pragma", "@co.dap.directive", "@co.dap.native", "@co.dap.class", "@co.dap.static", "@co.dap.instance", "@co.dap.object", "@co.dap.inline", "@co.dap.ctfe", "@co.dap.friend", "@co.dap.sealed", "@co.dap.extension", "@co.dap.override", "@co.dap.virtual", "@co.dap.abstract", "@co.dap.delegate", "@co.dap.dynamicscope", "@co.dap.lexicalscope", "@co.dap.staticscope", "@co.dap.mixedscope", "@co.dap.typeclass", "@co.dap.matcher", "@co.dap.constructor", "@co.dap.oops", "@co.dap.hokrt", "@co.dap.hokrlt", "@co.dap.indexer", "@co.dap.generic", "@co.dap.comptime", "@co.dap.typefromvalue", "@co.dap.local", "@co.dap.private", "@co.dap.public", "@co.dap.package", "@co.dap.protected", "@co.dap.internal", "@co.dap.export", "@co.dap.eager", "@co.dap.lazy", "@co.dap.packed", "@co.dap.declare", "@co.dap.simd", "@co.dap.reflection", "@co.dap.mop", "@co.dap.nested", "@co.dap.inner", "@co.dap.final", "@co.dap.const", "@co.dap.decorator", "@co.dap.specialize"	//mop => meta object programming
DECORATOR	"@co.dap.before", "@co.dap.after", "@co.dap.around", "@co.fx.onErrExcept", "@co.fx.InvokeAlways", "@co.fx.HandleEffect", "@co.dap.callback", "@co.dap.defer", "@co.dap.continuation", "@co.dap.event", "@co.dap.scale", "@co.dap.distributed", "@co.dap.concurrent", "@co.dap.parallel", "@co.dap.subroutine", "@co.dap.generator", "@co.dap.goroutine", "@co.dap.coroutine", "@co.dap.async", "@co.dap.promise", "@co.dap.future", "@co.dap.thread", "@co.dap.task", "@co.dap.fiber", "@co.dap.process", "@co.dap.spawn", "@co.dap.exec", "@co.dap.fork", "@co.dap.csp", "@co.dap.actor", "@co.dap.synthetic", "@co.dap.bridge", "@co.dap.greenlet", "@co.dap.channel", "@co.dap.callable", "@co.dap.iterator"	

Builtin Kinds

Kind	Purpose
------	---------

co.lang.type	
co.lang.struct	
co.lang.cstruct	
co.lang.realm	similar to loader where symbols reside
co.lang.loader	class, type, functions loader where objects reside loader takes realm as parameter
co.lang.class	
co.lang.interface	
co.lang.union	
co.lang.role	
co.lang.record	
co.lang.property	
co.lang.indexer	
co.lang.object	
co.lang.instance	
co.lang.matcher	
co.lang.trait	
co.lang.mixin	
co.lang.extension	
co.lang.delegate	
co.lang.typeclass	
co.lang.concept	
co.lang.typealias	
co.lang.module	
co.lang.unit	stateless file-level container; ordinary units merge into the package namespace and <code>*.comp.unit.fol</code> attaches to a struct
co.lang.macro	
co.lang.template	
co.lang.lambda	
co.lang.block	
co.lang.behavior	
co.lang.package	
co.lang.signature	
co.lang.function	
co.lang.method	
co.lang.namespace	
co.lang.stex	
co.lang.kind	
co.lang.level	
co.lang.order	
co.lang.rank	
co.lang.newtype	
co.lang.opaqueType	
co.lang.subtype	
co.lang.supertype	
co.lang.dependentType	
co.lang.refinementType	
co.lang.associatedtype	

co.lang.hokrlt	higer rank order kind level types
co.lang.data	
co.lang.enum	
co.lang.type	
co.lang.typekind	
co.lang.alias	
co.lang.value	
co.lang.just	
co.lang.nothing	
co.lang.library	
co.lang.symbol	
co.lang.reservedkeyword	

Builtin Operators

Arithmetic operators

+, -, *, /, %, **

Logical operators

&&, ||, !, &, |

Comparison operators

==, !=, <, >, <=, >=

Other operator and language-token spellings

@, #, !, ~, \$, ^, (,), _, ` , ? , { , [,] , } , \ , : , ; , " , ' , = , . , ?= , := , ::= , , , .. , ... , <.. , ..< , <..< , ==>> , ==>> , => , -> , <- , ->> , <-> , @@

Contiguous symbolic spellings that are absent from this inventory and from the active custom-operator table are unrecognized symbolic tokens; the lexer does not split them into shorter operators.

This inventory includes punctuation and reserved token spellings. Presence in this list does not make a spelling declarable as a `co.lang.operator`, nor usable with `mode=overload`, `mode=implements`, `mode=extends`, `mode=inherits` or `mode=override`.

Pre-Declared Operator Glyphs

λ, (o), ā, ī, ſ, ∃, o, ö, U, Š, Š, ħ, Π, ≡, f, T, v, F, ↓, ∂, ⊥, ↓, ↓

Every glyph in this list is language-owned and already has fixed parse properties. It cannot be redeclared with `co.lang.operator`, but it can receive implementations through `mode=overload` in a class, struct companion unit, or built-in extension package contribution. Until a matching implementation is visible, use of the glyph fails during resolution.

See [Pre-Declared Operator Glyphs](#).

Reserved words

co, let, this, self (contextual keyword), for, forall, fo (reserved word)

Difference between this and self

- `this` is for instances and objects
- `self` is for classes
- `static` — no shortcut; can be on variable or classname
- Both `self` and `this` can access member variables

Builtin Methods

method	Responsibilit
to_str	
to_int	
to_float	
to_double	
classof	

typeof	
new	
prototype	
proto	
make	
objectof	
instanceof	
is	
as	
iskindof	
has	
hasown	
uses	
match	
matchall	
matchany	
matchnone	
matchtype	
case	
default	
with	
print	
println	
printsp	
echo	
contains	
cast	
to	
dummy	
clone	
of	
for	
when	
where	
then	
callback	
getAttr	
inject	
isinstance	
cast_to	
cast_from	
do	
map	collection operation — admits a lambda callback
flatMap	
orElse	
filter	collection operation — admits a lambda callback
fold	
recover	
peek	

loop	
reduce	collection operation — admits a lambda callback
forEach	collection operation — admits a lambda callback
sortBy	collection operation — admits a lambda callback
groupBy	collection operation — admits a lambda callback
istruе	
isfalse	
if	
elif	
else	
return	
otherwise	
each	
containsVal	
in	
decltype	deduce the type at compile time
replace	
send	
receive	
submitToPool	
submitToEventLoop	
withTypes	will create appropriate object with types for generics (classes, structs, and functions/methods)
yield	
else	

Special methods

Method	Responsibility
@@new	Lifecycle method for class
@@init	Lifecycle method for class

Builtin Packages

co — root (reserved word)

The only package provided by default.

Sub-package	Responsibility
<code>co.lang</code>	All data types and kinds
<code>co.sys</code>	file, concurrent, parallel, goto, invoke, bind, call, apply, setTimeout, setInterval, scheduler, cron, event
<code>co.os</code>	signal, cmd, execute, run, env, getenv, setenv, sleep, exit, cwd, chdir, fork, wait, pipe, dup, dup2, close, readfd, writefd, random
<code>co.meta</code>	ast, instrument, transform, augment, reflect, introspect, patch, inject, create, runtime(eval,proto,prototype,etc), realm
<code>co.core</code>	list, set, map, tree, trie, sort, search, array, pointer, ref, address, ptr, matrix, word
<code>co.native</code>	load, register, asm, inline, emit, ffi, spawnnon[gpu,cpu,mpu,apu,fpga,asic,tpu,mki,mcu],arch[x86,x86-64,risc,arm,vliw]
<code>co.in</code>	read, readln
<code>co.out</code>	println, print
<code>co.regex</code>	stex, pattern, match, search
<code>co.crypto</code>	rsa, aes, hash, md5, rand, uuid, ssl, tls
<code>co.dap</code>	built-in decorators, annotations

<code>co.ddap</code>	built-in directives
<code>co.pdap</code>	built-in pragmas
<code>co.net</code>	tcp, udp, http
<code>co.const</code>	<code>true</code> , <code>false</code> , <code>none</code>
<code>co.encoding</code>	base64Encode, base64Decode, json, yml, bson
<code>co.utils</code>	makeImmutable, makeShared, copyOnWrite, toSnapshot — object behaviour policies
<code>co.dynamic</code>	dynamic capabilities
<code>co.runtime</code>	
<code>co.compiletime</code>	
<code>co.macro</code>	
<code>co.pattern</code>	
<code>co.control</code>	
<code>co.cpa</code>	concurrent, async, await, defer, lazy, parallel, process, thread, fiber, task, coroutine, continuation, cps, pool, channel ...
<code>co.hokrt1</code>	
<code>co.hokrt</code>	

FoLang Philosophy — Uniform Object Model

Core Principle

Everything in **FoLang** is an object. That is the reason for the name **FO — Functional Objects**.

FoLang gives the programmer one uniform object model across all types instead of forcing them to switch between separate type families for:

- ordinary values
- immutable values
- shared/concurrent values
- copy-on-write values
- literal values

The programmer writes against a single conceptual model and opts into the required object behaviour only when needed.

Uniform Object Principle

In FoLang, **everything is an object**.

Unless explicitly stated otherwise, the reference and mutation rules in this section apply to **managed FoLang objects**. `co.lang.cstruct` remains an object in the language model, but it is an explicitly value-semantic ABI representation: assignment and parameter passing copy its value rather than a managed object reference.

This includes:

- scalar objects
- collection objects
- user-defined objects
- ADT objects
- function objects

Functions are objects in the same sense as all other values.

This is true for both **named functions** and **anonymous functions**.

FoLang does not introduce a separate semantic model for functions, scalars, UDTs, or ADTs.

They all follow the same core object principles:

- assignment of managed objects copies references
- assignment of `co.lang.cstruct` copies its value
- `==` compares values
- mutation is applied through the object
- behaviour policies such as immutability, shared, copy-on-write, and literal conversion apply uniformly where meaningful

So in FoLang, the programmer does not need one mental model for data objects and another for function values.

The language treats them under one consistent object model.

1. Default Object Model

Every value in FoLang is an object and is **mutable by default**.

This includes:

```
co.lang.int, co.lang.float, co.lang.string → built-in scalar types
user-defined types (structs, classes, ADTs) → user-defined objects
co.core.list, co.core.map, co.core.array → built-in collections
```

Example:

```
positive_int co.lang.int = 10;
```

`positive_int` denotes an object.

Its current value is `10`.

By default, it is mutable.

2. Assignment, Aliasing, and Mutation

FoLang distinguishes three related ideas:

- **assignment**
- **aliasing**
- **mutation**

2.1 Assignment copies references

When one object variable is assigned to another, FoLang copies the **reference**, not the internal contents.

```
b := a
```

After this, `a` and `b` refer to the same object.

2.2 Mutation changes object contents

If two names refer to the same object, mutating through one name is visible through the other.

```
// Employee.fo1
_ co.lang.class = {
  Name co.lang.string
}

a Employee = { Name = "Kamesh" };
b := a;
c Employee = { Name = "Kamesh" };

a == b; // true
a == c; // true
b == c; // true

b.Name = "Ramesh"; // changes a.Name too
c.Name = "Ramesh"; // does not change a.Name
```

Why:

- `a` and `b` are the **same object**
- `c` is a **different object** with the same earlier value

2.3 `==` means value equality

In FoLang, `==` compares **values**, not object identity.

That rule is uniform across all object kinds, including built-in types and user-defined types.

So:

```
a == b
```

means:

- compare contents deeply
- return `true` when the values match
- even if `a` and `b` are not the same object

2.4 Mutation accessors

The accessor used for mutation depends on the object type:

scalar (int, float, bool, string, char, byte)	→ .value
struct field	→ .fieldname
class member	→ .membername
map entry	→ .key
array / list element	→ .index

Examples:

```
count.value = 30
emp.name = "Rao"
marks.math = 95
nums.0 = 42 or nums[0] = 42 //both forms supported
```

2.5 Function-call behaviour

Function parameters introduce a **new local binding**.

So:

- rebinding a parameter does **not** affect the caller
- mutating the object through the parameter **does** affect the caller if both still refer to the same object

```
checkPositive(a co.lang.int)->(co.lang.bool) = {
  a = 20;          // local rebinding only – caller unchanged
  a.value = 30;    // object mutation – caller sees 30
}
```

If `positive_int` is passed to `checkPositive(a)`:

<code>a = 20</code>	→ local rebinding only
<code>a.value = 30</code>	→ mutates the passed object

3. Literal Objects

All literals in FoLang create **Literal Objects**.

<code>"Kumar"</code>	→ Literal Object – string
<code>42</code>	→ Literal Object – int
<code>true</code>	→ Literal Object – bool

A Literal Object is an **anonymous object created from a literal expression**.

Literal Objects participate in value equality just like every other object in FoLang:

```
10 == 10 // true
```

In FoLang, `==` compares **values**, not object identity.

So two literal expressions with the same value compare equal, but that does not mean they are the same object.

Binding a Literal Object

When a literal is assigned to a named object, the name refers to the object created from that literal expression.

```
a co.lang.int = 10;
```

Here, `a` refers to the anonymous integer object created from literal `10`.

So:

```
a.value = 30;
```

mutates that same object, and `a` now has value `30`.

The important points are:

- literal expressions create anonymous objects
- literal-created objects are still ordinary objects unless another policy is applied
- `==` compares values, not identity
- once bound to a name, a literal-created object behaves like any other normal mutable object
- immutability applies only when explicitly requested, for example through `makeImmutable(...)`

Why Literal Objects Cannot Be Mutated Directly

Mutation can occur in two ways:

1. Through rebinding

```
a co.lang.int = 10; a = 20;
```

The above statement is valid because `a` is a valid identifier and named handle to `co.lang.int` type.

```
10 = 20; // ❌ invalid
```

The above statement is invalid because `10` is not a valid identifier and if `10` in literal object type there is no named handle.

The rebinding happens through named handles which are valid identifiers of `foLang`

2. Through a property or method

An object may also be mutated through one of its mutable properties or methods. `a co.lang.int = 10; a.value = 20;`

A literal object cannot be mutated directly because it does not provide properties/methods that can be accessed.

```
10.value = 20; // ❌ invalid
```

```
a co.lang.int = 10;
a.value = 30;
```

is valid after binding, a bare literal such as `10` cannot be mutated directly because there is no name or handle through which to perform mutation.

Not the Same as `toSnapshot(...)`

A literal object created by a literal expression is **not** the same thing as the internal snapshot representation produced by `co.utils.toSnapshot(...)`.

- literal expression → anonymous object
- `toSnapshot(x)` → internal snapshot representation used to reconstruct a fresh local object later

These are related concepts, but they are not the same mechanism.

Types of Literal objects:

1. Simple types
2. Complex or compound types (UDTS)

foLang Simple types are in literal form like `10`, `'A'`, `"A"` etc.,

```
k co.lang.int=10;
```

Compound types are in Json form

```
// Employee.fo _ co.lang.class{
```

```
id co.lang.string;
name co.lang.string;
```

```
}
```

```
k Employee = Employee { id: '10', name: "ABC" };
```

Even though compound literals are ended with brace block we need to end with semicolon as it is value not block.

```
'''
```

Summary

- literal objects are real objects
- literal expressions create anonymous objects
- identical literals compare equal by value
- identical literals are not automatically the same object
- literal-created objects are mutable by default once bound to a handle
- only `makeImmutable(...)` makes an object immutable

4. Object Behaviour Policies

Any object can be given a behaviour policy using `co.utils.*`.

All four policy calls are **in-place transformations**:

- the object itself changes behaviour kind
- there is no wrapper object
- there is no alternate binding to capture
- the original name now refers to the transformed object

All policies are **deep by default**.

They flow through nested structs, members, collection elements, and all reachable objects in the graph unless the specification later states otherwise.

4.1 Immutable

```
co.utils.makeImmutable(positive_int)
```

After this call the object itself is immutable.

This is an in-place transformation, not a wrapper.

Any attempt to mutate the object or any reachable object beneath it is a **compiler error** where detectable statically, and a **runtime error** otherwise.

```
positvie_int co.lang.int = 10;
positive_int.value = 30 // ✗ compiler error
positive_int = 30      // ✗ compiler error
```

For nested objects:

```
// Employee.fol
_ co.lang.struct = { address Address; }
Address co.lang.Address = {city co.lang.string; state co.lang.string; lane co.lang.string; pin co.lang.string;}

emp Employee = Employee{
  address = Address{
    city: "Pune";
  }
}
co.utils.makeImmutable(emp);

emp = Employee{
  address=Address{
    city: "ABC";
  }
} // ✗ compiler error

emp.address = newAddress // ✗ compiler error
emp.address.city.value = "Mumbai" // ✗ compiler error
```

Immutability is deep and total.

4.2 Value Immutable

```
positive_int co.lang.int = 20;

co.utils.makeValueImmutable(positive_int);

positive_int.value = 30; // ✗ current object is immutable
positive_int = 30;      // ✔ binding may reference a new object

co.utils.makeValueImmutable(emp);

emp.address.city = "ABC"; // ✗
emp.address.city.value = "ABC"; // ✗

emp = Employee{
  address= Address{
    city: "ABC";
  }
} // ✔
```

Difference between Immutable and Immutable Value

```
makeValueImmutable(x)
└─ current object graph cannot change

makeImmutable(x)
├─ current object graph cannot change
└─ binding cannot be reassigned
```

Table

Operation	Binding	Current value/object graph
<code>makeValueImmutable(x)</code>	Mutable	Immutable

<code>makeImmutable(x)</code>	Immutable	Immutable
-------------------------------	-----------	-----------

4.3 Shared

```
co.utils.makeShared(positive_int)
```

A Shared Object is safe for concurrent and multi-threaded use.

The programmer continues to use the same object model — same accessors, same syntax — while FoLang guarantees the required safety properties internally.

Shared behaviour is deep:

- all reachable objects within the shared object are also shared

Note on Analogies

Comparisons to things like Java's `AtomicInteger` or `ConcurrentHashMap` are **analogies only**.

They may help explain the concept, but they are **not part of the formal language contract**.

FoLang specifies the behavioural guarantee, not the exact runtime implementation.

A good explanatory statement is:

A shared `co.lang.int` may be thought of similarly to an atomic integer, and a shared map may be thought of similarly to a concurrent map. These comparisons are explanatory only. FoLang does not require any particular internal runtime representation.

4.4 CopyOnWrite

```
co.utils.copyOnWrite(positive_int)
```

A CopyOnWrite Object passes by reference like a normal object.

The copy is deferred.

It is created only when mutation is attempted in a context that must not affect the original source object.

```
process(a co.lang.int)->() = {
    a.value = 99;    // deep copy is made here – caller's object unchanged
}

co.utils.copyOnWrite(positive_int)
process(positive_int)

// positive_int still holds its old value
```

Copy-on-write is deep.

When a copy is made, the entire reachable object graph is copied.

Cyclic references must be handled by the runtime/compiler's structural clone logic with proper identity tracking.

4.5 toSnapshot

```
co.utils.toSnapshot(positive_int)
```

`toSnapshot` converts an object into a **snapshot representation** — a value descriptor, not a live object. The snapshot representation itself cannot be mutated.

When passed to a function, the compiler/runtime uses the snapshot representation to construct a fresh independent local variable bound to the parameter name. That local is a normal mutable object with no shared identity with the original. All of this happens automatically — the developer writes only

```
co.utils.toSnapshot(positive_int).
```

```
process(a co.lang.int)->() = {
    a.value = 99;    // mutates the fresh local – positive_int completely unaffected
}

process(co.utils.toSnapshot(positive_int))

// positive_int unchanged
```

The flow:

```

positive_int
  ↓ co.utils.toSnapshot(positive_int)
  snapshot representation ← value descriptor – immutable by nature, not by policy
  ↓ passed to function
  compiler constructs fresh local 'a' from the snapshot representation
  'a' is a new independent normal object – mutations never reach positive_int

```

`toSnapshot` is deep — the snapshot representation covers the entire reachable object graph.

5. Policy Summary

Object Kind	Mutation allowed	Caller sees mutation	Thread safe	Copy on write	Deep
Literal expression object	✗ directly in source — no handle	—	value-like use	—	✓
Normal (default)	✓ via accessor	✓	✗	✗	—
Immutable	✗	—	✓	—	✓
Shared	✓ via accessor	✓	✓	✗	✓
CopyOnWrite	✓ on own copy	✗	✓	✓	✓
toSnapshot result	✓ on reconstructed local object	✗	independent snapshot	—	✓

6. No Type Fragmentation

FoLang deliberately avoids special types for mutability or concurrency concerns.

There is no need for separate public type families such as:

```

ImmutableInt
SharedMap
CopyOnWriteList
AtomicInt
ConcurrentMap

```

Instead, FoLang keeps the type model uniform and applies behaviour policies through `co.utils.*`.

In many ecosystems, a programmer must constantly choose between:

```

normal integer    vs atomic integer
normal map        vs concurrent map
mutable value     vs immutable wrapper
raw structure     vs copy-on-write structure

```

FoLang instead aims to provide:

- one object model
- one programming style
- one mental model

while still allowing the programmer to opt into immutability, sharing, copy-on-write, or literal conversion when needed.

7. What Still Needs Precision

The philosophy is sound, but the formal specification still needs to define these precisely:

```

aliasing behaviour      → when two names refer to the same object
rebinding vs mutation interaction → edge-case coverage
deep policy propagation rules → exactly which objects are reachable
cyclic reference handling in COW → identity-tracking specification
visibility of mutation across calls → formal function-call model
identity vs value equality → what operators or built-ins expose identity
policy stacking         → can an object be both shared and COW?
                        → can an object be both shared and immutable?

```

8. Formal Philosophy Statement

All managed FoLang objects use reference semantics by default. `co.lang.cstruct` is an explicitly value-semantic ABI representation and is an exception to managed-object reference semantics.

In FoLang, everything is an object and managed objects are mutable by default.

Assignment of managed objects copies references, `co.lang.cstruct` assignment copies values, and `==` compares values deeply.

Developers may opt into immutability, shared behaviour, copy-on-write behaviour, or literal conversion depending on their needs.

All behaviour policies are deep — they apply to the entire reachable object graph unless stated otherwise by the formal specification.

This policy model is uniform across managed types, so programmers do not need separate type families for ordinary, immutable, concurrent, or snapshot-oriented use.

Familiar analogies such as atomic integers or concurrent maps may help explain the design, but they are not part of the formal implementation contract.

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- grammar productions and grammar notation;
- rules governing how FoLang syntax forms may be combined;
- semantic rules and behavioural descriptions;
- name-resolution and scope-resolution rules;
- type-system definitions and constraints;
- execution-model and control-abstraction descriptions;
- compiler-behaviour descriptions;
- source-code examples demonstrating FoLang syntax and semantics;
- diagrams, tables, illustrations, and formal notation;
- reference documentation;
- explanatory and instructional material.

This includes the documented expression and presentation of FoLang-specific constructs such as:

```
(booleanExpression).do({  
    ...  
}).otherwise.do({  
    ...  
});
```

and other original FoLang syntax, grammar, and semantic-rule descriptions contained in this document.

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- dependent types;
- concurrency and parallelism;
- coroutines and asynchronous execution. etc.,

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Appendix A - Complete FoLang EBNF Grammar

The following grammar is the normative lexical and syntactic grammar for FoLang.

(*

FoLang consolidated EBNF – decision-complete draft, revision 27
Derived from language-ref.md.

Revision 6 incorporated the clarified FoLang termination and identifier rules. Revisions 7 through 11 refined lexical disambiguation and selected a C++-compatible literal subset. Revision 12 formalizes the distinction between a body-closing brace and a brace that merely closes an expression. Revision 13 aligns dependent-index and method-activation semantics with the language reference. Revision 14 records the physical-nesting exceptions, ordinary local-function scope, @co.dap.inner call-site lexical context, and the package-level annotated-function constraint. Revision 15 recorded the initial project-local operator-source bootstrap and its later project-local, non-exported operator model.

Revision 16 synchronized the attached merged grammar with the finalized operator decisions, and revision 17 updated its decision metadata without changing production bodies. Revision 18 aligns type and container declarations with the then-current parser contract. Revision 19 made application-entry type declarations explicitly restricted to the reference allowlist, admitted contextual ``for`` member access, and recorded normalized companion/class operator ownership, mode, and operand rules. Revision 23 aligns the grammar with the filename-classified source model: file-backed primaries use ``_``, ordinary and companion units use reserved filename classes, ``package.fol`` uses a metadata body, entry files admit parameterized ``co.lang.type`` constructors, typeclasses have a dedicated higher-kinded parameter clause, matchers retain one declared subject type and one ``matchCase``, and type-producing function syntax is named a type-level function. Revision 24 adds an explicit source-filename metadata grammar: identifier-derived filename components are ASCII-only, begin with an ASCII letter, and permit only ASCII letters, digits, and isolated internal underscores. Filename-derived public declaration names and companion owners are normalized semantically to UpperCamelCase.

FoLang adopts a selected C++-compatible subset of built-in literal spellings, subject to these FoLang rules:

- the boolean literals are `co.const.true` and `co.const.false`
- the null/none literal is `co.const.none`
- numeric digit separators are not supported
- a floating point requires a digit on both sides of the decimal point
- the C++ pointer literal `nullptr` is not introduced
- the C++ user-defined-literal operator `""` mechanism is not introduced

A user-defined-type value is written with object-construction, for example `Employee{name: "Rao", id: 1}`, which is an expression rather than a literal token. There is therefore no separate user-defined-literal production.

Notation:

= defines a production
; ends a production
| alternative
[...] optional
{ ... } zero or more
(...) grouping
"..." terminal text
? ... ? a precisely described lexical or context-sensitive terminal

STATUS

The language reference remains authoritative. Where that document did not select a lexical or parsing rule, this revision makes an explicit design decision based on FoLang requirements and C++ backend compatibility. Every such decision is labelled DECISION-* in this grammar and in the companion decision register so it can be copied into language-ref.md.

PRINCIPAL DECISIONS

DECISION-SYN-001: A semicolon is mandatory after every simple statement whose production uses `statement-end`. Newlines never terminate statements and there is no semicolon insertion. Built-in directives are self-delimiting and do not take a trailing semicolon. A block-bodied declaration is not followed by a semicolon.

DECISION-SYN-006: TERMINATION MODEL, stated once and applied uniformly.

HARD ends. Exactly one is always required.

";" ends a simple statement, an expression-bodied or type-bodied declaration, and every forward declaration.

"}" ends a declaration body, a function body, a function-pattern body, or a standalone block statement. No ";" follows a body-selected "}".

SOFT end within one construct.

"," closes the current enum variant, map/object entry,

annotation item, parameter, argument, or grouped declarator while allowing another item to follow. It does not terminate the enclosing statement. A trailing "," is permitted where DECISION-COL-001 allows it.

NO terminator.

Built-in directives and annotations are self-delimiting; see DECISION-DIR-001.

EXPRESSION-BRACE RULE. A "}" that closes object construction, a map expression, an anonymous class expression, or another braced expression ends only that expression. The enclosing simple statement still needs its ";":

```
emp := Employee{ id: 1, name: "Rao" };
this.return Employee{ id: 1 };
cfg co.lang.map = { "a": 1, "b": 2 };
```

BODY-BRACE RULE. The following direct body forms end at their closing brace and take no following semicolon:

```
_ co.lang.struct = { id co.lang.int; } // Employee.fol
classify(n) => { this.return "positive"; }
_ co.lang.function = // SomeFArg.fol
  (a co.lang.int)->(co.lang.int) = {
    this.return a;
  }
```

DECISION-SYN-007: Body-versus-expression selection is encoded explicitly. A direct body alternative uses body-closure-guard, which rejects an immediately following semicolon. A competing expression alternative uses non-block-expression or non-anonymous-function-expression so that the same direct body cannot be reparsed as an expression plus semicolon. Parenthesized, postfix, or otherwise composed braced expressions remain expressions and require the enclosing statement terminator.

DECISION-SYN-008: PHYSICAL NESTING. Independent named type and container declarations remain package-owned primary declarations and cannot be physically declared inside another declaration, function, or executable block. Member functions, lifecycle methods, and signature/module type components are members or contract slots, not independently nested package declarations.

Two explicit syntactic exceptions exist:

local function	local-function-declaration is a named block-local function form. It requires a return-type clause and a block body.
anonymous form	anonymous-function-expression, anonymous-class-expression, block and other value expressions may appear wherever expression is admitted. forall-type is an anonymous polymorphic type expression wherever type-expression is admitted.

A lambda-expression is narrower: it is admitted only as a direct argument of map, filter, reduce, forEach, sortBy or groupBy. Nesting that call inside another expression does not remove the direct-argument relationship.

None of these anonymous forms creates an independently addressable package declaration identity. @co.dap.local, @co.dap.nested and @co.dap.inner annotate separately declared declarations and do not physically relocate them.

DECISION-SYN-009: ANNOTATED PRIMARY FUNCTIONS. Package source files forbid loose ordinary functions. annotated-function-primary is a syntactic envelope only for annotation-defined primary declaration kinds, such as a macro, decorator, native boundary declaration, or another annotation whose specification explicitly grants primary-declaration status. Adding an arbitrary annotation to an ordinary function does not make it legal at package-file scope. This legality check is semantic because annotation meaning is resolved after parsing.

DECISION-SCOPE-001: ORDINARY LOCAL FUNCTIONS. A local-function-declaration

has block-local identity and is not a package member. Its free runtime names resolve from the lexical declaration context in which the function is physically declared. Calling it does not replace that environment with the caller's runtime scope.

DECISION-SCOPE-002: `@co.dap.inner` is an association annotation, not local-function syntax and not physical nesting. An executable declaration annotated `@co.dap.inner` resolves parameters and locals first, then free runtime names through the lexical scope chain of the active attachment or call site. It does not search arbitrary caller frames as `@co.dap.dynamicscope` does. Types, imports, annotations and `co.*` names continue to use ordinary static resolution. These are semantic checks; the general annotation production intentionally remains unchanged.

DECISION-TYP-004: A dependent-type argument and an array dimension are INDEX positions. An index is an integer literal or a name; no operator, call or index expression may appear. A name must resolve to a type or value parameter in scope, or to a `@co.dap.const` compile-time constant. `@co.dap.final` denotes an immutable binding and does not qualify, because an immutable value need not be known at compile time.

An index is a NON-NEGATIVE integer. Enforcement is split across two phases, and both reject at compile time:

literal index	the grammar guarantees it. No prefix operator is reachable from dependent-index, so <code>-1</code> cannot be written. <code>->([-1])</code> and <code>Vector(-1)</code> are parse errors positioned at the <code>"-"</code> .
named index	the checker must verify it. After substituting a <code>@co.dap.const</code> name it rejects the program unless the resolved value is a non-negative integer:

```
@co.dap.const OFFSET co.lang.int = -1;
buf co.lang.int->([OFFSET]);
v Vector(OFFSET);
```

Both are compile errors. The checker already resolves the constant to compare types under DECISION-TYP-005, so the test costs nothing extra, and the diagnostic should name the constant and its declaration site.

The same check rejects a constant that resolves to a non-integer, and a name that is not a `@co.dap.const` and not a parameter in scope. Because both index positions share the dependent-index production, the rule applies identically to array dimensions and to dependent-type arguments. Zero is permitted; `language-ref.md` declares the zero-length array `co.lang.int->([0])`.

DECISION-TYP-005: DEPENDENT-TYPE EQUALITY, stated once. Two dependent types are equal when their constructors are the same and their indices are pairwise equal. An index comparison has exactly three cases:

integer literal	compared by value
<code>@co.dap.const</code>	substituted by its literal value, then compared by value
parameter	compared by name identity

Consequences, all decidable by inspection:

<code>Vector(3)</code>	vs <code>Vector(3)</code>	equal
<code>Vector(n)</code>	vs <code>Vector(n)</code>	equal
<code>Vector(n)</code>	vs <code>Vector(m)</code>	REJECTED
<code>Vector(SIZE)</code>	vs <code>Vector(1024)</code>	equal when <code>@co.dap.const SIZE = 1024</code>
<code>Vector(n+1)</code>	vs <code>Vector(1+n)</code>	not expressible; DECISION-TYP-004 rejects the index at parse time

FoLang deliberately does NOT decide index equality up to arithmetic. Accepting `n+1 == 1+n` requires symbolic reasoning, and there is no partial version of it: the next reasonable request is `2*n == n+n`, and the checker becomes

a theorem prover by accretion. Length-arithmetic signatures such as `concat(Vector(n), Vector(m)) -> Vector(n+m)` are therefore out of scope; return a dynamically sized type and check at run time instead.

What remains covers the common cases, because they need only same-parameter identity:

```
multiply(a Matrix(r, n), b Matrix(n, c)) -> Matrix(r, c)
dotProduct(a Vector(n), b Vector(n)) -> co.lang.int
zip(a Vector(n), b Vector(n)) -> Vector(n)
```

DECISION-TYP-006: Dependent types are CHECKED, never INFERRED. Every dependent type appears in a written signature, so the checker runs only in check mode. This is what keeps checking decidable without a constraint solver, and it is why Folang does not adopt Hindley-Milner style whole-program inference. Relaxing it would reintroduce the need to infer index VALUES, not merely types.

DECISION-SEM-002: ACTIVATION AND METHOD RESOLUTION. An instance may be activated by `@co.ddap.use`, making its functions callable as methods on the receiver. Activation is explicit and block-scoped, so importing a package activates nothing and can never change how an existing call resolves. For a call `xs.map(f)` the compiler tries, in order: a class method or companion-unit function on the receiver's type; an activated extension; an activated instance function whose typeclass declares the method with the receiver as first parameter; otherwise an error. First match wins, so a type's own declarations always outrank anything activated into scope. Within one scope a method name may be activated at most once per receiver type; a second activation is an error naming both sources. None of this is expressible syntactically; see the note on `use-directive`.

DECISION-SEM-001: INSTANCE SELECTION AND PLACEMENT. A typeclass instance is selected BY NAME; Folang never searches visible packages for an instance matching a type. Because nothing is inferred, no coherence check is needed and unique package names or aliases fully disambiguate. Separately, an instance is declared in the package defining the typeclass or the package defining the type, that exact package and not a sub-package. That placement rule is a convention enforced during name resolution, not a correctness requirement. Both are semantic constraints the grammar cannot express; instance-declaration accepts a misplaced instance and the compiler reports it. See the note on that production.

REVISION 25 DECISIONS

DECISION-PKG-005: REALMS ARE WITHDRAWN. The import fields "realm" and "parent-realm" and the kind `co.lang.realm` are removed. `language-ref.md` describes no realm concept and the decision register records no realm decision, so the grammar was accepting a construct that no normative document defines. Nothing referenced them: no other production, no decision, and no reference example.

DECISION-FILE-003: FILENAME-DERIVED NAMING DOES NOT APPLY EVERYWHERE. Six declaration productions deliberately take an identifier in the declaration head rather than filename-derived-name, and each has a reason that filename derivation cannot satisfy:

surface-struct-declaration	a library surface file
surface-cstruct-declaration	carries several declarations, so one filename cannot name them
data-declaration	an algebraic data type names its variants in the head
parameterized-type-declaration	a filename cannot
entry-parameterized-type-declaration	carry a generic parameter list such as (T)
entry-simple-type-declaration	the entry file is not file-backed

Every other primary declaration uses filename-derived-name and must spell "_" in the head. The "File-Backed Primary Declarations" section of `language-ref.md` lists the kinds

that require "_"; these six are its stated exceptions.

REVISION 26 DECISIONS

- DECISION-OPDECL-005: An operator body is a property list, not a declaration body. Its braces are map-shaped in the same way as package-alias-body, so under DECISION-SYN-006 the enclosing declaration takes ";" and body-closure-guard does not apply to it.
- DECISION-OPDECL-006: Operator property values are qualified constants under co.operator.*, consistent with co.const.true and co.const.none appearing in the same property list. The bare keyword spellings infix, left and binary are withdrawn, so no contextual keyword is introduced and none can collide with an ordinary identifier. co.operator is a built-in package supplying co.operator.fixity.*, co.operator.associativity.* and co.operator.arity.*.

REVISION 27 DECISIONS

- DECISION-DECL-001: primary-declaration is closed to the top-level kinds language-ref.md enumerates: struct, cstruct, class, module, signature, interface, enum, union, typeclass, instance, matcher, annotation/object, package, library and unit. The last three reach the grammar through their own source-file forms rather than through this production. Five alternatives were removed; the note on the production gives a reason for each.
- DECISION-DECL-002: A function object and a delegate are unit members, not file-backed primaries. Both take an ordinary identifier rather than "-" and both appear in unit-member.
- DECISION-DECL-003: A named block is a statement inside a function or method body. It takes an ordinary identifier and appears in statement.

CONFORMANCE NOTE, revision 27

language-ref.md declares a macro as an annotated function at file level, so annotated-function-primary is retained, but the fifteen-kind list in the reference names neither macros nor templates. Either that list should gain them or the reference should show them inside a unit. The grammar follows the examples.

- DECISION-DIR-001: Standalone built-in directives end at their complete directive form. A closing argument parenthesis, when the directive has arguments, is sufficient; no semicolon is accepted or required by the directive production.
- DECISION-OP-001: Built-in operators use the precedence table encoded in section 11. Assignment has the lowest built-in precedence.
- DECISION-OP-002: Runtime assignment operators are right-associative. Therefore a = b = c parses as a = (b = c). An assignment expression yields the assigned value. FoLang's separately specified target-first, left-to-right evaluation order is retained.
- DECISION-OP-003: := and ?= are statement-level definition operators, not general expression operators; they cannot be chained. ::= remains reserved and is not accepted by this grammar.
- DECISION-OP-007: A constant-expression may contain a registered custom operator at any declared precedence, but no runtime assignment operator may occur anywhere in its expression subtree. Grouping, arguments, literals, and other nested expression forms do not bypass this recursive guard.
- DECISION-LEX-001: Source files are UTF-8. FoLang identifiers are the ASCII subset [A-Za-z][A-Za-z0-9_]*, but may not contain consecutive underscores or end in an underscore. A lone _ is a dedicated contextual token, never an identifier.
- DECISION-LEX-002: // and non-nesting /* ... */ comments are supported. Line breaks are ordinary whitespace outside literals.
- DECISION-LEX-003: After comments, literals, and closed scanner-known composite spellings such as @@new are recognized, the lexer consumes each remaining complete maximal contiguous run of symbol characters as one symbolic token candidate. The run is never split into shorter operators as a fallback. It is classified as a structural spelling, contextual metadata, registered expression operator, or unrecognized token.
- DECISION-LEX-010: A registered expression operator containing more than one symbol character requires an explicit boundary on each operand-facing side. Whitespace, comments, and applicable delimiters establish a boundary. Structural and metadata spellings are not subject to this rule merely because they contain multiple symbols.
- DECISION-BACKEND-001: Each resolved user-defined FoLang identifier is lowered to C++ by appending the suffix _fo. Built-in names, keywords, and compiler-internal symbols use their own

compiler-defined lowering rules.

DECISION-LIT-000: FoLang accepts the selected C++-compatible numeric, character, and string literal spellings for the configured backend dialect and preserves their complete source lexemes. A C++ backend may emit those lexemes unchanged. `co.const.true`, `co.const.false`, and `co.const.none` use backend-defined lowering and are not raw C++ literals. The C++ pointer literal `nullptr` is not introduced.

DECISION-LIT-004: WITHDRAWN in revision 7. FoLang has no user-defined literal token. A user-defined-type value is built by object-construction, which is an ordinary expression.

DECISION-LIT-001: Integer literals use C++-compatible binary, leading-zero octal, decimal, and hexadecimal forms and standard integer suffixes. FoLang does not permit digit separators.

DECISION-LIT-002: Floating literals use the selected C++-compatible decimal and hexadecimal forms, exponent rules, and standard or backend-supported extended floating suffixes. FoLang does not permit digit separators and requires a digit on both sides of a decimal point.

DECISION-LIT-003: The alpha release accepts only unprefixed character and string literals without escapes. Encoding prefixes, escapes, universal character names, and raw strings are reserved and produce unsupported-feature diagnostics.

DECISION-COL-001: Commas separate enum variants, map entries, annotation-map entries, and object initializers; a trailing comma is allowed. In an enum, the comma is a soft item boundary and the closing brace is the hard structural end of the enum body. Object and annotation-map fields use colon.

DECISION-BLK-001: A block may end in one unterminated tail expression. That expression is the block value and is not a statement.

DECISION-EXT-001: The active precedence table is assembled from the language-owned built-in and pre-declared registrations plus the current compilation's custom registrations. Every implementation uses `mode=overload`; omission defaults to `overload`. `mode=override`, `mode=extends`, and `mode=define` are rejected. The alpha profile implements infix, prefix and postfix; other fixities are reserved.

REVISION 22 SYMBOLIC-RUN AND OPERATOR-BOUNDARY MODEL

DECISION-LEX-003 is revised: after comments, literals, and closed scanner-known composite spellings are recognized, the lexer consumes each remaining maximal contiguous run of symbol characters. The complete run is classified as a fixed structural spelling, contextual metadata spelling, registered expression operator, or unrecognized symbolic token. An unrecognized run is rejected without fallback splitting into shorter operators.

DECISION-LEX-010: A registered expression operator containing more than one symbol character requires an explicit operand boundary on every operand-facing side. Whitespace, comments, and applicable delimiters provide a boundary. Structural and metadata spellings are exempt unless parsed as expression operators.

DECISION-OP-004: WITHDRAWN in revision 24. `++` and `--` are removed from the built-in prefix/postfix grammar and precedence table. They receive no special prohibition; an unregistered contiguous `++` or `--` run fails under DECISION-LEX-003.

REVISION 21 UNIFORM OPERATOR REGISTRATION MODEL

DECISION-OPLIB-001: Registered built-in, pre-declared, and custom symbols all receive implementations through ordinary operator overload functions in a built-in extension unit, struct companion unit, or class. Operator functions cannot be loose package functions. Distinct normalized signatures are overloads; an equivalent signature is a duplicate. Operator override and extends modes remain unsupported.

DECISION-OPLIB-003: A custom symbol has exactly one registration in the fixed operator library, but may have zero or more distinct implementation overloads. Language-owned symbols cannot be registered locally.

DECISION-OPBOOT-001: `operator_library_folder` selects the project-local fixed operators.fol bootstrap surface. Missing configuration, folder, or file means no local custom registrations.

DECISION-OPBOOT-002: A dedicated lexer/parser accepts exactly the source-only marker `@co.dap.library(type=operator)` and the fixed `_ co.lang.library = { ... }` declaration. Its body admits only `co.lang.operator` registrations and emits no artifact.

DECISION-OPART-001/002/003: WITHDRAWN. Operator registrations and tables do not cross an ordinary library boundary and are never serialized into `.folib` or `.folenc` artifacts.

DECISION-OPBOOT-003: Before ordinary tokenization, the compiler combines the

language-owned registrations with the current compilation's local custom registrations and builds immutable maximal-munch, fixity, precedence, associativity, and arity tables. Imports contribute no operator metadata.

DECISION-OPDECL-001: A custom registration has required fixity, precedence, associativity, and arity properties and optional commutative, idempotent, identity, foldable, vectorizable, distributes_over, and desugar properties. Required keys occur once; optional keys occur at most once. Implementation annotations carry only symbol and optional mode.

DECISION-OPDECL-002: The documented mathematical/modifier glyph set is pre-declared by the language with complete parse properties and no required implementation. It cannot be registered locally and is implemented through mode=overload.

DECISION-OPDECL-003: Symbols are recognized globally in the compilation; implementations are selected later by owner, scope, activation, operand types, and normalized signature. One symbol/one concept is readability guidance, not a compiler restriction.

DECISION-OPDECL-004: The operator source uses a separate grammar. Both the operator-source lexer and ordinary lexer consume complete symbol runs with no fallback splitting. A custom symbol is recognized only by an exact whole-run match. Symbols that contain // or /* are invalid because comments take lexical priority. co.lang.operator is not an ordinary declaration kind.

REVISION 23 ALIGNMENT DECISIONS

DECISION-FILE-001: Package source filenames are classified longest suffix first as <Name>.comp.unit.fol, <Fragment>.unit.fol, then <Name>.fol. File-backed primary declarations use `` in the declaration-name slot. The filename supplies the public name; canonical identity uses normalized, case-folded filename stems.

DECISION-UNIT-001: Both ordinary and companion unit files contain `` co.lang.unit`. Ordinary unit members merge directly into the package namespace. Companion members attach only to the same-package struct named by the companion filename.

DECISION-PKG-001: The reserved package.fol source form contains exactly `` co.lang.package = { name: "segment" };`. The name field supplies the logical leaf segment and `` does not derive the literal name `package`.

DECISION-ENTRY-001: Entry/single-source files admit parameterized co.lang.type constructors but not ordinary functions, type-level functions, file-backed UDT/container declarations, units, instances, or matchers.

DECISION-TCLASS-001: A file-backed typeclass uses `` (parameters) co.lang.typeclass`. `` and the parameter clause are separate grammar components. Parameters may declare arity, and otherwise-unbound operation type variables are implicitly universally quantified per operation.

DECISION-MATCH-001: A matcher retains exactly one subject type in type= and exactly one protocol function named matchCase. The resolved first parameter type equals the declared subject type; the second parameter is co.lang.untyped and the result is (co.lang.int, co.lang.MatchBindings). matchCase overloading is forbidden.

DECISION-COMP-001: Companion ownership always comes from the filename. Ordinary non-operator parameters never establish ownership. For an operator function, a value receiver supplies the owner operand; a receiverless or type-receiver form must use the owner struct as its first ordinary parameter.

REVISION 24 FILENAME DECISION

DECISION-FILE-002: Every identifier-derived FoLang source filename component uses the ordinary ASCII identifier character shape: the first character is A-Z or a-z; later characters are A-Z, a-z, 0-9, or isolated underscores. A component cannot start or end with an underscore, contain consecutive underscores, or contain Unicode. Recognized suffixes such as .fol, .unit.fol, and .comp.unit.fol are structural and are not part of the component. Exact reserved filenames such as package.fol and operators.fol retain their context-defined classifications. Filename-backed public declaration names and companion owners are represented in UpperCamelCase. Underscores are explicit word boundaries; existing lower-to-upper boundaries in a mixed-case segment are preserved, while a single-case segment is case-folded and capitalized as one word. Multiword names written wholly in one case must therefore use underscores

when their word boundaries must be retained.

REVISION 5 DECISIONS

-
- DECISION-ANN-001: An annotation argument and an annotation-map entry both accept "=" or ":" as the key/value binder, and a bare key with no value is a flag. This admits the reference forms `@co.dap.oops(A: { inherit:true })` and `@co.dap.generic(type={T:{typename}}, R:{variance:invariant, bound=Number})`, which mix both binders freely.
- DECISION-TYP-001: Every type derivation, not only a pointer derivation, may carry a trailing attribute list. This admits the reference form `co.lang.int->(&, meta={type=out})`.
- DECISION-TYP-002: A type-level function has exactly one type-producing return item and its body binds a type-expression, so `Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);` is well formed. An algebraic ``|`` remains part of a data/type-expression binding, not a second return item. `co.lang.dependentType` is also a legal direct type declaration kind: a declaration's container kind is the kind produced by its defining expression. Where a token sequence satisfies both type-expression and expression, the type-expression reading is selected.
- DECISION-TYP-003: An array dimension may be elided in any position, so `->([,])` and `->([])` are both well formed.
- DECISION-GEN-001: Direct declaration-head parameters are restricted to parameterized `co.lang.type/co.lang.data` declarations, signature type components, and typeclass parameter clauses. Structs, classes, functions, and methods use `@co.dap.generic`. A typeclass parameter may declare arity, so unary and binary constructor parameters are written `F(_)` and `G(_, _)`.
- DECISION-KIND-001: An enabled built-in kind explicitly listed in `general-declarable-kind` is parsed by `general-kind-declaration` when it has no dedicated production. Reserved future kind names have no production and are rejected as unsupported until their syntax and semantics are specified.
- DECISION-SYN-003: The named local-function exception uses a statement-level local function declaration with both a return-type clause and a block body. This admits `someother()->()={...}` while keeping `foo()`; an expression statement rather than a forward declaration. Its lexical scope is defined by DECISION-SCOPE-001.
- DECISION-SYN-004: Annotations may prefix an expression statement, admitting `@co.dap.lazy applied to x = add(1, 2);`.
- DECISION-SYN-005: A block is a statement in its own right and takes no trailing semicolon.
- DECISION-FUN-001: The "=" before a function block body is optional, so both `add(a T)->(T) = { ... }` and `add(a T)->(T) { ... }` parse.
- DECISION-FUN-002: A closure declaration uses an equals sign before one or more parameter lists and `==>>` before its expression body. One parameter list declares an ordinary closure; two or more parameter lists declare a curried closure.
- DECISION-OP-005: `::=`, `->>`, `<->`, the backtick, and the backslash are reserved operator tokens. The lexer recognizes them and the parser rejects them, so they cannot be silently reused by a user-defined operator before the language assigns them meaning. The reserved-for-future glyph set is reserved the same way outside literals and declared operator symbols.
- DECISION-OP-006: `~`, `#` and `^` are reserved PREFIX spellings, refused in operand position exactly as DECISION-OP-005 refuses `::=`, `->>` and `<->`. Complement and length/count are candidate meanings; none is assigned yet. Every other role of each spelling is unaffected. `@` is not a prefix operator at all: it introduces a directive or annotation and marks the `->(@)` address derivation. Address, pointer and reference variables carry their kind in the TYPE derivation and are read and assigned like ordinary variables, so no use site ever restates the kind. See the note on prefix-operator.
- DECISION-LIT-005: `FoLang` boolean literals are `co.const.true` and `co.const.false` and the null literal is `co.const.none`. `true` and `false` are NOT reserved words and NOT literals. Inside an annotation argument, a bare `true`, `false`, or `True` is an ordinary annotation-value name.

REVISION 6 CLARIFICATIONS

-
- DECISION-LEX-001 is restored without a leading-underscore extension: ordinary identifiers must begin with an ASCII letter. `_x`, `_1`, `a_b`, and `a_` are not identifiers. A lone `_` remains available only through grammar productions that explicitly admit the contextual underscore token.

DECISION-DIR-001 records that built-in directives are self-delimiting and are therefore exceptions to simple-statement semicolon termination.

REVISION 7 LEXICAL DISAMBIGUATION

Revision 7 recorded four scanner interactions inherited from the initial C++-compatible literal draft. Later revisions simplified the literal set: DECISION-LEX-005 was withdrawn in revision 9 and DECISION-LEX-007 in revision 11. The active scanner constraints are DECISION-LEX-006 and DECISION-LEX-008 together with ordinary maximal munch.

DECISION-LIT-006: A FoLang floating literal carries at least one digit on EACH side of the decimal point. The C++ abbreviated forms 1. and .10 are rejected; write 1.0 and 0.10. The rule applies to hexadecimal floating literals too, so 0x1.8p3 is valid while 0x1.p3 and 0x.8p3 are not. The exponent form without a point, such as 1e5, is unaffected.

This narrows the C++ literal set. Every accepted FoLang floating literal remains C++ compatible, while the raw-lexeme policy of DECISION-LIT-000 remains applicable.

The rule is what makes the scanner simple. Because a numeric literal can no longer end at a point, plain maximal munch already yields the right tokens with no lookahead of any kind:

```
1 .. 10      -> 1 .. 10
3.to_str()   -> 3 . to_str ( )
3.14.to_str() -> 3.14 . to_str ( )
1.5 .. 2.5   -> 1.5 .. 2.5
1e5 .. 2e5   -> 1e5 .. 2e5
1 .... 10    -> unrecognized symbolic run
```

DECISION-LEX-005: WITHDRAWN in revision 9. It existed only to stop a numeric literal from swallowing a point that belonged to the range operator or to member access. DECISION-LIT-006 removes the abbreviated forms that made that possible, so the special scanning rule is no longer needed and DECISION-LEX-003 maximal munch applies unmodified.

DECISION-LEX-006: An identifier token must NOT be immediately followed by "_". This turns a trailing or doubled underscore into a lexical error instead of silently splitting a_ into the identifier a plus the contextual token _ , and a__b into a plus __b. Restated positively: an identifier starts with an ASCII letter, may contain single underscores between nonempty alphanumeric segments, never contains consecutive underscores, and never ends in an underscore.

DECISION-LIT-007: FoLang does NOT adopt the C++14 digit separator. A numeric literal contains digits only, so 1'000 and 0x1'a are rejected and 1000 and 0x1a are written plainly. The apostrophe therefore has exactly one meaning in FoLang source: it delimits a character literal. This is a narrowing of the C++ literal set. The numeric raw-lexeme policy of DECISION-LIT-000 still applies, and the choice is reversible because admitting a separator later would accept strictly more programs.

DECISION-LEX-007: WITHDRAWN in revision 11. The base-aware separator adjacency rule existed only to disambiguate the apostrophe between a digit separator and a character literal. DECISION-LIT-007 removes the separator, so the apostrophe is unambiguous and the rule is unnecessary.

DECISION-LEX-008: Adjacent encoding prefixes, raw-string introducers, and backslashes in quoted literals begin reserved post-alpha spellings. The scanner consumes the complete spelling and reports an unsupported feature. With intervening whitespace, those names remain ordinary identifiers.

DECISION-LEX-009: A SPECIAL METHOD is one complete spelling from a closed set, scanned whole and classified from a table the way a built-in method name is. "@@" is not a prefix operator over an arbitrary identifier: @@new and @@init are the members, and any other "@@" name is a lexical error that names the admissible set. Keeping the check in the scanner means the parser's lifecycle-name production never enumerates them, so the set is extended in one place. See special-method.

CONFORMANCE NOTE ON DECISION-SYN-001

Semicolons remain mandatory for productions that explicitly use statement-end, including ordinary declarations and executable statements. Built-in directives are a deliberate exception and are self-delimiting. Block-bodied declarations end structurally at their closing brace.

Context-sensitive rules such as source-file kind, declaration legality, type checking, visibility, operator ownership, capture, and definite initialization remain semantic constraints and are documented separately.

REVISION 5 CHANGE LOG

Constructs that revision 4 could not parse, now accepted:

1	inner/nested functions in a block	local-function-declaration
2	type-level function bodies	type-level-binding
3	@co.dap.oops(A: {...})	annotation-binder
4	bare flag keys such as {typename}	annotation-map-entry
5	"=" inside an annotation map	annotation-binder
6	co.lang.int->(&, meta={type=out})	attribute tails on every derivation form
7	co.lang.int->([,])	array-dimension-content
8	_ (F(_), G(_)) co.lang.typeclass	generic-parameter, generic-arity-clause
9	@co.dap.lazy over x = add(1, 2);	expression-statement
10	add(a T)->(T) { ... } without "="	function-definition
11	closure = (f int, x int) ==> x * f; curry = (f int)(v int) ==> f * v;	closure-declaration curried closure form
12	a bare block used as a statement	statement

Silent misparse removed:

~28 built-in kinds had no declaration production and were absorbed by variable-declaration. They are now parsed by general-kind-declaration.

Token inventory completed:

:= -> <-> backtick backslash are reserved-operator; the reserved glyph set is reserved-future-operator.

Reference divergences corrected or labelled:

booleans are co.const.true/false, co.const.none added, and true and false remain ordinary names rather than FoLang literals. Revision 6 restores ASCII-letter-first identifiers and makes built-in directives self-delimiting without removing any planned grammar production.

Hygiene:

resolved 10 formerly unreferenced productions. co-path, contextual-keyword, hard-reserved-word, octal-digit-sequence, result-binding, and white-space were wired into the grammar; declaration-prefix, external-variable-declaration, index-expression, and member-access-expression were deleted. The ambiguous word-type-specification was removed, and array-dimension plus type-or-value-argument were narrowed to drop subsumed alternatives.

REVISION 7 CHANGE LOG

- DECISION-LIT-004 withdrawn; production folang-user-defined-literal removed; literal now reduces to builtin-literal. A user-defined-type value is object-construction, already defined in section 11.
- DECISION-LEX-005 added: "." is not absorbed into a numeric literal when another "." follows, so the range operator survives maximal munch.

REVISION 8 CHANGE LOG

- DECISION-LEX-005 extended: "." is also not absorbed when an identifier-start character follows, so 3.to_str() scans as 3 . to_str and the parser resolves the dot as member access. The trailing-dot float 1. is retained, so revision 8 did not yet narrow the C++ literal set then in use. Revision 9 later removed that abbreviated form. The revision 7 residual note recommending (3).to_str() is withdrawn.

REVISION 14 CHANGE LOG

- Reconciled the grammar commentary and decision register with language-ref(36).md physical-nesting rules.
- DECISION-SYN-008 records that independent named type/container declarations cannot be physically nested, while ordinary local functions and anonymous expression/type-expression forms are explicit exceptions. Existing productions already encoded this distinction, so no executable production was widened.
- DECISION-SCOPE-001 records declaration-site lexical scope and block-local identity for ordinary local functions.
- DECISION-SCOPE-002 records call-site lexical-context resolution for separately declared executable @co.dap.inner declarations. The annotation remains parsed by the general annotation production; scope validation is semantic.
- DECISION-SYN-009 closes the annotated-function-primary loophole: arbitrary annotations do not legalize a loose package-level function. Only annotation-defined primary declaration kinds may use that envelope.
- Corrected the stale revision-12 production and reachability counts in the

grammar footer and regenerated validation from the actual revision-14 production graph.

REVISION 13 CHANGE LOG

- DECISION-SEM-002 added: @co.ddap.use activates extension and instance methods alike. use-directive gains a closed use-field list with the keys from and methods, matching import-field, so a mistyped key is a parse error. There is no separate "extensions" key: an extension already declares what it extends through @co.dap.extension on the method itself, so the activation site has nothing to restate. Resolution order and the one-activation-per-receiver rule are semantic and recorded as a note.
- DECISION-SEM-001 added: instance selection is by name, so no coherence check is required; instance placement is restricted to the typeclass's or the type's exact package as a convention. Both are semantic, so they are recorded as a note on instance-declaration. No production changes.
- DECISION-TYP-004 mechanized: type-or-value-argument and array-dimension now take the new production dependent-index instead of constant-expression and expression. Arithmetic, calls and index expressions are rejected syntactically in both positions.
- DECISION-TYP-005 added: the dependent-type equality rule, its three index comparison cases, and an explicit statement of what is out of scope and why.
- DECISION-TYP-006 added: dependent types are checked, never inferred.
- A constant-expression is retained; it is still used for enum variant values, where ordinary constant arithmetic remains permitted.
- DECISION-TYP-004 records that an index is non-negative, that a literal index is guaranteed by the grammar because no prefix operator is reachable, and that a named index must be verified by the checker after @co.dap.const substitution. The rule applies identically to array dimensions and dependent-type arguments, since both share dependent-index. Zero remains permitted.
- Verified against language-ref.md: every dependent-type argument and array dimension in the document is a literal or a name, so no example changes.

REVISION 12 CHANGE LOG

- DECISION-SYN-007 added. Direct body forms and expression forms are now formally separated with zero-width contextual guards rather than only relying on ordered choice.
- Named UDT/container bodies use body-close, which rejects an immediate semicolon after the body-closing brace.
- A direct pattern block, function definition, function-kind inline body, named block, labeled block, or standalone block statement uses body-closure-guard.
- Object construction, map literals, anonymous class expressions, and other braced expressions remain expressions; their enclosing simple statements still require semicolons.
- Stale literal-decision wording was corrected after removal of numeric digit separators in revision 11.

REVISION 11 CHANGE LOG

- DECISION-LIT-007 added: the C++14 digit separator is not adopted. The production digit-separator is deleted and the five digit-sequence productions that referenced it are simplified to digits only.
- DECISION-LEX-007 WITHDRAWN. With no separator, the apostrophe has a single meaning and needs no adjacency test.
- Net effect across revisions 9 and 11: two of the four revision 7 scanning rules are now gone. What remains is DECISION-LEX-006, the identifier trailing-underscore guard, and DECISION-LEX-008, the reserved-literal adjacency rule.

REVISION 10 CHANGE LOG

- DECISION-SYN-006 added: the termination model is now stated once, covering the ";" and "}" hard ends, the "," soft separator, the self-delimiting directives and annotations, and the nuance that an expression-closing "}" does not terminate its statement.
- function-object-declaration no longer demands ";" after a block body, which contradicted every sibling declaration. It gains function-object-binding, which also admits the reference form _ co.lang.function = add; that revision 9 could not parse.
- pattern-result gains a hard end: an expression-bodied function-pattern clause now takes ";", while a block-bodied clause still ends at "}". Previously such a clause had no terminator at all.
- Audited declaration- and statement-level productions against the intended model. Revision 12 later encoded the remaining direct body-versus-expression priority explicitly.

REVISION 9 CHANGE LOG

- DECISION-LIT-006 added: a floating literal needs a digit on each side

- of the point, so 1. and .10 are rejected in favour of 1.0 and 0.10. fractional-constant and hexadecimal-fractional-constant lose their abbreviated alternatives. This narrows the C++ literal set while preserving the numeric raw-lexeme policy of DECISION-LIT-000.
- DECISION-LEX-005 WITHDRAWN. With the abbreviated forms gone, a numeric literal can never end at a point, so DECISION-LEX-003 maximal munch produces the correct token stream unaided. The scanner needs no numeric lookahead and the parser never re-lexes.
- Net effect: one scanning rule removed, two productions simplified, and every ambiguity discussed in revisions 7 and 8 eliminated at source rather than worked around.
- DECISION-LEX-006 added: an identifier may not be followed by "_", so a trailing or doubled underscore is an error rather than a token split.
- DECISION-LEX-007 added: base-aware digit-separator adjacency rule.
- DECISION-LEX-008 added: reserved-literal adjacency rule.
- Production count corrected.

Verified after revision 23: 350 unique productions; no duplicate production; no undefined nonterminal reference; no unterminated production; balanced EBNF grouping delimiters; 312 productions reachable from compilation-unit; and 38 intentional secondary/lexical/informative roots with no unclassified unreachable production.

```

*)

(* ===== *)
(* 1. Compilation units *)
(* ===== *)

compilation-unit = package-source-file
                  | application-entry-file
                  | library-surface-file ;

(*)
DECISION-FILE-001 / DECISION-FILE-002 / DECISION-UNIT-001 /
DECISION-PKG-001:
The compiler validates and classifies the external source filename before
selecting a source-text root. source-filename is an intentional secondary
metadata root; it is not concatenated with the source text. Classification
uses the longest recognized suffix first. Filename-to-UppercamelCase
conversion, owner lookup, and package-index merging are semantic/indexing
checks.

(*)
(*)
DECISION-FILE-002:
source-filename is an external metadata grammar. Identifier-derived filename
components reuse the ASCII identifier shape but do not undergo source-token
reserved-word classification. The alternatives are classified contextually
and longest suffix first.

*)
source-filename = companion-unit-filename
                  | ordinary-unit-filename
                  | package-metadata-filename
                  | operator-bootstrap-filename
                  | ordinary-source-filename ;

companion-unit-filename = filename-identifier, ".comp.unit.fol" ;

ordinary-unit-filename = filename-identifier, ".unit.fol" ;

package-metadata-filename = "package.fol" ;

operator-bootstrap-filename = "operators.fol" ;

ordinary-source-filename = filename-identifier, ".fol" ;

filename-identifier = identifier-head, { "_", identifier-segment } ;

package-source-file = package-primary-source-file
                      | ordinary-unit-source-file
                      | companion-unit-source-file
                      | package-metadata-source-file ;

package-primary-source-file = file-preamble, primary-declaration ;

ordinary-unit-source-file = file-preamble, unit-declaration ;

companion-unit-source-file = file-preamble, unit-declaration ;

package-metadata-source-file = file-preamble, package-alias-declaration ;

application-entry-file = file-preamble, { entry-item } ;

library-surface-file = file-preamble, library-declaration ;

```

```

file-preamble = { file-directive } ;

file-directive = import-directive
                | alias-directive
                | use-directive
                | dynamic-runtime-directive
                | pragma-directive
                | generic-directive ;

entry-item = file-directive
            | entry-type-declaration
            | bare-function-pattern-clause
            | capturing-function-pattern-clause
            | statement ;

(*)
DECISION-SYN-008 / DECISION-SYN-009:
This is the complete package-source primary-declaration entry point.
Independent named type/container declarations are reachable here, not from
statement or block-item. annotated-function-primary is reserved for
annotation-defined primary declaration kinds; it does not permit an
arbitrary annotated ordinary function at package-file scope.
*)
primary-declaration = struct-declaration
                    | cstruct-declaration
                    | enum-declaration
                    | union-declaration
                    | class-declaration
                    | interface-declaration
                    | signature-declaration
                    | module-declaration
                    | typeclass-declaration
                    | object-declaration
                    | instance-declaration
                    | matcher-instance-declaration
                    | annotated-function-primary
                    | forward-type-declaration ;

(*)
DECISION-DECL-001, revision 27: primary-declaration is CLOSED to the
top-level kinds language-ref.md enumerates. Five alternatives were removed:

function-object-declaration    a unit member, see unit-member
delegate-declaration           a unit member, see unit-member
named-block-declaration        a statement; the reference states a block
                                cannot live outside a function or method
annotated-contract-declaration a pre-co.lang.typeclass spelling,
                                superseded by typeclass-declaration
general-kind-declaration        admitted about twenty-eight further
                                co.lang.* kinds that the reference lists
                                only in its built-in kind table and gives
                                no declaration form

forward-type-declaration is retained as the forward spelling of a struct or
class, not an additional kind.
*)

(*) ===== *)
(*) 2. Directives, annotations, and metadata *)
(*) ===== *)

annotations = { annotation } ;

one-or-more-annotations = annotation, { annotation } ;

(*)
DECISION-SCOPE-002:
@co.dap.inner uses this ordinary annotation syntax. Whether it is legal on
the annotated declaration, how an attachment is established, and how free
runtime names resolve are semantic properties; the lexer/parser does not
create a physically nested declaration node.
*)
annotation = "@", qualified-name,
            [ "(", [ annotation-argument-list ], ")" ] ;

annotation-argument-list = annotation-argument,
                          { ",", annotation-argument }, [ ",", ] ;

(*)
DECISION-ANN-001: "=" and ":" are interchangeable binders.
The optional group is written as a unit so a recursive-descent or PEG
parser backtracks it cleanly: for a bare value such as co.lang.int the
group tries "co" then finds no binder, abandons the group as a whole, and
the value is matched by annotation-value.

```

```

*)
annotation-argument = [ annotation-key, annotation-binder ], annotation-value ;

annotation-binder = "=" | ":" ;

annotation-key = identifier, { "-", identifier } ;

annotation-value = literal
    | type-expression
    | qualified-name
    | declaration-reference
    | annotation-list
    | annotation-map
    | annotation-arrow-pair ;

annotation-list = "[", [ annotation-value,
    { ",", annotation-value }, [ ",", ] ], "]" ;

annotation-map = "{", [ annotation-map-entry,
    { ",", annotation-map-entry }, [ ",", ] ], "}" ;

(*
    DECISION-COL-001 / DECISION-ANN-001:
    A map entry uses either binder, and a bare key is a flag whose value is the
    boolean true. This admits {T:{typename}} and {variance:invariant,
    bound=Number} and {type=out}.
*)
annotation-map-entry = annotation-key, annotation-binder, annotation-value
    | annotation-key ;

annotation-arrow-pair = string-literal, ">", string-literal ;

(* DECISION-DIR-001: standalone directives are self-delimiting. *)
import-directive = "@co.ddap.import", "(", import-field,
    { ",", import-field }, [ ",", ], ")" ;

(*
    DECISION-PKG-REALM-WITHDRAWN, revision 25: "realm" and "parent-realm" were
    removed. language-ref.md documents no realm concept, the decision register
    records no realm decision, and an import field the reference never describes
    cannot be validated by any conformance check.
*)
import-field = ( "package" | "library" | "src-library" | "expect"
    | "as" ), "=", annotation-value ;

(* The alias target must be a co.* path, so co-path is used directly. *)
alias-directive = "@co.ddap.alias", "(", co-path, ",",
    "as", "=", string-literal, [ ",", ], ")" ;

(*
    DECISION-SEM-002, activation. A closed field list, matching import-field, so
    a mistyped key such as "method" is a parse error rather than a silently
    ignored argument.

    "from" names a DECLARATION, unlike "package" which names a package only:

        from="stringextension"      bare, resolved in the current package
        from="tc.ListFuncutor"      alias + instance
        from="ext.stringextension"  alias + unit
        from="abc.tc.ListFuncutor"  full package + instance
        from="abc.ext.stringextension" full package + unit

    "methods" is the only list attribute. It activates named functions whatever
    "from" resolves to, an extension unit or a typeclass instance. An extension
    already declares what it extends through @co.dap.extension on the method
    itself, so the activation site has nothing to restate. The list may be
    omitted to activate everything the source provides.

    These are semantic constraints; this production accepts any combination.
*)
use-directive = "@co.ddap.use", "(", use-field, { ",", use-field },
    [ ",", ], ")" ;

use-field = ( "from" | "methods" ), "=", annotation-value ;

dynamic-runtime-directive = "@co.ddap.dynamicruntime",
    [ "(", [ annotation-argument-list ], ")" ] ;

pragma-directive = ( "@co.pdap.compiler" | "@co.pdap.scale" ),
    [ "(", [ annotation-argument-list ], ")" ] ;

generic-directive = "@co.ddap.", identifier,
    [ "(", [ annotation-argument-list ], ")" ] ;

```

```

(* ===== *)
(* 3. Names and references *)
(* ===== *)

filename-derived-name = "_" ;

qualified-name = ( identifier | "co" ), { ".", identifier } ;

co-path = "co", ".", identifier, { ".", identifier } ;

declaration-reference = qualified-function-reference | qualified-name ;

qualified-function-reference = qualified-name, "(", [ type-list ], ")",
    return-type-clause ;

(*
    DECISION-LEX-009: a special method is a CLOSED set of complete spellings, not the
    "@@" prefix applied to an arbitrary identifier. The scanner resolves the whole
    spelling against that set and emits one token, exactly as it resolves a built-in
    method name from its table, so "@@" never reaches the parser as a prefix and a name
    outside the set is a lexical error naming the admissible ones.

    The set is listed under "Special methods" in language-ref.md. Adding a special
    method means adding its spelling here and to the scanner's table; nothing in the
    parser enumerates them.
*)
lifecycle-name = special-method ;

special-method = "@@new" | "@@init" ;

special-binding = result-binding | self-binding ;

(* $ alone is the self-referential let binding. *)
self-binding = "$" ;

(* $1, $2, ... capture the previous result in a =>> delegation chain. *)
result-binding = "$", digit, { digit } ;

wildcard = "_" ;

(* ===== *)
(* 4. Type syntax *)
(* ===== *)

type-expression = forall-type | union-type-expression ;

(*
    DECISION-SYN-008:
    forall-type is an anonymous polymorphic TYPE EXPRESSION. It is not a
    declaration prefix and creates no package-owned declaration identity.
*)
forall-type = "forall", "(", type-parameter-list, ")", ".", type-expression ;

type-parameter-list = identifier, { ",", identifier } ;

union-type-expression = arrow-type-expression,
    { "|", arrow-type-expression } ;

arrow-type-expression = type-postfix-expression,
    [ "->", arrow-type-tail ]
    | "(", [ function-type-parameter,
        { ",", function-type-parameter } ],
        ")", "->", arrow-type-tail ;

arrow-type-tail = type-derivation
    | parenthesized-type-list
    | type-expression ;

type-postfix-expression = type-atom, { type-argument-list } ;

(* A parenthesized type atom groups exactly one type. `(A, B)` remains a valid
    tuple/grouping expression in section 11, but it is not silently reclassified
    as one type atom. A comma in a function-type head must introduce another
    parameter; `(A,)->(B)` is therefore invalid. *)
type-atom = qualified-name
    | "(", type-expression, ")" ;

type-argument-list = "(", [ type-or-value-argument,
    { ",", type-or-value-argument } ], ")" ;

(*
    DECISION-TYP-004:
    A dependent-type argument is an INDEX, not a general expression. It admits
    an integer literal or a name only. Arithmetic, calls, indexing and every

```

other operator are rejected here. That restriction is what keeps dependent-type equality decidable by inspection instead of by a solver.

A name used as an index must resolve either to a type or value parameter in scope, or to a `@co.dap.const` compile-time constant. `@co.dap.final` marks an immutable binding and is NOT sufficient, because an immutable value need not be known at compile time while an index must be.

```
@co.dap.const SIZE co.lang.int = 1024;
buf Vector(SIZE);           legal, SIZE substitutes to 1024
@co.dap.final n co.lang.int = readInput();
bad Vector(n);              rejected, immutable but not constant
```

Because the restriction is syntactic, the diagnostic can name the offending operator: "arithmetic is not permitted in a dependent index".

An index is non-negative. A literal index cannot be negative because no prefix operator is reachable here, so `-1` is a parse error. A named index must be verified by the checker after substitution:

```
@co.dap.const OFFSET co.lang.int = -1;
buf co.lang.int->([OFFSET]);  compile error, resolves to -1
v Vector(OFFSET);            compile error, resolves to -1
```

Zero is permitted, as in the zero-length array `co.lang.int->([0])`.

```
*)
type-or-value-argument = type-expression | dependent-index ;

dependent-index = integer-literal | qualified-name ;

type-list = type-expression, { ",", type-expression } ;

parenthesized-type-list = "(", [ type-list ], ")" ;

type-derivation = "(", derivation-specification, ")" ;

(*
  DECISION-TYP-001:
  Every derivation form may carry a trailing attribute list. The bare
  derivation-attribute-list alternative covers the repr/sign/region word and
  address forms, so the former word-type-specification production, which was
  textually identical to it, has been removed.
*)
derivation-specification = pointer-specification
                          | array-specification
                          | reference-specification
                          | range-type-specification
                          | slice-type-specification
                          | thunk-type-specification
                          | address-type-specification
                          | derivation-attribute-list ;

pointer-specification = pointer-stars,
                        [ ",", derivation-attribute-list ] ;

pointer-stars = ? one contiguous symbolic run consisting only of one or
                more "*" characters; its length is the pointer degree ? ;

reference-specification = ( "&" | "&&" | "~" ),
                          [ ",", derivation-attribute-list ] ;

address-type-specification = "@", [ ",", derivation-attribute-list ] ;

thunk-type-specification = "^", [ ",", derivation-attribute-list ] ;

slice-type-specification = "[:]", [ ",", derivation-attribute-list ] ;

range-type-specification = "..", [ ",", derivation-attribute-list ] ;

array-specification = array-dimension-group, { array-dimension-group },
                      [ ",", derivation-attribute-list ] ;

array-dimension-group = "[", array-dimension-content, "]" ;

(* DECISION-TYP-003: any dimension may be elided, including the first. *)
array-dimension-content = [ array-dimension ], { ",", [ array-dimension ] } ;

(* integer-literal and identifier are already reachable through expression. *)
(*
  DECISION-TYP-004 applies to array dimensions too. The array derivation is
  the representation underlying a dependent type, as in

  Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);

```

```

    so admitting arithmetic here would reintroduce it behind the dependent
    type. Every array dimension in language-ref.md is a literal or a name.
*)
array-dimension = "... " | "." | dependent-index ;

derivation-attribute-list = derivation-attribute,
    { ",", derivation-attribute } ;

derivation-attribute = annotation-key, "=", annotation-value ;

return-type-clause = "->", "(", [ return-item-list ], ")" ;

return-item-list = return-item, { ",", return-item } ;

return-item = [ identifier ], type-expression ;

(* ===== *)
(* 5. Common declaration components *)
(* ===== *)

(*
    DECISION-GEN-001:
    A generic parameter may declare arity, giving higher-kinded parameters such
    as Transformer(F(_), G(_)).
*)
generic-parameter-clause = "(", generic-parameter,
    { ",", generic-parameter }, ")" ;

generic-parameter = identifier, [ generic-arity-clause ] ;

generic-arity-clause = "(", generic-arity-slot,
    { ",", generic-arity-slot }, ")" ;

generic-arity-slot = "_" | identifier ;

kind-options = "->", "(", [ annotation-argument-list ], ")" ;

field-declaration = annotations, identifier, type-expression,
    [ "=", expression ], statement-end ;

embedded-field-declaration = annotations, type-expression, statement-end ;

value-specification = annotations, identifier, type-expression, statement-end ;

(* DECISION-SYN-002: comma-separated variable declarations are one statement. *)
variable-declaration = annotations, typed-variable-declarator,
    { ",", typed-variable-declarator }, statement-end ;

typed-variable-declarator = identifier, type-expression,
    [ "=", expression ] ;

inferred-variable-declaration = annotations, inferred-variable-declarator,
    { ",", inferred-variable-declarator },
    statement-end ;

inferred-variable-declarator = identifier, definition-operator, expression ;

definition-operator = ( ":@" | "?@" ),
    multi-symbol-infix-operator-boundary-guard ;

(* ===== *)
(* 6. Data and type declarations *)
(* ===== *)

struct-declaration = annotations, filename-derived-name,
    "co.lang.struct", "=", struct-body ;

surface-struct-declaration = annotations, identifier,
    "co.lang.struct", "=", struct-body ;

struct-body = "{", { struct-member }, body-close ;

(*
    language-ref.md, "Struct Rules": structs cannot have default values to
    fields/members – the struct declaration remains pure data, and all behaviour
    lives in the companion unit. A cstruct is a further-restricted C-compatible
    data representation, so the same restriction applies there. Both therefore
    use pure-field-declaration, which is field-declaration WITHOUT the
    initializer option; the initializer remains available to the other
    containers that reference field-declaration directly.
*)
pure-field-declaration = annotations, identifier, type-expression,
    statement-end ;

```



```

struct-member = pure-field-declaration | embedded-field-declaration ;

cstruct-declaration = annotations, filename-derived-name,
    "co.lang.cstruct", "=", cstruct-body ;

surface-cstruct-declaration = annotations, identifier,
    "co.lang.cstruct", "=", cstruct-body ;

cstruct-body = "{", { pure-field-declaration }, body-close ;

enum-declaration = annotations, filename-derived-name,
    "co.lang.enum", "=", enum-body ;

enum-body = "{", [ enum-variant,
    { enum-separator, enum-variant }, [ enum-separator ] ],
    body-close ;

(*
    DECISION-COL-001: enum variants are comma-separated. The comma is a soft
    boundary between variants; the closing brace is the hard structural end of
    the enum body. A trailing comma is permitted.
*)
enum-separator = "," ;

enum-variant = annotations, identifier,
    [ "(", [ type-list ], ")" ],
    [ "=", constant-expression ] ;

union-declaration = annotations, filename-derived-name,
    "co.lang.union", "=", union-body ;

union-body = "{", { field-declaration }, body-close ;

data-declaration = annotations, identifier,
    [ generic-parameter-clause ], "co.lang.data", "=",
    data-variant, { "|", data-variant }, statement-end ;

data-variant = qualified-name,
    [ "(", [ type-list ], ")" ] ;

type-declaration = parameterized-type-declaration
    | simple-type-declaration ;

parameterized-type-declaration = annotations, identifier,
    generic-parameter-clause, "co.lang.type",
    [ kind-options ], [ "=", type-expression ],
    statement-end ;

simple-type-declaration = annotations, identifier, type-declaration-kind,
    [ kind-options ], [ "=", type-expression ],
    statement-end ;

type-declaration-kind = "co.lang.type"
    | "co.lang.typealias"
    | "co.lang.newtype"
    | "co.lang.opaquetype"
    | "co.lang.subtype"
    | "co.lang.supertype"
    | "co.lang.associatedtype"
    | "co.lang.refinementType"
    | "co.lang.dependentType"
    | "co.lang.typekind" ;

(*
    DECISION-ENTRY-001:
    Entry files admit the same non-structural type declarations as the
    single-source application form, including parameterized co.lang.type
    constructors. They do not admit ordinary or type-level function
    declarations, file-backed UDT/container primaries, units, instances, or
    matchers.
*)
entry-type-declaration = entry-parameterized-type-declaration
    | entry-simple-type-declaration ;

entry-parameterized-type-declaration = annotations, identifier,
    generic-parameter-clause,
    "co.lang.type",
    [ kind-options ],
    [ "=", type-expression ],
    statement-end ;

entry-simple-type-declaration = annotations, identifier,
    ( "co.lang.type"

```

```

        | "co.lang.typealias"
        | "co.lang.newtype"
        | "co.lang.opaquetype"
        | "co.lang.subtype"
        | "co.lang.supertype"
        | "co.lang.dependentType" ),
    [ kind-options ],
    [ "=", type-expression ],
    statement-end ;

forward-type-declaration = annotations, filename-derived-name,
    forward-declarable-kind, [ kind-options ],
    statement-end ;

forward-declarable-kind = "co.lang.struct"
    | "co.lang.cstruct"
    | "co.lang.class"
    | "co.lang.interface"
    | "co.lang.signature"
    | "co.lang.module"
    | "co.lang.enum"
    | "co.lang.union"
    | "co.lang.data"
    | "co.lang.object"
    | "co.lang.instance"
    | "co.lang.function" ;

package-alias-declaration = filename-derived-name, "co.lang.package", "=",
    package-alias-body, statement-end ;

package-alias-body = "{", "name", ":", string-literal, "}" ;

(*)
    DECISION-KIND-001:
    Only enabled kinds enumerated by general-declarable-kind are routed here.
    Reserved future names such as co.lang.typeconstructor and
    co.lang.typefunction intentionally have no source production and are
    rejected as unsupported. Ordered choice still gives dedicated declarations
    and ordinary variable declarations priority in their contexts.
*)
(* "co.lang.realm" was removed in revision 25 with the realm import fields. *)
(* ===== *)
(* 7. Containers and behavioral declarations *)
(* ===== *)

unit-declaration = annotations, filename-derived-name,
    "co.lang.unit", "=", unit-body ;

unit-body = "{", { unit-member }, body-close ;

unit-member = function-declaration
    | data-declaration
    | type-declaration
    | type-level-function-declaration
    | function-object-declaration
    | delegate-declaration
    | annotated-function-primary ;

(*)
    DECISION-COMP-001:
    The same unit grammar is selected for ordinary and companion unit files.
    Filename classification determines whether members merge into the package or
    attach to a struct. Explicit value/type receivers in a companion must resolve
    to that owner. Non-operator receiverless members need no owner-typed
    parameter. Operator members add the operand rule described by
    DECISION-COMP-001; it is checked after receiver and type resolution.
*)

class-declaration = annotations, filename-derived-name,
    "co.lang.class", [ kind-options ], "=", class-body ;

class-body = "{", { class-member }, body-close ;

class-member = field-declaration
    | function-declaration
    | lifecycle-method-declaration ;

lifecycle-method-declaration = annotations, lifecycle-name,
    parameter-list, [ return-type-clause ],
    function-definition ;

interface-declaration = annotations, filename-derived-name,
    "co.lang.interface", "=", interface-body ;

```

```

interface-body = "{", { function-specification }, body-close ;

signature-declaration = annotations, filename-derived-name,
                        "co.lang.signature", "=", signature-body ;

signature-body = "{", { signature-member }, body-close ;

signature-member = value-specification
                  | function-specification
                  | signature-type-component ;

signature-type-component = annotations, identifier,
                           [ generic-parameter-clause ], "co.lang.type",
                           [ "=", type-expression ], statement-end ;

module-declaration = annotations, filename-derived-name,
                     "co.lang.module", [ kind-options ], "=", module-body ;

module-body = "{", { module-member }, body-close ;

module-member = variable-declaration
               | inferred-variable-declaration
               | function-declaration
               | signature-type-component ;

library-declaration = annotations, filename-derived-name,
                      "co.lang.library", "=", library-body ;

library-body = "{", { library-member }, body-close ;

library-member = import-directive
               | surface-struct-declaration
               | surface-cstruct-declaration
               | function-declaration ;

object-declaration = annotations, filename-derived-name,
                     "co.lang.object", [ kind-options ], "=", object-body ;

object-body = "{", { field-declaration | function-declaration }, body-close ;

(*
  DECISION-SEM-001, instance coherence. Recorded here because the grammar
  cannot express it and an implementer must not infer its absence.

  An instance is declared in the package that defines the typeclass, or the
  package that defines the type. That exact package: sub-packages are distinct
  packages and do not qualify. A typeclass may live in any package, so the
  rule is structural rather than tied to co.*.

      abc.tc.ListFunctor      for=Functor, type=List      legal
      myapp.ab.TreeFunctor   for=Functor, type=myapp.ab.Tree legal
      other.util.ListFunctor for=Functor, type=List      MISPLACED

  This placement rule is a semantic correctness requirement even though
  instance selection is BY NAME and never uses implicit search. A misplaced
  instance parses successfully but is rejected during name resolution. Folang
  does not require global uniqueness for a typeclass/type pair; explicit names
  and package qualification disambiguate legal instances.

  Method syntax such as xs.map(f) is unrelated. It calls an associated
  function in a companion unit, which lives in the type's own package and is
  therefore unambiguous by construction.

  A misplaced instance PARSES correctly; this production accepts it. The check
  belongs to name resolution, where the diagnostic can name the typeclass, the
  represented type, and the two packages in which the instance would have been
  legal.
*)
instance-declaration = annotations, filename-derived-name,
                      "co.lang.instance", [ kind-options ], "=",
                      instance-body ;

instance-body = "{", { function-declaration | variable-declaration }, body-close ;

(*
  DECISION-TCLASS-001:
  ``_` occupies the filename-derived declaration-name slot. The following
  parenthesized clause is a separate typeclass-parameter clause, and its
  parameters may carry arity such as F(_) or G(_, _). Otherwise-unbound type
  variables in each operation specification are implicitly universally
  quantified within that operation.
*)
typeclass-declaration = annotations, filename-derived-name,
                       typeclass-parameter-clause, "co.lang.typeclass", "=",

```

```

        contract-body ;

typeclass-parameter-clause = generic-parameter-clause ;

(*
  DECISION-MATCH-001:
  A matcher declares exactly one subject type in type=. Its body contains
  exactly one protocol function named matchCase. The resolved first parameter
  type must equal the declared subject type; the second parameter is
  co.lang.untyped and the result is exactly
  (co.lang.int, co.lang.MatchBindings). Overloaded matchCase declarations are
  rejected semantically; helper functions may use other names.
*)
matcher-instance-declaration = annotations, filename-derived-name,
                              "co.lang.matcher", matcher-options, "=",
                              matcher-body ;

matcher-options = "->", "(", "type", annotation-binder, type-expression, ")" ;

matcher-body = "{", { function-declaration }, body-close ;

contract-body = "{", { function-specification | value-specification }, body-close ;

(*
  DECISION-DECL-003: a named block is a STATEMENT inside a function or method
  body. language-ref.md states blocks cannot live outside functions and that
  inner blocks for a class, struct, typeclass or module are prohibited, so it
  is not a file-backed primary and takes an ordinary identifier.
*)
named-block-declaration = annotations, identifier,
                          "co.lang.block", "=", block, body-closure-guard ;

(* DECISION-DECL-002 applies to a delegate as well: a unit member, not a
  file-backed primary, so it takes an ordinary identifier. *)
delegate-declaration = annotations, identifier,
                      "co.lang.delegate", "=", function-type, statement-end ;

(*
  DECISION-SYN-006:
  An inline function body is a block and ends at "}" with no semicolon; a
  binding to an existing callable is an expression and ends at ";". Both
  forms appear in the reference:

      _ co.lang.function = (a co.lang.int)->(co.lang.int) = {
          this.return a * 2;
      }
      _ co.lang.function = add;
*)
(*
  DECISION-DECL-002: a function object is a UNIT MEMBER, not a file-backed
  primary, so it carries an ordinary identifier rather than "_". The reference
  presents it as Syntax 3 of the function-type feature, beside the named type
  alias, which has always taken an identifier.
*)
function-object-declaration = annotations, identifier,
                              "co.lang.function", "=",
                              function-object-binding ;

(*
  DECISION-SYN-007:
  A direct anonymous function is the inline body of this function-kind
  declaration and ends at its closing brace. Any other expression binding,
  including object construction or a callable reference, ends with ";".
*)
function-object-binding = anonymous-function-expression,
                        body-closure-guard
                        | non-anonymous-function-expression,
                        statement-end ;

(*
  DECISION-SYN-009:
  Parsing this envelope does not by itself make the function legal. Semantic
  analysis must confirm that at least one resolved annotation defines a legal
  primary declaration kind. Otherwise the source is a forbidden loose
  package-level function.
*)
annotated-function-primary = one-or-more-annotations, function-declaration ;

(*
  DECISION-TYP-002:
  A type-level function returns exactly one type-producing result, so it does not
  use the ordinary return-item-list (which permits zero or multiple results).
  A `|` combines type-producing result kinds into that one union result; a
  comma cannot introduce another result. Its body binds a type-expression. The

```

type-expression alternative precedes the expression alternative, so
 Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);
 is parsed as a type, not as an unparseable expression.

The algebraic form Option(T) co.lang.data = Some(T) | None(); is handled by
 data-declaration. Its `|` similarly belongs to one union type binding rather
 than spelling multiple function returns.

```

*)
type-level-function-declaration = annotations, function-name, parameter-list,
                                { parameter-list }, type-level-return-clause,
                                type-level-binding ;

type-level-return-clause = "->", "(", type-level-result-kind,
                          { "|", type-level-result-kind }, ")" ;

type-level-result-kind = "co.lang.dependentType"
                       | "co.lang.type"
                       | "co.lang.typetype"
                       | "co.lang.typekind"
                       | "co.lang.kind" ;

type-level-binding = function-definition
                   | function-delegation
                   | "=", type-expression, statement-end
                   | "=", non-block-expression, statement-end
                   | statement-end ;

(* ===== *)
(* 8. Functions *)
(* ===== *)

function-declaration = annotations, [ receiver-clause ], function-name,
                          parameter-list, { parameter-list },
                          [ return-type-clause ], function-binding ;

(* Lifecycle names have their own class-member production and are not ordinary
   function names in a unit, module, local block, or package declaration. *)
function-name = identifier ;

receiver-clause = "(", ( type-expression
                       | identifier, type-expression ), ")" ;

parameter-list = "(", [ parameter,
                      { ",", parameter }, [ ",", " ] ], ")" ;

parameter = [ "...", [ "~" ], identifier, [ "?" ],
             [ type-expression ], [ "=", expression ] ;

function-binding = function-definition
                 | function-delegation
                 | function-alias-binding
                 | statement-end ;

(* DECISION-FUN-001: the "=" before a block body is optional. *)
function-definition = [ "=" ], block, body-closure-guard ;

function-delegation = ( ">" | ">>" ), expression,
                     { ">>", expression }, statement-end ;

function-alias-binding = "=", non-block-expression, statement-end ;

function-specification = annotations, [ receiver-clause ], function-name,
                          parameter-list, { parameter-list },
                          [ return-type-clause ], statement-end ;

function-type = "(", [ function-type-parameter,
                     { ",", function-type-parameter } ], ")",
               return-type-clause ;

(* Reference examples permit both `(co.lang.int)` and `(value co.lang.int)`. *)
function-type-parameter = type-expression
                       | identifier, type-expression ;

anonymous-function-expression = [ "forall", "(", type-parameter-list, ")", ".", " ]",
                                parameter-list, return-type-clause,
                                [ "=" ], block ;

lambda-expression = "|", [ lambda-parameter,
                          { ",", lambda-parameter } ], "|", ">",
                      ( expression | block ) ;

lambda-parameter = identifier, [ type-expression ] ;

(*

```

DECISION-FUN-002:

The equals sign makes this a named closure declaration. It is followed by one or more parameter lists and the four-character `==>>` marker. One list is an ordinary closure and two or more lists make the closure curried. This is distinct from `=>`, which introduces lambdas and bare function-pattern clauses, and from `=>>`, which performs function delegation/chaining.

```
*)
closure-declaration = annotations, identifier, "=", parameter-list,
                      { parameter-list }, "==>>", expression,
                      statement-end ;

(*)
DECISION-SYN-003 / DECISION-SYN-008 / DECISION-SCOPE-001:
This is FoLang's sole named local-function syntax. It requires a return-type
clause and a block body, which keeps foo(); an expression statement. It has
block-local identity, is never a package member, and resolves free runtime
names from its lexical declaration context. Source-form restrictions, such
as the application entry profile, remain semantic checks.
*)
local-function-declaration = annotations, function-name, parameter-list,
                             { parameter-list }, return-type-clause,
                             function-definition ;

(* ===== *)
(* 9. Function-pattern groups and patterns *)
(* ===== *)

(*)
A function-pattern group is the maximal set of entry-file clauses with the
same name. All clauses in one group have the same arity and use the same
form: either every clause is bare or every clause begins with let.

The bare form is non-capturing. The let form must capture at least one
definitely initialized surrounding entry-file runtime binding. Capture,
compatible result types, clause reachability, overlap, and exhaustiveness
are semantic checks over the complete group.

`=>` introduces a bare pattern clause. `==>>` instead introduces the body
of a named closure declaration after its parameter lists, and `=>>` is the
function-delegation operator from section 8.
*)

bare-function-pattern-clause = annotations, identifier, pattern-parameter-list,
                              [ where-clause ], "=>", pattern-result ;

capturing-function-pattern-clause = annotations, "let", identifier,
                                     pattern-parameter-list,
                                     [ where-clause ], "=", pattern-result ;

pattern-parameter-list = "(", [ pattern,
                              { ",", pattern } ], ")" ;

where-clause = ".where", "(", expression, ")" ;

(*)
A where guard is evaluated only after every parameter pattern has matched.
Bindings introduced by those patterns are in scope in the guard and result.
The guard is required semantically to have type co.lang.bool.
*)

(*)
DECISION-SYN-006: a block-bodied clause ends at "}"; an expression-bodied
clause takes the semicolon.

    fib(0) => 0;
    fib(n) => fib(n-1) + fib(n-2);
    classify(n).where(n > 0) => { this.return "positive"; }
*)
pattern-result = block, body-closure-guard
                | non-block-expression, statement-end ;

(*)
Expression-bodied clauses end with `;`, including capturing let clauses.
Block-bodied clauses end at `}` and do not take `;`. A newline is never a
clause terminator.
*)

pattern = wildcard
        | literal-pattern
        | binding-pattern
        | constructor-pattern
        | record-pattern
        | tuple-pattern
        | qualified-name ;
```

```

literal-pattern = literal
    | ( "+" | "-" ),
      ( integer-literal | floating-literal ) ;

binding-pattern = identifier ;

constructor-pattern = qualified-name, "(" , [ pattern,
    { ",", pattern } ], ")" ;

(* Record-pattern fields have no trailing comma: every comma must introduce the
   following record-pattern-field before the closing brace. *)
record-pattern = qualified-name, "{", [ record-pattern-field,
    { ",", record-pattern-field } ], "}" ;

record-pattern-field = identifier, [ ":", pattern ] ;

tuple-pattern = "(", pattern, ",", pattern,
    { ",", pattern }, ")" ;

match-case = ".case", "(", match-case-body, ")" ;

match-case-body = pattern, [ ":", expression ], "=>",
    ( expression | block ) ;

match-default = ".default", "(", ( expression | block ), ")" ;

(* ===== *)
(* 10. Statements and blocks *)
(* ===== *)

(*
  DECISION-SYN-001:
  - Every simple statement whose production uses statement-end ends with ";".
  - Built-in directives are self-delimiting and are not simple statements.
  - A newline is whitespace and never terminates a statement.
  - A block statement and a block-bodied declaration do not take a trailing
    semicolon merely because their final token is ";".

  DECISION-BLK-001:
  A final expression without a semicolon is a block tail expression, not an
  expression statement. It supplies the block's value.
*)
block = "{", { block-item }, [ block-tail-expression ], "}" ;

block-item = statement ;

block-tail-expression = expression ;

statement = named-block-declaration
    | variable-declaration
    | inferred-variable-declaration
    | grouped-variable-declaration
    | let-value-declaration
    | local-function-declaration
    | closure-declaration
    | multiple-assignment-statement
    | return-statement
    | expression-statement
    | labeled-block
    | block-statement
    | empty-statement ;

(* Grouped declarators have no trailing comma: every comma must introduce the
   following typed-variable-declarator before the closing parenthesis. *)
grouped-variable-declaration = "(", typed-variable-declarator,
    { ",", typed-variable-declarator }, ")",
    statement-end ;

let-value-declaration = "let", identifier, [ type-expression ], "=",
    expression, statement-end ;

(*
  DECISION-OP-003:
  := and ?= occur only in inferred-variable-declaration. They are not
  assignment-expression operators and cannot participate in a = b = c-style
  chain. ::= remains reserved for a future feature and is rejected.
*)

(* Multiple assignment is a statement because it has multiple destinations. *)
multiple-assignment-statement = assignment-target, ",", assignment-target,
    { ",", assignment-target }, "=",
    expression-list, statement-end ;

```

```

assignment-target = postfix-expression
                  | tuple-assignment-target ;

tuple-assignment-target = "(" , assignment-target , "," , assignment-target ,
                          { "," , assignment-target } , ")" ;

return-statement = ( "this" | "self" ) , ".return" ,
                   [ expression-list ] , statement-end ;

(* DECISION-SYN-004: an expression statement may carry annotations. *)
expression-statement = annotations , non-block-expression , statement-end ;

labeled-block = identifier , ":", block , body-closure-guard ;

empty-statement = ";" ;

expression-list = expression , { "," , expression } ;

statement-end = ";" ;

(*)
DECISION-SYN-007 – zero-width contextual guards:

body-closure-guard rejects a semicolon when the enclosing production has
selected a declaration/function/pattern/block BODY. It examines the next
significant token after whitespace and comments and consumes no token.

non-block-expression rejects an expression when its complete source span
is admissible as an unparenthesized block. When the same span could be read
both as a block and as another braced expression (for example `{}`), the
direct body/block reading has priority. A grouped block, a block with a
postfix suffix, or an operator expression containing a block is still an
expression.

non-anonymous-function-expression applies the same priority rule when the
complete source span is admissible as an unparenthesized
anonymous-function-expression.
*)
body-close = "}" , body-closure-guard ;

body-closure-guard =
    ? zero-width condition: the next significant token is not ";", or there is no next token ? ;

block-statement = block , body-closure-guard ;

non-block-expression = expression , non-block-expression-guard ;

non-block-expression-guard =
    ? zero-width condition: the complete source span is not admissible as an
      unparenthesized block production; when both block and another braced
      expression are possible, the block reading has priority ? ;

non-anonymous-function-expression =
    expression , non-anonymous-function-expression-guard ;

non-anonymous-function-expression-guard =
    ? zero-width condition: the complete source span is not admissible as an
      unparenthesized anonymous-function-expression production; that direct
      body reading has priority over the general expression reading ? ;

(* ===== *)
(* 11. Expressions and built-in operator precedence *)
(* ===== *)

(*)
DECISION-OP-001 – built-in precedence, highest to lowest:

100 postfix: calls, indexing, member access, postfix !           left
 90 exponentiation: **                                           right
 80 prefix: +, -, ! (~, #, ^ reserved; @ is not a prefix)       right
 70 multiplicative: *, /, %                                       left
 60 additive: +, -                                               left
 55 ranges: .., <.., <.<, <.<<                                   none
 50 relational: <, <=, >, >=                                       left
 45 equality: ==, !=                                              left
 40 bitwise AND: &                                               left
 38 bitwise XOR: ^                                               left
 36 bitwise OR: |                                                left
 30 logical AND: &&                                              left
 20 logical OR: ||                                              left
 10 assignment: =, +=, -=, *=, /=, %=, **=, &=, ^=, |=         right

Operands are still evaluated according to FoLang's normative left-to-right
and target-first evaluation rules. Associativity determines grouping, not

```


the order in which operand subexpressions are evaluated.

DECISION-EXT-001 / DECISION-OPBOOT-003:

Before operator-dependent source is tokenized, the compiler parses the fixed local operator library with its dedicated lexer/parser, combines those custom registrations with the language-owned built-in and pre-declared registrations, and builds one immutable operator table. Imports contribute no operator metadata. Existing and custom implementations use mode=overload and add no precedence entries. Operator mode=override is rejected.

```
*)
expression = assignment-expression
            | extended-operator-expression ;

(* DECISION-OP-002: right recursion makes assignment right-associative. *)
assignment-expression = logical-or-expression,
                        [ runtime-assignment-operator,
                          assignment-expression ] ;

runtime-assignment-operator = "="
                             | compound-assignment-operator ;

compound-assignment-operator = ( "+=" | "-=" | "*=" | "/=" | "%="
                                | "***=" | "&=" | "^=" | "|=" ),
                                multi-symbol-infix-operator-boundary-guard ;

constant-expression = ( logical-or-expression
                        | extended-operator-expression ),
                        ? zero-width condition: no runtime-assignment-operator
                          occurs anywhere in this constant-expression subtree ? ;

logical-or-expression = logical-and-expression,
                        { logical-or-operator, logical-and-expression } ;

logical-or-operator = "||", multi-symbol-infix-operator-boundary-guard ;

logical-and-expression = bitwise-or-expression,
                        { logical-and-operator, bitwise-or-expression } ;

logical-and-operator = "&&", multi-symbol-infix-operator-boundary-guard ;

bitwise-or-expression = bitwise-xor-expression,
                        { "|", bitwise-xor-expression } ;

bitwise-xor-expression = bitwise-and-expression,
                        { "^", bitwise-and-expression } ;

bitwise-and-expression = equality-expression,
                        { "&", equality-expression } ;

equality-expression = relational-expression,
                        { equality-operator, relational-expression } ;

equality-operator = ( "==" | "!=" ),
                    multi-symbol-infix-operator-boundary-guard ;

relational-expression = range-expression,
                        { relational-operator, range-expression } ;

relational-operator = "<" | ">" | multi-symbol-relational-operator ;

multi-symbol-relational-operator = ( "<=" | ">=" ),
                                   multi-symbol-infix-operator-boundary-guard ;

(* A range expression contains at most one range operator. *)
range-expression = additive-expression,
                  [ range-operator, [ additive-expression ] ]
                  | range-operator, additive-expression ;

range-operator = ( ".." | "<.." | "..<" | "<.<" ),
                 multi-symbol-range-operator-boundary-guard ;

additive-expression = multiplicative-expression,
                     { additive-operator, multiplicative-expression } ;

additive-operator = "+" | "-" ;

multiplicative-expression = unary-expression,
                           { multiplicative-operator, unary-expression } ;

multiplicative-operator = "*" | "/" | "%" ;

unary-expression = { prefix-operator }, power-expression ;

(*)
```

DECISION-OP-004 is withdrawn: ++ and -- are absent from the built-in prefix/postfix grammar. An unregistered contiguous occurrence is handled by the general symbolic-run rule rather than by a special operator rule.

DECISION-OP-006: "~", "#" and "^" are RESERVED prefix spellings. They are listed here so no user-defined operator can claim them, and the parser refuses them in operand position with a reserved-operator diagnostic, the same treatment DECISION-OP-005 gives ::= , -> and <-> . Candidate meanings include complement and length/count; none is assigned yet. Each spelling keeps every other role it already has: "~" marks a named parameter and the ->(~) heap reference, "^" is bitwise xor and the ->(^) thunk derivation.

"@" is NOT a prefix operator. It introduces a directive or an annotation, and it marks the ->(@) address derivation. A variable of an address, pointer or reference kind is read and assigned like any other variable – the kind is carried by the type derivation and never restated at the use site – so there is no address-of prefix:

```
someInt co.lang.int = 10;
someAdd co.lang.int->(@);
someAdd = someInt;      no "@" at the use site
co.out.println(someAdd);
```

*)

```
prefix-operator = "+" | "-" | "!"
                | reserved-prefix-operator ;
```

```
reserved-prefix-operator = "~" | "#" | "^" ;
```

(* Right recursion makes exponentiation right-associative. *)

```
power-expression = postfix-expression,
                  [ power-operator, unary-expression ] ;
```

```
power-operator = "**", multi-symbol-infix-operator-boundary-guard ;
```

```
postfix-expression = primary-expression,
                   { postfix-suffix | postfix-operator } ;
```

```
postfix-operator = "!" ;
```

```
postfix-suffix = call-suffix
               | index-suffix
               | member-suffix
               | match-suffix ;
```

```
call-suffix = "(", [ argument-list ], ")" ;
```

```
argument-list = argument, { ",", argument }, [ ",", ] ;
```

```
argument = ( [ identifier, "=" ], expression )
          | block
          | lambda-expression
          | wildcard ;
```

(* Lambda and wildcard are contextual method-call arguments. A lambda must be a direct argument of a receiver-qualified map/filter/reduce/forEach/sortBy/groupBy call. Wildcard is admitted only in the first iterator-index slot of a receiver-qualified each call. Transparent grouping around the member callee does not change either rule. contains and containsVal require an actual comparison value and reject wildcard. *)

```
index-suffix = "[", [ expression-list ], "]" ;
```

(* "for" is hard-reserved for comprehensions but remains contextual after a member dot because the built-in method table includes `.for`. *)

```
member-suffix = ".", ( identifier | "for" | lifecycle-name ) ;
```

(* Member access and indexing are already expressed by postfix-expression with member-suffix and index-suffix; the former duplicate productions were removed in revision 5.

*)

```
primary-expression = literal
                  | special-binding
                  | "this"
                  | "self"
                  | qualified-name
                  | grouped-expression
                  | tuple-expression
                  | array-literal
                  | map-literal
                  | object-construction
                  | anonymous-class-expression
                  | block
```

```

    | anonymous-function-expression
    | let-expression
    | comprehension-expression ;

grouped-expression = "(" , expression , ")" ;

tuple-expression = "(" , expression , "," , expression ,
    { "," , expression } , ")" ;

array-literal = "[" , [ expression ,
    { "," , expression } , [ "," ] ] , "]" ;

map-literal = "{" , [ map-entry ,
    { "," , map-entry } , [ "," ] ] , "}" ;

map-entry = expression , ":" , expression ;

(* DECISION-COL-001: object fields use colon and comma. *)
object-construction = type-postfix-expression , "{",
    [ object-field-initializer ,
    { "," , object-field-initializer } , [ "," ] ] , "}" ;

object-field-initializer = identifier , ":" , expression ;

(*
    DECISION-SYN-008:
    The forms below are expressions, not independent named declarations.
    Their syntactic containment does not create a package-level nested identity.
*)
anonymous-class-expression = "co.lang.class" , "{",
    { class-member } , "}" ;

let-expression = "let" , "(" , "{", let-binding ,
    { "," , let-binding } , "}" , ")" ,
    ".in" , "(" , "{", expression , "}" , ")" ;

let-binding = ( identifier | special-binding ) , "=" , expression ;

comprehension-expression = "for" , "(" , comprehension-binding , ")" ,
    ".yield" , "(" , expression-list , ")" ;

comprehension-binding = pattern , "<-" , expression ;

(*
    At least one case is required semantically. The grammar accepts zero so a
    malformed chain produces a semantic diagnostic rather than a parse error.
*)
match-suffix = ".match" , [ "(" , [ expression ] , ")" ] ,
    { match-case } , [ match-default ] ;

multi-symbol-infix-operator-boundary-guard =
    ? zero-width condition: a multi-symbol infix operator has an explicit
    boundary on both operand-facing sides; a boundary is whitespace, a
    comment, or an applicable delimiter, checked before separators are
    discarded ? ;

multi-symbol-range-operator-boundary-guard =
    ? zero-width condition: a multi-symbol range operator has an explicit
    boundary on every side for which an operand is present ? ;

extended-operator-expression =
    ? expression containing a registered non-built-in operator, parsed by
    precedence climbing from its declared fixity, precedence, associativity,
    and arity; all built-in subexpressions obey the table above, and every
    registered multi-symbol operator satisfies the operand-facing boundary
    rule for its fixity ? ;

(* ===== *)
(* 11a. Control-flow chain shapes (informative) *)
(* ===== *)

(*
    FoLang has no imperative if, else, while, or foreach keyword. `for` is
    reserved for comprehension-expression; the associated-function control
    forms below are otherwise parsed as
    postfix-expression with member-suffix and call-suffix, where argument
    admits a block. The productions below are INFORMATIVE. They document the
    canonical shapes the semantic analyzer enforces; they are not a second
    parse path and must not be entered from expression.

    Enforcing these shapes in the parser instead would require unbounded
    lookahead to distinguish a control chain from any other method chain, so
    the check belongs to semantic analysis.
*)

```

```

informative-condition-chain =
    "(", expression, ")", ".do", "(", block, ")",
    { ".otherwise", "(", expression, ")", ".do", "(", block, ")" },
    [ ".otherwise", ".do", "(", block, ")" ] ;

informative-loop-chain =
    "(", expression, ")", ".loop", "(", block, ")",
    { ".otherwise", "(", expression, ")", ".loop", "(", block, ")" },
    [ ".otherwise", ".loop", "(", block, ")" ] ;

informative-mixed-chain =
    "(", expression, ")", informative-branch-verb, "(", block, ")",
    { ".otherwise", "(", expression, ")",
        informative-branch-verb, "(", block, ")" },
    [ ".otherwise", informative-branch-verb, "(", block, ")" ] ;

informative-branch-verb = ".do" | ".loop" ;

informative-ternary-chain =
    "(", expression, ")", ".return", "(", expression, ")",
    { ".otherwise", "(", expression, ")", ".return", "(", expression, ")" },
    ".otherwise", ".return", "(", expression, ")" ;

informative-each-chain =
    postfix-expression, ".each", "(", ( identifier | "_" ), ",", identifier,
    ")", ".do", "(", block, ")" ;

informative-contains-chain =
    postfix-expression, ".contains", "(", expression, ")",
    ".do", "(", block, ")",
    [ ".otherwise", ".do", "(", block, ")" ] ;

informative-pipeline-chain =
    postfix-expression,
    { ( ".filter" | ".map" | ".reduce" | ".forEach"
        | ".sortBy" | ".groupBy" | ".fold" ),
        "(", argument-list, ")" } ;

```

```

(* ===== *)
(* 12. Literals and lexical grammar *)
(* ===== *)

```

```

(*)
DECISION-LEX-001:
Source text is UTF-8. A U+FEFF byte-order mark is permitted only as the
first code point and is otherwise an error.

```

A FoLang identifier begins with an ASCII alphabetic character. Remaining characters are ASCII alphabetic characters, decimal digits, or isolated underscores. Consecutive underscores and a trailing underscore are lexical errors. The spelling "_" is a dedicated contextual token used for discard, wildcard, or filename-derived declaration names; it is never an identifier.

DECISION-BACKEND-001:
A resolved user-defined FoLang identifier is lowered to C++ by appending the suffix "_fo". The no-consecutive-underscore and no-trailing-underscore rules ensure this lowering never creates a C++-reserved double underscore.

DECISION-LEX-002:
Horizontal whitespace, line terminators, line comments, and non-nesting block comments are discarded between tokens. A line terminator has no statement-termination meaning.

DECISION-LEX-003 / DECISION-LEX-010:
After comments, literals, and closed scanner-known composite spellings are recognized, the lexer consumes each remaining complete maximal contiguous symbol run. The run is never split into shorter valid operators as a fallback. It is classified by grammar context as a fixed structural spelling, contextual metadata spelling, registered expression operator, or unrecognized symbolic token. Comment introducers are recognized before symbolic-run scanning. Parentheses and other delimiters terminate a run; therefore +(+) contains two separate + tokens.

A registered expression operator containing more than one symbol character requires an explicit boundary on every operand-facing side. Whitespace, a comment, or an applicable delimiter supplies that boundary. Structural forms such as T->(**) are exempt because -> and the pointer-star run are not parsed as expression operators in that context.

DECISION-LIT-000:
The lexer stores the COMPLETE original numeric, character, or string literal lexeme in the AST. A C++ backend may emit that raw lexeme unchanged. The productions below define the selected C++-compatible numeric, character,

and string forms used by this revision. Backend-conditionally-supported suffixes are accepted only when the configured C++ compiler supports them.

co.const.true, co.const.false, and co.const.none are FoLang-defined literal forms and use backend-defined lowering rather than raw C++ lexeme emission.

This decision does not import the C++ nullptr literal or the C++ user-defined-literal operator"" mechanism.

DECISION-LIT-004 (WITHDRAWN in revision 7):

FoLang has no user-defined literal token. A value of a user-defined type is written with object-construction, for example Employee{name: "Rao", id: 1}, which is an expression in section 11 rather than a literal in section 12.

DECISION-LEX-005 was withdrawn in revision 9. DECISION-LIT-006 removes the abbreviated floating forms that formerly required special dot scanning.

*)

literal = builtin-literal ;

```
builtin-literal = integer-literal
                  | floating-literal
                  | string-literal-sequence
                  | character-literal
                  | boolean-literal
                  | none-literal ;
```

(*

DECISION-LIT-006 – floating literal shape, normative:

A floating literal carries at least one digit on each side of the point. The abbreviated C++ forms are rejected:

1.0	valid	1.	rejected, write 1.0
0.10	valid	.10	rejected, write 0.10
0x1.8p3	valid	0x1.p3	rejected
1e5	valid	0x.8p3	rejected

Because a numeric literal can no longer terminate at a point, the point that follows a number always belongs to some other token. DECISION-LEX-003 maximal munch therefore produces the correct token stream with no numeric lookahead, no parser re-lexing, and no special case:

1 .. 10	-> 1 .. 10	range over integers
1.0 .. 0.10	-> 1.0 .. 0.10	range over floats
1.5 .. 2.5	-> 1.5 .. 2.5	
1e5 .. 2e5	-> 1e5 .. 2e5	
3.to_str()	-> 3 . to_str ()	member access on an integer
3.14.to_str()	-> 3.14 . to_str ()	float, then member access
1 ... 10	-> 1 ... 10	... is not an infix operator, so the parser rejects it
1 10	-> unrecognized symbolic run; no fallback splitting	

A diagnostic for the rejected literal forms should name the replacement, for example "floating literal needs a digit on both sides of the point; write 1.0". A diagnostic for the multi-dot range forms should suggest 1.0 .. 0.10.

The narrowing keeps every FoLang floating literal inside the C++ floating literal set, so DECISION-LIT-000 raw-lexeme passthrough is unaffected. Range operators still require the operand-facing boundaries defined by DECISION-LEX-010; numeric maximal munch does not waive that requirement.

*)

(* DECISION-LIT-001: C++-compatible built-in integer literal spelling. *)

```
integer-literal = ( binary-integer-literal
                  | octal-integer-literal
                  | decimal-integer-literal
                  | hexadecimal-integer-literal ),
                  [ integer-suffix ] ;
```

```
binary-integer-literal = ( "0b" | "0B" ), binary-digit-sequence ;
```

```
octal-integer-literal = "0", [ octal-digit-sequence ] ;
```

```
decimal-integer-literal = nonzero-digit, { decimal-digit } ;
```

```
hexadecimal-integer-literal = hexadecimal-prefix,
                             hexadecimal-digit-sequence ;
```

```
hexadecimal-prefix = "0x" | "0X" ;
```

```
binary-digit-sequence = binary-digit, { binary-digit } ;
```

```

octal-digit-sequence = octal-digit, { octal-digit } ;

decimal-digit-sequence = decimal-digit, { decimal-digit } ;

hexadecimal-digit-sequence = hexadecimal-digit, { hexadecimal-digit } ;

(*)
    DECISION-LIT-007 – no digit separator, normative:

    FoLang numeric literals contain digits only. The C++14 digit separator is
    NOT adopted, so an apostrophe never occurs inside a numeric literal and
    always delimits a character literal.

        1000  0xFFFFFFFF  0b101100100110      valid
        1'000  0x1'a  0b1011'0010              rejected

    Consequences:
    - The apostrophe has exactly one meaning in FoLang source: it opens or
      closes a character literal. No base-aware adjacency test is needed and
      DECISION-LEX-007 is withdrawn.
    - A character literal admits any translation character except the
      apostrophe, the backslash, CR and LF, so a space, a semicolon and a
      comma are ordinary c-characters. A line break is written with an
      escape such as \n.
    - This is a NARROWING of the C++ literal set, so every FoLang numeric
      literal remains a valid C++ numeric literal and DECISION-LIT-000
      numeric raw-lexeme passthrough remains valid.
    - It is also REVERSIBLE. Admitting a separator later accepts strictly
      more programs, so no FoLang source written against this revision would
      break. Reinstating it would require restoring the withdrawn
      DECISION-LEX-007 adjacency rule.

*)

integer-suffix = unsigned-suffix,
    [ long-suffix | long-long-suffix | size-suffix ]
    | long-suffix, [ unsigned-suffix ]
    | long-long-suffix, [ unsigned-suffix ]
    | size-suffix, [ unsigned-suffix ] ;

unsigned-suffix = "u" | "U" ;

long-suffix = "l" | "L" ;

long-long-suffix = "ll" | "LL" ;

size-suffix = "z" | "Z" ;

(* DECISION-LIT-002: C++-compatible decimal and hexadecimal floating forms. *)
floating-literal = decimal-floating-literal
    | hexadecimal-floating-literal ;

decimal-floating-literal = fractional-constant,
    [ exponent-part ],
    [ floating-point-suffix ]
    | decimal-digit-sequence,
    exponent-part,
    [ floating-point-suffix ] ;

(*)
    DECISION-LIT-006: a floating literal carries at least one digit on EACH
    side of the point. 1. and .10 are rejected; write 1.0 and 0.10.

*)
fractional-constant = decimal-digit-sequence, ".",
    decimal-digit-sequence ;

hexadecimal-floating-literal = hexadecimal-prefix,
    ( hexadecimal-fractional-constant
    | hexadecimal-digit-sequence ),
    binary-exponent-part,
    [ floating-point-suffix ] ;

hexadecimal-fractional-constant = hexadecimal-digit-sequence, ".",
    hexadecimal-digit-sequence ;

exponent-part = ( "e" | "E" ), [ sign ], decimal-digit-sequence ;

binary-exponent-part = ( "p" | "P" ), [ sign ], decimal-digit-sequence ;

sign = "+" | "-" ;

floating-point-suffix = "f" | "F" | "l" | "L"
    | "f16" | "F16"
    | "f32" | "F32"
    | "f64" | "F64"

```

```

        | "f128" | "F128"
        | "bf16" | "BF16" ;

(* DECISION-LIT-003: alpha character and string literal subset. *)
character-literal = single-quote, alpha-basic-c-character, single-quote ;

alpha-basic-c-character =
    ? any translation character except apostrophe, backslash,
      carriage return, or line feed ? ;

string-literal-sequence = string-literal, { string-literal } ;

string-literal = double-quote, { alpha-basic-s-character }, double-quote ;

alpha-basic-s-character =
    ? any translation character except double quote, backslash,
      carriage return, or line feed ? ;

(*
    DECISION-LEX-008 – reserved post-alpha literal spellings, normative:

    An adjacent u8, u, U, or L prefix, an R raw-string introducer, or a
    backslash within a quoted literal starts a reserved post-alpha spelling.
    The scanner consumes the complete spelling and reports an unsupported
    feature. With intervening whitespace, u8, u, U, L, and R remain ordinary
    identifiers.
*)
double-quote = ? Unicode scalar value U+0022 ? ;

single-quote = ? Unicode scalar value U+0027 ? ;

backslash = ? Unicode scalar value U+005C ? ;

(*
    DECISION-LIT-005:
    Folang booleans are co.const.true and co.const.false, and the null literal
    is co.const.none. These spellings also match qualified-name; the literal
    reading is selected. A bare true, false, or True inside an annotation
    argument is an ordinary annotation-value name, not a literal.
*)
boolean-literal = "co.const.true" | "co.const.false" ;

none-literal = "co.const.none" ;

(*
    The revision 6 production folang-user-defined-literal was removed in
    revision 7. Folang has no user-defined literal token. A value of a
    user-defined type is written with object-construction in section 11, for
    example Employee{name: "Rao", id: 1} or Point{x: 0.0, y: 0.0}.
*)

(*
    DECISION-LEX-001 with DECISION-LEX-006:

    An ordinary identifier begins with an ASCII letter a-z or A-Z. An
    underscore may occur only between nonempty alphanumeric segments. An
    identifier therefore never begins with an underscore, never contains
    consecutive underscores, and never ends in an underscore.

    The shape of the production alone does not enforce the last two rules: a
    greedy scanner would match a_ as the identifier a followed by the
    contextual token _ , and a__b as a followed by __b, silently producing the
    wrong tokens instead of an error. DECISION-LEX-006 supplies the missing
    negative constraint, expressed here as the trailing guard: the character
    immediately after a complete identifier must not be "_".

    name, myVar2, v1_hr, a_b_c    -> identifiers
    _x, _1                        -> errors, leading underscore
    a_, a__b                      -> errors, DECISION-LEX-006 guard
    _                             -> never an identifier; it is the
                                   contextual token admitted only by
                                   productions that spell "_" directly

    Reserved-word rejection remains a token-class check applied after the
    character sequence is recognized, using hard-reserved-word.

    DECISION-BACKEND-001 appends _fo to a resolved user identifier. Because no
    identifier ends in an underscore or contains a doubled underscore, the
    lowered C++ name never contains "__" and never matches a C++ reserved
    identifier.
*)
(*
    DECISION-FILE-002 reuses identifier-head and identifier-segment for external

```

```

filename-identifier validation. Filename canonical display spelling is not
a source token transformation and is therefore checked by package/library
indexing rather than by this lexical production.
*)
identifier = identifier-head, { "_", identifier-segment },
            identifier-trailing-guard ;

identifier-trailing-guard =
    ? a zero-width assertion that the next character is not "_" ? ;

identifier-head = ascii-letter, { ascii-alphanumeric } ;

identifier-segment = ascii-alphanumeric, { ascii-alphanumeric } ;

ascii-alphanumeric = ascii-letter | decimal-digit ;

ascii-letter = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H"
              | "I" | "J" | "K" | "L" | "M" | "N" | "O" | "P"
              | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X"
              | "Y" | "Z"
              | "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h"
              | "i" | "j" | "k" | "l" | "m" | "n" | "o" | "p"
              | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x"
              | "y" | "z" ;

binary-digit = "0" | "1" ;

octal-digit = "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" ;

hexadecimal-digit = decimal-digit
                  | "a" | "b" | "c" | "d" | "e" | "f"
                  | "A" | "B" | "C" | "D" | "E" | "F" ;

digit = decimal-digit ;

decimal-digit = "0" | nonzero-digit ;

nonzero-digit = "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9" ;

(* DECISION-LIT-005 removed true and false from this set. *)
hard-reserved-word = "co" | "let" | "this" | "for" | "forall" | "fo" ;

contextual-keyword = "self" ;

(*
DECISION-OP-005:
These tokens appear in the reference operator table but have no assigned
meaning. The lexer recognizes each as one token and the parser rejects it,
so a user-defined operator cannot silently claim it before the language
assigns it a meaning.
*)
reserved-operator = "::~=" | "->>" | "<->" | "`" | backslash ;

reserved-future-operator = ? one character from the hard-reserved future
                           syntax set, explicitly excluding every
                           language-predeclared operator glyph; unavailable
                           for use, declaration, or overload ? ;

(*
Informative token-class summary. The parser consumes tokens; separators are
discarded between them.
*)
token = identifier
      | keyword-token
      | literal
      | delimiter-token
      | symbolic-token
      | reserved-future-operator ;

keyword-token = hard-reserved-word | contextual-keyword ;

delimiter-token = "(" | ")" | "{" | "}" | "[" | "]"
                | "," | ";" | double-quote | single-quote ;

symbolic-token = ? the complete maximal contiguous run of one or more symbol
                  characters after comments, literals, and closed composite
                  spellings are recognized; the run is preserved whole for
                  contextual classification and is never split as a
                  fallback ? ;

token-separator = white-space ;

line-comment = "//", { ? any Unicode scalar value except CR or LF ? } ;

```



```

block-comment = "/*", { block-comment-character }, "*/" ;

block-comment-character = ? any Unicode scalar value that does not begin the
                           two-character sequence */ ? ;

line-break = "\r\n" | "\n" | "\r" ;

horizontal-white-space = " " | "\t" | "\f" ;

white-space = horizontal-white-space | line-break | line-comment | block-comment ;

(* ===== *)
(* 13. Operator source grammar (separate start symbol) *)
(* ===== *)

(*
  DECISION-OPDECL-004.

  This is a SEPARATE grammar with its own start symbol. It is not reachable
  from compilation-unit and must not be entered from it. The compiler parses
  the configured operator source area with this grammar first, builds the
  operator table, and only then parses ordinary source with the main grammar.

  The operator source parser accepts one fixed source-only library declaration.
  Its body contains only operator registrations. Its lexer reads each declaration
  name as one complete maximal symbol run. The ordinary lexer later applies the
  same whole-run rule after loading the completed operator table. Neither lexer
  falls back to splitting an unknown run into shorter operators.

  arithmetic-symbol-character is any character that is not an ASCII
  letter, digit, underscore, whitespace, or one of the delimiters
  ( ) { } [ ] , ; " '

  A registration whose symbol is language-owned, whether a built-in or a
  pre-declared glyph, is rejected per DECISION-OPDECL-002. Semantic validation
  additionally requires fixity, precedence, associativity, and arity exactly
  once; optional metadata keys may occur at most once.

  operator-source-file is an INTENTIONAL SECOND ROOT. It is deliberately
  unreachable from compilation-unit and is listed as such in the validation
  manifest, alongside the informative-* companions and the lexical roots.
*)

operator-source-file = operator-library-declaration ;

operator-library-declaration = operator-library-marker, "_",
                              "co.lang.library", "=",
                              operator-library-body ;

operator-library-marker = "@co.dap.library", "(", "type", "=",
                          "operator", ")" ;

operator-library-body = "{", { operator-declaration }, body-close ;

operator-declaration = operator-symbol, "co.lang.operator", "=",
                      operator-body, statement-end ;

(*
  DECISION-OPDECL-005, revision 26: an operator body is a PROPERTY LIST, not a
  declaration body. Its braces are map-shaped, exactly like package-alias-body,
  so DECISION-SYN-006 requires the enclosing declaration to be terminated by
  ";" and body-closure-guard does not apply. The guard was removed here.
*)
operator-body = "{", operator-property, { ",", operator-property },
               [ ",", ], "}" ;

operator-property = "fixity", annotation-binder, operator-fixity
                  | "precedence", annotation-binder,
                    ( "0" | decimal-integer-literal )
                  | "associativity", annotation-binder,
                    operator-associativity
                  | "arity", annotation-binder, operator-arity
                  | "commutative", annotation-binder, boolean-literal
                  | "idempotent", annotation-binder, boolean-literal
                  | "identity", annotation-binder, operator-identity-value
                  | "foldable", annotation-binder, boolean-literal
                  | "vectorizable", annotation-binder, boolean-literal
                  | "distributes_over", annotation-binder,
                    operator-symbol-list
                  | "desugar", annotation-binder, string-literal ;

(*
  DECISION-OPDECL-006, revision 26: operator property values are QUALIFIED
  constants under co.operator.*, matching co.const.true / co.const.none used in

```

```

the same property list. Bare keywords were withdrawn, so infix, left and
binary are never contextual keywords and cannot collide with an identifier.
*)
operator-fixity = "co.operator.fixity.infix"
                  | "co.operator.fixity.postfix"
                  | "co.operator.fixity.prefix"
                  | "co.operator.fixity.circumfix"
                  | "co.operator.fixity.postcircumfix"
                  | "co.operator.fixity.precircumfix"
                  | "co.operator.fixity.mixfix"
                  | "co.operator.fixity.ternary"
                  | "co.operator.fixity.distfix" ;

operator-associativity = "co.operator.associativity.left"
                        | "co.operator.associativity.right"
                        | "co.operator.associativity.none" ;

operator-arity = "co.operator.arity.unary"
                 | "co.operator.arity.binary"
                 | "co.operator.arity.ternary"
                 | decimal-integer-literal ;

operator-identity-value = literal ;

operator-symbol-list = "[", [ operator-symbol-reference,
                             { ",", operator-symbol-reference }, [ ",", ] ], "]" ;

operator-symbol-reference = character-literal | string-literal ;

operator-symbol = ? a maximal run of one or more symbol characters, where a
                  symbol character is any character that is not an ASCII
                  letter, digit, underscore, whitespace, or one of the
                  delimiters ( ) { } [ ] , ; " ' ; the run must not be a
                  language-owned or hard-reserved symbol and must not contain
                  // or /* ? ;

```

Appendix B - Grammar Decisions and Rationale

FoLang Grammar and Semantic Decision Register — Revision 30

- Grammar: `foLang.ebnf`
- Grammar SHA-256: `17ca00bdf588bcd372328ae0122de731f9d295f161a279077bd9d012261380`
- Language reference basis: `../language-ref.md`
- Language reference SHA-256: `7405b691b7e775c1613163328d4c3060a1c28dfca08abcaafd00fc0e5f896a72`
- Status: decision-complete grammar and semantic register aligned with the current language reference
- Planned syntax policy: productions described as planned in `../language-ref.md` remain in the complete grammar unless explicitly removed. Release-specific availability is handled by the parser/compiler conformance profile.
- Revision 24 adds the whole-symbol-run and operator-boundary model. A contiguous symbolic run is preserved as one candidate and is never split into shorter operators as a fallback. Grammar context distinguishes structural spellings, metadata spellings, and registered expression operators. Every multi-symbol expression operator requires explicit operand-facing boundaries. `++` and `--` are removed from the built-in prefix/postfix grammar; when unregistered they fail through the general symbolic-run rule.
- Revision 25 aligns source-file classification and declaration heads with the current reference: filename-backed primaries use `_`; ordinary and companion units have dedicated filename-selected roots; `package.fo1` uses `_ co.lang.package = { name: "..."};`; entry files admit parameterized `co.lang.type` constructors; typeclasses use a separate higher-kinded parameter clause; matchers retain one declared subject type and one non-overloaded `matchCase`; companion operator operand rules are separated from ownership; and type-producing function syntax is recorded as a type-level function.
- Revision 26 adds a normative source-filename grammar. Identifier-derived source filename components are ASCII-only and follow the ordinary identifier character shape; reserved suffixes are structural. Filename-backed public declaration names and companion owners are canonicalized to `UpperCamelCase`, with underscores serving as explicit word boundaries when case alone cannot preserve them.

Termination model

FoLang distinguishes **body braces** from **expression braces** by the enclosing production. The brace character alone does not determine termination.

```

emp := Employee{ id: 1, name: "Rao" }; // object construction expression: ; required

classify(n) => {                                // function-pattern body: no ; after }
  this.return "positive";
}

someFArg co.lang.function =                    // inline function-kind body: no ; after }
  (a co.lang.int, b co.lang.int)->(co.lang.int) = {
    this.return a + b;
  }

_ co.lang.struct = {                            // Employee.fol; UDT body: no ; after }
  id co.lang.int;
  name co.lang.string;
}

```

A comma is a soft end inside an enum or other grouped construct. It closes the current item but not the enclosing statement. Built-in directives are self-delimiting and take no semicolon.

Physical nesting and scope model

FoLang distinguishes an independently named declaration from a construct that is merely nested syntactically.

```

outer()->() = {
  value co.lang.int = 10;

  inner()->() = {                                // named local function: permitted
    co.out.println(value);
  }

  operation := (x co.lang.int)->(co.lang.int) = {
    this.return x * 2;                          // anonymous function expression
  };

  worker := co.lang.class {                    // anonymous class expression
    run(x co.lang.int)->(co.lang.int) = {
      this.return operation(x);
    }
  }.init();
}

transformer co.lang.type = forall(T).(T)->(T); // anonymous type expression

```

Independent package-owned classes, structs, cstructs, enums, unions, modules, units, interfaces, signatures, type declarations, instances, matchers, macros, templates, and similar primary declarations cannot be physically nested. Member methods and module/signature type components are members or contract slots rather than independent nested package declarations.

An ordinary local function is physically declared in its enclosing executable block and uses declaration-site lexical scope. `@co.dap.inner` is different: it annotates a separately declared association and executable declarations use call-site lexical-context resolution as defined by [DECISION-SCOPE-002](#).

Package-level function envelope

Ordinary loose functions are forbidden in package source files. The `annotated-function-primary` production exists for annotation-defined primary declaration kinds. Parsing that envelope does not establish legality; semantic analysis must confirm that a resolved annotation explicitly grants primary-declaration status.

Filename-classified package sources

The compiler selects the package-source grammar from the filename before parsing, using the longest recognized suffix first:

```

<Name>.comp.unit.fol  -> companion-unit-source-file
<Fragment>.unit.fol  -> ordinary-unit-source-file
package.fol          -> package-metadata-source-file
<Name>.fol            -> package-primary-source-file

```

A file-backed primary uses `_` in the declaration-name slot; the filename supplies its public name. Filename-derived primary and companion-owner identity use normalized, case-folded stems for duplicate detection and lookup. Ordinary unit filenames are organizational only and create no namespace. Companion ownership comes from the filename and is validated against a same-package `co.lang.struct`.

Every identifier-derived source filename component follows this ASCII-only shape:

```

first character: A-Z or a-z
remaining characters: A-Z, a-z, 0-9, or isolated internal _

```

A component cannot begin or end with `_`, contain `_`, or contain Unicode, whitespace, hyphens, or other punctuation. Recognized suffixes such as `.fol`, `.unit.fol`, and `.comp.unit.fol` are structural and are excluded from component validation. The exact reserved filenames `package.fol` and `operators.fol` retain their context-defined classifications.

Filename-backed public declaration names and companion owners use canonical UpperCamelCase display spelling. The compiler splits at underscores, preserves existing lower-to-upper boundaries in a mixed-case segment, normalizes single-case segments as one word, capitalizes each word, and concatenates the words. Consequently:

```
employee.fol      -> Employee
EMPLOYEE.fol      -> Employee
employee_service.fol -> EmployeeService
EMPLOYEE_SERVICE.fol -> EmployeeService
employeeService.fol -> EmployeeService
EmployeeService.fol -> EmployeeService
EMPLOYEESERVICE.fol -> Employeeservice
```

The final case cannot recover an absent word boundary; a multiword name written wholly in one case must use underscores when its word boundaries must be retained. All case and canonically equivalent forms still use one normalized, case-folded identity key for duplicate detection.

`package.fol` has the exact declaration shape:

```
_ co.lang.package = {
  name: "emp"
};
```

The `name` field replaces only the current logical package segment. In this source form `_` occupies the declaration-name slot but does not derive a declaration named `package`.

Typeclass, matcher, and companion specialization

A file-backed typeclass uses a separate declaration-name placeholder and parameter clause:

```
_ (F(_), G(_)) co.lang.typeclass = {
  ...
}
```

`_` is the filename-derived name slot; `(F(_), G(_))` is the typeclass parameter clause. Arity-bearing parameters describe higher-kinded inputs. Otherwise-unbound type variables in an operation signature are implicitly universally quantified within that operation.

A matcher declares one subject type with `type=` and exactly one protocol function named `matchCase`. The compiler resolves and compares the declared subject type with the first `matchCase` parameter type. The second parameter is `co.lang.untyped`, and the result is exactly `(co.lang.int, co.lang.MatchBindings)`. `matchCase` overloading is forbidden; a different subject type uses a different matcher declaration.

Companion ownership always comes from `<StructName>.comp.unit.fol`. A non-operator receiverless function needs no owner-typed parameter. For an operator function, a value receiver supplies the owner instance; a receiverless or type-receiver function must use the owner struct as its first ordinary parameter so operator lowering can pass the instance.

Operator bootstrap and artifact model

FoLang separates operator registration from operator implementation.

```
language-owned built-in or pre-declared symbol
  -> symbol and parse properties already registered by the language

project-local custom symbol
  -> registered once inside the fixed operators.fol operator library
  -> carries parse and optional optimization metadata

all registered symbols
  -> implementations use mode=overload in a legal function owner
  -> multiple distinct normalized operand signatures are permitted
  -> mode=override and mode=extends are unsupported
```

`operator_library_folder` in `fol-conf.yaml` identifies a source-only bootstrap area excluded from package discovery. Its fixed `operators.fol` is parsed by a dedicated lexer/parser and must have this outer shape:

```
@co.dap.library(type=operator)
_ co.lang.library = {
  // co.lang.operator declarations only
}
```

The operator library is not imported and produces no artifact. Operators do not cross ordinary library boundaries. Both lexers preserve one complete contiguous symbol run and never fall back to shorter operators. Grammar context classifies the whole spelling as structural syntax, contextual metadata, a registered expression operator, or an unrecognized token. A multi-symbol expression operator requires explicit boundaries on every operand-facing side, so a registered `+-` is written `a +- b`; `a + -b` denotes separate operators.

Operator implementations are normal functions and cannot be loose at package scope:

- built-in operand type: `@co.dap.extension` function inside a unit;
- struct: same-package companion unit;
- class: class operator method;

- module, enum, union, interface, signature, and cstruct: unsupported.

A duplicate custom symbol declaration is an error. A duplicate normalized implementation signature is also an error, but one registered custom symbol may have multiple distinct overload signatures. Using one symbol consistently for one concept is recommended for readability and is not compiler-enforced.

Pre-declared mathematical/modifier glyphs are language-owned registrations with no required built-in implementation. They cannot be declared locally, but they can be implemented through `mode=overload` exactly like `+` or `*`.

Decision index

Decision	Status	Normative decision
DECISION-ANN-001	Active	Annotation arguments and annotation-map entries accept <code>=</code> or <code>:</code> as binders. A bare key is a flag. Mixed binders are permitted within one annotation.
DECISION-BACKEND-001	Active	Each resolved user-defined FoLang identifier is lowered to C++ by appending <code>_fo</code> . Built-ins, keywords, and compiler-generated names use separate compiler-defined lowering.
DECISION-BLK-001	Active	A block may end with one unterminated tail expression. That final expression is the block value and is not an expression statement.
DECISION-COL-001	Active	Commas separate enum variants, map entries, annotation-map entries, object initializers, parameters, arguments, and other grouped items. A comma is a soft end inside the enclosing construct; a permitted trailing comma does not terminate the enclosing statement. Object and annotation-map fields use <code>:</code> .
DECISION-DECL-001	Active	<code>primary-declaration</code> is closed to the top-level declaration kinds <code>language-ref.md</code> enumerates: struct, cstruct, class, module, signature, interface, enum, union, typeclass, instance, matcher, annotation/object, package, library and unit. The last three reach the grammar through their own source-file forms. Five alternatives were removed in grammar revision 27: <code>function-object-declaration</code> and <code>delegate-declaration</code> are unit members; <code>named-block-declaration</code> is a statement; <code>annotated-contract-declaration</code> was a pre- <code>co.lang.typeclass</code> spelling superseded by <code>typeclass-declaration</code> ; and <code>general-kind-declaration</code> admitted about twenty-eight further <code>co.lang.*</code> kinds that the reference lists only in its built-in kind table and gives no declaration form. <code>forward-type-declaration</code> is retained as the forward spelling of a struct or class, not an additional kind.
DECISION-DECL-002	Active	A function object and a delegate are unit members, not file-backed primaries. Both take an ordinary identifier rather than <code>_</code> , and both appear in <code>unit-member</code> . <code>language-ref.md</code> presents the function object as Syntax 3 of the function-type feature, beside the named type alias, which has always taken an identifier.
DECISION-DECL-003	Active	A named block is a statement inside a function or method body, not a file-backed primary. <code>language-ref.md</code> states that a block cannot live outside a function or method and that inner blocks for a class, struct, typeclass or module are prohibited. It takes an ordinary identifier and appears in <code>statement</code> .
DECISION-DIR-001	Active	Built-in directives are self-delimiting. Their complete directive form ends the directive; no trailing semicolon is accepted or required.

DECISION-EXT-001	Active	The contextual registered precedence table is built from the language-owned operator registrations plus the current compilation's custom declarations from the configured operator library. Built-in, pre-declared, and custom symbols all receive implementations through <code>mode=overload</code> ; omitted mode defaults to overload. <code>mode=override</code> , <code>mode=extends</code> , <code>mode=define</code> , and other explicit modes are rejected. A custom declaration supplies complete parse metadata. The alpha profile implements infix, prefix, and postfix; other fixities remain reserved until their delimiter/slot grammars are defined.
DECISION-PKG-005	Active	Realms are withdrawn. The import fields <code>realm</code> and <code>parent-realm</code> and the kind <code>co.lang.realm</code> are removed from the grammar. <code>language-ref.md</code> describes no realm concept and this register records no realm decision, so the grammar was accepting a construct no normative document defines. Nothing referenced them.
DECISION-FILE-003	Active	Filename-derived naming does not apply to every primary declaration. Six productions keep an explicit name in the head: <code>surface-struct-declaration</code> and <code>surface-cstruct-declaration</code> , because a library surface file carries several declarations; <code>data-declaration</code> , because the head names the variants; <code>parameterized-type-declaration</code> and <code>entry-parameterized-type-declaration</code> , because a filename cannot carry a generic parameter list such as <code>(T)</code> ; and <code>entry-simple-type-declaration</code> , because the application entry file is not file-backed. Every other primary declaration uses <code>filename-derived-name</code> and must spell <code>_</code> .
DECISION-FILE-004	Active	The canonical file key is <code>asciiCaseFold(stem with underscores removed)</code> . A stem is ASCII by construction, so no Unicode normalization applies and the earlier <code>normalize()</code> step is withdrawn. Folding must be locale-invariant: a locale-tailored fold maps <code>I</code> to <code>i</code> under a Turkish locale and would index the same filename differently on different machines. Dropping underscores before folding makes <code>employee_service.fo1</code> , <code>EmployeeService.fo1</code> and <code>employeeService.fo1</code> share one key and therefore conflict, consistent with all three deriving <code>EmployeeService</code> under <code>DECISION-FILE-002</code> .
DECISION-FUN-001	Active	The <code>=</code> before a function block body is optional. Both <code>f()->T = { ... }</code> and <code>f()->T { ... }</code> are valid.
DECISION-FUN-002	Active	A named closure uses <code>name = (parameters) ==>> expression;</code> . Additional adjacent parameter lists make it curried: <code>name = (first)(second) ==>> expression;</code> .
DECISION-COMP-001	Active	Companion ownership is determined only by <code><StructName>.comp.unit.fo1</code> . Explicit value/type receivers must resolve to the owner. Non-operator receiverless members need no owner-typed parameter. For an operator member, a value receiver supplies the owner operand; a receiverless or type-receiver member must use the owner struct as its first ordinary parameter.
DECISION-ENTRY-001	Active	Application-entry and single-source files admit non-structural type declarations, including parameterized <code>co.lang.type</code> constructors such as <code>Option(T)</code> , but do not admit ordinary functions, type-level functions, file-backed UDT/container primaries, units, instances, or matchers.

DECISION-FILE-001	Active	Package filenames are classified longest suffix first as companion unit, ordinary unit, reserved <code>package.f01</code> , then file-backed primary. A file-backed primary uses <code>_</code> ; its public name comes from the validated filename stem and is canonicalized under <code>DECISION-FILE-002</code> . Canonically equivalent case variants conflict on every filesystem.
DECISION-FILE-002	Active	Every identifier-derived FoLang source filename component is ASCII-only, begins with <code>A-Z</code> or <code>a-z</code> , and thereafter contains only ASCII letters, digits, or isolated internal underscores. Leading, trailing, and consecutive underscores, Unicode, spaces, hyphens, and other punctuation are invalid. Recognized suffixes are structural. Filename-backed public declaration names and companion owners use canonical UpperCamelCase display spelling; underscores are explicit word boundaries, existing lower-to-upper boundaries are preserved, and a wholly single-case multiword spelling must use underscores to preserve its boundaries.
DECISION-GEN-001	Active	Direct declaration-head parameters are restricted to parameterized <code>co.lang.type</code> / <code>co.lang.data</code> declarations, signature type components, and typeclass parameter clauses. Structs, classes, functions, and methods use <code>@co.dap.generic</code> . Typeclass parameters may declare arity, such as <code>F(_)</code> or <code>G(_, _)</code> ; an arity slot is <code>_</code> or a named placeholder. Entry files admit parameterized <code>co.lang.type</code> constructors but not <code>@co.dap.generic</code> behavioral declarations.
DECISION-KIND-001	Active, revised in revision 30	A <code>co.lang.*</code> name that the built-in kind table lists but that no production admits as a declaration is rejected as an unsupported kind. It is not treated as an identifier and not absorbed by a variable declaration. This covers reserved future names such as <code>co.lang.typeconstructor</code> and <code>co.lang.typefunction</code> . Grammar revision 27 removed <code>general-kind-declaration</code> and <code>general-declarable-kind</code> under <code>DECISION-DECL-001</code> , so no general fallback remains and rejection is now the only behaviour; the earlier wording naming those two productions is superseded. Dedicated declarations and ordinary variable declarations retain priority. The reserved-name diagnostic continues to cite this ID.
DECISION-LEX-001	Active	Source is UTF-8, but ordinary identifiers use ASCII letters, digits, and isolated internal underscores. An identifier begins with an ASCII letter, cannot contain consecutive underscores, cannot end in an underscore, and has no minimum-length requirement. Lone <code>_</code> is contextual, not an identifier.
DECISION-LEX-002	Active	FoLang supports <code>//</code> line comments and non-nesting <code>/* ... */</code> block comments. Line breaks are whitespace outside literals.
DECISION-LEX-003	Active	After comments, literals, and closed scanner-known composite spellings such as <code>@@new</code> are recognized, the lexer consumes each remaining complete maximal contiguous run of symbol characters as one candidate. The whole run is classified by grammar context as a fixed structural spelling, contextual metadata spelling, registered expression operator, or unrecognized symbolic token. It is never split into shorter operators as a fallback. Comment openers are recognized before symbolic-run scanning.

DECISION-LEX-010	Active	A registered expression operator containing more than one symbol character requires an explicit boundary on every operand-facing side. Whitespace, comments, and applicable delimiters supply boundaries, and boundary presence is checked before separators are discarded. Infix operators require both sides, prefix operators the operand side after the symbol, and postfix operators the operand side before it. Structural and metadata spellings are exempt unless parsed as expression operators.
DECISION-LEX-005	Withdrawn in revision 9	The special numeric dot-scanning rule was removed after abbreviated floating forms such as <code>1.</code> and <code>.10</code> were rejected. Numeric-literal recognition and the current whole-symbol-run classifier now distinguish decimal points, range spellings, and member access without that rule.
DECISION-LEX-006	Active	After recognizing an identifier, the scanner verifies that the next character is not <code>_</code> . This converts trailing or doubled underscores into lexical errors instead of silently splitting them into multiple tokens.
DECISION-LEX-007	Withdrawn in revision 11	The apostrophe digit-separator adjacency rule was removed because FoLang no longer supports numeric digit separators.
DECISION-LEX-008	Active	Adjacent encoding prefixes, raw-string introducers, and backslashes in quoted literals begin reserved post-alpha spellings. The scanner consumes the complete spelling and reports an unsupported feature; separated names remain identifiers.
DECISION-LEX-009	Active	A special method is one complete spelling from a closed scanner-known set. <code>@@</code> is not a prefix operator over an arbitrary identifier: spellings such as <code>@@new</code> and <code>@@init</code> are classified as whole tokens, and any unrecognized <code>@@</code> spelling is a lexical error that reports the admissible set.
DECISION-LIT-000	Active	FoLang accepts a selected C++-compatible subset of numeric, character, and string literal spellings. The frontend preserves their complete raw lexemes, and a C++ backend may emit those lexemes unchanged. <code>co.const.true</code> , <code>co.const.false</code> , and <code>co.const.none</code> use backend-defined lowering. <code>nullptr</code> is not introduced.
DECISION-LIT-001	Active	Integer literals support C++-compatible binary, leading-zero octal, decimal, hexadecimal, and standard integer suffix forms. Numeric digit separators are not supported.
DECISION-LIT-002	Active	Floating literals support selected C++-compatible decimal and hexadecimal forms, exponents, and configured suffixes. Digit separators are not supported, and a decimal point requires a digit on both sides.
DECISION-LIT-003	Active	The alpha release accepts only unprefixed character and string literals without escapes. Prefixes, escapes, universal character names, and raw strings are reserved and rejected with unsupported-feature diagnostics.
DECISION-LIT-004	Withdrawn in revision 7	FoLang has no separate user-defined-literal token or C++ <code>operator""</code> mechanism. A value of a user-defined type is created through ordinary object construction.
DECISION-LIT-005	Active	Boolean literals are <code>co.const.true</code> and <code>co.const.false</code> ; none/null is <code>co.const.none</code> . Bare <code>true</code> , <code>false</code> , and <code>none</code> are not FoLang literals.

DECISION-LIT-006	Active	A floating literal has at least one digit on each side of its decimal point. Write <code>1.0</code> , <code>0.10</code> , and <code>0x1.8p3</code> ; forms such as <code>1.</code> , <code>.10</code> , <code>0x1.p3</code> , and <code>0x.8p3</code> are rejected.
DECISION-LIT-007	Active	FoLang does not adopt C++14 apostrophe digit separators or underscore digit separators. Numeric digit sequences contain digits only.
DECISION-OP-001	Active	Built-in operators use the precedence table encoded in the grammar. Runtime assignment has the lowest built-in precedence.
DECISION-OP-002	Active	Runtime assignments are right-associative, so <code>a = b = c</code> groups as <code>a = (b = c)</code> . Assignment expressions yield the assigned value; FoLang evaluation-order rules remain separate.
DECISION-OP-003	Active	<code>:=</code> and <code>?=</code> are statement-level definition operators, not general expression operators, and cannot be chained. Each requires an explicit boundary on both the name-facing and value-facing side. <code>:=</code> remains reserved.
DECISION-OP-004	Withdrawn in revision 24	Earlier drafts recognized <code>++</code> and <code>--</code> as built-in prefix and postfix operators. They are removed from the operator inventory, precedence table, and grammar. There is no special rejection rule: an unregistered contiguous <code>++</code> or <code>--</code> spelling is rejected by the general whole-symbol-run rule in DECISION-LEX-003, while separate operators must be written with a boundary, such as <code>+ +a</code> or <code>+(+a)</code> .
DECISION-OP-005	Active	<code>::=</code> , <code>->></code> , <code><-></code> , backtick, backslash, <code>#</code> , and comment openers remain hard-reserved and cannot be declared or overloaded. The documented mathematical/modifier glyph set is not in this rejected category: those glyphs are pre-declared language-owned operators and may receive <code>mode=overload</code> implementations.
DECISION-OP-006	Active	<code>~</code> , <code>#</code> , and <code>^</code> are reserved prefix spellings and are rejected in operand-prefix position. <code>#</code> has no semantics and cannot be used, defined, overloaded, or overridden. <code>^</code> remains the built-in infix XOR symbol and <code>~</code> retains its named-parameter and <code>-></code> (<code>~</code>) roles. <code>@</code> is not a prefix expression operator.
DECISION-OP-007	Active	A constant expression may use a registered custom operator at any declared precedence, but runtime assignment is recursively forbidden throughout the complete constant-expression subtree. Grouping, call arguments, collection/object elements, and other nested expression forms cannot hide an assignment from this guard. Constant evaluation and custom-operator foldability remain semantic checks after parsing.
DECISION-OPDECL-005	Active	An operator body is a property list, not a declaration body. Its braces are map-shaped in the same way as <code>package-alias-body</code> , so under DECISION-SYN-006 the enclosing declaration is terminated by <code>;</code> and <code>body-closure-guard</code> does not apply. <code>operator-body</code> lost the guard and <code>operator-declaration</code> gained <code>statement-end</code> , matching the examples in <code>language-ref.md</code> .

DECISION-OPDECL-006	Active	Operator property values are qualified constants under <code>co.operator.*</code> , consistent with <code>co.const.true</code> and <code>co.const.none</code> appearing in the same property list. The bare spellings <code>infix</code> , <code>left</code> and <code>binary</code> are withdrawn, so no contextual keyword is introduced and none can collide with an ordinary identifier. <code>co.operator</code> is a built-in package supplying <code>co.operator.fixity.*</code> , <code>co.operator.associativity.*</code> and <code>co.operator.arity.*</code> .
DECISION-OPLIB-001	Active	Operator <code>mode=overload</code> is supported for every registered symbol—built-in, pre-declared, or project-local custom. Because operator implementations are normal functions, they cannot be declared loose at package scope: built-in operands use an <code>@co.dap.extension</code> function inside a unit, structs use their same-package companion unit, and classes use class operator methods. Receiver normalization determines the complete operand signature and its count must match the registered arity. Equivalent normalized signatures are duplicates; distinct signatures are overloads. Modules, enums, unions, interfaces, signatures, and cstructs cannot own operator implementations. Operator <code>mode=override</code> and <code>mode=extends</code> are unsupported; ordinary class-method overriding through <code>@co.dap.override</code> remains separate.
DECISION-OPLIB-002	Withdrawn in revision 17	The separately distributable <code>operator</code> library-kind model was removed. <code>type=operator</code> remains only as the source marker for the configured project-local bootstrap surface and never produces an independently imported artifact.
DECISION-OPLIB-003	Active	A project-local custom operator symbol has exactly one registration in the compilation's fixed operator library. Duplicate declarations, aliases, merges, selection, and remapping are rejected. The registered symbol may have zero or more implementation overloads in legal owners; each distinct normalized operand signature is permitted and an equivalent signature is a duplicate error. Language-owned built-in and pre-declared symbols cannot be registered locally and are implemented through the same overload mechanism.
DECISION-OPLIB-004	Withdrawn in revision 17	Operator-specific imports and activation were removed. Historical operator-specific/imported metadata activation was removed; ordinary imports do not supply operator metadata.
DECISION-OPLIB-005	Withdrawn in revision 17	Historical operator manifest/export models were removed; no operator table is embedded in an ordinary library artifact.
DECISION-OPLIB-006	Withdrawn in revision 17	Historical imported operator-table bootstrap was removed. Only language-owned registrations and the current compilation's local operator library build the contextual operator table.
DECISION-OPBOOT-001	Active	<code>operator_library_folder</code> in <code>fol-conf.yaml</code> identifies the project-local operator source area. A relative path is resolved from the project root. The compiler checks the fixed file <code><operator_library_folder>/operators.fol</code> ; absent configuration, folder, or file means no local new operators. The area is excluded from ordinary package discovery.

DECISION-OPBOOT-002	Active	The configured <code>operators.fol</code> is parsed first by a dedicated operator-source lexer and parser. It must contain exactly <code>@co.dap.library(type=operator)</code> followed by <code>co.lang.library = { ... }</code> , whose body contains only <code>co.lang.operator</code> declarations. It is a source-only bootstrap surface compiled into its owning project, produces no independent artifact, and is not ordinary FoLang/package source.
DECISION-OPART-001	Withdrawn in revision 22	Library operator export was removed. An operator implementation is bound to a class, a companion unit, or an extension unit, and <code>library-member</code> admits only import directives, struct declarations, cstruct declarations, and function declarations. No production can therefore place an operator in a library surface file, and no operator table is emitted into a <code>.folib</code> or <code>.folenc</code> artifact.
DECISION-OPART-002	Withdrawn in revision 22	Operator-table loading on import was removed. A direct library import loads the projected symbol table only. An importer that wants operator notation for an imported boundary struct declares its own symbol in its own operator source area and implements it in an extension unit.
DECISION-OPART-003	Withdrawn in revision 22	The exported operator entry schema was removed with <code>DECISION-OPART-001</code> . Nothing operator-related is serialized into a library artifact.
DECISION-OPBOOT-003	Active	The frontend reads configuration, parses and validates the local operator library, combines its custom registrations with the language-owned built-in and pre-declared registrations, and builds immutable symbol/fixity/precedence tables before ordinary source tokenization. Imports contribute no operator metadata. Both lexers preserve a complete contiguous symbol run and perform exact whole-run classification without fallback splitting. The parse is single-pass and import-order independent.
DECISION-OPDECL-001	Active	A new custom symbol is registered only by <code>SYMBOL co.lang.operator = { ... }</code> inside the fixed <code>co.lang.library</code> operator-source body. The declaration carries required <code>fixity</code> , <code>precedence</code> , <code>associativity</code> , and <code>arity</code> , plus the optional <code>commutative</code> , <code>idempotent</code> , <code>identity</code> , <code>foldable</code> , <code>vectorizable</code> , <code>distributes_over</code> , and <code>desugar</code> metadata. Required properties occur exactly once; optional properties occur at most once; duplicates, omissions, unknown keys, invalid types, and invalid ranges are errors. <code>@co.dap.operator</code> at implementation sites carries only <code>symbol</code> and optional <code>mode</code> .
DECISION-OPDECL-002	Active	The documented mathematical/modifier glyph set is pre-declared by the language with symbol, fixity, precedence, associativity, and arity but no required implementation. These glyphs are language-owned registrations: they parse everywhere, cannot be redeclared in the local operator library, and may receive any number of distinct <code>mode=overload</code> implementation signatures under <code>DECISION-OPLIB-001</code> . A missing matching implementation is a resolution error.
DECISION-OPDECL-003	Active	Operator symbols are global to a compilation while implementation availability is resolved by owner, scope, activation, operand types, and normalized signature. The tokenizer recognizes each registered symbol throughout the compilation; an unavailable implementation therefore produces a resolution diagnostic rather than a syntax error. Using one symbol for one recognizable concept across overloads is recommended for readability but is not a compiler-enforced rule.

DECISION-OPDECL-004	Active	The operator source area is parsed by a separate grammar rooted at <code>operator-source-file</code>. The exact outer declaration is <code>@co.dap.library(type=operator) _</code> <code>co.lang.library = { ... }</code>, and the body admits only operator registrations. <code>co.lang.operator</code> is absent from ordinary declaration kinds. The dedicated and ordinary lexers preserve complete contiguous symbol runs; a custom symbol is recognized only by an exact whole-run match and an unknown run is never split into shorter operators. Symbols containing <code>//</code> or <code>/*</code> are rejected because a comment opener terminates the preceding symbolic run and takes lexical priority.
DECISION-SCOPE-001	Active	An ordinary local/inner function has block-local identity and resolves free runtime names from its lexical declaration context. Calling it does not replace that environment with the caller's runtime scope.
DECISION-SCOPE-002	Active	<code>@co.dap.inner</code> is an association annotation on a separately declared declaration, not physical nesting. Executable inner-associated declarations resolve free runtime names through the lexical scope chain of the active attachment/call site, without unrestricted runtime caller-chain lookup. Compile-time names remain statically resolved.
DECISION-SEM-001	Active	Typeclass instances are selected explicitly by name; FoLang performs no implicit instance search. An instance must be declared in the exact package defining either the typeclass or the represented type. Selection and placement are semantic name-resolution checks.
DECISION-SEM-002	Active	<code>@co.ddap.use</code> explicitly and block-scopingly activates extension-unit or typeclass-instance functions as methods. Receiver-owned declarations take priority, followed by activated extensions and then activated instances. A method name may be activated at most once per receiver type in one scope. Parser <code>CallKind</code> values remain provisional; any early control-chain lowering must preserve the original member-call chain so resolution can select an overriding user declaration without reconstructing lost syntax. Contextual lambda/wildcard admission requires a receiver-qualified method and is unchanged by transparent grouping of that member callee.
DECISION-MATCH-001	Active	A <code>co.lang.matcher</code> declares exactly one subject type with <code>type=</code> and exactly one protocol function named <code>matchCase</code>; overloading <code>matchCase</code> is forbidden. After alias/type resolution, the first parameter type equals <code>type=</code>, the second parameter is <code>co.lang.untyped</code>, and the result is exactly <code>(co.lang.int, co.lang.MatchBindings)</code>. Other subject types require separate matcher declarations.
DECISION-PKG-001	Active	The reserved <code>package.fo1</code> source form contains <code>_</code> <code>co.lang.package = { name: "segment" };</code>. At most one exists per package folder. Its required string <code>name</code> field changes only the logical leaf package segment; <code>_</code> does not derive the literal filename stem <code>package</code>.
DECISION-TCLASS-001	Active	A file-backed typeclass uses <code>_</code> (parameters) <code>co.lang.typeclass;</code> <code>_</code> and the parenthesized parameter clause are distinct grammar components. Parameters may carry arity such as <code>F(_)</code>. Otherwise-unbound type variables in each operation signature are implicitly universally quantified within that operation.

DECISION-UNIT-001	Active	Ordinary <code><Fragment>.unit.fo1</code> and companion <code><StructName>.comp.unit.fo1</code> files both contain <code>_ co.lang.unit</code> . Ordinary unit members merge directly into the package namespace and the fragment creates no symbol. Companion members attach to the same-package struct derived from the filename; only structs can own companions.
DECISION-SYN-001	Active	Every simple statement whose production uses <code>statement-end</code> requires <code>;</code> . Newlines never terminate statements and FoLang performs no semicolon insertion. Built-in directives are exceptions because they are self-delimiting.
DECISION-SYN-002	Active	Comma-separated variable declarators form one declaration statement and share one final semicolon.
DECISION-SYN-003	Active	The sole named local-function syntax requires a return-type clause and a block body. This preserves <code>foo()</code> as an expression statement. The local function has block-local identity and declaration-site lexical scope.
DECISION-SYN-004	Active	Annotations may prefix an expression statement.
DECISION-SYN-005	Active	A standalone block is a statement and takes no trailing semicolon.
DECISION-SYN-006	Active	Termination depends on syntactic role, not merely on the final character. <code>;</code> ends simple, expression-bodied, type-bodied, and forward forms. A body-selected <code>}</code> ends UDT/container bodies, function bodies, function-pattern bodies, and standalone block statements. A brace that closes object construction, a map, an anonymous class, or another braced expression does not end the enclosing statement.
DECISION-SYN-007	Active	Body-versus-expression selection is explicit. Direct body branches use <code>body-closure-guard</code> , which rejects an immediately following semicolon. Competing expression branches use <code>non-block-expression</code> or <code>non-anonymous-function-expression</code> , preventing a body from being reparsed as an expression plus <code>;</code> . Grouped, postfix, or otherwise composed braced forms remain expressions.
DECISION-SYN-008	Active	Independent named type and container declarations cannot be physically nested. Ordinary named local functions are the explicit named exception. Anonymous functions, lambdas/callback blocks, anonymous classes, ordinary value expressions, and <code>forall</code> type expressions may be nested wherever their expression/type-expression grammar permits and create no package-level declaration identity.
DECISION-SYN-009	Active	<code>annotated-function-primary</code> is only a syntactic envelope for annotation-defined primary declaration kinds. An arbitrary annotation does not legalize a loose ordinary function at package-file scope; that legality is checked semantically after annotation resolution.
DECISION-TYP-001	Active	Every type derivation may carry a trailing attribute list, not only pointer derivations.

DECISION-TYP-002	Active	A type-level function has exactly one type-producing result. Its result may be one of <code>co.lang.dependentType</code> , <code>co.lang.type</code> , <code>co.lang.typeType</code> , <code>co.lang.typeKind</code> , or <code>co.lang.kind</code> , or one union joined by <code> </code> ; commas and named/multiple result items are rejected. Its body may bind a type expression, with the type-expression reading taking priority. <code>co.lang.dependentType</code> , <code>co.lang.typeType</code> , and <code>co.lang.typeKind</code> are also direct <code>type-declaration-kind</code> alternatives and are excluded from general-kind routing.
DECISION-TYP-003	Active	An array dimension may be elided in any position, including <code>->{[]}</code> and <code>->{[,]}</code> .
DECISION-TYP-004	Active	A dependent-type argument and an array dimension are index positions, not general expressions. An index is a non-negative integer literal or a name resolving to an in-scope parameter or <code>@co.dap.const</code> compile-time constant. Arithmetic, calls, indexing, and other operators are rejected.
DECISION-TYP-005	Active	Dependent types are equal when their constructors match and indices are pairwise equal. Literal and substituted <code>@co.dap.const</code> indices compare by value; parameter indices compare by declaration identity. Equality is not decided modulo arithmetic.
DECISION-TYP-006	Active	Dependent types are checked against written signatures and are never inferred. FoLang does not infer index values or perform whole-program dependent-type inference.

Current lexical examples

```
Valid identifiers:  a, x, id, name, myVar2, v1_hr, a_b_c
Invalid identifiers: _x, _1, a_, a__b
Contextual token:  _

Valid numbers:      1000, 0b11110000, 0xFFFF0000, 3.141592
Invalid numbers:    1_000, 1'000, 1., .10
```

Planned syntax retention

Package aliasing, comprehensions, and other planned constructs remain in the complete grammar. Their availability in alpha, 0.x, 1.0, or later compiler profiles is a version-conformance decision, not a reason to delete their productions from the complete grammar.